Tools to Design a Biomimetic Humanoid Robot Actuated by Pneumatic Artificial Muscles and a Synthetic Neural Network

by

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Abstract

Biomimetic robots offer a platform for studying neuromechanical control through experiments that would be impractical or unethical with living subjects. This dissertation advances a long-running effort at Portland State University to build a humanoid robot, from the trunk down, actuated by pneumatic artificial muscles and controlled by a synthetic nervous system. The isometric force of Festo braided pneumatic actuators (BPAs) was characterized over a wide range of resting lengths, contractions, and pressures using a custom test jig. The analysis produced novel equations for the maximum isometric force as a function of resting length and a three-coefficient nondimensional force surface that predicts isometric force from relative strain and relative pressure. The actuator model was carried to the joint level through isometric knee torque experiments on two test stands: a simplified pinned knee and a biomimetic knee with a migrating instantaneous center of rotation. This approach isolated the sources of prediction error, revealing constant length offsets, series compliance in brackets and artificial tendons, and a loss of usable actuator length when BPAs wrap around the joint. The biomimetic knee driven by optimized BPA placements was shown to meet or exceed human isometric knee torque over most of the range of motion. Tools were created for curating relevant neuromechanical research, as well as simulating neuromechanical models based on human and robot biomechanical models.

Dedication

*

Darian

azīzam, jānam,

nūr-e cheshmam — the light of my eyes

qorbānat beram

Bābā kheyli dūset dāre

↘ ↘ ↘

"Los corazones obreros que laten

por nuestra causa, felices serán.

Si entusiasmados y unidos combaten,

de la victoria, la palma obtendrán."

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Finally, I thank my family for their support throughout this long process.

Contents

Abstract **ii**

Dedication **iii**

Acknowledgments **iv**

1 Introduction **1**

1.1 Motivation 1

1.2 The Biomimetic Humanoid Robot Project 4

1.3 Research Objectives 5

1.4 Dissertation Organization 6

2 Background **11**

2.1 Biomimetic and Musculoskeletal Robots 11

2.2 Robot Actuation Technologies 12

2.2.1 Overview 12

2.2.2 Braided Pneumatic Actuators 15

2.2.3 BPA Force–Length–Pressure Models 18

2.3 Human Lower-Limb Biomechanics 20

2.4 Biomimetic Knee and Bipedal Robot Model 23
2.5 Neural Control of Locomotion: Central Pattern Generators 26
2.6 Sensory Feedback in Locomotion 29
2.7 Computational Neuroscience of Locomotor Circuits 32
2.8 Synthetic Nervous Systems 38
2.9 Neuromechanical Simulation 39
2.9.1 Simulation Tooling 41
2.10 Testing Neuromechanical Hypotheses on Robots: The Ethical Case 43

3 Isometric Force Characterization of Braided Pneumatic Actuators 47
3.1 Methods 48
3.1.1 Actuator Force Model 48
3.1.2 BPA Isometric Force Characterization Experiment 49
3.2 Results 51
3.2.1 BPA Force Characterization 51
3.3 Discussion 60
3.3.1 Effects of Resting Length 60
3.3.2 Nondimensionalized Force Prediction 63
3.3.3 Comparison with Other Published BPA Models 64
3.3.4 Implications for Real-Time Control and Simulation 68
3.3.5 Biomimetic Artificial Muscle Analogy 69

4 Isometric Knee Torque 72
4.1 Methods 73

4.1.1 Robot Leg Model and Knee Test Stands 73
4.1.2 Knee Torque Data Collection and Equipment 77
4.1.3 Hybrid Torque Calculation Method 78
4.1.4 Improved Torque Model 79
4.1.5 Optimization Algorithms and Cost Functions 95
4.2 Results 99
4.2.1 Flexor on Pin Joint 100
4.2.2 Extensor on Pin Joint 105
4.2.3 Extensor on Biomimetic Knee 107
4.2.4 Flexor on Biomimetic Knee 108
4.3 Discussion 110
4.3.1 Sources of Error in Isometric Torque Generation 110
4.3.2 Follow-Up Identification and Route Redesign 114
4.3.3 Mono- and Bi-Pulley Tendon Routing 122

5 The Sensory Afferent Database 125
5.1 A Structured Afferent Resource 125
5.2 Audited Corpus and Curation Coverage 130
5.3 AI-Assisted Literature Triage 131
5.4 Toward Robot and Simulation Use 134

6 Synthetic Nervous System Simulation 139
6.1 The AnimatLab Bipedal Walker 140
6.2 Methods: A Neuromechanical Simulation Toolchain 142

6.2.1 Existing Tools and Data Resources 142
6.2.2 New Tools 144
6.2.3 AnimatLab Body and Controller Organization 145
6.2.4 MuJoCo Conversion and Synthetic BPA Knee 146
6.2.5 Simulink SNS Library and Reflex Topology 152
6.2.6 Controller and Body Ports into Simulink and Simscape Multibody . 157
6.2.7 Cross-Platform Verification of the Contact-Driven Rhythm Core . . . 158
6.2.8 Spiking-Neuron Mirrors of the Simulation Models 159
6.2.9 Tuning, Robustness, and Reference Data 162
6.3 Results: Tool Demonstrations and Experiments 164
6.3.1 AnimatLab Phase-1 Recordings 164
6.3.2 Simulink Library Demonstrations and Cross-Program Ports 165
6.3.3 Cross-Engine Verification of the Contact-Driven Rhythm Core 168
6.3.4 Headless Validation of the AnimatLab Reference Models 168
6.3.5 Gait-Validation Sweep against the Locomotion Reference Library . . 170
6.3.6 Stimulus and Deletion Experiments on the Walker Networks 172
6.4 Spiking-Neuron Mirror Results 175
6.4.1 MuJoCo Spinal-Network Mirror 176
6.4.2 Simulink Spiking Demonstration Twins 177
6.4.3 AnimatLab Spiking Conversions 183
6.5 Discussion 185
6.5.1 What Spiking Changes and Does Not 185
6.5.2 The Five-Stage Training Curriculum and Its Scoring 188

6.5.3 Integration of the BPA Force Model 190
6.5.4 The BPA-Actuated Robot in MuJoCo 190
6.5.5 Controller Comparison on the Robot Models and the Walking Status 192
6.5.6 The New Tools and the Program They Enable 193

7 Discussion **195**
7.1 The Work as a Whole 195
7.2 Future Work 200
7.2.1 Expanding the BPA Force Characterization 200
7.2.2 Expanding Attachment and Routing Design 201
7.2.3 Expanding the Sensory Afferent Database 203
7.2.4 Closing the Loop in Simulation 204
7.2.5 Extending the Sensorium 206
7.2.6 Real-Time Hardware Realization 208
7.2.7 Toward Dynamic Whole-Body Maneuvers 210

8 Conclusion **211**

Bibliography **214**

Appendices **232**
A Appendix A: Muscle Force and Joint Torque Calculations 233
A.1 Human muscle force calculations 233
A.1.1 OpenSim Thelen2003 and Millard2012 Formulations 233
A.1.2 OpenSim Moment Arms and Human Torque Targets 234

A.2 Muscle Model Formulations and Cross-Platform Force Comparison 235
A.2.1 MuJoCo Muscle Actuator (as converted) 235
A.2.2 Cross-Platform Force Comparison 235
A.3 BPA Force Model Implementation 237
A.3.1 Goodness-of-Fit Metrics 237
A.3.2 Strain Measures 239
A.3.3 Normalized Force Surface 239
A.3.4 Maximum Force 240
A.4 Muscle Path Length and Moment Arm 240
A.5 Compliance Model Implementation 241
A.6 Wrapping-Loss Implementation 243
B Appendix B: Neuron Model Equations 244
B.1 Spiking Neuron Model 244
B.2 Non-Spiking (Graded) Chemical Synapses 245
B.3 Relation to the SNS-Toolbox Implementation 246
B.4 Spiking-Mirror Implementation, Calibration, and Verification 247
B.4.1 Adapting Integrate-and-Fire Cells and Hybrid Synapses 247
B.4.2 Calibration of Record 249
B.4.3 Verification Gates of the MuJoCo Mirror 250
B.4.4 AnimatLab Conversion Variants 250
B.5 Connectomes of the Spinal-Network Models 255
C Appendix C: Model Geometry, Bracket Frames, and Compliance Parameters . 259
C.1 Static BPA Pathway Data 259

C.2 Joint Transformation Matrices 261
C.3 Bracket Reference Points and Frames 264
C.4 Stiffness Arrays and Compliance 266
C.5 Adopted Correction-Term Values 268
C.6 Redesigned Extensor Route Geometry 270

List of Tables

2.1 Coefficients of the 10 mm Festo BPA pressure model of Equation (2.1), corrected from Hunt et al. [2017a]. 20

3.1 Maximum force equation coefficient values with confidence intervals (CI) and goodness-of-fit measures. Equation (3.5) is compared to data taken at 620 kPa. Equations (3.7) and (3.8) are compared against maximum force from using the Festo tool. It was not necessary to fit an adjusted R^2 in these cases. 56
3.2 Normalized isometric BPA force equation (Equation (3.4)) coefficient values and goodness-of-fit measures. 56
3.3 Goodness-of-fit comparison for the Bolen, Sárosi, and Martens and Boblan models at different resting lengths (in mm) for 10 mm and 20 mm BPAs. RMSE and maximum error are reported in newtons, and FVU denotes the fraction of variance unexplained. 65

4.1 Pre-optimization pinned-knee flexor torque-prediction goodness of fit per test, comparing the three pre-optimization torque predictions of Figure 4.6: the rigid-body prediction, the unoptimized prediction of the improved BPA characterization, and the back-calculated (hybrid) torque. The rigid-body prediction is undefined near full extension, where its modeled strain falls below zero, so the rigid-body and improved-characterization columns are scored head-to-head on the measured points of each test at which the rigid-body prediction is defined (75 of 90, 17 of 20, 14 of 19, and 15 of 22 points, in panel order); the back-calculated torque is scored on all measured points. RMSE and maximum absolute residual (Max.) are in $\mathrm{N \cdot m}$, and FVU denotes the fraction of variance unexplained. Training and validation follow the adopted cross-validation fold. 102

4.2 Pinned-knee flexor torque-prediction goodness of fit per test. The zero-correction prediction uses the improved BPA force characterization alone (rigid attachments, no length offset, and no wrapping loss); the corrected model adds the adopted correction terms of the two-stage identification (Section 4.1.4). RMSE and maximum absolute residual (Max.) are in $\mathrm{N \cdot m}$. In the adopted cross-validation fold the $l_0 = 48.5\,\mathrm{cm}$ and $41.0\,\mathrm{cm}$ tests formed the training set and the $l_0 = 45.7\,\mathrm{cm}$ and $40.6\,\mathrm{cm}$ tendon tests were held out for validation; the $l_0 = 47.9\,\mathrm{cm}$ test was not used in the identification and is reported in Appendix C. 103

4.3 Pinned-knee extensor torque-prediction goodness of fit per test, at the adopted correction terms (stiffness pair locked to the flexor-identified values). The zero-correction prediction uses the improved BPA force characterization alone; the corrected model adds the extensor-specific terms (χ_0, χ_3). RMSE and maximum absolute residual (Max.) are in $\mathrm{N \cdot m}$. The extensor terms were trained on the $l_0 = 43.6\,\mathrm{cm}$, 43.2 cm tendon, and 45.7 cm tests and validated on the $l_0 = 40.0\,\mathrm{cm}$, 40.0 cm tendon, and 46.5 cm tests; the three remaining extensor tests were not used (Appendix C). 107

4.4 Goodness of fit for the improved torque model across test configurations. RMSE and maximum absolute residual are in $\mathrm{N \cdot m}$. 112

4.5 Adopted correction terms of the two-stage identification. The extensor refit locked the effective-stiffness pair (χ_1, χ_2) to the flexor-identified values, so a single stiffness pair is shared across the flexor and extensor configurations, and only χ_0 and χ_3 were re-identified for the extensor. The pair was adopted from a family of near-equivalent fits by a requirement external to the identification, namely that both route-redesign configurations meet the required 5 % torque-overshoot target simultaneously. The stiffness terms are effective system stiffnesses that lump the compliance of the brackets, fixtures, and winch cable; χ_3 is unitless and scales the fictive-strain wrapping loss. 114

4.6 Bi-pulley batch on five muscle sites of the robot-body model's OpenSim leg. Objective 101 is the batch's infeasibility sentinel; all five runs closed with a nonpositive torque margin. The soleus and tibialis-anterior runs converged feasibly (pattern-search exit flag 1), and their objectives are the normalized torque shortfalls; the remaining three runs exited with flag −2. 121

5.1 Afferent classes represented in the Sensory Afferent Database, the signal each class carries, and the pathway roles the corpus records for it. Anchor references are corpus papers linked to the corresponding records; the signal summary for the Ia, II, and Ib rows follows the ensemble afferent recordings of Prochazka and Gorassini [1998]. 136

5.2 Principal feedback-pathway records of the database's Feedback table, named by the fixed grammar of afferent, action, and phase. Linked papers are counted over the full audited corpus of 943 papers; for every pathway listed, the count is unchanged under the retained corpus of 561. Demonstrating references are the corpus papers linked to each record, as curated in the database. 137

6.1 Gait-validation sweep of the four production walking configurations against the 43-cycle locomotion reference library. The score is the production pipeline's cycle-shape kinematic score (0 denotes perfect agreement); the tuning-reference score is each configuration's evaluation on the tuning pipeline's walking reference, and the library mean averages the same score over all 43 references (among genuine gait cycles, higher scores indicate closer agreement). 171

6.2 Pathway-removal sweep on the tuning-reference evaluation. The score is the production pipeline's cycle-shape kinematic score (0 denotes perfect agreement; -25 is the fixed no-cycle fallback, a failed evaluation that scores above every genuine gait in this table). The full-model rows repeat each configuration's unmodified production evaluation. 174

6.3 Untuned spiking mirror versus its non-spiking twin at the same winner parameters, ground evaluation (16 s). The cycle knee minimum and stance fractions are computed over detected gait cycles; the kinematic score is the cycle-shape error metric of the production pipeline (0 denotes perfect agreement; higher scores indicate closer agreement). The spiking mirror carries mapped parameters only, with zero re-tuning. 180

6.4 Knee-reflex demonstration, committed baseline versus spiking twin at the matched operating point (5 s runs; settle statistics over the last 2 s). 181

6.5 Beverage-holding demonstration, committed baseline versus spiking twin (10 s pour; sag measured after 2 s). 181

6.6 AnimatLab spiking conversions versus their graded baselines (5.1 s headless runs; joint ranges in degrees where the rhythm survives). The thresholds-only ablation converts the neurons but keeps the graded synapses. 184

List of Figures

1.1 Dissertation roadmap 10

2.1 Kengoro human-mimetic musculoskeletal humanoid 13
2.2 Representative robotic actuator technologies 16
2.3 Festo braided pneumatic actuator assembly 17
2.4 Overview of the BPA force-characterization study 21
2.5 Steele biomimetic knee mechanism 24
2.6 Laboratory biped skeleton and OpenSim muscle model 25
2.7 Master's-study routing models and moment-arm construction 27
2.8 Biceps femoris long-head torque comparison from the master's study 28
2.9 Locomotor circuit architecture synthesized from the literature 30
2.10 Deng two-layer CPG and sensory–motor network 32
2.11 Di Russo spinal network with afferent feedback 33
2.12 Shinohara bilateral neuromusculoskeletal model 34
2.13 Rybak two-level locomotor CPG concept 35

2.14 Shevtsova extended CPG model with two-joint-muscle circuits 36
2.15 Decerebrate cat treadmill experiment, restored footage frame 44

3.1 A picture of a BPA in the isometric force test stand with components labeled. 52
3.2 Results for finding the relationship between F_{620} and l_0. (**a**) $F_{620_{10}}$ as a function of l_0, at P_{620}. Dashed line is the $F_{620_{10}}$ data from Festo. Solid line is the fit from Equation (3.7). (**b**) $F_{620_{20}}$ as a function of l_0 at P_{620}. Dashed line is $F_{620_{20}}$ data from Festo. Solid line is the fit from Equation (3.8). 54
3.3 Results for finding the relationship between F_{620}, l_0, P_{620}, and $\epsilon_{620_{10}}$. (**a**) Isoclines of the surface fit for $F_{620_{10}}(l_0, P_{620})$. (**b**) $\epsilon_{620_{10}}$ versus l_0 at P_{620}. Although there is a general trend of longer resting lengths producing more contraction, no conclusive relationship between ϵ_{620} and l_0 could be deduced from this experiment. The error bars show the effect of being ± 1 mm in measuring l_0 and l_{620}. 58
3.4 Surface fit for $F^*(\epsilon^*, P^*)$. Solid lines are from our model. Dashed lines from Festo supplied data. In total, 537 data points were collected and used to fit the 3D surfaces (321 data points for ∅10 mm BPAs and 216 data points for ∅20 mm BPAs). In total, 80% of these data were used for training, and 20% were used for validation. For figure clarity, not all data is included and plotted circles represent collected data at ± 10 kPa of the stated pressure. (**a**) Fit data for ∅10 mm Festo BPAs. (**b**) Fit data for ∅20 mm Festo BPAs. 59

3.5 Comparison between measured and predicted BPA force at varying amounts of contraction and pressure. A $y = x$ perfect fit line is added for reference. The BPAs compared are: (**a**) $\varnothing 10\,\text{mm}$, $l_0 = 257\,\text{mm}$, (**b**) $\varnothing 10\,\text{mm}$, $l_0 = 233\,\text{mm}$, (**c**) $\varnothing 20\,\text{mm}$, $l_0 = 300\,\text{mm}$, and (**d**) $\varnothing 20\,\text{mm}$, $l_0 = 450\,\text{mm}$. 66

4.1 **(A)** CAD solid model assembly of the revolute joint knee in the test stand. **(B)** 1-DoF pinned-joint robot knee in the test apparatus with important components labeled. Leg is shown with 30 degrees flexion, i.e. $\theta_\text{k} = -30^\circ$. **(C)** Picture from isometric torque testing. Setup shows the revolute knee configured for a test of the extensor BPA. The knee is flexed with $\theta_\text{k} = -120^\circ$. Also note the compressed shape of the inflated BPA on the knee during this high degree of flexion. 74

4.2 ICR for biomimetic robot knee as a function of knee angle. Measured from θ_1 on the tibia. Plot is overlayed on model of the knee at $\theta_\text{k} = 0^\circ$ (solid line), and -60° and -120° (dotted dash lines). White circles represent measured points in -15° increments from -150° to -15°, -5° increments from -15° to 0°, and then at 2°, 5°, and 10°. 76

4.3 Two-rotation construction of the bracket frame 83

4.4 Bracket frame orientations for: **(A)** flexor configuration on the pinned knee, **(B)** extensor configuration on the pinned knee, **(C)** flexor and extensor configurations on the biomimetic knee with $\varnothing 10\,\text{mm}$ BPAs, and **(D)** flexor configuration on the biomimetic knee with the $\varnothing 20\,\text{mm}$ BPA. 84

4.5 Two-bracket deflection and force-path projection 91

4.6 Pre-optimization pinned-knee torque versus knee angle for the four cross-validation flexor tests. Knee angle is negative in flexion, and flexion torque is negative. Gold circles are the hybrid calculation, which combines the measured muscle length and moment arm with the modeled force; indigo filled circles are experimentally measured torque; the gray dashed line is the rigid-body prediction; and the light orange dash-dot line is the unoptimized prediction, the improved BPA force characterization applied to the same rigid kinematics. Panel titles give the nominal resting length; the actual resting lengths are $l_0 = 48.5\,\text{cm}$ **(A)**, $l_0 = 41.0\,\text{cm}$ **(B)**, $l_0 = 45.7\,\text{cm}$ **(C)**, and $l_0 = 40.6\,\text{cm}$ with tendon **(D)**. Panels **(A)** and **(B)** are the training tests and panels **(C)** and **(D)** the validation tests of the adopted cross-validation fold. 101

4.7 Optimized pinned-knee torque versus knee angle for the four cross-validation flexor tests, at the adopted correction terms (Table 4.5). Knee angle is negative in flexion, and flexion torque is negative. Blue filled circles are experimentally measured torque; the dashed light orange line is the unoptimized prediction of the improved BPA characterization; the magenta line is the optimized prediction of the corrected model. Panel titles give the nominal resting length; the actual resting lengths are $l_0 = 48.5\,\text{cm}$ **(A)**, $l_0 = 41.0\,\text{cm}$ **(B)**, $l_0 = 45.7\,\text{cm}$ **(C)**, and $l_0 = 40.6\,\text{cm}$ with tendon **(D)**. Panels **(A)** and **(B)** are the training tests and panels **(C)** and **(D)** the validation tests of the adopted cross-validation fold; the goodness-of-fit values are given in Table 4.2. 104

4.8 Pinned-knee torque versus knee angle for the extensor BPAs: the six identification-pool tests at the adopted correction terms, with the extensor-specific terms (χ_0, χ_3) on the shared stiffness pair (Table 4.5). Knee angle is negative in flexion, and extension torque is positive. Gold circles are the hybrid calculation; indigo filled circles are experimentally measured torque; the dashed light orange line is the unoptimized prediction of the improved BPA characterization; the magenta line is the optimized prediction of the corrected model. Panel titles give the nominal resting length; the actual resting lengths are $l_0 = 43.6\,\text{cm}$ **(A)**, $l_0 = 43.2\,\text{cm}$ with tendon **(B)**, $l_0 = 45.7\,\text{cm}$ **(C)**, $l_0 = 40.0\,\text{cm}$ **(D)**, $l_0 = 40.0\,\text{cm}$ with tendon **(E)**, and $l_0 = 46.5\,\text{cm}$ **(F)**. Panels **(A)–(C)** are the training tests and panels **(D)–(F)** the held-out validation tests of the adopted cross-validation fold; the legend is given in panel **(A)**. 106

4.9 Measured versus expected isometric torque results using a $\varnothing 10\,\text{mm}$, $l_0 = 51.8\,\text{cm}$ extensor BPA on the biomimetic knee. Measured results are blue dots. Original prediction is a dashed black line. Optimized prediction is a solid magenta line. Hybrid calculation shown in yellow dots. 108

4.10 A configuration that is particularly hard to model. The BPA passes between the femoral condyles close to ICR. 109

4.11 Biomimetic-knee torque versus knee angle for the flexor BPAs. Knee angle is negative in flexion, and flexion torque is negative. **(A)** ∅10 mm, $l_0 = 41.5$ cm actuator with a 21 mm artificial tendon at 604 kPa. **(B)** ∅20 mm, $l_0 = 41.5$ cm actuator with a 14 mm artificial tendon at 620 kPa. Indigo filled circles are measured torque; the magenta line is the optimized prediction of the corrected model using the flexor terms identified on the pinned knee; the dashed light orange line is the unoptimized prediction of the improved BPA characterization alone. In **(B)**, the black dotted line is the maximum human isometric torque, and the pink circles are measured points at the reduced pressures 560 kPa, 421 kPa, 325 kPa and 281 kPa (from left to right, respectively), recorded because full pressure would load the printed femur beyond its ultimate strength. 111
4.12 Route geometry of the redesigned extensor route 118
4.13 Pulley-position sweep surfaces for the reeved flexor route 120

5.1 The SADb Explorer offline corpus tool 132
5.2 The SADb Explorer bubble map view 133
5.3 The SADb Explorer pivot view 135

6.1 Simulation-to-robot pipeline 143
6.2 AnimatLab walker and controller hierarchy 147
6.3 AnimatLab controller reference schematics 148
6.4 AnimatLab working-project interface 149
6.5 MuJoCo model representations 153

6.6 Simulink SNS library blocks and neuron implementation 155
6.7 Reduced-order knee-reflex circuit topology 156
6.8 AnimatLab phase-1 hip, knee, and ankle recordings 166
6.9 W2L rhythm-core traces across engines 169
6.10 Constant-drive rhythm, non-spiking versus spiking 178
6.11 Ground gait, non-spiking winner versus untuned spiking mirror 179
6.12 Simulink knee-reflex twin, baseline versus spiking 182
6.13 Five-stage training curriculum and stage scores 189

7.1 The first-pass design workflow the enablers make possible 199
7.2 Construction of the braided pneumatic actuator 202
7.3 Control stack with the planned supraspinal additions 207

1 Introduction

1.1 Motivation

Biomimetic robots provide a platform for investigating how mechanical structure and control interact to generate movement, offering complementary insights to simulation-based studies [Andrews, 1985, Millard et al., 2013, Arnold et al., 2010]. Wheeled robots remain the platform of choice in structured environments, where their speed, efficiency, and payload capacity are difficult to match; in the unstructured terrain of the natural and built world, legged machines hold the advantage, and biomimetic legs that reproduce human force capabilities have the potential to excel where wheels cannot [Frey et al., 2026, Ha et al., 2025]. Musculoskeletal humanoids and limb systems driven by tendon-like actuators have been shown to support biologically meaningful experiments on motion generation, internal force transmission, and neural control hypotheses [Asano et al., 2017, 2019, Hitzmann et al., 2018, Liu et al., 2018, Chen et al., 2019]. The dual perspective of using robots to learn about biology, and using biology to design more adaptive robots, has been emphasized in both biomimetic robotics and biomechanics [Ijspeert, 2020, Asano et al., 2017]. With such robotic systems, experiments can be performed that would not be practical or ethical with human test subjects, such as removing a specific muscle from the system or blocking a sensory feedback pathway. Experiments of this kind allow direct testing of how the components of the neuromechanical system interact, and they offer new opportunities for testing neural control theories that are faster and cheaper than human

observation studies [Shin et al., 2018, Goldsmith et al., 2021, Schilling et al., 2022].

Artificial muscles are a key enabling technology for this class of robot. Among the available actuation technologies, McKibben-style braided pneumatic actuators (BPAs) combine low mass, high force-to-weight ratio, and intrinsic compliance with a force–length curve that qualitatively resembles that of skeletal muscle [Liang et al., 2020, Hunt et al., 2017a]. BPAs have been used in quadrupedal robots [Aschenbeck et al., 2006, Hunt et al., 2017a, Scharzenberger, 2019], musculoskeletal humanoids [Asano et al., 2017, Hitzmann et al., 2018], and dexterous manipulation systems [Chen et al., 2019]. However, BPAs also differ from biological muscle in important ways: their maximum tensile force occurs at resting length rather than at an optimal fiber length, their maximum contraction ratio is substantially lower than that of human muscle, and their force output is a highly nonlinear function of internal pressure, contraction, and loading state [Hunt et al., 2017a, Sárosi et al., 2017, Liang et al., 2020]. The author's experience with 10 mm actuators made this gap concrete: manufacturer specifications overstate the force and contraction available at the short resting lengths a leg design requires, and designs based on those specifications delivered less range of motion and joint torque than intended. Nor could the published empirical models repair the discrepancy, because each is fitted to measurements at a single resting length [Sárosi et al., 2017, Martens and Boblan, 2017, Hunt et al., 2017a]; none accurately predicts how force changes when a designer chooses a different actuator length. Because muscle resting length is one of the primary degrees of freedom available to the robot designer, the absence of a force model valid at arbitrary resting length blocked rational actuator selection. Designing a biomimetic robot around BPAs therefore requires quantitative, predictive models of the actuator and of the joint torques it can produce.

Actuator force alone, however, does not determine the torque a joint produces. The robot knee transmits BPA force to the skeleton through mounting brackets, artificial tendons, and paths that wrap over the joint, and each of these elements departs from the rigid-body idealization: brackets and fixtures deflect under load, tendons stretch, and a wrapping actuator loses usable length as the knee flexes. At the start of this work it was unknown how much these transmission effects reduce the torque delivered at the knee relative to the rigid-body prediction, and no method existed for correcting that prediction in a form that would transfer between joint configurations. The biomimetic knee's optimized muscle routing [Morrow et al., 2020] compounded the problem: it had never been validated experimentally or compared against human isometric torque data, so even the nominal torque envelope of the robot leg rested on unverified calculations.

The remaining gaps concern the controller. No simulation pipeline existed for closing the loop between a musculoskeletal robot model of this kind and a biologically realistic neural controller. The neuromechanical modeling literature provides the ingredients, including two-level spinal central pattern generator architectures [Rybak et al., 2006, McCrea and Rybak, 2008], neuromechanical simulation frameworks [Markin et al., 2010], and synthetic nervous system tooling [Nourse et al., 2023], but it does not provide their assembly for a BPA-driven bipedal robot with human-like muscle routing. Wiring realistic feedback into such a controller exposes a fourth gap, this one in the literature rather than in methods: the physiological properties of the group Ia, group Ib, and group II afferents that should inform the feedback design are scattered across hundreds if not thousands of primary papers and reviews [Van de Crommert et al., 1998, Windhorst, 2007, 2022], with no structured resource from which a robot designer or an animal researcher could draw

them.

Four gaps thus stood between the existing robot design and a robot that can walk under neural control: no actuator force model valid at arbitrary resting length, no quantified correction for the torque lost through brackets, tendons, and wrapping, no closed-loop simulation pipeline joining the robot's musculoskeletal model to a biologically realistic neural controller, and no consolidated resource for the afferent data that such a controller requires.

1.2 The Biomimetic Humanoid Robot Project

This dissertation is part of a long-running effort at Portland State University to build a biomimetic humanoid robot from the trunk down, actuated by pneumatic artificial muscles and controlled by a synthetic nervous system (SNS). The project is deliberately iterative. Knowledge from biomechanics and neuroscience guides the design of robots intended to perform complex tasks in unstructured environments with improved efficiency, balance, adaptability, and resilience. Those robots then become physical models with which biological hypotheses can be tested, validated, and refined through controlled changes that would be impractical or unethical in living subjects. The resulting biological insight informs the next generation of robot design, and lessons from both directions can be translated into improved prosthetic and orthotic devices.

The present work builds directly on a series of studies by the author and collaborators. In the author's master's project, the kinematics and dynamics of a biomimetic robot leg were matched against the Gait2392 OpenSim model, artificial muscle attachment and wrap-

ping locations were determined for the robot legs, and MATLAB tooling was developed for evaluating muscle lengths, moment arms, and isometric joint torques over the full range of motion [Bolen, 2019, Bolen and Hunt, 2019]. That work showed that simply transplanting human muscle attachment locations onto a BPA-actuated robot does not reproduce human torque profiles; the placement optimization that closes this gap was completed during this dissertation and is summarized in Section 7.1. In parallel, a biomimetic four-bar knee linkage with a migrating instantaneous center of rotation was designed and validated by Steele [Steele et al., 2017, Steele, 2018, Steele et al., 2018], and a canine-inspired quadruped platform, Muscle Mutt, formerly called DoggyDeux, was developed with antagonistic BPA pairs and synthetic nervous system control [Scharzenberger, 2019]. That platform extended an earlier AARL demonstration of central pattern generator control of a pneumatically actuated robot [Hunt et al., 2017b].

1.3 Research Objectives

This dissertation addresses the first two gaps experimentally, resolves the fourth by consolidating the afferent literature into the Sensory Afferent Database [Bolen et al., 2023, Bolen and Hunt, 2024], and develops the tools and framework that close the third. The objectives are:

1. **Characterize the isometric force of Festo BPAs** with uninflated diameters of 10 mm and 20 mm over a wide range of resting lengths, contractions, and pressures; derive improved maximum-force and normalized force–strain–pressure models; and compare them against existing models (Chapter 3).

2. **Validate and improve the prediction of isometric knee torque** produced by BPAs on pinned-joint and biomimetic four-bar knees; identify and model losses caused by constant length offsets, series compliance, and actuator wrapping; and compare the resulting torque envelopes with human capabilities (Chapter 4).

3. **Consolidate the sensory afferent literature into a structured afferent resource**, the Sensory Afferent Database, by auditing and curating the published record of afferent properties across animal species and their role in motor control, and by evaluating AI-assisted literature triage for expanding the corpus (Chapter 5).

4. **Develop tools and a framework for neuromechanical control** of the humanoid robot by converting its OpenSim musculoskeletal model to MuJoCo, implementing an SNS-Toolbox two-level spinal CPG with proprioceptive and heel/toe contact feedback, and establishing a staged tuning and verification workflow (Chapters 6 and 7).

1.4 Dissertation Organization

Chapter 2 establishes the technical and scientific context for the dissertation. It reviews biomimetic and musculoskeletal robots, robot actuation technologies with an emphasis on braided pneumatic actuators and published models of their force–length–pressure behavior, and the human lower-limb biomechanics and OpenSim benchmark models that serve as design targets. It then covers the computational neuroscience of locomotor control, from two-level central pattern generator architectures and sensory afferent feedback to the reference circuits from which the dissertation's controller architecture is

drawn. Two further topics frame the simulation work: neuromechanical simulation as a concept, and the tooling used to execute it, from OpenSim and its SimTK ecosystem to MuJoCo, SNS-Toolbox, and AnimatLab. The chapter closes with the ethical case for testing neuromechanical hypotheses on robots rather than living subjects.

Chapter 3 develops and validates the actuator force models. It presents the isometric force characterization of Festo BPAs with 10 mm and 20 mm uninflated diameters over a wide range of resting lengths, contractions, and pressures; derives improved maximum-force and normalized force–strain–pressure models; and compares them against the manufacturer's tool and published models. Later sections examine the effects of resting length, the implications of the validated models for real-time control and simulation, and the biomimetic analogy between braided pneumatic actuators and skeletal muscle. The chapter's text and figures are adapted from the author's publication of this work [Bolen et al., 2026].

Chapter 4 carries the actuator models to the joint. It describes the robot leg model, the pinned-joint and biomimetic knee test stands, and the equipment used to collect isometric torque data; formulates the hybrid torque calculation method and the improved torque model whose correction terms account for a constant length offset, series compliance, and actuator wrapping; and documents the optimization algorithms used both to identify those terms and to redesign actuator routes. The chapter then reports the measured torque on both knees, discusses the physical sources of prediction error, presents the follow-up identification and the redesigned routes evaluated on the Gait2392 robot-body model, and compares the resulting torque envelopes with human capabilities, before closing on the transmission-design implications of mono- and bi-pulley routing.

Chapter 5 presents the Sensory Afferent Database, the structured afferent resource built to answer the afferent-literature gap. It describes the audited corpus and its curation coverage, in which papers carry structured curation notes, afferent classifications, and notes translating each sensory pathway toward robot or simulation use; the packaging of the corpus with its citation graph as a self-contained laboratory tool; and the evaluation of AI-assisted literature triage for expanding the database, before closing by mapping the corpus onto the controller's sensory channels.

Chapter 6 builds the tools for neuromechanical simulation, with a stated goal: a closed-loop environment in which a synthetic nervous system controls the experimentally characterized robot model and supports the lesion, stimulation, and sensory-deletion experiments that cannot be performed on living subjects. It opens with the preliminary AnimatLab bipedal walker results, whose hand-tuned two-layer CPG produced suspended stepping, with neither weight-bearing walking nor any role for afferent feedback in weight-bearing motion demonstrated; that outcome set the verification-first approach the tools follow. The tools themselves are the OpenSim-to-MuJoCo conversion of the robot's musculoskeletal model with its synthetic BPA knee, the Simulink SNS library built to prototype reflex topologies, the spiking-neuron mirrors, and the cross-platform verification protocol that ties the implementations together. Each tool is demonstrated on the fundamental exercises that establish what it does and does not show: cross-engine reproduction of the contact-driven rhythm core, headless re-runs of the archived AnimatLab reference models, the BPA-actuated walker in MuJoCo, a gait-validation sweep against a locomotion reference library, and a comparison of the production walker variants. The stimulus and deletion experiments complete the demonstrations, and the chapter closes by discussing

the five-stage training curriculum, the spiking-neuron mirrors, and the new tools the program leaves other researchers.

Chapter 7 discusses the work as a whole. It opens with the outcome, that improved first-pass robot design is now achievable, and attributes it: the design factors that remove the rigid-body idealization, the improved BPA characterization, the tiered attachment and routing optimization, the sensory afferent database, and the new neuromechanical simulation tools. Its remaining subsections scope the future work these results enable for other researchers, from BPA characterization and routing expansion through database and simulation tooling to real-time neural control of the physical robot. Chapter 8 consolidates the principal contributions of the dissertation. Figure 1.1 summarizes the flow.

Appendix A documents the muscle force formulations and their cross-platform comparison, the BPA force model implementation, the muscle-path length and moment-arm calculations, and the compliance and wrapping-loss implementations. Appendix B provides the neuron and synapse equations, spiking and non-spiking alike, used by the synthetic nervous system tools, relates those equations to their SNS-Toolbox implementation, and records the spiking-mirror calibration and the connectomes of the spinal-network models. Appendix C records the static BPA pathways, joint transformation matrices, bracket reference frames, stiffness and compliance parameters, adopted correction-term values, and the redesigned extensor route geometry.

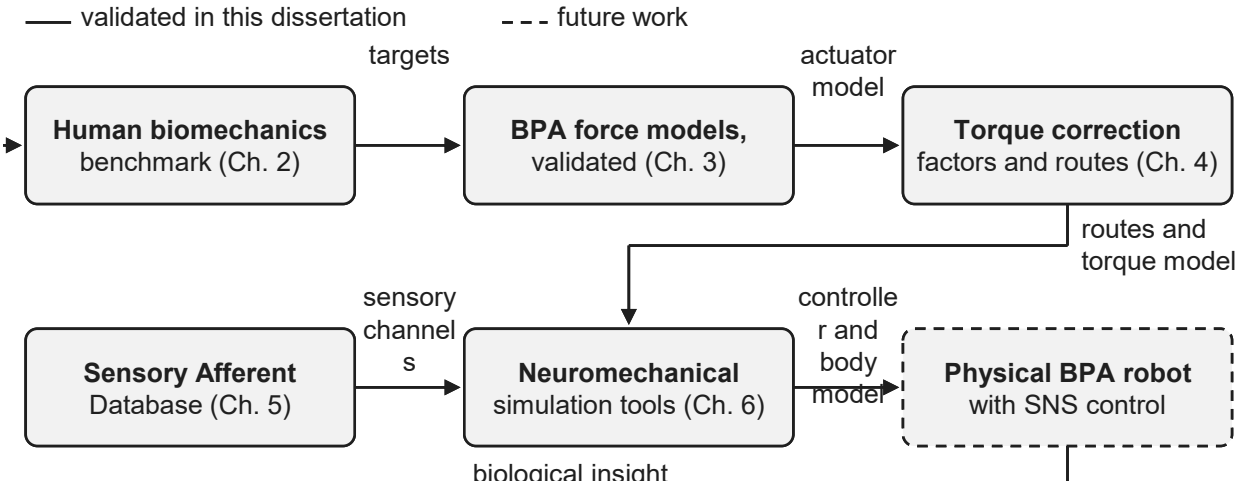


Figure 1.1: The program's flow. The biomechanics benchmark of the background chapter feeds the actuator force characterization, whose correction factors and redesigned routes carry the models to the joint; the Sensory Afferent Database fixes the controller's sensory channels; and the simulation tools close the loop between converted body and synthetic nervous system. Solid blocks are built and validated in this dissertation, and the dashed block is the physical robot the tools define the path toward, which feeds biological insight back to the benchmark.

2 Background

This chapter reviews the established knowledge against which the dissertation's contributions are defined: biomimetic and musculoskeletal robotics, the actuation technologies available to them, the human lower-limb biomechanics they imitate, the neuroscience of locomotor control they implement, and the simulation and ethical context in which they are tested. Work from the author's own program that goes beyond this background, including the prior AnimatLab bipedal walker, the Sensory Afferent Database, and the neuromechanical simulation pipeline, is presented as dissertation work in Chapters 5 and 6 rather than repeated here.

2.1 Biomimetic and Musculoskeletal Robots

Biomimetic robots offer many potential applications, ranging from assisting in factories and environmental exploration to serving as physical models of human movement. By developing high-fidelity biomimetic robots, we can perform tests that help improve our understanding of both biomechanics and the underlying neuromechanical systems that control movement [Shin et al., 2018, Asano et al., 2019, Ijspeert, 2020, Goldsmith et al., 2021, Hunt et al., 2017b, Schilling et al., 2022]. A few such biomimetic robots exist, such as the impressive Kengoro and Kenshiro robots of Asano et al. [2017] and Asano et al. [2019], which have proportionally human bodies, anatomically correct musculotendon arrangements, rolling-sliding joints, and joint ranges and output power comparable with human-measured values; their artificial muscles are implemented with brushless DC

motors, wires as tendons, gears, and integrated tension measurement. Kengoro, the more recent of the two, stands 1670 mm tall, weighs 56.5 kg, and carries 174 degrees of freedom when the end effectors are included [Asano et al., 2017]. A dog-like robot developed by Aschenbeck et al. [2006] and actuated with Festo BPAs was designed from a planar kinematic model scaled to an adult greyhound, with three sagittal degrees of freedom per leg driven by antagonistic BPA pairs; its front-leg pair alone lifted a 13.5 kg payload, almost double the complete robot's predicted mass. That platform was used to show how central pattern generators provide robust locomotion patterns that are adaptive to perturbations in ground speed [Hunt et al., 2017b].

Robots built with artificial muscles thus provide opportunities to investigate how muscles are controlled to produce desired motions; however, artificial muscles do not exactly replicate muscle properties, and existing robots do not capture all of the degrees of freedom and joint torques of the animal limbs they mimic. The biomimetic leg developed and tested in this dissertation sits squarely in this lineage: a human-proportioned skeleton, muscle-like actuators, and the goal of human-matched joint torques.

Figure 2.1 illustrates the human-scale skeletal structure and tendon routing of this approach.

2.2 Robot Actuation Technologies

2.2.1 Overview

A wide range of actuation technologies have been explored in robotics, including electric motors, hydraulic systems, smart-material actuators, and artificial muscles [He and Gao,

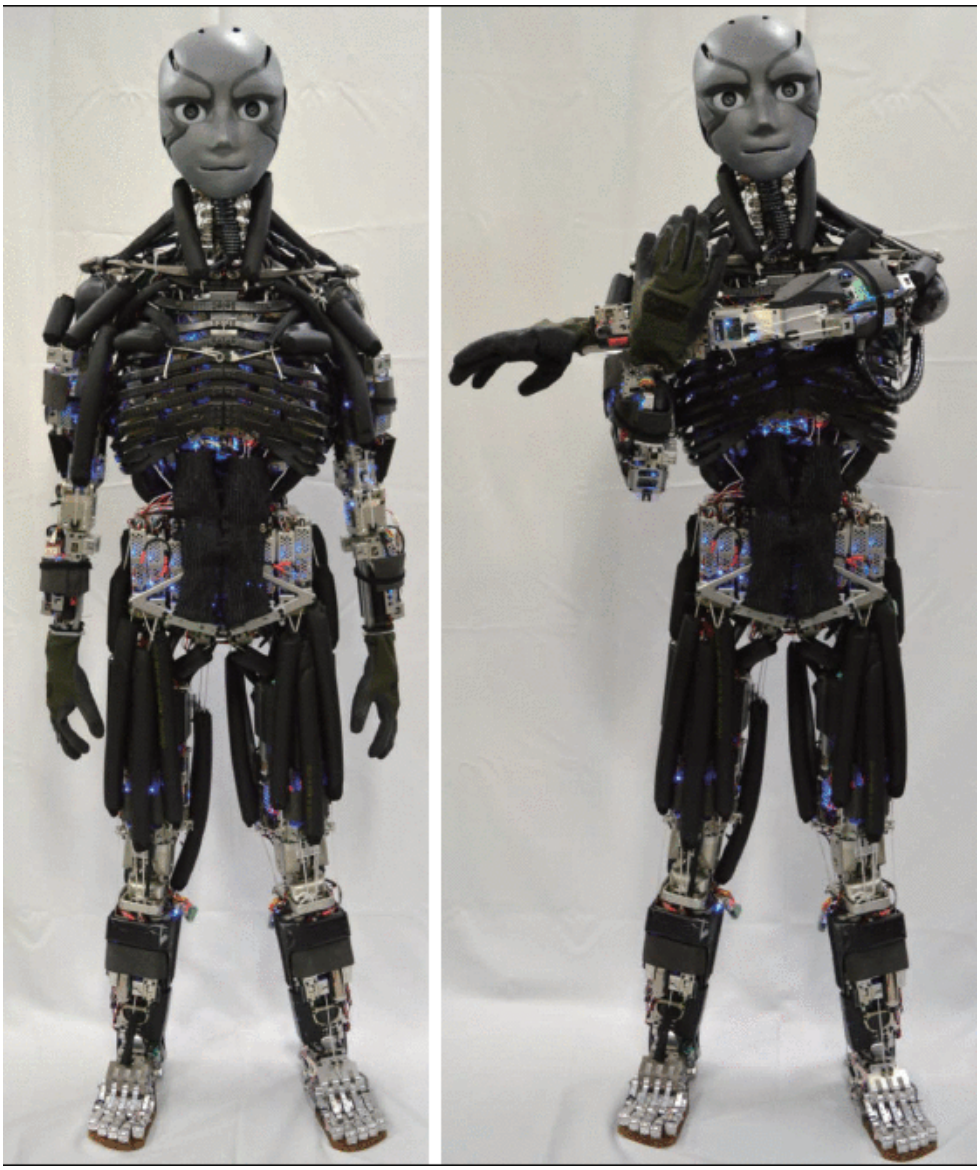


Figure 2.1: Kengoro human-mimetic musculoskeletal humanoid in neutral and upper-limb poses, illustrating its human-scale skeletal structure and densely routed tendon-driven musculature [Asano et al., 2017].

2020, Liang et al., 2020]. Electric motors are widely used due to their controllability and maturity, but they are limited by thermal constraints, gearing requirements, and reduced compliance compared to muscle-like actuators [Veale and Xie, 2016, Suzumori and Faudzi, 2018]. Hydraulic actuators offer exceptionally high power density and have enabled highly dynamic legged systems, but they remain relatively heavy, less efficient than biological muscle, and require a dedicated fluid infrastructure [Suzumori and Faudzi, 2018].

Among smart-material actuators, dielectric elastomer actuators (DEAs), a class of electronic electroactive polymers, contract by a different mechanism: an applied voltage squeezes a soft polymer film between compliant electrodes through electrostatic stress, and the thinning and radial spreading of the film produces strains that can exceed biological muscle [Madden et al., 2004]. Comparative studies report high specific work and good bandwidth for DEAs [Liang et al., 2020], but kilovolt-range driving voltages, electromechanical instability, and the compliant-electrode requirement have kept them out of legged robots. Thermally activated artificial muscles reach the most extreme performance figures of any technology surveyed by Liang et al. [2020]: twisted-fiber muscles support loads more than one hundred times those of a biological muscle of the same length and weight, with output stresses of 22 MPa. Their activation route, however, is shared with shape-memory alloys, whose heating is fast but whose cooling is limited by heat transfer away from the element, and that asymmetry sets the cycle time.

Closer to muscle, series-elastic and pneumatic actuators trade peak force and bandwidth for compliance, a trade that wearable and rehabilitation robotics exploit because force control and backdrivability matter most when a machine interacts with a human [Aksöz et al., 2019, Leibach, 2020, Li et al., 2020]. A recent review of pneumatic artificial

muscle technology documents continued progress in actuator design and in assistive applications [Zhagiparova et al., 2025]. None of these technologies yet combines muscle-like force, strain, and compliance in one actuator, which is the combination a biomimetic humanoid leg demands and that braided pneumatic actuators approximate most closely.

Figure 2.2 compares the operating mechanisms of these actuator technologies.

2.2.2 Braided Pneumatic Actuators

In contrast to the technologies above, McKibben-style braided pneumatic actuators (BPAs) combine low mass, high force-to-weight ratio, and intrinsic compliance with a force–length curve that qualitatively resembles that of skeletal muscle [Liang et al., 2020, Hunt et al., 2017a]. A BPA consists of an inflatable bladder inside a braided sleeve; pressurizing the bladder expands the bladder radially and contracts the sleeve axially, producing tensile force. As shown by Chou and Hannaford, the force–length relationship of McKibben actuators is governed by geometric constraints imposed by the braided shell, leading to a nonlinear dependence of force on actuator length through the braid angle [Chou and Hannaford, 1996]: actuator force is proportional to pressure and to a nonlinear geometric term $(3\cos^2\theta - 1)$, which governs both the maximum contraction limit and the curvature of the force–length relationship. Predicting their force as a function of length and pressure is therefore the first improvement of the torque tools developed in this dissertation, and the state of that prediction is the subject of the next subsection.

Figure 2.3 shows the commercial actuator assembly used in the laboratory.

However, BPAs exhibit several important differences from biological muscle. First,

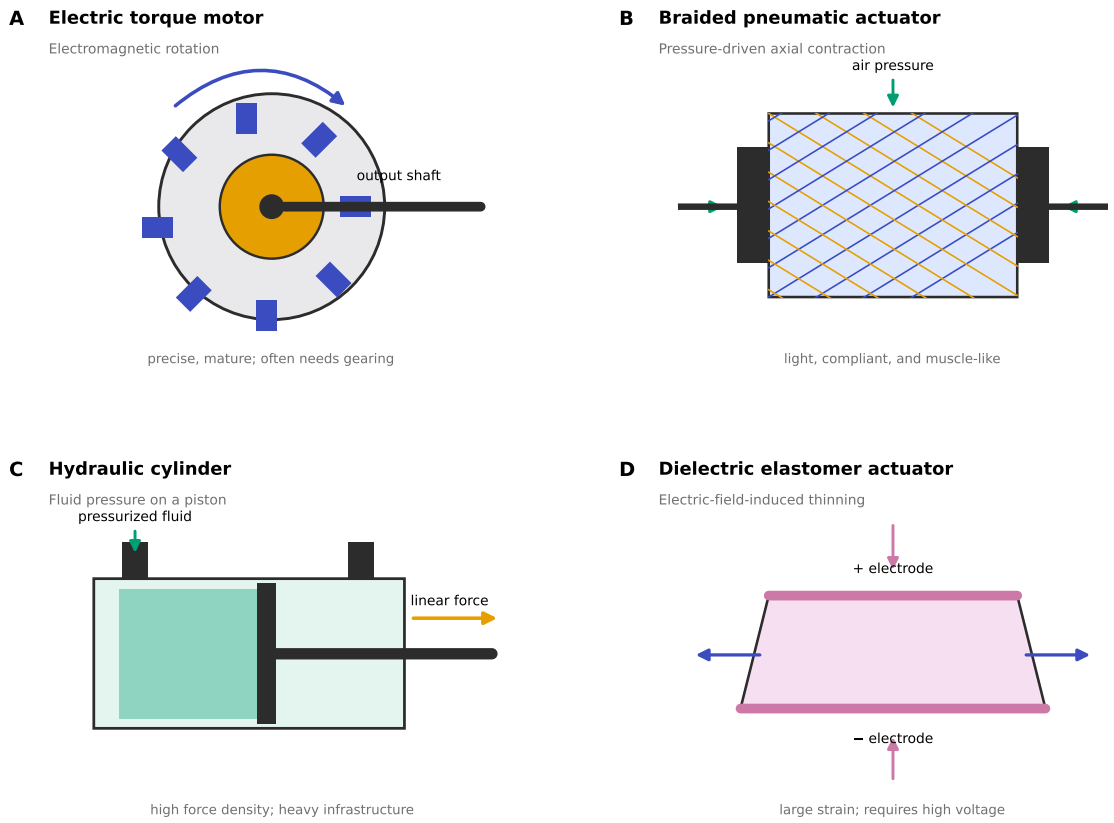


Figure 2.2: Representative operating mechanisms for robotic actuators. **(A)** An electric torque motor produces controlled shaft rotation. **(B)** A braided pneumatic actuator shortens as internal pressure expands its sleeve radially. **(C)** A hydraulic cylinder converts fluid pressure into linear piston force. **(D)** A dielectric elastomer thins and expands laterally under an applied electric field. The panels use a consistent schematic style to emphasize the distinct energy-conversion mechanisms and design tradeoffs summarized in the text [Suzumori and Faudzi, 2018, Liang et al., 2020].

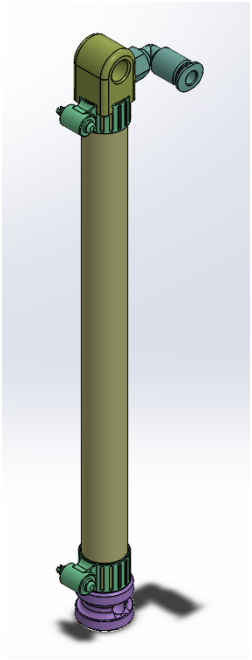


Figure 2.3: CAD representation of a Festo DMSP braided pneumatic actuator with the mounting and pneumatic connections used in the laboratory assembly [Festo AG & Co. KG, 2026]. The long cylindrical body contains the bladder and braided sleeve; pressurization through the end fittings converts radial expansion into axial contraction.

BPA maximum tensile force occurs at resting (i.e., uninflated) length (l_0) rather than at an optimal fiber length [Liang et al., 2020]. Second, their maximum contraction ratio is substantially lower than that of human muscle, presenting challenges for achieving biomimetic joint ranges of motion [Hunt et al., 2017a, Sárosi et al., 2017, Liang et al., 2020]. Third, BPAs have a highly nonlinear dependence on internal pressure (P), contraction (ϵ), and loading state [Hunt et al., 2017a, Martens and Boblan, 2017], which complicates control and motivates high-fidelity force–length–pressure modeling. Despite these differences, their muscle-like properties have made them attractive for biomimetic quadrupeds [Aschenbeck et al., 2006, Scharzenberger, 2019], humanoids [Asano et al., 2017, Hitzmann et al., 2018], and musculoskeletal arms [Kurumaya et al., 2016, Chen et al., 2019].

2.2.3 BPA Force–Length–Pressure Models

Previous studies have generated empirical models of the force–length–pressure relationship of BPAs. Sárosi et al. [2017] discuss several high-fidelity BPA force models. They present a static force model with a 21-coefficient polynomial function for a Festo MAS-20-200N (i.e., ∅20 mm, $l_0 = 200$ mm) with an impressive $R^2 = 0.9994$, as well as Sárosi's static force model for a Festo DMSP-20-400N-RM-RM (i.e., ∅20 mm, $l_0 = 400$ mm), with six coefficients and a $R^2 = 0.9995$ [Sárosi, 2012, Sárosi et al., 2013]. Martens and Boblan present an even more accurate model, in terms of absolute error, using five coefficients for a DMSP-10-250 (i.e., ∅10 mm, $l_0 = 250$ mm) and a DMSP-20-300 (i.e., ∅20 mm, $l_0 = 300$ mm) [Martens and Boblan, 2017]. However, as demonstrated in this work, different resting lengths produce different force–length curves, preventing the use of these single-length

models for designing systems with different resting lengths.

Hunt et al. [2017a] looked at six resting lengths of ∅10 mm Festo BPAs, accounting for differences in maximum contractile percentages, and elucidated the force-length-pressure relationship. In particular, for a given contraction percent and force F (in Newtons), the scalar pressure P (in kPa) required to produce this force for a ∅10 mm Festo artificial muscle can be determined by solving the equation:

$$P = P_0 + k_F F + k_S S + P_t \tan\left(c_a \left[\frac{\epsilon}{\epsilon_{\max} - k_E F} - c_b \right] \right) \tag{2.1}$$

where F is the required force, ϵ is the contraction, S encodes hysteresis ($S = 1$ shortening, $S = -1$ lengthening, $S = 0$ static), and the seven coefficients are defined in Table 2.1. The coefficients have been updated with corrected values, as those reported in Hunt et al. [2017a] contained typographical errors. The contraction ϵ and the maximum free-load contraction at 620 kPa, which appears as $\epsilon_{\max}$ in Equation (2.1), are defined in Section 3.1.1, together with the relative contraction $\epsilon^* = \epsilon / \epsilon_{620}$ on which the force surface developed in this dissertation is built. Because Equation (2.1) expresses pressure in closed form as a function of force and contraction, it can also be solved onto a grid of pressures and contractions to build a force–pressure–relative-contraction lookup table for design studies.

Festo provides a manufacturer tool, MuscleSIM, for predicting force as a function of pressure and contraction [Lang, Robert, 2005], but experimental comparisons indicate that it does not adequately account for variation in l_0 or actuator-specific contraction limits. Comparisons with experimental data indicate that none of the existing models effectively take into account differences in maximum force due to initial actuator length, and the Hunt

Table 2.1: Coefficients of the 10 mm Festo BPA pressure model of Equation (2.1), corrected from Hunt et al. [2017a].

Symbol	Value	Role
P_0	254 kPa	pressure offset
k_F	1.23 kPa N^{-1}	linear force-to-pressure gain
k_S	15.6 kPa	hysteresis offset (sign of S)
P_t	192 kPa	tangent-term amplitude
c_a	2.03	contraction-shape coefficient
c_b	0.46	contraction-shift coefficient
k_E	0.331×10^{-3} N^{-1}	force-dependent reduction of usable contraction

model was taken with low external forces ($\leqslant$106 N), so it is unclear how well it captures actuator behavior at higher forces. Chapter 3 compares these models with data collected for this dissertation and develops the improved characterization on which the torque and controller work of the later chapters depends.

Figure 2.4 summarizes the force-characterization workflow developed in this dissertation.

2.3 Human Lower-Limb Biomechanics

Designing a biomimetic leg requires a quantitative benchmark of human muscle–joint behavior. This dissertation uses OpenSim [Delp et al., 2007, Seth et al., 2011, 2018] and the Gait2392 model, which descends from the interactive lower-extremity model of Delp et al. [1990] and includes musculoskeletal geometry from Hoy et al. [1990], planar knee kinematics from Yamaguchi and Zajac [1989], and a lower-limb model with 23 degrees of freedom and 54 musculotendon actuators [Arnold et al., 2010]. Muscle moment arms, force-generating properties, and functional roles in such models have been characterized

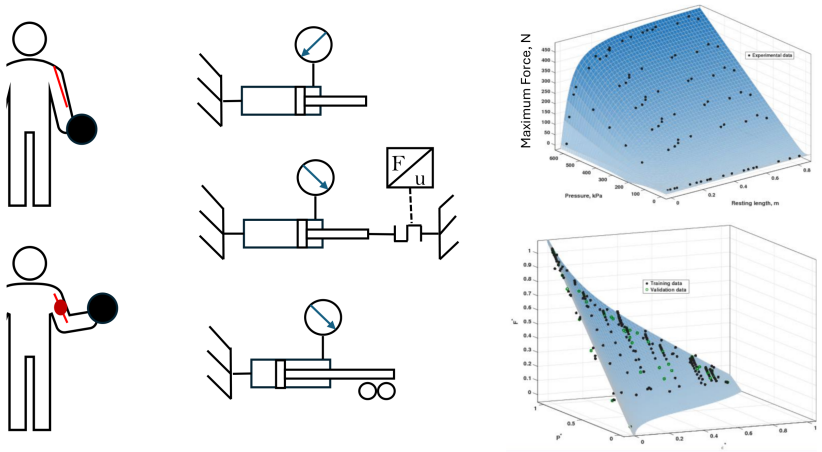


Figure 2.4: Overview of the BPA force-characterization study. Biological-muscle inspiration and the laboratory test apparatus lead to measured force–length–pressure data and fitted force surfaces used for actuator and robot design.

extensively [Zajac and Gordon, 1989, Brand et al., 1982, Menegaldo et al., 2004, Millard et al., 2013, Sherman et al., 2013, Suderman and Vasavada, 2012], and standard reporting conventions for kinematics and joint mechanics are established by the International Society of Biomechanics [Wu and Cavanagh, 1995, Wu et al., 2002, 2005, Derrick et al., 2020].

The moment arm is the quantity that turns muscle force into joint torque, and two methods of obtaining it anchor the literature. The geometric method computes the arm directly from attachment-point data: the arm about a joint axis is the perpendicular distance from that axis to the muscle's line of action, evaluated at the configuration of interest. Brand et al. [1982] assembled the canonical lower-extremity attachment anatomy from cadaver measurements so that such computations could be made at all, and reported that anatomical variation and marking errors rarely shift calculated moment arms by more than twenty percent, usually by less than ten, while attachment data taken from

a single unscaled specimen can bias the resulting force predictions substantially. The tendon-excursion method instead defines the arm as the sensitivity of musculotendon length to joint angle, $r = \partial l / \partial \theta$, obtained by perturbing a joint and differencing path lengths. Menegaldo et al. [2004] fitted polynomial regressions of musculotendon lengths and three-dimensional moment arms across the joint space of a lower-limb model so that either quantity could be evaluated cheaply inside large simulations. Both methods deliver configuration-dependent quantities rather than constants, and the dependence compounds for multi-articular muscles, whose arm about one joint varies with the configuration of every joint they cross [Zajac and Gordon, 1989, Young et al., 2019]; the full-muscle rat hindlimb model of Young et al. [2019] makes that adjacent-joint dependence a first-class result of its attachment-point technique. OpenSim evaluates moment arms as muscle-length sensitivities of this kind, so the arms used throughout this dissertation are properties of the Gait2392 geometry rather than measured inputs [Delp et al., 2007, Seth et al., 2018].

The human knee is the joint where these definitions are stressed hardest. Its instantaneous center of rotation migrates posteriorly with flexion, a behavior captured by four-bar linkage descriptions of the joint [Romero et al., 2020, Steele, 2018], and Romero et al. [2020] treat locating that moving center as a kinematic problem in its own right, solved with natural coordinates so that the answer stays cheap to evaluate when an optimization must recompute it many times. That difficulty is visible at the repository level: the SimTK repository that distributes OpenSim also hosts multiple projects dedicated specifically to the human knee, and they differ in how they formalize joint torques, centers of rotation, and reaction forces for the same anatomy. A hinge knee, a four-bar knee, and a contact-defined tibiofemoral joint give different moment arms for identical muscle paths, so the

modeling choice must be matched to the question asked of it. The ISB joint coordinate system recommendations were written against exactly this definitional freedom, to keep reports of joint motion comparable across laboratories [Wu et al., 2002]. Throughout this dissertation, Gait2392 serves as the reference against which the robot's muscle routing, moment arms, and torque capabilities are designed and judged.

2.4 Biomimetic Knee and Bipedal Robot Model

Steele's biomimetic knee joint supplies the physical knee used in this dissertation [Steele et al., 2017, Steele, 2018, Steele et al., 2018]. It is a one-degree-of-freedom four-bar linkage whose crossed links reproduce the posterior migration of the human knee's instantaneous center of rotation (ICR) during flexion, so the rolling-and-sliding behavior of the human joint emerges from the link geometry itself rather than from cams or gear profiles; the design was calculated to reproduce an average sliding-to-rolling ratio of 0.079 over a 151° range of motion [Steele et al., 2018]. Because the ICR migrates, muscle moment arms about the knee change with flexion angle in a human-like way, which is exactly the behavior a torque prediction tool must capture. The biomimetic knee tests of Chapter 4 therefore exercise the corrected torque model against this geometry, and Figure 2.5 summarizes the mechanism.

In the author's master's work, muscle lengths, moment arms, and isometric joint torques were computed over the range of motion for both the Gait2392 model and a PAM-actuated robot leg model, demonstrating that several robot muscle torque profiles could closely match the human model while many fell far short for part of the RoM [Bolen, 2019,

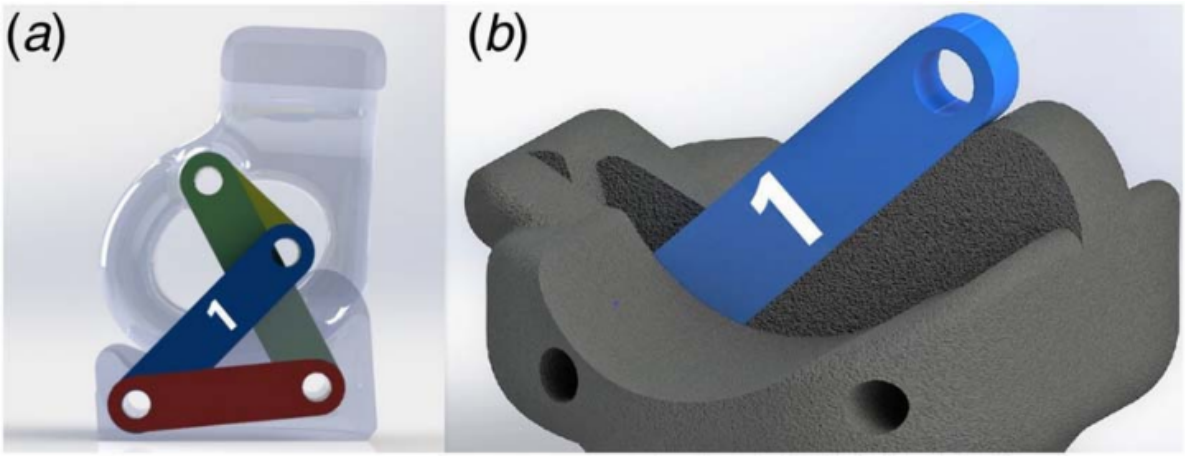


Figure 2.5: Biomimetic knee mechanism. **(a)** Sagittal cutaway at full extension showing the crossed four-bar linkage. **(b)** Exploded view of the mechanical stop. Reproduced from Steele et al. [2018, Fig. 6] under CC BY 4.0.

Bolen and Hunt, 2019]. Hand placement of muscle origins and insertions was a painstaking and iterative task, and the placement optimization that replaced it is dissertation work [Morrow et al., 2020], discussed in Chapter 4. Figure 2.6 places the laboratory's bipedal skeleton beside the OpenSim muscle-routing reference developed for it.

The mechanics of that master's study are worth recording because the dissertation's torque work reuses them. Moment arms were computed geometrically from attachment points in the body frame, as the cross product of a joint-axis unit vector with the vector from the joint center to the muscle's line of action, the same construction described in Section 2.3. PAM routings were then placed by hand in OpenSim beside the Gait2392 paths they imitate, moved to respect Festo BPA contraction limits and to keep actuators from interfering with one another, and the result was saved as the gait2392_robotbody model that the later chapters of this dissertation continue to refine [Bolen, 2019]. Isometric torque followed as muscle force multiplied by moment arm, evaluated across each joint's range

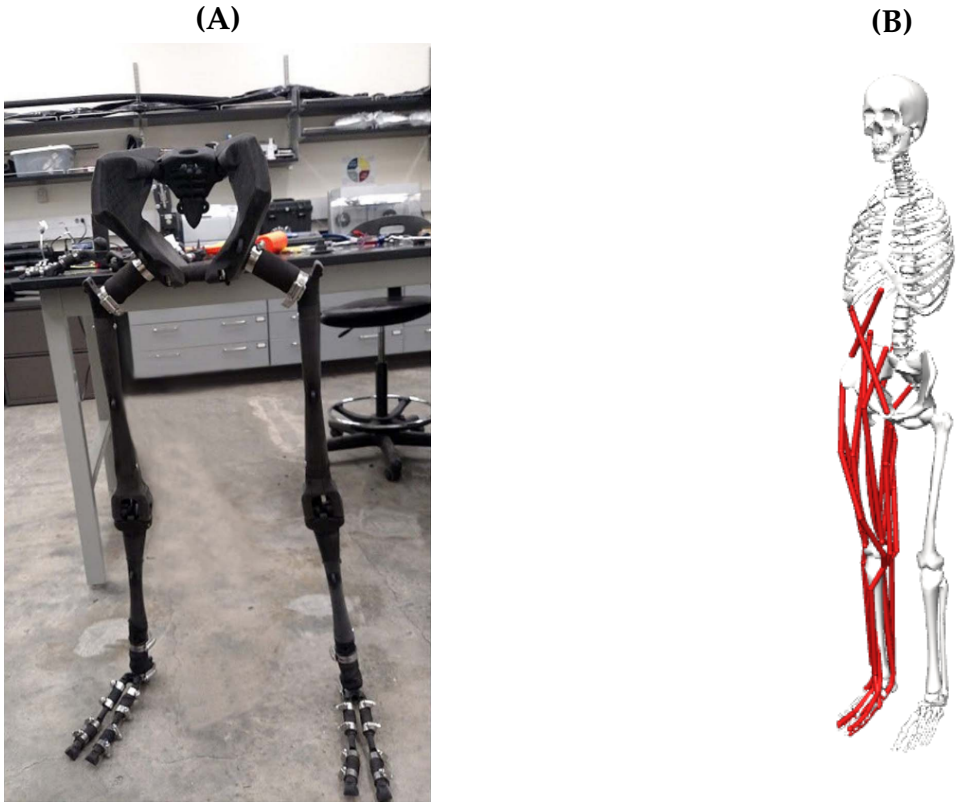


Figure 2.6: **(A)** The laboratory biped skeleton (actuators not installed in this view). **(B)** The Gait2392-derived OpenSim model used to develop its muscle routing; the view exposes the retained lower-limb musculotendon paths used as biomechanical design references [Bolen and Hunt, 2019, Morrow et al., 2020].

of motion. The biceps femoris long head shows what the comparison exposed: its path differs visibly between the two models as the knee flexes, and the robot model's knee- and hip-axis torque surfaces reach only a fraction of the human model's peaks over the hip–knee range. Profiles of this kind, read across the full muscle set, defined the matching problem that the placement optimization later formalized [Morrow et al., 2020]. Figure 2.7 shows the routing models and the moment-arm construction, and Figure 2.8 shows the biceps femoris comparison.

2.5 Neural Control of Locomotion: Central Pattern Generators

Locomotion in mammals is driven by spinal central pattern generators (CPGs), neural circuits that can produce alternating rhythmic activity of flexor and extensor motoneurons in the absence of rhythmic input and proprioceptive feedback [Brown, 1911, Rybak et al., 2006]. The modern architectural understanding of mammalian locomotor CPGs comes largely from the modeling work of the Rybak group and collaborators. Rybak et al. [2006] proposed that the locomotor CPG consists of two levels: a half-center rhythm generator (RG) and a pattern formation (PF) network, with reciprocal inhibitory interactions between antagonist neural populations at each level. This two-level organization explains, among other phenomena, spontaneous deletions of motoneuron activity during fictive locomotion in which the locomotor period and phase are maintained, and it permits independent control of the level of motoneuron activity and of step-cycle timing. McCrea and Rybak [2008] reviewed the electrophysiological support for this organization, and Rybak et al. [2015] related the model architecture to genetically identified spinal interneuron classes.

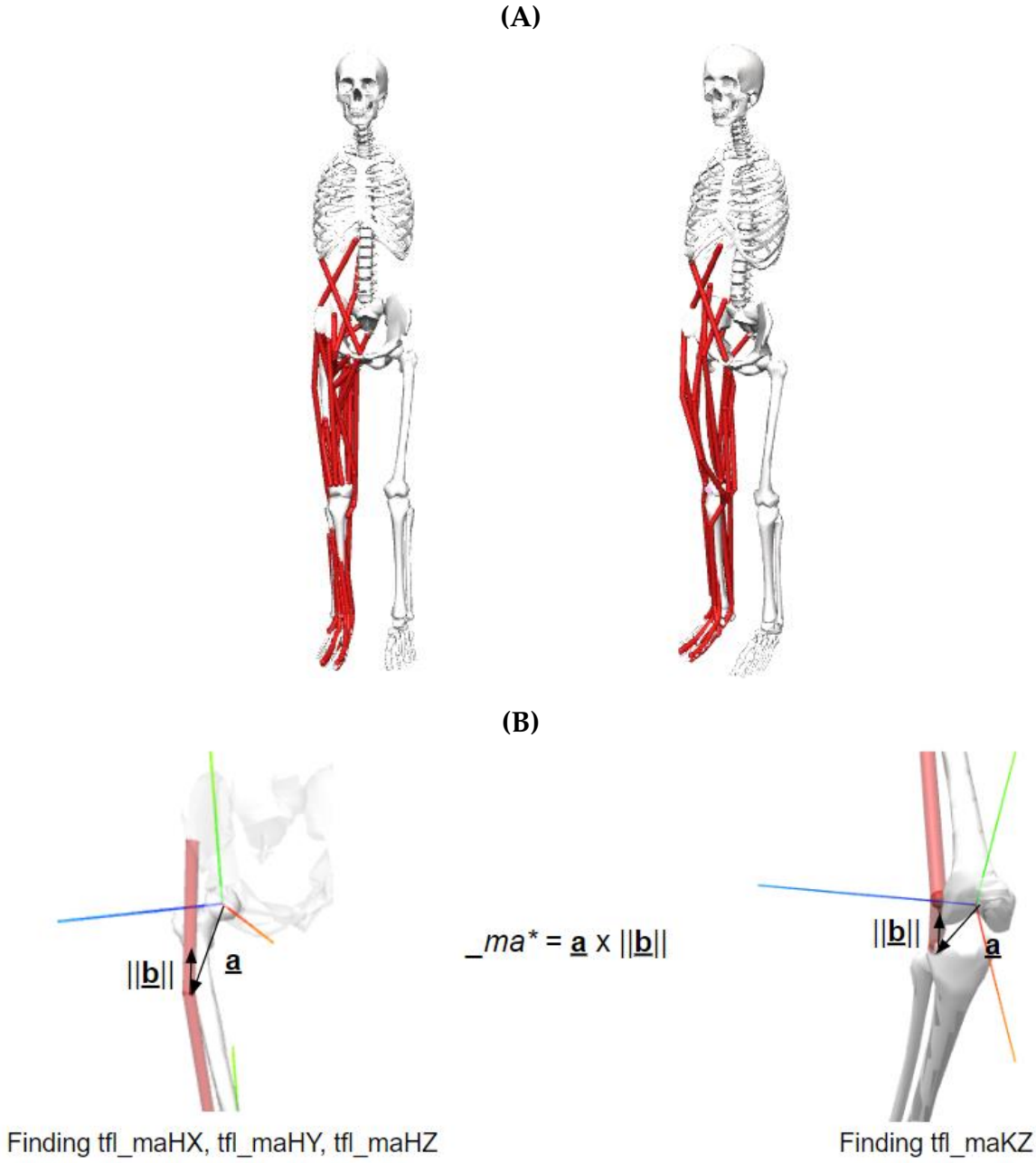


Figure 2.7: From the author's master's study. **(A)** Muscle-path models: OpenSim Gait2392 paths (left) and the robot leg's PAM routings (right). **(B)** Moment arms computed in the body frame as the cross product of the joint-axis unit vector with the vector from the joint center to the muscle's line of action, shown for the tensor fasciae latae about the hip (left) and the knee (right) [Bolen, 2019, Figs. 4 and 6].

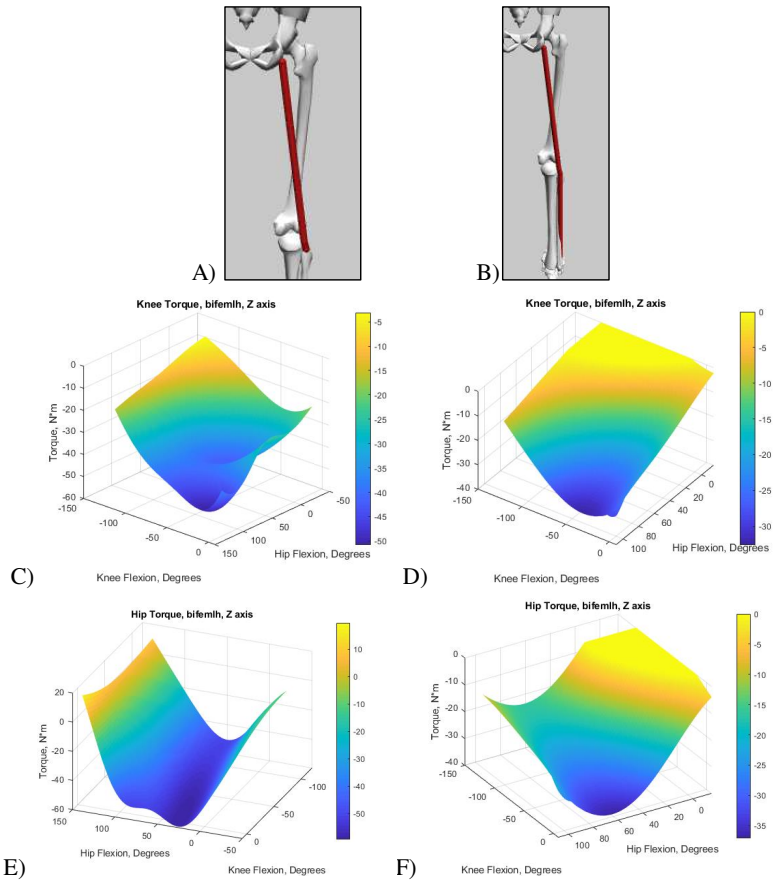


Figure 2.8: Biceps femoris long head in the master's study: muscle paths in the OpenSim model (A, C, E) and the robot model (B, D, F), with isometric torque surfaces about the knee (C, D) and the hip axis (E, F) over the hip–knee range [Bolen, 2019, Fig. 8].

Left–right coordination is provided by commissural pathways between the two half-centers [Molkov et al., 2015, Danner et al., 2016], and the rhythmogenic core of the CPG continues to be refined experimentally [Rancic and Gosgnach, 2021]. How this architecture is expressed as simulable circuit models is the subject of Section 2.7; the locomotor controller developed in Chapter 6 is built directly on this two-level organization.

2.6 Sensory Feedback in Locomotion

Sensory feedback is not a passive correction layer but an integral part of locomotor pattern generation. Reviews of afferent control of the CPG [Van de Crommert et al., 1998] and of muscle proprioceptive feedback and spinal networks [Windhorst, 2007, 2022] summarize the classical picture: group Ia afferents from muscle spindles signal muscle length and velocity, group Ib afferents from Golgi tendon organs signal force, and group II afferents signal length, with load-related feedback playing a role distinct from length-related feedback during locomotion [Dietz and Duysens, 2000]. Removing proprioceptive feedback degrades locomotor patterns in mice [Akay et al., 2014]; proprioceptive input regulates and adapts locomotor activity [Lam and Pearson, 2002]; and the relative contributions of proprioceptive and vestibular systems to locomotion are the subject of ongoing study [Akay and Murray, 2021, Akay, 2020]. Of particular relevance to the tests planned in Chapter 7, tonic central and sensory stimuli have been shown to facilitate involuntary air-stepping in humans [Selionov et al., 2009], demonstrating that simple graded stimulation of motoneuronal and rhythmic circuits can elicit stepping-like output, an experiment readily reproduced in simulation by tonic stimulation of pattern-formation and rhythm-generator

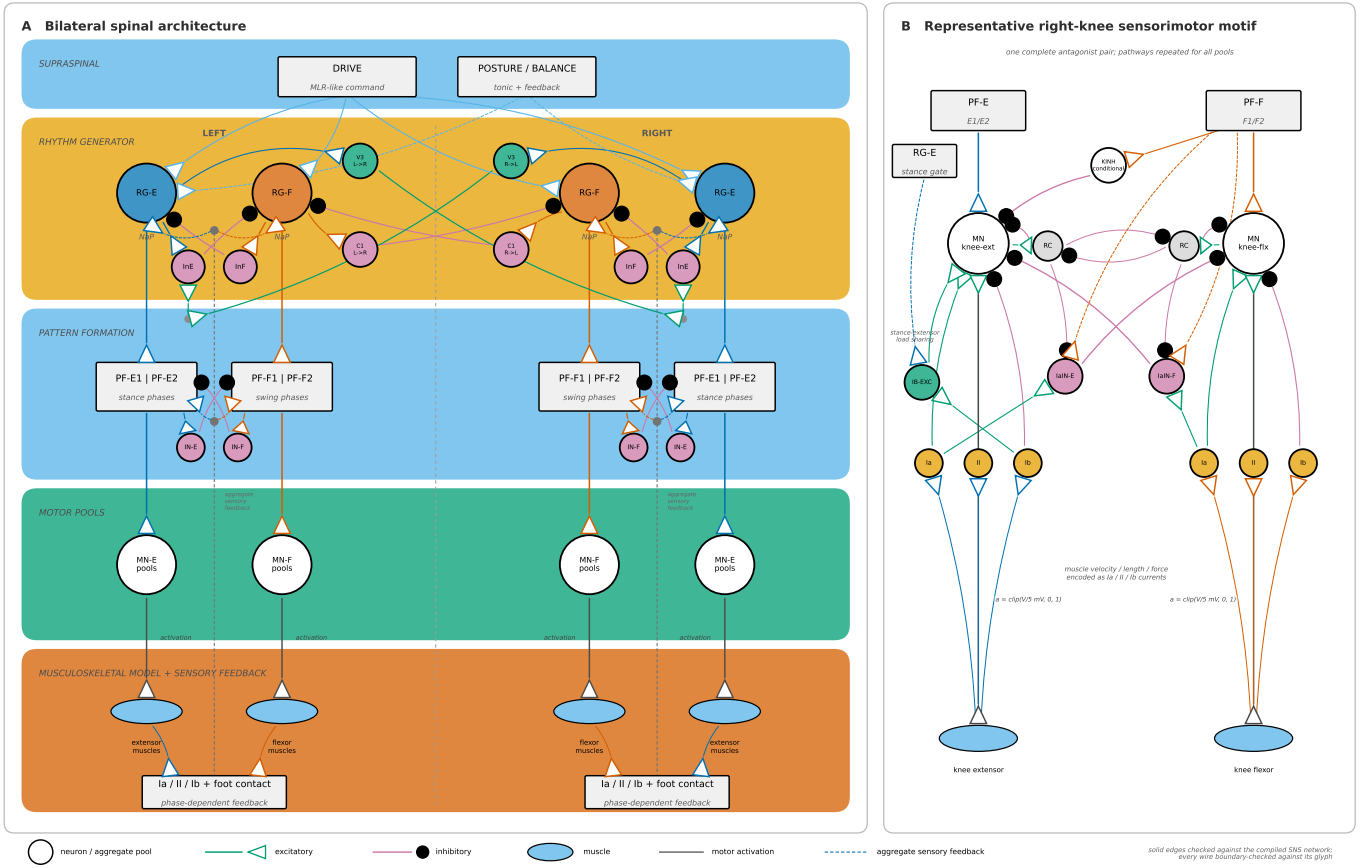


Figure 2.9: Locomotor circuit architecture synthesized from the literature. The bilateral organization combines rhythm-generation, pattern-formation, motoneuron, commissural, and proprioceptive pathways used across two-level CPG and neuromechanical models [Rybak et al., 2006, 2015, Shevtsova et al., 2016, Deng et al., 2019]. The detailed knee-control motif makes the reciprocal antagonist and sensory pathways explicit.

populations. The proprioceptive and contact feedback channels designed in Chapter 6 follow this afferent organization.

Three published circuits, reproduced in Figures 2.10, 2.11, and 2.12, place these afferent channels in their circuit context. The rat hindlimb controller of Deng et al. [2019] is organized exactly as Section 2.5 describes: a rhythm-generator half-center and a pattern-formation layer sit above a sensory–motor network in which extensor and flexor muscle receptors feed back through interneurons and Renshaw cells onto the same motoneurons the CPG drives. The human-scale model of Di Russo et al. [2023] carries the idea to a full lower limb: each agonist–antagonist muscle pair carries Ia, II, and Ib reflex populations with Renshaw cells, and pattern-formation nodes from the locomotor controller terminate on the same motoneuron pools. The cat interlimb model of Shinohara et al. [2025] adds the left–right half of the problem, coupling the two rhythm generators through V3 and commissural interneuron populations while force, length, and velocity feedback returns from every muscle. The schematics differ in animal, scale, and emphasis, but they wire the same afferent classes of this section into the circuit at the level each hypothesis concerns, whether that is the rhythm generator, the pattern-formation layer, or the motor pool.

The two-level organization these circuits share descends from the modeling program of the Rybak group, and its foundational schematics are reproduced in Figures 2.13 and 2.14. In the concept model of Rybak et al. [2006], afferent feedback ascends to both the rhythm generator and the pattern-formation network, so a sensory perturbation can reset the rhythm at the generator or delete motoneuron activity at the pattern level without necessarily doing both, precisely the deletion phenomenology that motivated the two-level split. The extended model of Shevtsova et al. [2016] carries the same organization to

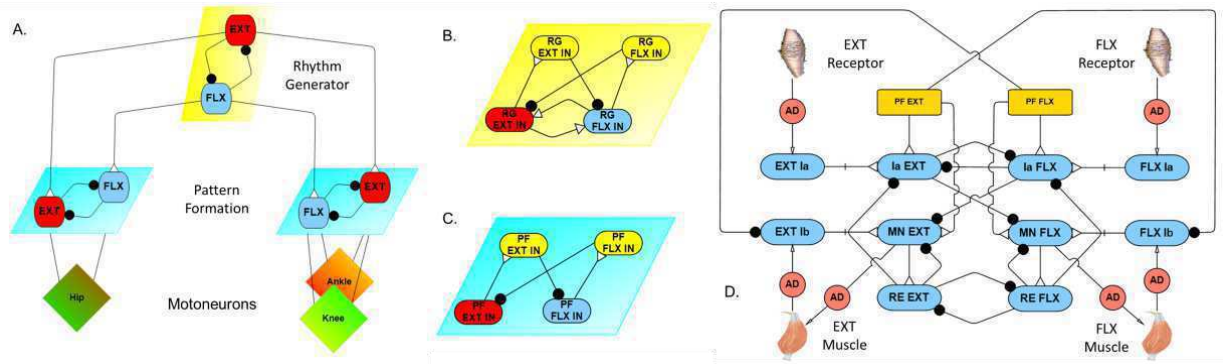


Figure 2.10: Synthetic nervous system of the rat hindlimb model: **(A)** single limb control hierarchy, **(B)** rhythm-generator configuration, **(C)** pattern-formation network, and **(D)** sensory–motor network. Reproduced from Deng et al. [2019, Fig. 4] under CC BY 4.0.

the two-joint muscles, embedding dedicated pattern-formation populations for the bi-functional posterior biceps–semitendinosus and rectus femoris pools, and that biarticular convergence problem is the one a robot muscle routed across two joints presents to the controllers developed in this dissertation.

2.7 Computational Neuroscience of Locomotor Circuits

Computational neuroscience expresses hypotheses about locomotor circuitry as executable models. A locomotor model is a system of coupled differential equations in which every element of the schematic corresponds to an identified neural population or pathway; the wiring diagram is not an illustration of the model, it is the model. The founding hypothesis is Brown's half-center: alternating flexor and extensor activity arises from two reciprocally inhibitory central populations with opposing bias [Brown, 1911]. Modern instantiations of the half-center endow each neuron with persistent sodium channels, whose slow inactivation terminates each burst so that reciprocal inhibition switches the

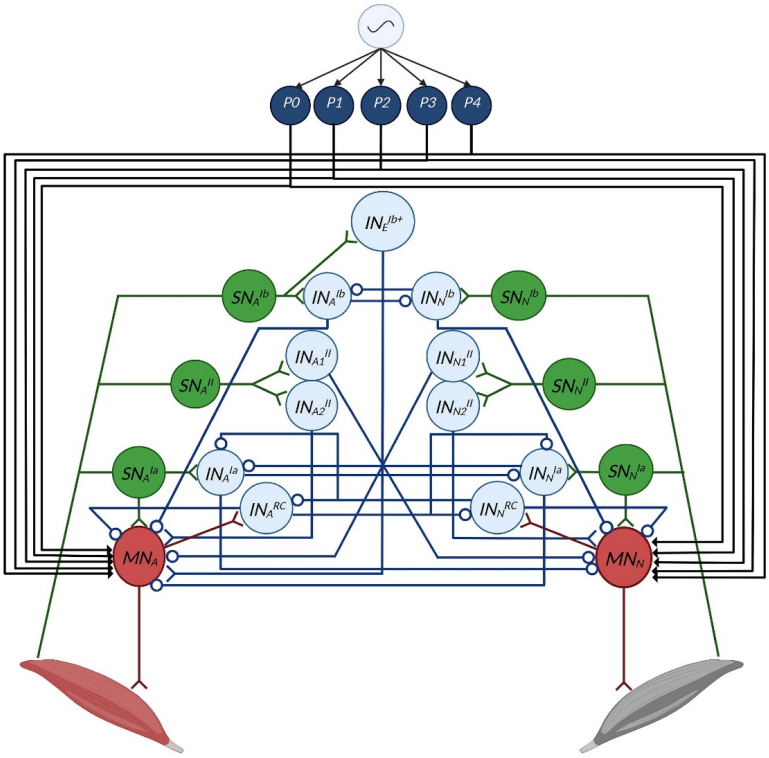


Figure 2.11: Spinal network between a muscle and its antagonist in the human locomotion model of Di Russo et al. [2023]: Ia, II, and Ib reflex populations with Renshaw cells, and pattern input from the locomotor controller onto the same motoneuron pools. Reproduced from Di Russo et al. [2023, Fig. 5] under CC BY 4.0.

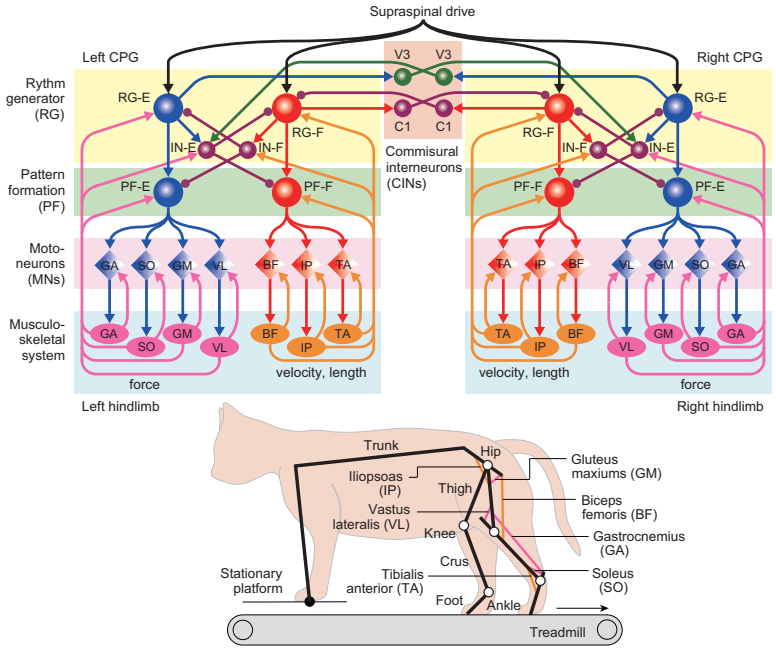


Figure 2.12: Neuromusculoskeletal cat model of Shinohara et al. [2025]: bilateral rhythm-generator and pattern-formation layers coupled by V3 and commissural interneurons, motoneuron pools for seven muscles per hindlimb, and force, length, and velocity afferent feedback. Reproduced from Shinohara et al. [2025, Fig. 1] under CC BY 4.0.

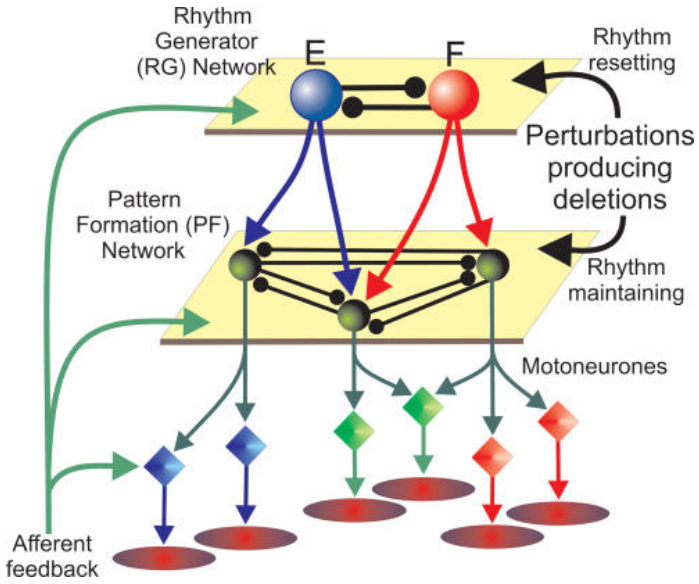


Figure 2.13: Two-level concept of the locomotor central pattern generator: a half-center rhythm generator (RG) sets the locomotor rhythm above a pattern-formation (PF) network whose interneuron populations drive synergist motoneuron pools, and afferent feedback ascends to both levels. Reproduced from Rybak et al. [2006, Fig. 1].

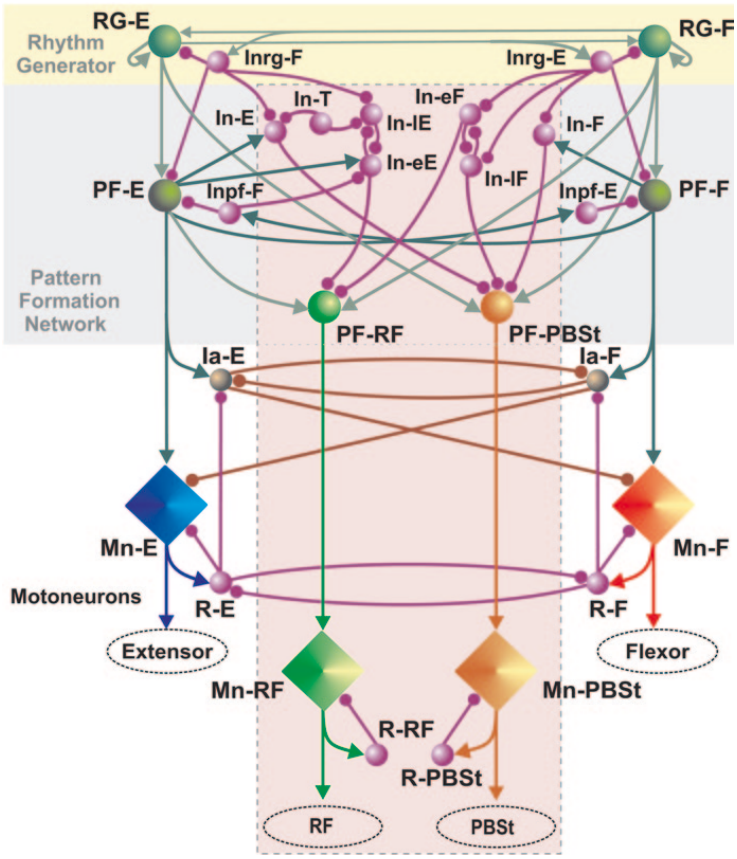


Figure 2.14: Extended model of the locomotor CPG of Shevtsova et al. [2016]: the rhythm generator and pattern-formation network of the basic model operate above extensor and flexor motoneuron and Renshaw populations, augmented (outlined) by pattern-formation circuitry and bifunctional motoneuron pools for the two-joint muscles posterior biceps–semitendinosus (PBSt) and rectus femoris (RF). Reproduced from Shevtsova et al. [2016, Fig. 5.8].

populations without any phasic input [Deng et al., 2019]. Rhythmogenesis, in this tradition, is a property of the intrinsic membrane dynamics, not of a clock elsewhere in the network.

The two-level organization described in Section 2.5 is the field's current working architecture, and its models account for specific electrophysiological observations: the fictive-locomotion deletions reproduced by Rybak et al. [2006], the genetically identified interneuron classes mapped onto the model by Rybak et al. [2015], and the organization of circuits controlling two-joint muscles reviewed by Shevtsova et al. [2016]. Coupling such circuits to musculoskeletal models shows how afferent feedback shapes CPG output [Markin et al., 2010] and how biomechanics and sensory feedback regularize the behavior of different CPG architectures [Deng et al., 2022]. Physiologically plausible models of human locomotion use the same machinery to dissect the relative contributions of reflexes and CPGs, with locomotor primitives, Ia, II, and Ib pathways, and Renshaw cells represented explicitly [Di Russo et al., 2023], and recent models address interlimb coordination under abrupt perturbations such as the sudden loss of ground support [Shinohara et al., 2025].

For a mechanical engineer, the productive way to read this literature is as a controls tradition in which the block diagram is the controller itself: each block is an identifiable neural population with conductance-based dynamics, and each connection is a synaptic conductance with a physiologically interpretable weight. The connection schematics published with these models are therefore worth reading as carefully as their equations. The controller developed in Chapter 6 comes from this tradition: its two-layer organization follows the rat hindlimb model of Deng et al. [2019], and its verification tests are drawn from the deletion and stimulation phenomena that the models of Rybak et al. [2006] and the air-stepping experiments of Selionov et al. [2009] explain.

2.8 Synthetic Nervous Systems

For robots, these biological architectures can be implemented as synthetic nervous systems (SNSs): networks of conductance-based spiking and non-spiking neurons that are small enough to run in real time yet biologically grounded enough to be compared against animal data. The approach has been pursued most persistently by the insect-biorobotics lineage of the Quinn laboratory at Case Western Reserve University and by laboratories closely connected to it, the author's own included. Works in this lineage span several platforms: CPG-based neural control of a pneumatically actuated dog robot [Hunt et al., 2017b], analytically designed dynamic neural networks for human balance control [Hilts et al., 2019], the full-muscle rat hindlimb model and its moment-arm analysis [Young et al., 2019], the two-layer CPG neuromechanical walker built on it [Deng et al., 2019], which supplies this dissertation's controller architecture, and the Muscle Mutt quadruped, a canine-inspired BPA robot designed at the author's laboratory, the Agile and Adaptive Robotics Laboratory (AARL) at Portland State University, for synthetic nervous system control [Scharzenberger, 2019]. Related controllers from the wider neuromechanical community include sensory-entrained CPG control of bipedal walking [Li et al., 2017], inter-leg coordination analyses in stick-insect-inspired controllers [Nourse et al., 2019], reinforcement-reflex hybrids in insects [Goldsmith et al., 2021], and adaptive hindlimb walking controllers [Schilling et al., 2022].

The SNS-Toolbox [Nourse et al., 2023] makes such networks practical: it is an open-source Python package that simulates hundreds to thousands of spiking and non-spiking neurons in real time or faster, supports multiple software and hardware backends including

GPUs and embedded platforms, and couples directly to physics simulators such as MuJoCo for musculoskeletal control. The field it serves is active and growing: the perspective review of Szczecinski et al. [2023] traces neuromorphic legged-locomotion control from its insect-robot beginnings through current platforms to open problems, and physical testbeds in this lineage now extend from insect-scale robots to a robotic cat hindlimb [Armstead et al., 2025].

The boundary between background and contribution runs through this section as well. The prior AnimatLab walker with which the author's program first applied these architectures to a biped [Bolen and Morrow, 2020] is presented as prior work of this dissertation in Section 6.1, the Sensory Afferent Database that organizes the afferent data such networks require [Bolen et al., 2023] is the subject of Chapter 5, and the simulation pipeline that couples the controller to the robot model is built in Chapter 6. None of it is repeated as background here.

2.9 Neuromechanical Simulation

Neuromechanical simulation, the coupling of a musculoskeletal model to a model of the neural controller that drives it, has advanced along three partially overlapping lines. The first line embeds reflex physiology directly into the controller. The muscle-reflex walking model of Geyer and Herr [2010] demonstrated that positive force, length, and velocity feedback around each muscle, organized with simple interlimb coordination, is sufficient to produce self-stabilizing human-like walking dynamics and realistic muscle activations without any central pattern generator. That model became the reference point

for a generation of reflex-based neuromechanical controllers, including the cat hindlimb models of Section 2.7 [Markin et al., 2010, Shevtsova et al., 2016] and their extensions. Hybrid work makes the complementarity of the two strategies explicit: Haeufle et al. [2018] combined feed-forward muscle activations with neuronal reflex feedback in simulated human walking and found that the robustness benefit of each depends on which muscle carries it.

The second line replaces hand-designed circuitry with machine learning. Song et al. [2021] trained deep reinforcement learning policies on a physiologically realistic musculoskeletal model in OpenSim, reproducing human kinematics and muscle activations across locomotion tasks, with the learned policy re-usable for experiments that would be difficult in the laboratory. This approach inherits the fidelity of the musculoskeletal model but encodes control in opaque function approximations: the trained policy can reproduce behavior, but the individual "neurons" of the controller do not correspond to identifiable physiological circuits, so the class of scientific questions it can answer, such as lesions, deletions, and pathway stimulation, is narrower than its behavioral repertoire. Recent perspectives on predictive musculoskeletal simulation [De Groote and Falisse, 2021] place both learned and optimal-control approaches in the same frame: the models are mature, and the open problems are the cost of solving them and the biological interpretability of the control they find.

The third line, in which this dissertation sits, is biologically grounded synthetic nervous systems: controllers whose neurons and synapses are explicit, conductance-based, and mapped to identified physiological circuits, simulated at speeds sufficient for real-time robotics [Nourse et al., 2023]. Hierarchical CPG control is meanwhile spreading

to bipedal-capable musculoskeletal models, including motor-module bipedal walkers [Ichimura and Yamazaki, 2019], two-layer rhythm-generator and pattern-formation control of a musculoskeletal macaque model [Nishizaki et al., 2026], and two-level CPG control of muscle-driven robotic hind legs [Fukuoka et al., 2021], though always in simulation with generic musculature or on non-humanoid platforms; a two-layer SNS driving a custom-anatomy, artificial-muscle humanoid leg remains unclaimed. What distinguishes the present program from all three lines is the combination it attempts: an anatomically grounded controller (two-layer CPG with Ia, Ib, II, and contact feedback), a biomechanically validated pneumatic model (Chapter 4), and an eventual physical robot.

2.9.1 Simulation Tooling

All three lines depend on a software substrate that is itself part of the method: musculoskeletal geometry too elaborate to integrate by hand, joined to neural models too large to wire by hand. The tools used in this dissertation, and the ones adjacent to it in the same ecosystem, are the following.

OpenSim [Delp et al., 2007, Seth et al., 2018], distributed through the SimTK (Simulation Toolkit) repository, provides the biomechanical modeling side: musculoskeletal models with detailed bone geometry, muscle routing, and force-generating properties, along with tools for analyzing moment arms and isometric torque. These models are approximations with known limits rather than exact anatomies: muscle paths are interpolations over discrete attachment and wrapping data, and near the ends of a joint's range the routed paths of sparsely wrapped muscles can bunch up or pass through bone, an interference

documented within Gait2392-derived routings themselves [Bolen, 2019]. Their geometry is accurate for a purpose, namely moment arms and torque predictions over the functional range of motion, and that is the standard against which the robot model of this program is compared. The same SimTK infrastructure hosts SCONE, an open-source package for predictive simulation and optimization of neuromuscular control of biological motion [Geijtenbeek, 2019]. AnimatLab [Cofer et al., 2010] provides a graphical environment in which physical bodies and synthetic neural systems are constructed and co-simulated; the AARL's prior walker (Section 6.1) was built in it. MuJoCo [Todorov et al., 2012] provides the physics-engine side: fast, accurate simulation of rigid-body dynamics with contacts, along with muscle-like actuators that include activation dynamics. MyoConverter bridges the biomechanical and physics-engine worlds by converting OpenSim musculoskeletal models to MuJoCo format with optimized muscle kinematics and kinetics [Caggiano et al., 2022, Wang et al., 2022]. Combined with the SNS-Toolbox's MuJoCo backend [Nourse et al., 2023], I used these tools to convert the laboratory's modified OpenSim leg model to MuJoCo and couple it to a synthetic nervous system, a pipeline assembled and evaluated in Chapter 6. Simulation speed is the practical enabler: a single overnight run can sweep thousands of parameter combinations or execute hundreds of simulated gait cycles per condition, throughput that no human-subject or animal protocol could match, and that transforms lesion and stimulation studies from bespoke demonstrations into statistically powered experiments.

2.10 Testing Neuromechanical Hypotheses on Robots: The Ethical Case

The experiments this research program is designed to enable, including deleting a muscle from a limb, blocking a sensory pathway, lesioning part of a rhythm generator, and applying tonic stimulation to spinal circuits, cannot be performed on healthy humans. Invasive intervention on the human neuromuscular system is limited to clinical populations, is tightly constrained by ethics review, and permits neither the within-subject control nor the reversibility that neuromechanical hypotheses demand. The same experiments in animals also carry real moral cost, and their justification is governed by the Three Rs framework: replacement of animal subjects with non-animal methods, reduction of the numbers required, and refinement of the remaining procedures [Russell and Burch, 1959, Tannenbaum and Bennett, 2015]. A robot that implements a validated neuromechanical hypothesis is precisely such a replacement method: every deletion, lesion, and stimulation experiment that runs on the synthetic nervous system is an experiment that does not require a living subject, and the robot population can be duplicated exactly, so that "subjects" are statistically identical across conditions, a form of reduction no breeding colony can offer. Refinement follows as well: the failure modes of these experiments damage hardware, not tissue.

Figure 2.15 shows what that accounting leaves out. The image is a frame from footage of a real decerebrate-cat experiment, the class of preparation on which the field's understanding of spinal locomotion has been built since its beginnings [Brown, 1911, WhiteCoatWaste, n.d.]: a living cat, its forebrain disconnected by a brainstem transection, walks on the treadmill because the neurons distal to the laceration are still living, still connected, and

still firing action potentials. The published record of this work is drawn almost exclusively in schematic, labeled boxes for half-centers and color-coded reflex arcs that are honest about the circuitry and silent about the animal. The figure is included as the counterweight, documenting what the experiments looked like, and therefore what a replacement method stands in for.

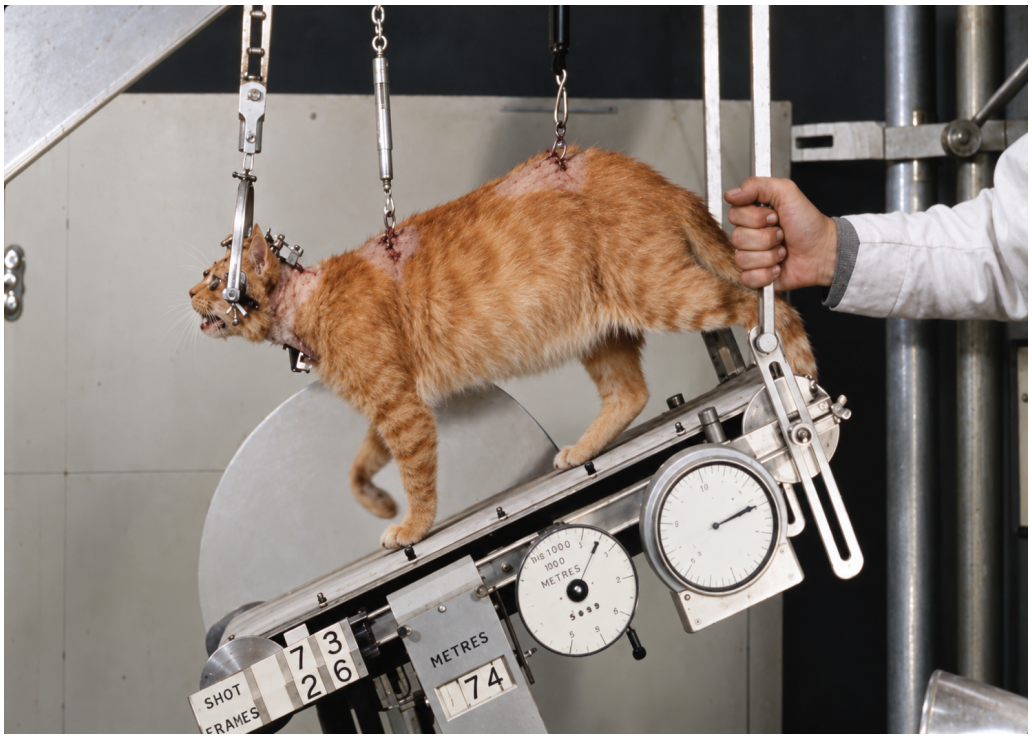


Figure 2.15: Frame from archival footage of a decerebrate-cat treadmill experiment [WhiteCoatWaste, n.d.], restored and machine colorized; color and fine detail are inferred by the restoration rather than recovered from the source. The cat walks on the treadmill belt while held in a stereotactic head holder whose fixation pin passes through or beside the eye orbit, with skull clamps and slung support completing the frame. Shown as documentary context for the replacement argument of this section, not as gait evidence.

The procedures behind that record are documented in films that remain publicly available. A 50 min classroom recording posted by the UTHealth Houston School of

Dentistry first shows an intact cat playing with its handler, righting itself when lowered and placing its paws accurately on the edge of a table, and then documents the decerebration performed for the class [UTHealth Houston School of Dentistry, 2025]: the animal is anesthetized with open-drop ether, more than the horse in the same demonstration series received, the carotid arteries are ligated and a tracheal cannula installed, and the skull is opened with a trephine before the brainstem is transected between the superior and inferior colliculi. The treadmill preparations that established locomotor neurophysiology are likewise preserved in the historical literature: Stuart [2007] reproduce the arrangement of the high-decerebrate cat treadmill experiments, with electromyographic recording from the hindlimb muscles, from the original Russian-language reports [Severin et al., 1967].

The honest accounting is murkier than the Three Rs framing alone suggests, because the models a robot implements are themselves products of animal experimentation. The half-center organization, the two-level rhythm and pattern architecture, and the length, force, and load signaling of the afferents were each established in preparations like the ones described above [Brown, 1911, Severin et al., 1967, Rybak et al., 2006], and the databases of afferent properties that parameterize sensory feedback are compiled from animal recordings [Bolen et al., 2023]. A synthetic nervous system therefore replaces future experiments; it does not replace, and cannot repay, the animal work on which its architecture is built. Recognizing this inheritance sharpens rather than undermines the case for robotic subjects: the knowledge purchased at that cost is precisely what makes a robotic replacement scientifically meaningful, and the robot is the instrument that keeps that knowledge productive without adding to the ledger.

For a robot to serve as a replacement subject, however, it must first earn standing as

a scientific model. Webb's influential analyses of bio-robotic methodology [Webb, 2001, 2002] argue that a robot is a good model of biological behavior when the mechanism it implements is tested at the same level of organization as the target phenomenon, matching dynamics, control structure, and sensory ecology rather than surface appearance. The validation chain of this dissertation is constructed to that standard: the actuator force model is validated against bench data (Section 3.2.1), the joint torque model against isometric measurements and human torque envelopes (Section 4.2), and the neuromechanical controller is verified against published tonic stimulation and deletion phenomena [Selionov et al., 2009, Rybak et al., 2006], the verification step that precedes locomotion tuning in this program. Only then can a robot experiment be read as evidence about the hypothesized biological circuit rather than about the robot.

The argument is ultimately bidirectional, completing the loop described in Section 1.2. Experiments that are unethical, impractical, or statistically unreachable in humans and animals become routine on the robot; the resulting insights into neuromechanical control feed back into biology, into the design of the next robot iteration, and directly into prosthetic and orthotic devices that face the same problem of controlling muscle-like actuators for human users, whose central pattern generator controllers are an active topic in lower-limb exoskeleton research [Duan et al., 2024, Pery et al., 2026]. The ethical case and the scientific case are the same case: the robot is the instrument that makes these experiments possible.

3 Isometric Force Characterization of Braided Pneumatic Actuators

Publication note: Sections 3.1.1–3.1.2, Section 3.2.1, and Sections 3.3.1–3.3.5, with their associated figures and tables, appeared in Bolen et al. [2026] (Bolen, Elzein, Pang, and Hunt, *Actuators*, 2026). For that work, B.B. performed the conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, and original draft preparation; M.E. and L.P. contributed methodology, software, investigation, and data curation; and A.H. supervised the work, administered the project, and acquired funding. All co-authors contributed to writing, review, and editing.

This chapter develops and validates the actuator force models on which the rest of the dissertation builds, and it reads as one self-contained study, from methods through results to discussion. Sections 3.1.1 and 3.1.2 present the methods: the force equations used throughout this work, and the test jig and measurement procedure used to characterize Festo BPAs with 10 mm and 20 mm uninflated diameters over a wide range of resting lengths, contractions, and pressures. Section 3.2.1 reports the resulting maximum-force and normalized force–strain–pressure models and compares them with the manufacturer's tool. The remaining sections discuss the effects of resting length, the nondimensionalized force prediction, the comparison with other published BPA models, the implications for real-time control and simulation, and the analogy between these artificial muscles and biological muscle. The validated force model is the one used for the knee-torque predictions of Chapter 4 and for the BPA actuation of the neuromechanical simulations in

Chapter 6.

3.1 Methods

3.1.1 Actuator Force Model

The actuator force model used throughout this dissertation is described here. First, the amount of contraction ε in a BPA is calculated as

$$\varepsilon = \frac{(l_0 - l_{\mathrm{m}})}{l_0} \tag{3.1}$$

where l_0 is uninflated resting length of the BPA measured from hose clamp to hose clamp. Current muscle length is l_{m}.

Similarly, ε_{620} is the amount of contraction in a BPA without external load when inflated at 620 kPa, calculated as

$$\varepsilon_{620} = \frac{(l_0 - l_{620})}{l_0} \tag{3.2}$$

where l_{620} is the muscle length measured at 620 kPa and free from external loads.

The amount of relative contraction of a BPA ε^* can then be calculated by dividing equation (3.1) by (3.2), such that

$$\varepsilon^* = \frac{\varepsilon}{\varepsilon_{620}} \tag{3.3}$$

describes the BPA's current contraction as a normalized percentage of its total.

The amount of normalized isometric force F^* in a Festo BPA is then calculated as

$$F^*(\varepsilon^*, P^*) = \begin{cases} c_0 \cdot \left(\exp(-c_1 \cdot \varepsilon^*) - 1\right) + P^* \cdot \exp\left(-c_2(\varepsilon^*)^2\right) & \text{for } F^* > 0 \\ 0 & \text{for } F^* \leqslant 0 \end{cases} \tag{3.4}$$

where P^* is pressure P normalized by 620 kPa. BPA diameter dependent coefficients are determined by the actuator characterization experiments reported in Section 3.2.1 (Table 3.2).

The maximum force F_{max} in a BPA is calculated as

$$F_{max}(l_0, P_{\max}) = a_1 \cdot P \cdot \arctan\left(a_2 \cdot P \cdot (l_0 - 0.0075)\right) \tag{3.5}$$

with coefficients determined by the actuator characterization experiments reported in Section 3.2.1 (Table 3.1). When maximum system pressure is set to $P = 620\,\text{kPa}$, $F_{\max}(l_0, 620) = F_{620}(l_0)$. Multiplying (3.4) by (3.5) yields

$$F(\varepsilon^*, P^*, l_0) = F^*(\varepsilon^*, P^*) \cdot F_{620}(l_0) \tag{3.6}$$

as the equation for isometric force F in a Festo BPA.

3.1.2 BPA Isometric Force Characterization Experiment

To measure artificial muscle force as a function of length and pressure, we built a test jig from extruded aluminum and 3D printed parts. Festo BPAs with 10 mm and 20 mm diameter were tested with various resting lengths and different amounts of contraction.

These data are used to develop an improved model that more accurately predicts maximum muscle force based on l_0, current muscle length l_m, and pressure P. The results are compared with existing models and the new data are used to create an updated model.

A test jig was made of 80/20 brand 1515 series extruded aluminum (Figure 3.1). Artificial muscles were placed vertically in the jig one at a time. The upper end was attached to a load cell. Force data was collected using a CALT DYLY-103 100 kg S-shaped load cell for the ∅10 mm BPAs and a Loadstar RAS1-2HKS-S for the ∅20 mm BPAs. The load cell was used in conjunction with a HX711 Load Cell Amplifier. Pressure data was obtained from a Freescale MPX5700 GP 5 V pressure sensor. Building air supply pressure was controlled with two pressure regulators in series. The first was a Parker model 20R113GC $0 - 120$ psi pressure regulator. The second was a Husky 3/8 in. High Performance Air Regulator HDA72200. A Festo VTUG-10-MRCR-S1T-26V20-T516LA-UL-532S-8K valve manifold VTUG-G was used to deliver air from the pressure regulator to the actuator. This manifold is comprised of eight two-in-one bidirectional normally closed Festo VUVG-S10-T32C-AZT-M5-1T1L valves.

Pressure and load cell amplifier data were sent to MATLAB 2021b via an Arduino Uno style Sparkfun BlackBoard C microcontroller. The computers that used MATLAB were running either Windows 10 or Windows 11. During phases when the Arduino was collecting force and pressure data to send to MATLAB, the Arduino would also trigger (via MATLAB) an Onsemi 2N4401 NPN transistor to make the valve manifold open or close the valve. For other data collection, the valve was manually opened and closed.

The inner distance between the hose clamps on each BPA was measured with a FANUC tape measure to determine the muscle's length l and resting length (l_0). This is how

Festo AG & Co. KG [2026] defines l_0, although in Hunt et al. [2017a] it was measured to also include end cap length. We then inflated each BPA to $P_{620} = 620\,\text{kPa}$, with one end allowed to move freely in the axial Degree of Freedom (DoF), and measured the length l_{620} to calculate maximum contraction at 620 kPa using Equation (3.2). The distance between the crossmembers was then controlled to obtain different amounts of contraction, the muscles were inflated to various pressures, and the contractile force was recorded at the pressure–contraction pairs. This was done for ∅10 mm BPA resting lengths (l_0) of 112 mm–518 mm. For ∅20 mm BPAs, resting lengths (l_0) of 300 mm–509 mm were used.

3.2 Results

3.2.1 BPA Force Characterization

This section reports the results of the isometric force characterization experiments described in Section 3.1.2. First, maximum isometric force at 620 kPa is modeled as a function of resting length. The remaining data are then normalized to build a predictive force surface, and the resulting model is compared with the manufacturer's tool and with other published BPA models.

Maximum Force at 620 kPa

Data of the maximum force from BPA characterization tests of the 10 mm and 20 mm diameters at 620 kPa show a dependency on resting length (Figure 3.2). This is a previously unreported characteristic of these artificial muscles. Detailed analysis of the Festo

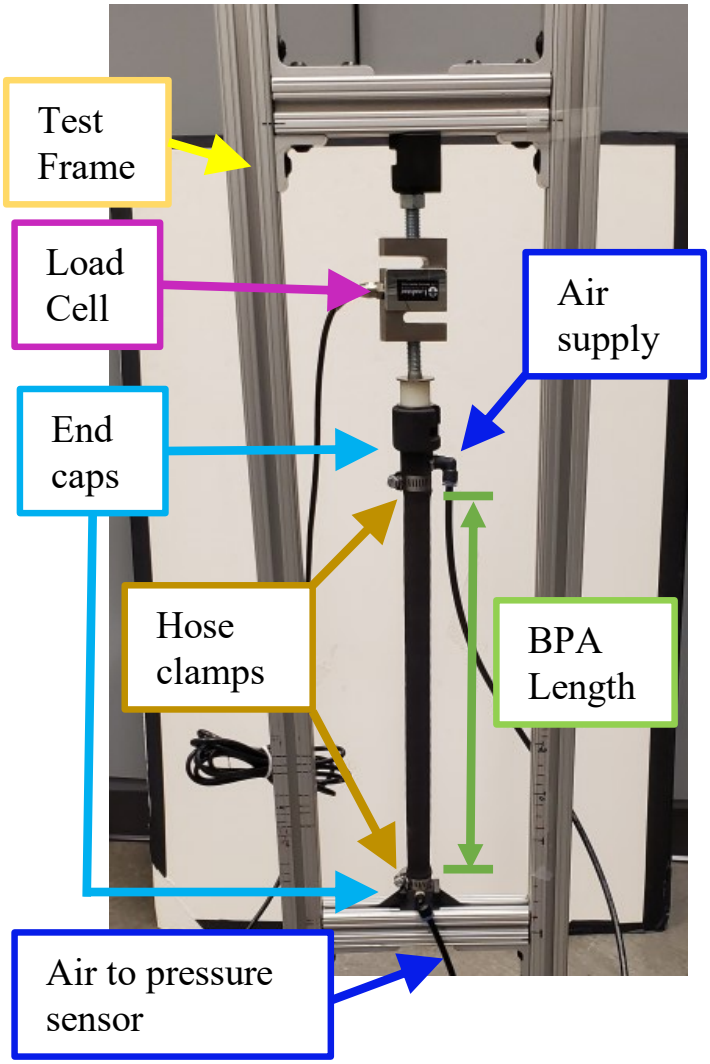


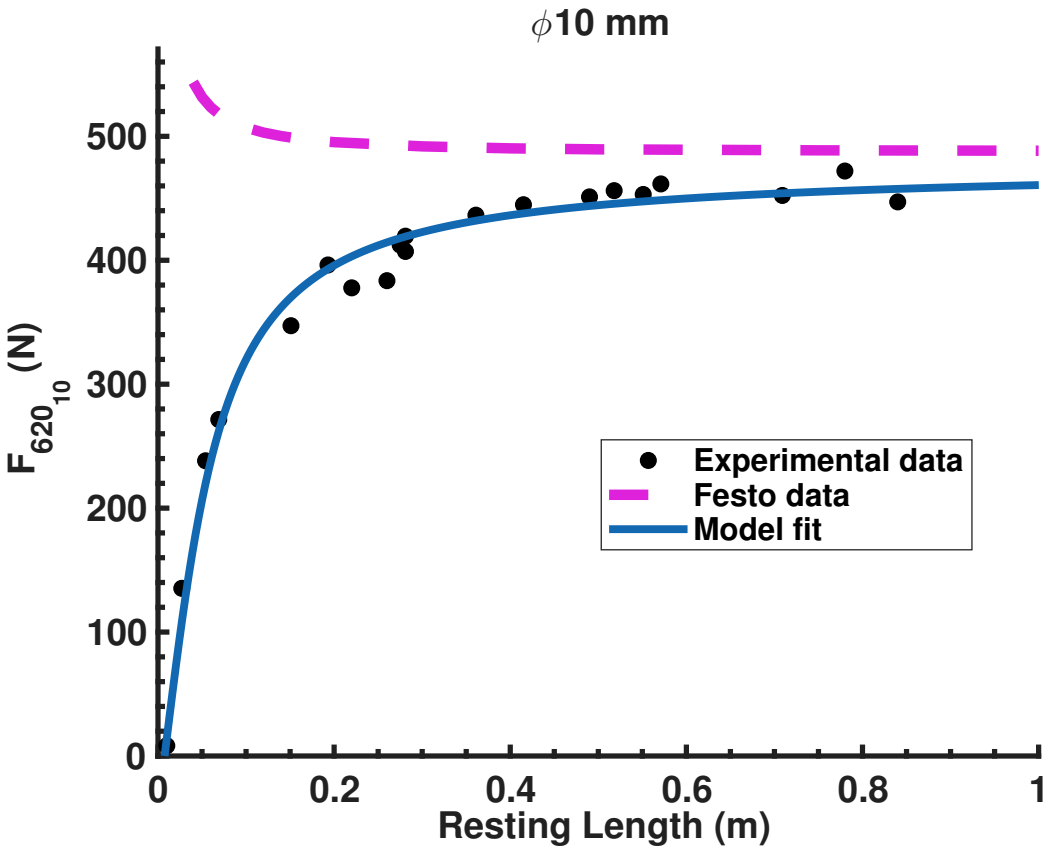
Figure 3.1: A picture of a BPA in the isometric force test stand with components labeled.

tool [Festo AG & Co. KG, 2026] does in fact predict a change in maximum force with the resting length; however, the Festo tool predicts increased force with shorter lengths, while our collected data indicate decreasing force with shorter lengths. The data show a force response resembling an arctan curve along the l_0 dimension. Using the Nonlinear Least Squares method and a Least Absolute Residual robustness, we fit an arctan curve to the data to get the maximum force at 620 kPa given a resting length as:

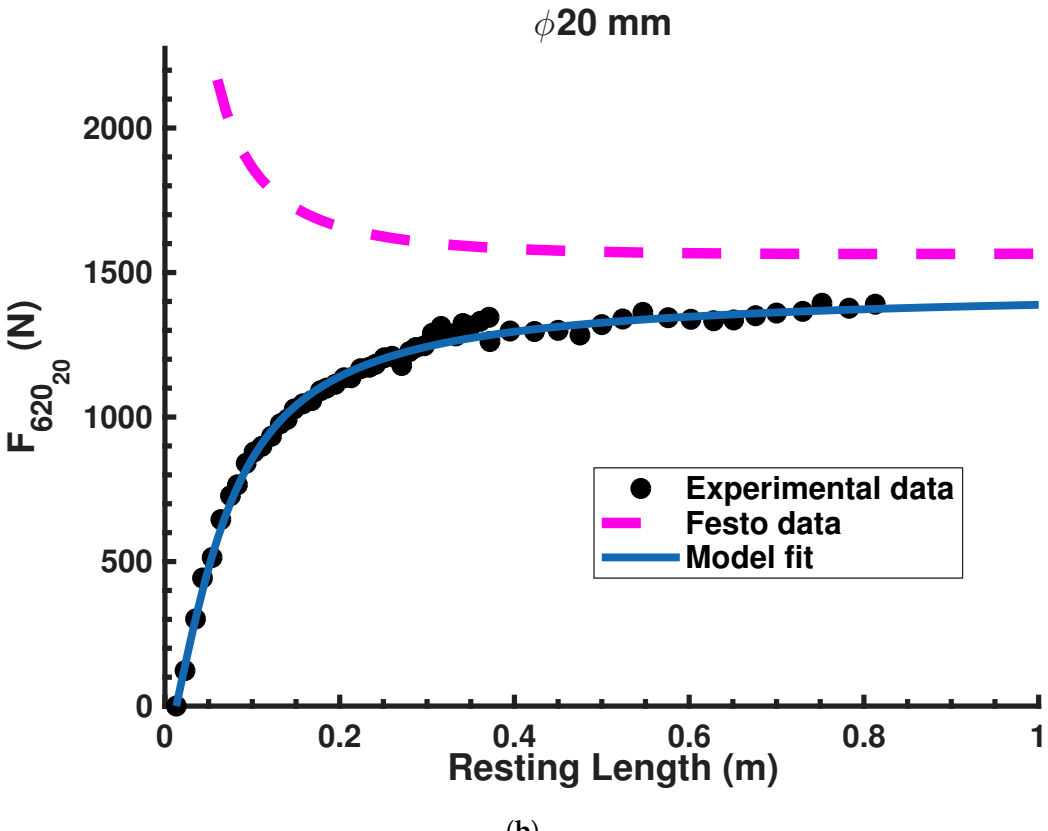
$$F_{620_{10}} = 303.5\,\mathrm{N} \cdot \arctan\left(19.03\,\mathrm{m}^{-1} \cdot (l_0 - 0.0075)\right) \tag{3.7}$$

$$F_{620_{20}} = 922.4\,\mathrm{N} \cdot \arctan\left(15.37\,\mathrm{m}^{-1} \cdot (l_0 - 0.013)\right) \tag{3.8}$$

The length is offset by 0.0075 m and 0.013 m because solid modeling showed that the end caps contact each other at these lengths. At these lengths, the actuator would not be able to contract to produce force.


ϕ10 mm
500
400
300
200
100
0
$F_{620_{10}}$ (N)
0
0.2
0.4
0.6
0.8
1
Resting Length (m)
Experimental data
Festo data
Model fit

(a)


ϕ20 mm
2000
1500
1000
500
0
$F_{620_{20}}$ (N)
0
0.2
0.4
0.6
0.8
1
Resting Length (m)
Experimental data
Festo data
Model fit

(b)

Force as a Function of Pressure and Resting Length

Data from BPA characterization tests of the 10 mm muscles resulted in force–pressure pairings for different muscle resting lengths (l_0) (Figure 3.3a). Similar to the maximum force data, these data show a force response resembling an arctan curve along the l_0 dimension with a more linear response to changes in pressure. Using the Nonlinear Least Squares method and a Least Absolute Residual robustness, we fit an arctan curve to the data to get the maximum force given a resting length and pressure as described by the model form given in Section 3.1.1, where $a_1 = 0.4895\,\mathrm{N\,kPa^{-1}}$ and $a_2 = 0.030\,68\,\mathrm{kPa^{-1}\,m^{-1}}$ for the 10 mm actuator, and $a_1 = 1.49\,\mathrm{N\,kPa^{-1}}$, $a_2 = 0.0248\,\mathrm{kPa^{-1}\,m^{-1}}$ for the 20 mm actuator. Goodness-of-fit measures are given in Table 3.1; throughout this dissertation, goodness of fit is reported as the root-mean-square error (RMSE), the adjusted fraction of variance unexplained (FVU, for which $R^2 = 1 - \mathrm{FVU}$), and the maximum absolute residual, as defined alongside the reference implementation in Appendix A.

Maximum Contraction at 620 kPa

Figure 3.3(b) shows an attempt at a linear fit for maximum contraction at 620 kPa (ϵ_{620}) as a function of resting length (l_0). Error bars have been included to show the effect of ± 1 mm accuracy with measurements of l_0 and l_{620}. There was a large amount of variance in the data, with the linear fit giving an adjusted $\mathrm{R}^2 = 0.4124$ and an RMSE $= 0.0083$. Since there is no direct, predicable relationship between maximum contraction and l_0, this value should be recorded in each muscle used on a robot to best predict the force it may produce at different pressures and contraction.

Table 3.1: Maximum force equation coefficient values with confidence intervals (CI) and goodness-of-fit measures. Equation (3.5) is compared to data taken at 620 kPa. Equations (3.7) and (3.8) are compared against maximum force from using the Festo tool. It was not necessary to fit an adjusted R^2 in these cases.

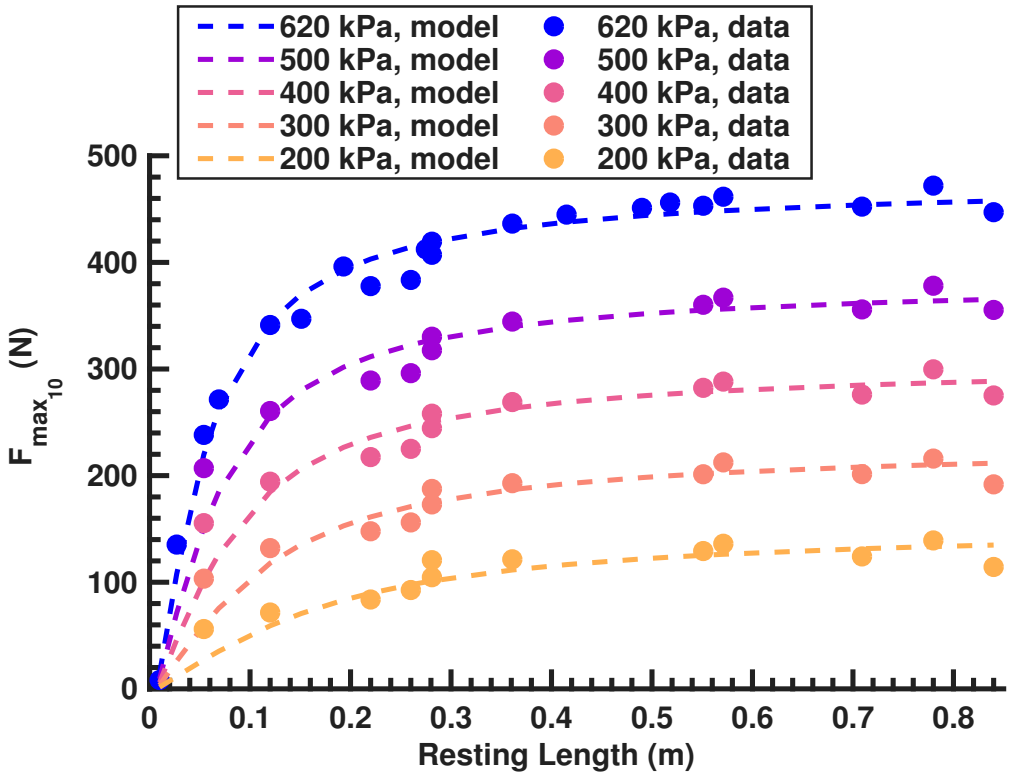
Equation	Coefficient	CI (95%)	Our Model			Festo Tool		
			Adj. R^2	RMSE	Max. Error	Adj. R^2	RMSE	Max. Error
(3.5)	$a_1 = 0.4895\,\mathrm{N\,kPa^{-1}}$ $a_2 = 0.030\,68\,\mathrm{kPa^{-1}\,m^{-1}}$	(0.4822, 0.4968) (0.0282, 0.03317)	0.9945	11.61 N	55.3 N	0.9854	14.7 N	30.9 N
(3.7)	$b_1 = 303.5\,\mathrm{N}$ $b_2 = 19.03\,\mathrm{m^{-1}}$	(300, 308) (17.48, 20.57)	0.9854	14.72 N	30.9 N	–	189.9 N	375.6 N
(3.8)	$b_1 = 922.4\,\mathrm{N}$ $b_2 = 15.37\,\mathrm{m^{-1}}$	(914.2, 930.7) (14.75, 15.98)	0.9945	23.83 N	62.1 N	–	668.4 N	1590.1 N

Table 3.2: Normalized isometric BPA force equation (Equation (3.4)) coefficient values and goodness-of-fit measures.

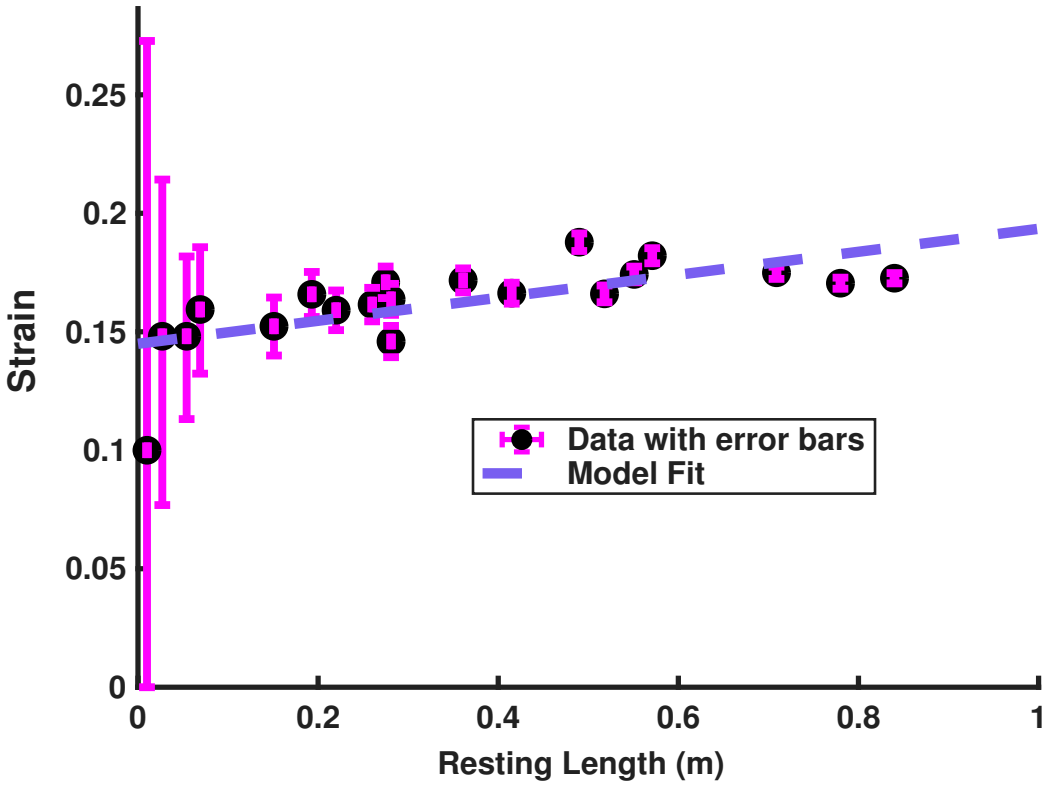
BPA	Coefficient	CI (95%)	Model			Validation		
			Adj. R^2	RMSE	Max. Error	Adj. R^2	RMSE	Max. Error
∅10 mm	$c_0 = 0.5682$	(0.5584, 0.578)	0.9998	0.005118	10.3%	0.9994	0.0245409	10.6%
	$c_1 = 4.254$	(4.126, 4.383)						
	$c_2 = 0.5597$	(0.5429, 0.5766)						
∅20 mm	$c_0 = 0.2579$	(0.2401, 0.2756)	0.992	0.02294	6.8%	0.9943	0.0231303	5.7%
	$c_1 = 6.477$	(5.558, 7.396)						
	$c_2 = 1.321$	(1.239, 1.403)						

Force as a Function of Pressure and Contraction

In addition to being a function of the pressure and resting length, the force produced by the actuator is also a function of the amount of actuator contraction, with less force being applied as the actuator contracts. To build this model, the collected data of force, pressure, and contraction are normalized by dividing by F_{620}, P_{620}, and ϵ_{620}, respectively. This had the effect of compressing the data into a 3D surface ranging from 0 to 1 on all axes. Normalized data are compared with pressure isoclines of force predicted by the Festo tool divided by the maximum force equation described in the previous section in Figure 3.4. With this comparison, it is clear that the Festo tool over-predicts the expected force, especially at lower pressures, contraction, and resting lengths.

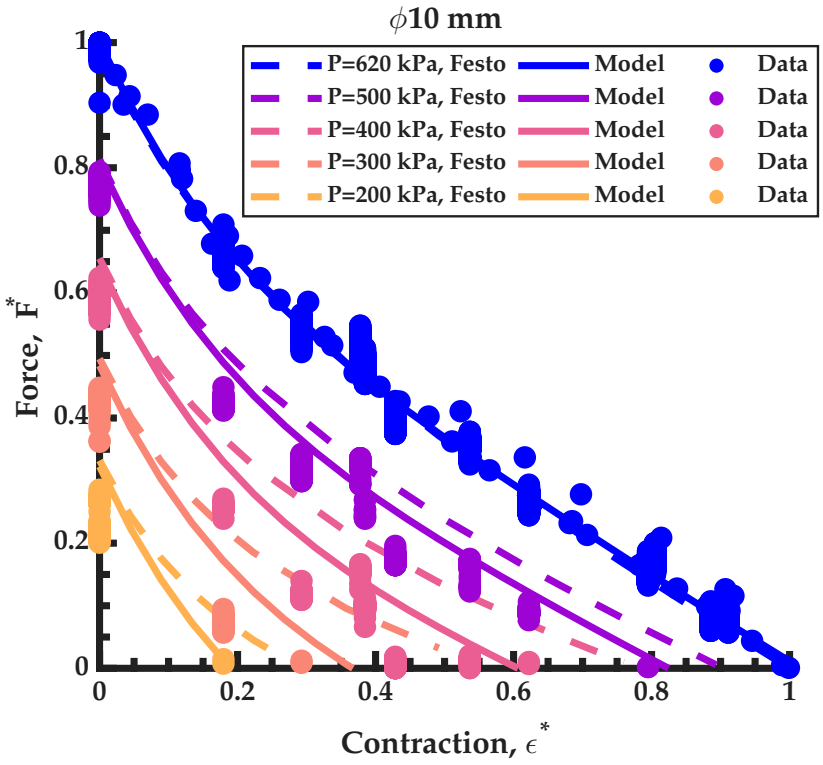


(a)

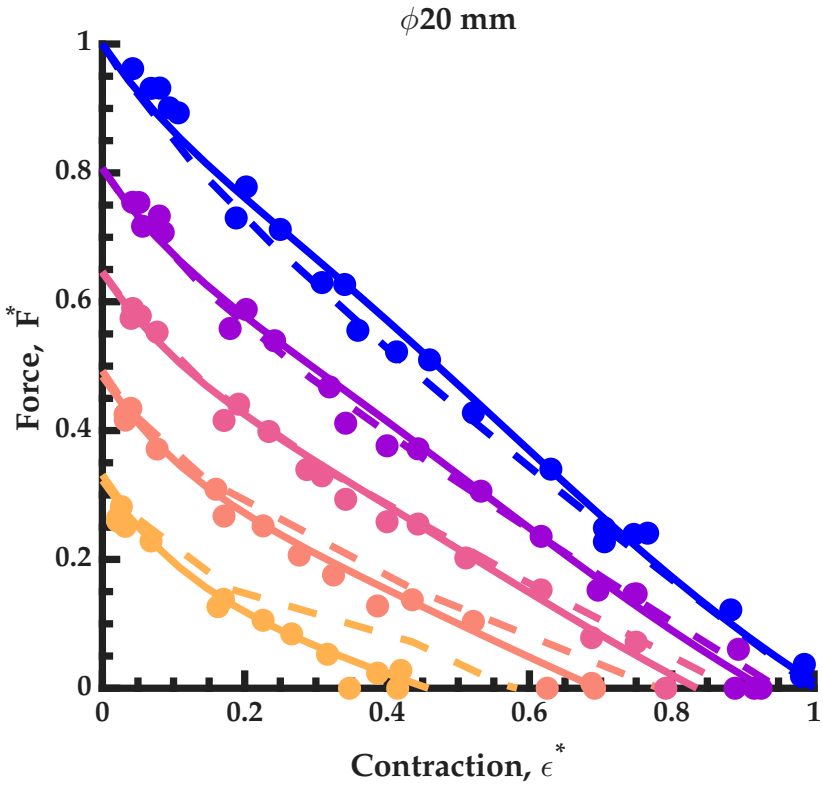


(b)

Figure 3.3: Results for finding the relationship between F_{620}, l_0, P_{620}, and $\epsilon_{620_{10}}$. (**a**) Isoclines



(a)



(b)

Figure 3.4: Surface fit for $F^*(\epsilon^*, P^*)$. Solid lines are from our model. Dashed lines from

We therefore derived our own equation for isometric force in the BPA as a function of pressure and contraction. Visual analysis of the experimental data shows an exponential relationship between ϵ^* and F^*, and a linear relationship between P^* and F. We fit a surface to the data using Nonlinear Least Squares and a Least Absolute Residual robustness such that it is described by Equation (3.4). For the ∅10 mm BPAs, the result of the improved fit can be seen in Figure 3.4a. The adjusted $R^2 = 0.9998$, a RMSE $= 0.004537$, and a maximum absolute residual of 10.6%. Solving (3.4) for the ∅20 mm BPA, yields different coefficients, and results are seen in Figure 3.4b. Coefficient values and goodness-of-fit statistics are found in Table 3.2.

3.3 Discussion

3.3.1 Effects of Resting Length

We have developed new F_{620} equations for 10 mm and 20 mm diameter Festo BPAs that capture the change in maximum force produced by the actuators at 620 kPa as a function of their resting length. When examining Equation (3.7) and measured data in Figure 3.2b, it can be seen that as l_0 goes to infinity, $F_{620_{10}}$ goes to 470 N. When the Festo tool [Lang, Robert, 2005] was queried at $l_0 = 1$ m as an approximation of infinity, it predicts the ∅10 mm BPA can produce $F = 490$ N at P_{620}. This is within the 10% variability that Festo states may occur in manufacturing tolerances. Similarly, the Festo data predict a ∅20 mm BPA will produce $F = 1570$ N. Our results for Equation (3.8) show that $F_{620_{20}}$ goes to 1460 N as l_0 approaches infinity. Although the difference is much larger, it is also within the 10% manufacturing tolerance.

Our measured data differ significantly from the Festo model as l_0 goes to zero. For the measured data, as the length gets smaller, the force does as well, whereas the Festo tool predicts exponential force increase (see Figure 3.2). As a specific example, when specifying $l_0 = 112\,\text{mm}$ for the 10 mm diameter BPAs, the Festo predicted F_{620} is 498.6 N. However, using Equation (3.7), F_{620} is calculated to be 335.3 N. Actual force in the $l_0 = 112\,\text{mm}$ was measured at 350.9 N. Therefore, the error in predicted F_{620} for the ∅10 mm BPA is 4.5% for our model and 42.1% using the Festo tool. Researchers using BPA resting lengths under 300 mm should take note of these results.

Festo AG & Co. KG [2026] (p. 12) states that force will be reduced by approximately 10% for a minimum nominal length. In Festo AG & Co. KG [2018] (p. 14) they say the force can deviate by 10% based on factors such as production variation, material variation, and deviation from nominal length (l_0). Festo AG & Co. KG [2013] (p. 13) notes that testing was done with BPAs that are ten times the uninflated diameter, and that theoretical force can increase by up to 10%. In each datasheet, the characteristic curves appear to be the same. As demonstrated above, the 100 mm ∅10 mm BPA should have much less force than what is shown in the curve. This is our experience and the reason for the testing. It is unclear why the MuscleSIM tool [Lang, Robert, 2005] then predicts higher force as resting length decreases. It could be a reversed sign coefficient in their internal curve-fit model. We have shown here that as the nominal length goes to zero, so does force.

BPA models commonly predict higher forces than are measured experimentally because they do not account for end effects in regions of the pressure vessel where those effects dominate. Nominally, the clamped ends fix the inner and outer diameter while allowing axial and radial displacement. In the isometric case, there is only radial displacement.

The boundary conditions force steep gradients in stress and strain near the ends. Near the boundary conditions, axial, radial, and circumferential strains deviate from analytic solutions for a cylindrical model. For a constant pressure, the end effect region appears to have the same length. Therefore, in shorter BPAs, the end effects dominate. The axial length scale of the boundary section is comparable to l_0, and the active force differs substantially from that of an assumed perfect cylinder.

Manufacturing variations seem to play an important role in the behavior of these actuators. It remains to be seen how ϵ_{620} can be known a priori. We still suspect that it is a function of l_0, but it might also be a function of other factors such as product batch number or UV light exposure [Festo AG & Co. KG, 2026] (p. 15); [Festo AG & Co. KG, 2018] (p. 6); [Festo AG & Co. KG, 2013] (p. 13). We found there to be larger variations in the 10 mm BPAs than in the 20 mm BPAs. Modeling work on ideal McKibben actuators describes how the relationship between contraction and initial length is a function of the total number of twists and the angle of twist in the usable muscle area [Chou and Hannaford, 1996]. It is unclear if there are variations in the angle and number of twists in the Festo BPAs due to manufacturing. Uncovering the relationship between ϵ_{620} and l_0 for the Festo DMSP artificial muscles, if it exists in a meaningful way, will require additional controlled tests.

Similarly, it is not clear how the maximum force might change with manufacturing variations. This was reasonably corrected for by normalizing the equations to each individual's maximum contraction, but more nuanced details might emerge with more data collection. Not knowing these relationships a priori means there is still a need for robots and other systems to go through an iterative design stage if it is discovered that the system does not produce the torque or have the RoM that the design team expects.

However, the data and models provided here should enable faster redesigns with fewer additional measurements.

3.3.2 Nondimensionalized Force Prediction

We have also introduced the concept of nondimensionalized isometric force that is a function of relative strain and relative pressure (Equation (3.4)). This elegant equation has only three coefficients with low error (see Table 3.2). In the work presented here, we introduce the relative pressure term, $P^* = P/P_{620}$. Our data show that by normalizing force, contraction, and pressure, we are able to create a simplified force equation as a function of contraction and pressure that scales well with initial actuator length, as expressed by Equation (3.6) in Section 3.1.1.

For example, given $l_0 = 54\,\text{mm}$ and $P = 300\,\text{kPa}$, the measured force was $F_{actual} = 103\,\text{N}$. Force prediction using only the Festo tool predicts the force to be $F_{predict} = 270\,\text{N}$. This is 161% greater than the measured force. Using Equations (3.7) and (3.4) with Equation (3.6) gives $F_{predict} = F^*(0, 0.48) \cdot F_{620}(0.054) = 0.49 \cdot 230\,\text{N} = 112\,\text{N}$. This is an error of 8%, which is much more accurate than using the Festo tool only.

The presented work is capable of predicting force output for 10 mm and 20 mm muscles with different initial muscle lengths, internal pressure, and contraction amount with minimal initial data collection. However, it remains to be seen if these relationships are maintained over the full lifecycle of the muscles. There are not yet any peer reviewed studies investigating the long-term behavior of these artificial muscles, though the Festo datasheet provides some indication of what can be expected [Festo AG & Co. KG, 2026]

(p.19). They report that the 40 mm BPA can complete 1 million cycles at 6000 N (maximum rated load), or 10 million cycles at 4000 N. The service life of the other size BPAs is not reported; however, it is mentioned that thermal stressing may reduce the lifespan and they should be inspected every 500,000 cycles for signs of aging [Festo AG & Co. KG, 2018] (pp. 10–11). Future research should look at the behavior of these muscles over time.

In our work, $P_{620} = 620\,\text{kPa}$ is the maximum supply pressure for our system, and other users of this actuator may use a different supply pressure. However, this supply pressure is not required, and normalizing by this value was chosen as a method for improving the optimization techniques by reducing the sensitivity of the term associated with pressure. This equation will work for any pressure supplies below 620 kPa. We also anticipate the equation will work for pressures up to 800 kPa (the maximum rated pressure for Festo ∅10 mm BPAs); however, we do not have the data to confirm this.

3.3.3 Comparison with Other Published BPA Models

The proposed model was compared with the formulations by Sárosi [2012], Sárosi and Fabulya [2012], and Martens and Boblan [2017], to evaluate their predictive performance across actuator sizes and operating conditions (see Table 3.3 and Figure 3.5). For the ∅20 mm BPA, parameters from [Sárosi, 2012] were used, corresponding to actuators with resting lengths of 200 mm and 400 mm. These were compared against the experimental data at $l_0 = 300\,\text{mm}$ and $l_0 = 450\,\text{mm}$, respectively. For the ∅10 mm BPA with resting lengths of 257 mm and 233 mm, parameters from [Sárosi and Fabulya, 2012] were applied. In each case, the maximum contraction parameter (k_{max}) was determined numerically

by solving $F(6.2\,\text{bar}, k_{max}) = 0$, ensuring consistency with the experimental maximum pressure condition.

Table 3.3: Goodness-of-fit comparison for the Bolen, Sárosi, and Martens and Boblan models at different resting lengths (in mm) for 10 mm and 20 mm BPAs. RMSE and maximum error are reported in newtons, and FVU denotes the fraction of variance unexplained.

Diameter	Length	Bolen			Sarosi			Martens and Boblan		
		RMSE	FVU	Max. Error	RMSE	FVU	Max Error	RMSE	FVU	Max. Error
10 mm	257	22.9	0.0353	46.0	143.8	1.3903	235.8	73.8	0.3668	122.9
	233	16.8	0.0187	30.3	132.7	1.1655	222.6	62.6	0.2590	101.2
20 mm	300	35.6	0.0144	114.0	132.1	0.1992	313.4	72.8	0.0605	137.8
	450	67.6	0.0258	166.6	177.7	0.1785	294.8	862.1	4.2022	1321.1

While the Sárosi [2012] and Sárosi and Fabulya [2012] models capture general trends in force–contraction behavior, discrepancies were observed when applying parameters derived from different actuator lengths. In particular, the model tends to over-predict force at higher pressures and low amounts of contraction, suggesting sensitivity to geometric scaling effects not explicitly accounted for in the formulation.

The Martens and Boblan [2017] model is derived from a geometric and material-based framework and provides a more physically grounded representation of actuator behavior. However, its accuracy depends strongly on precise geometric parameters, including braid angle, wall thickness, and fiber length. Small deviations in these parameters can lead to significant differences in predicted force, particularly at higher pressures.

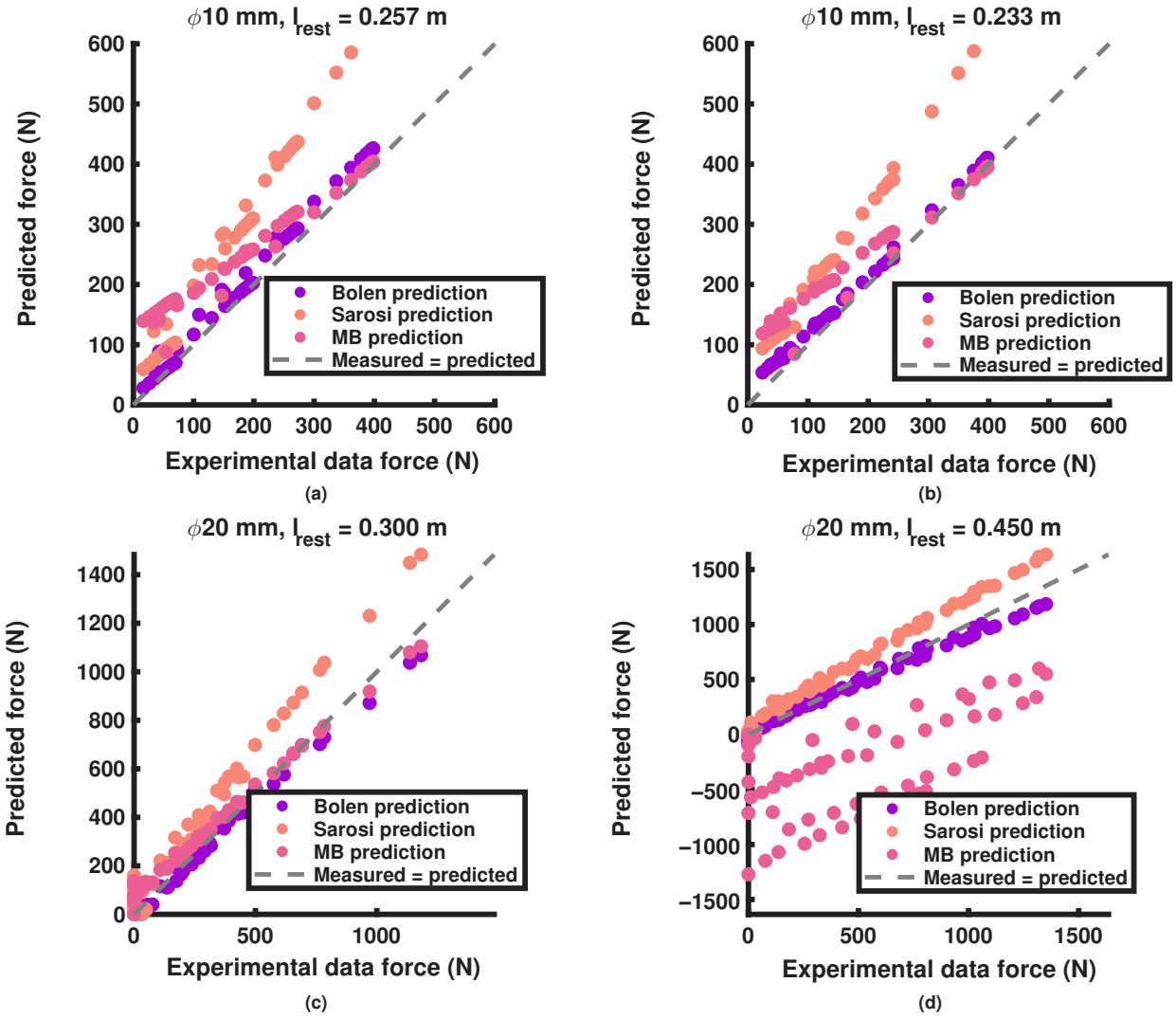


Figure 3.5: Comparison between measured and predicted BPA force at varying amounts of contraction and pressure. A $y = x$ perfect fit line is added for reference. The BPAs compared are: (**a**) ∅10 mm, $l_0 = 257$ mm, (**b**) ∅10 mm, $l_0 = 233$ mm, (**c**) ∅20 mm, $l_0 = 300$ mm, and (**d**) ∅20 mm, $l_0 = 450$ mm.

The Martens and Boblan model was first evaluated at the same pressure–contraction points as in their experimental data [Martens and Boblan, 2017] (Tables A1 and A2),

indicating that it reproduces the general published force map shape but is sensitive to the assumed initial geometry. Examining their ∅20 mm BPA, for example, using the explicitly stated $l_0 = 300\,\text{mm}$ and initial diameter $D_0 = 20\,\text{mm}$ significantly under-predicted the results compared to using the implied $l_0 = 296\,\text{mm}$ ([Martens and Boblan, 2017] (Table A1)) and $D_0 + H_0 = 21.8\,\text{mm}$, where $H_0 = 1.8\,\text{mm}$ is the material thickness. The diameter adjustment can be inferred from [Chou and Hannaford, 1996]. This adjusted diameter was also included for the ∅10 mm BPA to give a better fit. As can be seen in Figure 3.5 and Table 3.3, the highest accuracy is observed when ∅20 mm and $l_0 = 300\,\text{mm}$, which is at or very near their experimental measurements. For the ∅10 mm BPAs the accuracy decreases even though their experimentally measured $l_0 = 250\,\text{mm}$ is close to our $l_0 = 257\,\text{mm}$ and $l_0 = 233\,\text{mm}$. The largest error occurs at ∅20 mm and $l_0 = 450\,\text{mm}$ and the model starts to predict negative force. We have calculated that the $F_{PE} \cdot \frac{dD}{DL}$ term becomes very large compared to the others [Martens and Boblan, 2017] (p. 5). This suggests that the model is sensitive to geometric parameter selection and interpretation, particularly with respect to initial length and diameter definitions.

In comparison, the present normalized model demonstrates improved consistency across actuator lengths by incorporating experimentally derived scaling through F_{620} and ϵ_{620}. This normalization reduces sensitivity to geometric variation and enables a unified representation of actuator behavior, while maintaining predictive accuracy over the tested range of pressures and contractions.

3.3.4 Implications for Real-Time Control and Simulation

Although pneumatic muscle actuators exhibit hysteresis, thermodynamic effects, and other dynamic behavior, accurate and high-performance controller designs can still be achieved using a static force map approach when the underlying static force model is accurate. Martens and Boblan explicitly note that BPA dynamics are often dominated by the air dynamics inside the actuator and the mass flow behavior of the valve, such that precise modeling of the static force characteristic remains crucial for torque control accuracy [Martens and Boblan, 2017]. This view is consistent with more recent BPA control work, including a sensor-less torque control interface for BPA-driven joints [Martens et al., 2018], stable force/position/stiffness control of antagonistic pneumatic-muscle joints [Ugurlu et al., 2019, Hunt et al., 2017b], and sensor-less angle stiffness control of antagonistic BPA systems [Shin and Kogiso, 2022].

In this context, the present model is well suited for real-time use because it reduces the static isometric force prediction to a low-order normalized surface, $F^*(\epsilon^*, P^*)$, scaled by the resting-length-dependent term $F_{620}(l_0)$. This structure is computationally inexpensive to evaluate online and is also suitable for lookup-table implementation. While the present work does not model the full actuator–valve dynamics, it provides the static force relationship needed as a feedforward or inner-model component in real-time torque control, simulation, and preliminary design workflows. It is anticipated that improved real-time control for high-speed and high-precision systems will benefit more from improved modeling of airflow dynamics than from modeling actuator hysteresis, damping, and other dynamic behavior. Characterizing the actuators under dynamic operating conditions,

such as isokinetic (constant-velocity), isobaric, and isotonic (constant-load) contractions, is a natural next step beyond the isometric characterization reported here, for which the pressure–force dynamics model of Elzein et al. [2025] provides the modeling foundation.

3.3.5 Biomimetic Artificial Muscle Analogy

BPA artificial muscles are often said to be analogous to biological muscles because they have force–length curves and can produce force only in tension. This analogy is worth closer inspection. For example, the passive force–length curve has been described as an exponential function arising from connective tissue elements, while the active force–length relationship is typically modeled as a bell-shaped curve reflecting actin–myosin overlap [Winters, 1990]. These formulations emphasize that muscle force generation is governed by nonlinear geometric and material interactions rather than simple linear elasticity. McKibben-type pneumatic artificial muscles (of which Festo is an example) exhibit analogous nonlinear behavior, though arising from fundamentally different physical mechanisms. As shown by Chou and Hannaford, the force–length relationship of McKibben actuators is derived from geometric constraints imposed by the braided shell, leading to a nonlinear dependence of force on actuator length through the braid angle [Chou and Hannaford, 1996]. In their formulation, actuator force is proportional to pressure and a nonlinear geometric term $(3\cos^2\theta - 1)$, which governs both the maximum contraction limit and the curvature of the force–length relationship. The work presented here demonstrates that this nonlinear curve can be well approximated by an exponential as well.

However, the behavior of these artificial muscles differs distinctly from that of biological muscles. For example, the optimal fiber length of a biological muscle, l_{OFL}, is not equivalent to l_0 in an artificial muscle. Artificial muscles produce their maximum active force at their resting length, which is also their maximum length. In contrast, the maximum active force that a biological muscle can produce is somewhere in the middle of the total contractile length of the muscle.

Furthermore, biological muscles can adapt the optimal operating length through structural remodeling (e.g., sarcomere addition or tendon compliance) while BPAs have a fixed geometric configuration. This allows biological systems to self-optimize for the particular tasks the animal is subjected to. In order to re-optimize a robot for new tasks, new actuators would have to be swapped into the system. This distinction highlights a key limitation of BPAs in biomimetic applications, while also reinforcing the importance of accurate nonlinear modeling for control and design.

The analogy also suggests two design observations. First, the force–length penalty is shared by both families: force falls as the actuator operates away from its optimum, so a limb design must route each actuator such that its highest-force operating range coincides with the joint angles where torque is needed most, because the force lost at the extremes of the range of motion cannot be recovered by pressure alone. Second, the two families gain moment arm in different ways. Skeletal muscle attaches to the skeleton itself, through tendons or directly over broad areas of bone, so the moment arms of large animals follow from skeletal geometry. The robot's BPAs instead act through brackets that protrude from the limb segments; that hardware is mechanically exposed, and its series compliance is one of the loss mechanisms quantified in Chapter 4.

The analogy finally raises a question that this dissertation can pose but not settle. Every muscle-like actuator loses force as it operates away from its optimal length, and a soft actuator working against soft structure has no rigid skeleton from which to gain a moment arm. Given those two facts, it may be no accident that nearly all large land animals are built on rigid skeletons and that large soft-bodied land animals are essentially absent. Whether the force–length penalty contributes to that absence, or whether it reflects other constraints such as support and protection, is a question for comparative biomechanics rather than for a single actuator study. For the designer, the lesson holds regardless of the answer: some moment arm must come from somewhere, and on this robot it comes from protruding brackets whose behavior must be modeled rather than ignored.

4 Isometric Knee Torque

Publication note: This chapter is the subject of an unpublished manuscript in preparation by Bolen, Burgess, Morrow, Elzein, Pang, Scharzenberger, and Hunt [Bolen et al., 2025]. For that work, B.B. performed the conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, and original draft preparation; M.E., L.P., L.B., and C.M. contributed methodology, software, investigation, and data curation; and A.H. supervised the work, administered the project, and acquired funding. All co-authors contributed to writing, review, and editing.

This chapter connects the actuator force characterization of Chapter 3 to torque about the robot knee. The rigid-body torque model of the earlier robot work assumed inextensible muscle paths on rigid links, and the isometric tests reported here show where that assumption fails and by how much. The chapter proceeds from the simple pinned knee to the biomimetic joint, identifies the correction terms that remove the rigid-body idealization from the torque prediction, validates them on held-out tests, and closes with the attachment-route redesign that brings the actuator configurations to human-scaled torque targets. The methods, results, and discussion follow the manuscript noted above.

4.1 Methods

4.1.1 Robot Leg Model and Knee Test Stands

The primary components of the robot leg are the knee joint, femur, tibia, BPA assemblies, and artificial tendons (Fig. 4.1**A–C**). The artificial bone components are 3D printed using a combination of Onyx, carbon fiber, and high-strength high-temperature fiberglass on Markforged Onyx One and Mark Two printers, as well as machined links for the knee mechanism. Onyx is a proprietary Markforged material that consists of chopped carbon fiber in nylon. Artificial tendons are made with Shimano bicycle brake cable.

The leg assembly geometry and dimensions are based on bones scans of a 6 ft (1.83 m) tall person [Bolen and Hunt, 2019, Seth et al., 2018, Delp et al., 1990, 2007]. Two knee joints were tested, one pure rotational joint, and one biomimetic joint with rotation and translation. Each joint is driven by two antagonistic Festo actuators. Both robot joints have the same center of rotation in the femur frame at $\theta_{\mathrm{k}} = 0\,\mathrm{deg}$.

Details on the biomimetic robot knee used in this study can be found in [Steele, 2018, Steele et al., 2017, 2018]. In brief, this 1-DoF joint (Fig. 4.2) uses a four-bar linkage to change the effective moment arm during rotation. The intersection point of the internal crossed links define the instantaneous center of rotation (ICR). This causes the joint ICR location in the femur frame to translate posteriorly in the X direction and distally in the Y direction except for a brief proximal move near full flexion. Reference coordinate frames are given by Wu and Cavanagh [1995], Wu et al. [2002]. This is similar to how the human knee behaves [Bolen and Hunt, 2019, Morrow et al., 2020, Seth et al., 2018, Delp et al., 2007, 1990]. The correct location of the ICR is important for both theoretical and experimental

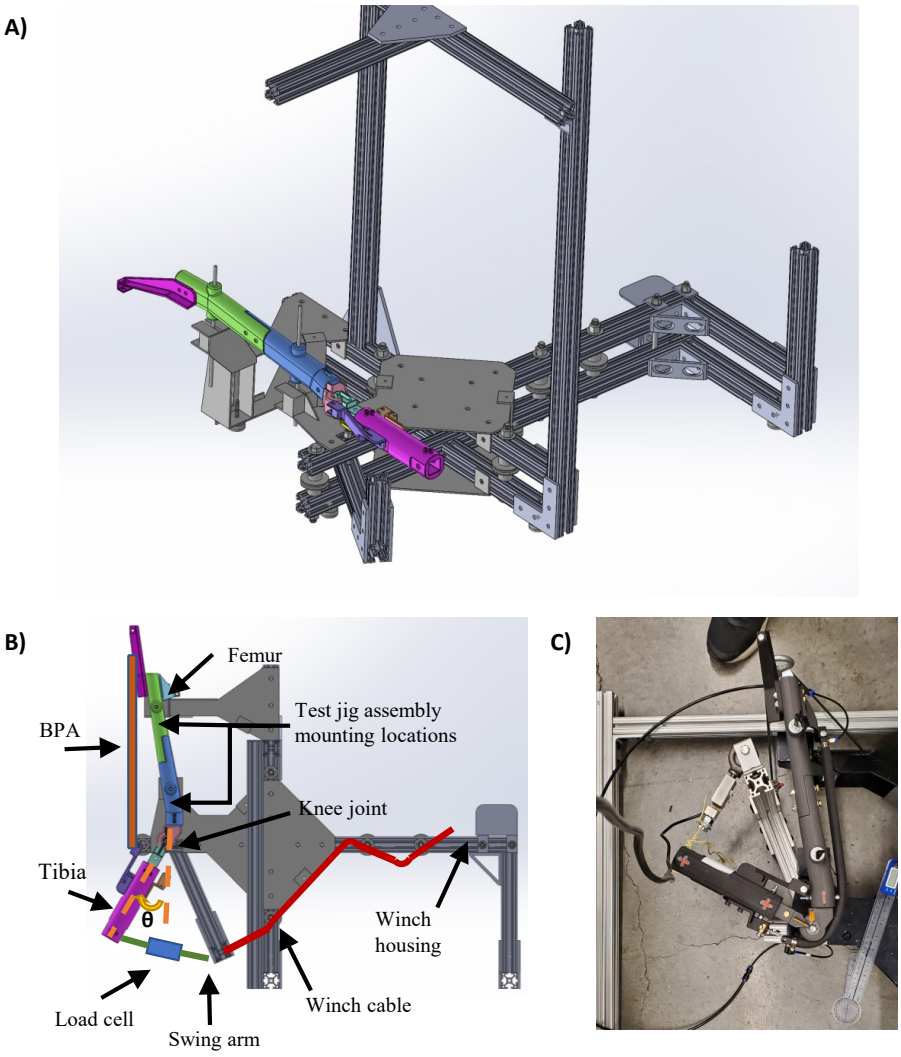


Figure 4.1: **(A)** CAD solid model assembly of the revolute joint knee in the test stand. **(B)** 1-DoF pinned-joint robot knee in the test apparatus with important components labeled. Leg is shown with 30 degrees flexion, i.e. $\theta_k = -30^\circ$. **(C)** Picture from isometric torque testing. Setup shows the revolute knee configured for a test of the extensor BPA. The knee is flexed with $\theta_k = -120^\circ$. Also note the compressed shape of the inflated BPA on the knee during this high degree of flexion.

torque calculations.

Uniarticular knee extensor and knee flexor are used to actuate the joint. Each knee joint type was tested with a single ∅10 mm Festo BPA per muscle. The biomimetic knee was additionally tested with ∅20 mm Festo BPAs in a paired configuration: the extensor with two BPAs in parallel, and the flexor with two BPAs whose pathways are mirrored about the Z axis. Each artificial muscle is pinned at the muscle origin location on the femur while the other end is either pinned or attached to the muscle insertion location on the tibia via an artificial tendon.

Torque about the knee is generated by the moment of the BPA force about the joint's instantaneous center of rotation (ICR). Two vectors enter that moment: the position vector $\mathbf{d}$, drawn from the ICR to the point where the actuator force is applied, and the force vector $\mathbf{F} = \|\mathbf{F}\|\hat{\mathbf{u}}$, whose direction $\hat{\mathbf{u}}$ is the line of action of the actuator. The moment of the force about the joint is $\mathbf{m} = \mathbf{d} \times \mathbf{F}$, and its magnitude is the force magnitude multiplied by the moment arm r, the perpendicular distance from the joint axis to the line of action,

$$r = \|\mathbf{d} \times \hat{\mathbf{u}}\| \tag{4.1}$$

The moment arm is therefore a scalar, and it is the same quantity whether it is computed from the position vector with Eq. (4.1) or measured directly with a tape measure. In this study it is calculated using the method specified by Young et al. [2019]. The scalar moment arm r_k about the joint axis $\hat{\mathbf{z}}_k$ (i.e., the Z axis in the knee frame) is calculated as

$$r_k = \mathbf{p}_{proj,xy} \cdot \frac{\mathbf{p}_f \times \hat{\mathbf{z}}_k}{\|\mathbf{p}_f \times \hat{\mathbf{z}}_k\|} \tag{4.2}$$

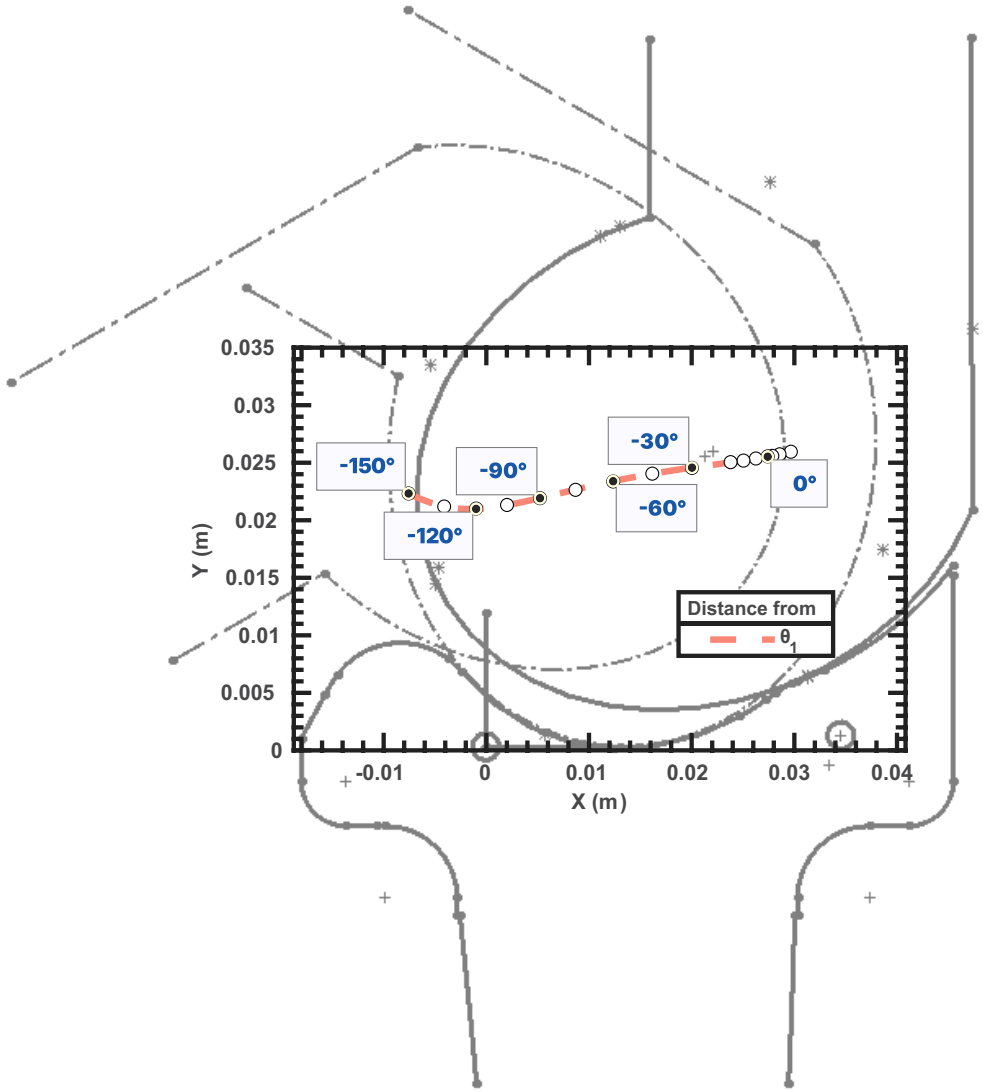


Figure 4.2: ICR for biomimetic robot knee as a function of knee angle. Measured from θ_1 on the tibia. Plot is overlayed on model of the knee at $\theta_k = 0^\circ$ (solid line), and -60° and -120° (dotted dash lines). White circles represent measured points in -15° increments from -150° to -15°, -5° increments from -15° to 0°, and then at 2°, 5°, and 10°.

A muscle path may be modeled as a single segment running from origin to insertion, or as a chain of segments joined at via points. The segment that crosses from one reference frame into the other, here from the femur frame to the tibia frame, is the free muscle segment: because its endpoints ride on different bodies, its length and its direction respond directly to joint rotation, whereas a segment contained within one body's frame rotates rigidly with that body and changes neither. In Eq. (4.2), $\mathbf{p}_{proj,xy}$ is the free muscle segment projected onto the XY–plane, which is normal to the joint axis, and $\mathbf{p}_f$ is the projected muscle segment vector. Moment arm is calculated in the tibial body frame.

Details of the existing model and the torque calculation methodology in the modeled robot are given by Bolen and Hunt [2019], Morrow et al. [2020].

4.1.2 Knee Torque Data Collection and Equipment

A test jig was built to collect knee torque measurements at different knee angles over its range of motion. The test stand frame is made predominantly out of 80/20 1515 series extruded aluminum components. The knee joint is free to rotate the tibia while the femur is fixed to the frame. A force sensor has one end connected to the tibia and the other is connected to a swing arm. The swing arm is tied to a winch with 3/16 inch Kevlar rope from Quality Nylon Rope routed over intermediate idler pulleys (see Fig. 4.1**B** and 4.1**C**). At each measured position of θ_k the winch would be locked and the BPA would be inflated and then deflated. This test procedure kept the measured results as isometric as possible. The knee joint is co-axial with gravity to eliminate gravitational torque (see Fig. 4.1**A** and Fig. 4.1**C**).

Force data was collected using one of two different sensors. The first is a MODERN STEP 300 kg digital crane scale. The second force sensor is a CALT DYLY-103 100 kg S shaped load cell. The load cell is used in conjunction with a HX711 Load Cell Amplifier. Pressure is measured with a Freescale MPX5700 GP 5 V pressure sensor. Building air supply pressure is controlled with two pressure regulators in series. The first is a Parker model 20R113GC $0 - 120$ psi pressure regulator. The second is a Husky $3/8$ inch High Performance Air Regulator HDA72200. A Festo VTUG-10-MRCR-S1T-26V20-T516LA-UL-T532S-8K valve manifold was used to supply air to the actuator. Pressure and load cell amplifier data are sent to MATLAB via an Arduino Uno style Sparkfun BlackBoard C microcontroller.

The length measurements were made using a tape measure. When there was sufficient curvature in the BPA during knee extensor torque tests, a flexible piece of string was used to mark the axial length and then transferred to the tape measure. Angle measurements were taken with a Medigauge digital electronic goniometer or with MALENOO analog goniometers of 6, 8, or 12 inch lengths.

4.1.3 Hybrid Torque Calculation Method

A Hybrid Torque calculation method was developed to help uncover inconsistencies between the model and recorded torque values. This method measures the actual muscle length and angles of the robot to calculate force and joint torque. With this method, it is easier to determine which parts of the torque equation did not achieve their predicted value and provide a correction to that part of the equation. The classical mechanics way to

calculate the torque about a joint is to take the cross product of the position vector **d** and the force vector **F**, both defined in Section 4.1.1,

$$\mathbf{m} = \mathbf{d} \times \mathbf{F} \tag{4.3}$$

Knee torque is the scalar component of this moment about the joint axis, written M_z to make its axis explicit. Because the moment arm is the perpendicular distance from that axis to the line of action, the vector cross product of Eq. (4.3) reduces to an ordinary scalar product,

$$M_z = r\,F \tag{4.4}$$

where r is the scalar moment arm of Eq. (4.1), here measured directly with a tape measure as the perpendicular distance from the knee axis to the actuator's line of action, and F is the scalar force magnitude, calculated by measuring the BPA's P, l_0, l_{620} and l_{m} values and using the model fit equations determined in the following section. The hybrid method therefore replaces the model-predicted moment arm r_k of Eq. (4.2) and the model-predicted muscle length with their measured counterparts, leaving the force model as the only predicted quantity.

4.1.4 Improved Torque Model

The classical kinematic torque model assumed rigid links and inextensible muscle paths. Data collected during the first flexor experiments (Section 4.2) showed that this assumption

does not hold for a pneumatic system: measured muscle lengths were consistently shorter than predicted, and measured torque deviated from prediction in a load-dependent manner. This section describes the corrections that were developed to account for a constant length offset, series compliance in the attachment hardware and artificial tendons, and the loss of usable muscle length when a BPA wraps around the joint.

These corrections are deliberately formulated to be identified from experiment rather than derived analytically, because the physical effects they represent are distributed and not separately measurable. The lumped compliance absorbs the cantilevered three-dimensional-printed attachment brackets, whose geometry is a printed CAD part rather than a prismatic beam with a tractable second moment of area, together with the bone fixtures, the cable winch of the test stand, and the artificial tendon. Predicting that aggregate stiffness from first principles would require finite-element models of each component with material properties (anisotropic printed polymer, braided sleeve, construction-dependent tendon stiffness) that are themselves unmeasured. The correction terms therefore parameterize only what the torque model needs: χ_0 absorbs the constant length discrepancy observed at zero load, χ_1 and χ_2 are effective stiffnesses in the bracket frame's principal directions, and χ_3 scales the wrapping-length loss. Each is identified by global optimization against the measured torque rather than solved in closed form, for two reasons. First, the terms enter the prediction through an implicit force balance, because the compliance displacement depends on the force that it itself modifies, so the inverse problem has no analytic solution. Second, the parameters are only jointly identifiable: preliminary fits showed that widely different stiffness pairs can reproduce the training tests nearly equally well, so the adopted solution must be selected by cross-validated prediction on

held-out tests rather than by residual minimization alone, which is what the multiobjective Pareto-front procedure of Section 4.1.5 provides.

Length Offset and System Stiffness

Analysis of measured muscle length data shows that it was consistently shorter than the predicted muscle length at all joint angles. This shorter length results in less muscle force, a shifted RoM, and less torque in the experimental setup. That this shorter length occurs when $\varepsilon = 1$ and $F = 0$ indicates that a constant length offset should be used. Therefore, to produce an improved torque prediction model, l_{m} was recalculated as

$$l_{\mathrm{m}} = l_{LMT} - 2\, l_{fitting} - \chi_0 \tag{4.5}$$

where l_{LMT} is the path length of the musculotendon from origin to insertion point, $l_{fitting}$ is the length of the end caps, and χ_0 is a constant length offset.

Additionally, as the muscles produce more force, compliant components in the limb segments and attachment hardware deform. This deformation changes the effective braided pneumatic actuator (BPA) length, which alters contraction and therefore changes the force predicted by the BPA model. This effect is most apparent at high forces and low contractions. The purpose of the compliance model developed next is predictive, not structural: it exists so that the torque calculation can correct the actuator length for load-dependent deformation, and it makes no claim about the stiffness of any one component in isolation. Accordingly, the stiffness values that populate the model, χ_1 and χ_2, are not measured on the hardware directly; they are identified by the optimization of Section 4.1.5

from the measured torque, and every stiffness value quoted later in this chapter is an optimizer output. The combined hardware compliance is approximated using Hooke's law in the bracket's principal directions.

A local bracket frame was established at the attachment bracket. The $\hat{\mathbf{x}}_{br}$ axis was aligned with the vector from the bracket reference point to the BPA attachment point and represents the stiffest direction of the bracket. The $\hat{\mathbf{y}}_{br}$ axis was chosen to point in the posterior direction with respect to the tibia frame, corresponding to the primary cantilever bending direction (see Fig. 4.4), while the $\hat{\mathbf{z}}_{br}$ axis completes the right-handed frame. In this frame the bracket is modeled as three independent linear springs along the principal axes, with stiffnesses k_x, k_y, and k_z in the x_{br}, y_{br}, and z_{br} directions.

For the two-rotation evaluator, the bracket frame was constructed from the vector joining the bracket origin to attachment point $\mathbf{p}_A$ or $\mathbf{p}_B$. The first rotation, through θ_z about the fixed space-frame axis $\hat{\mathbf{z}}_s$, points the intermediate axis $\hat{\mathbf{x}}_1$ toward the sagittal-plane projection, $\mathbf{p}_A^{\text{sag}}$ for the insertion point $\mathbf{p}_A$ and $\mathbf{p}_B^{\text{sag}}$ for the origin point $\mathbf{p}_B$. The second rotation, through $-\theta_y$ about the current axis $\hat{\mathbf{y}}_1 = \hat{\mathbf{y}}_{br}$, brings the final bracket axis $\hat{\mathbf{x}}_{br}$ into alignment with the full three-dimensional point vector. Figure 4.3 shows these geometric targets. The resulting orientation and homogeneous transform are

$$\mathbf{R}_{sb} = \mathbf{R}_z(\theta_z)\mathbf{R}_y(\theta_y)^T, \qquad \mathbf{T}_{sb} = \begin{bmatrix} \mathbf{R}_{sb} & \mathbf{p}_{br} \\ \mathbf{0}^T & 1 \end{bmatrix}. \tag{4.6}$$

$$\mathbf{K}_{br} = \operatorname{diag}\big(k_x,\ k_y,\ k_z\big) \tag{4.7}$$

These three principal stiffnesses are not independent in this model, and the reason

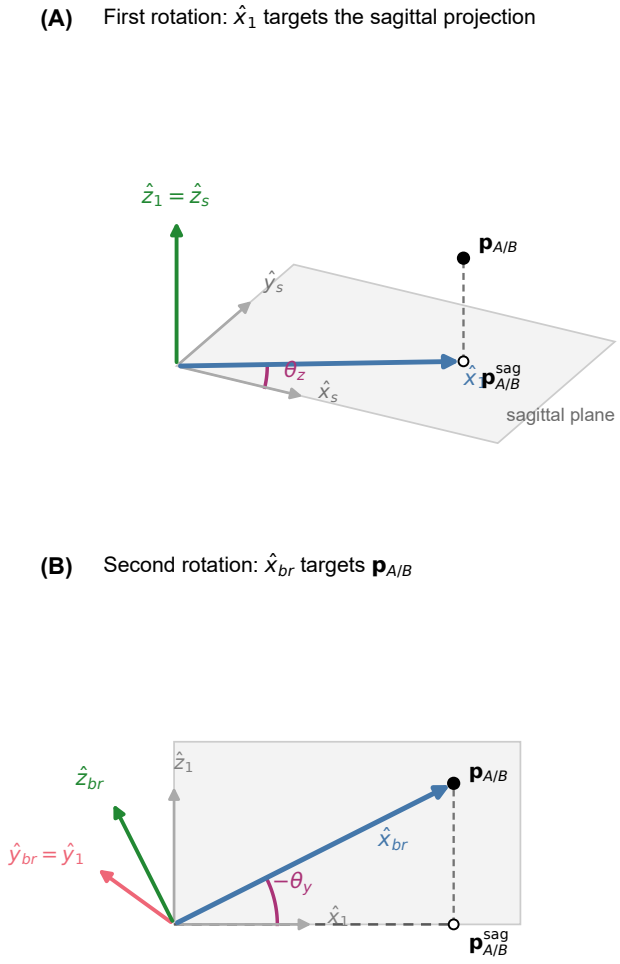


Figure 4.3: Two-rotation construction of the bracket frame from attachment point $\mathbf{p}_A$ or $\mathbf{p}_B$. **(A)** Rotation through θ_z about fixed $\hat{\mathbf{z}}_s$ aligns $\hat{\mathbf{x}}_1$ with the point's sagittal-plane projection. **(B)** Rotation through $-\theta_y$ about current $\hat{\mathbf{y}}_1 = \hat{\mathbf{y}}_{br}$ aligns $\hat{\mathbf{x}}_{br}$ with the full point vector.

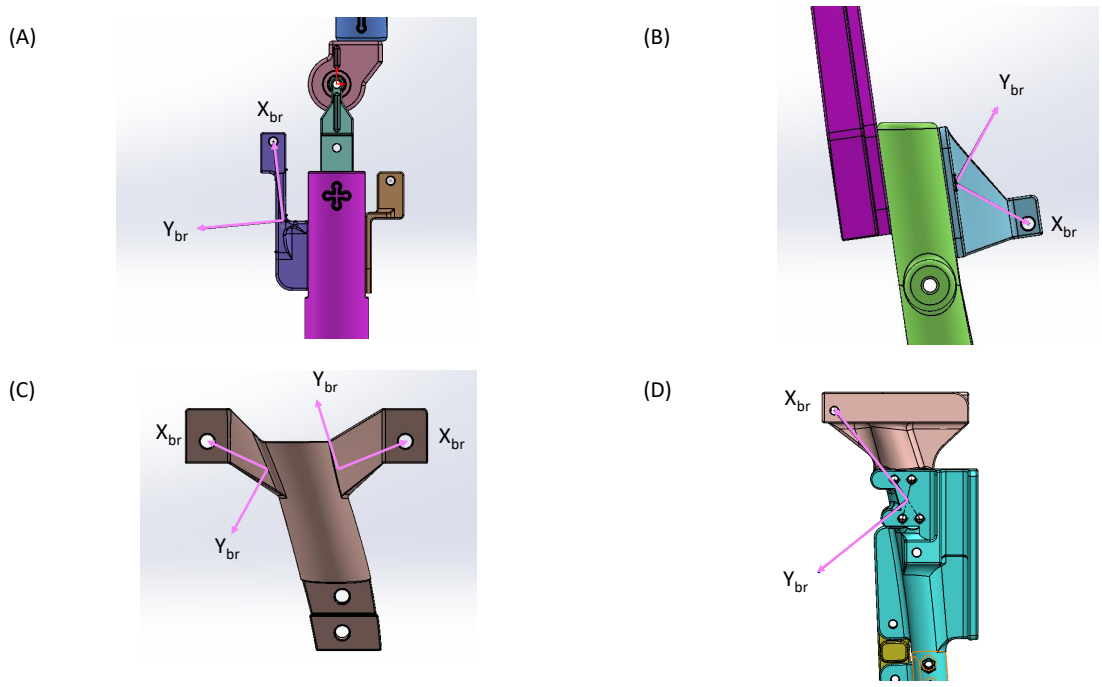


Figure 4.4: Bracket frame orientations for: **(A)** flexor configuration on the pinned knee, **(B)** extensor configuration on the pinned knee, **(C)** flexor and extensor configurations on the biomimetic knee with ∅10 mm BPAs, and **(D)** flexor configuration on the biomimetic knee with the ∅20 mm BPA.

follows from the frame construction itself. The first rotation of Eq. (4.6) is fixed by the sagittal-plane projection of the attachment point, but the orientation of the bracket frame about its own $\hat{\mathbf{y}}_{br}$ axis is a modeling choice: the pitch-only convention (no second rotation) and the two-rotation convention of Fig. 4.3 describe the same physical bracket and differ by exactly a rotation about $\hat{\mathbf{y}}_{br}$. A diagonal stiffness array is invariant under such a rotation, $\mathbf{R}_y(\theta)\,\mathbf{K}_{br}\,\mathbf{R}_y(\theta)^T = \mathbf{K}_{br}$ for all θ, if and only if the two stiffnesses in the plane of the rotation are equal, $k_x = k_z$. Adopting this y-symmetric form therefore makes the compliance prediction independent of the frame convention, and the redundant third stiffness is eliminated: with this array the two frame conventions produce identical torque predictions to numerical precision.

The change of notation that follows is deliberate. The symbols k_x, k_y, and k_z name the stiffness components of the general bracket-frame array, Eq. (4.7); χ_1 and χ_2 are the two identified stiffness parameters, the quantities the optimization of Section 4.1.5 solves for, and each enters the diagonal array in a fixed axis assignment. Two stiffness values remain, χ_1 in the x_{br} and z_{br} directions and χ_2 in the cantilever bending direction y_{br}:

$$\mathbf{K}_{br} = \begin{bmatrix} \chi_1 & 0 & 0 \\ 0 & \chi_2 & 0 \\ 0 & 0 & \chi_1 \end{bmatrix} \tag{4.8}$$

The corresponding local compliance matrix is

$$\mathbf{C}_{br} = \mathbf{K}_{br}^{-1} \tag{4.9}$$

The force applied to the bracket was expressed in the bracket frame as $\mathbf{F}_{br}$. This force

vector can be written in terms of its magnitude and unit direction as

$$\mathbf{F}_{br} = F\,\hat{\mathbf{u}} \tag{4.10}$$

where $F = \|\mathbf{F}\|$ is the scalar force magnitude of Eq. (4.4) and $\hat{\mathbf{u}} = [u_x, u_y, u_z]^T$ is the BPA force direction expressed in the bracket frame. The bracket deflection vector, $\boldsymbol{\epsilon}_{br}$, was calculated from Hooke's law as

$$\mathbf{F}_{br} = \mathbf{K}_{br}\,\boldsymbol{\epsilon}_{br} \tag{4.11}$$

Solving Eq. 4.11 for the bracket deflection gives

$$\boldsymbol{\epsilon}_{br} = \mathbf{K}_{br}^{-1}\,\mathbf{F}_{br} = \mathbf{C}_{br}\,\mathbf{F}_{br} \tag{4.12}$$

Substituting Eq. 4.10 into Eq. 4.12 gives

$$\boldsymbol{\epsilon}_{br} = F\,\mathbf{K}_{br}^{-1}\,\hat{\mathbf{u}} \tag{4.13}$$

For the diagonal stiffness matrix in Eq. 4.8, Eq. 4.12 is equivalent to the component-wise expression

$$\boldsymbol{\epsilon}_{br} = \begin{bmatrix} F\,u_x/\chi_1 \\ F\,u_y/\chi_2 \\ F\,u_z/\chi_1 \end{bmatrix} \tag{4.14}$$

Although $\boldsymbol{\epsilon}_{br}$ is a three-dimensional deflection vector, the BPA force model depends

on the scalar change in effective BPA length (δ). Therefore, only the component of $\boldsymbol{\epsilon}_{br}$ along the BPA line of action directly changes the effective BPA length. This scalar bracket deflection is the projection of $\boldsymbol{\epsilon}_{br}$ onto $\hat{\mathbf{u}}$:

$$\delta_{br} = \hat{\mathbf{u}}^T \boldsymbol{\epsilon}_{br} \tag{4.15}$$

Substituting Eq. 4.12 into Eq. 4.15 gives

$$\delta_{br} = \hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \mathbf{F}_{br} \tag{4.16}$$

Using Eq. 4.10, Eq. 4.16 becomes

$$\delta_{br} = F \, \hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \hat{\mathbf{u}} \tag{4.17}$$

The scalar stiffness along the BPA line of action due to the bracket compliance is therefore

$$k_{\hat{u},br} = \frac{F}{\delta_{br}} = \left(\hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \hat{\mathbf{u}} \right)^{-1} \tag{4.18}$$

This projected-compliance form is used because the force is applied along $\hat{\mathbf{u}}$, while the bracket is allowed to deflect independently along its local $\hat{\mathbf{x}}_{br}$, $\hat{\mathbf{y}}_{br}$, and $\hat{\mathbf{z}}_{br}$ directions. The reduced-order Hooke's law expression for the bracket deflection along the BPA line of action is therefore

$$F = k_{eq} \, \delta \tag{4.19}$$

Here k_{eq} is the series sum of every compliant element in the actuator path, and the bracket-only case is the special case with no tendon. When the BPA attaches directly to the bracket, k_{eq} excludes the tendon and equals $k_{\hat{u},br}$ of Eq. (4.18). When an artificial tendon was used, it was modeled as an axial spring with stiffness k_{ten}: bracket deflection and tendon stretch both shorten the effective BPA length along the actuator line of action, the two elements carry the same force, and their scalar compliances add in series,

$$k_{eq} = \left(\frac{1}{k_{\hat{u},br}} + \frac{1}{k_{ten}} \right)^{-1} \tag{4.20}$$

This series reduction is exact only under a specific set of conditions, each of which corresponds to a way the physical hardware could violate it. The bracket and the tendon must carry the same force, which holds for two massless elements coupled in series under quasi-static loading; a compliant coupling between them would break this. Their displacements must be kinematically additive along the line of action, with no displacement lost to off-axis geometry. Within a single iteration of the force balance the line of action $\hat{\mathbf{u}}$ is treated as fixed, which is an approximation, and one the results justify: once the balance returns the compliance displacement, the calculation updates the attachment locations with the deflected bracket positions and re-evaluates $\hat{\mathbf{u}}$ and the moment arm at the deflected geometry, as described below. The elements are assumed to carry no moment, so rotational compliance and force–moment coupling at the attachments are neglected. Both elements are linear, with stiffness independent of deflection, and the tendon is modeled without preload, so its tangent stiffness does not stiffen with load state. Finally, compliance deflection shortens the actuator, as Eq. (4.21) assumes, because a BPA produces force only

in tension: under its own pulling force the compliant attachments give way in the direction that relieves that tension. A deflection that would lengthen the actuator can be produced by an antagonistic pull, but that loading lies outside this treatment.

Since δ is the scalar compliance displacement along the BPA line of action, then the compliance-corrected active BPA length used in the force balance is

$$l_m(\delta) = l_m(0) - \delta \tag{4.21}$$

where $l_m(0)$ is the initial muscle length calculated from (4.5).

For a specified set of correction parameters, the following calculation solves the attachment-compliance displacement at each knee angle; it is distinct from the optimization that identifies those parameters from experimental torque data (Section 4.1.5). The corresponding relative contraction is $\varepsilon^*(\delta)$. Since the BPA force depends on contraction, and contraction depends on the compliance displacement, δ is the root of the nonlinear force-balance equation

$$F_{\max} \cdot F_{\mathrm{BPA}}(\varepsilon^*(\delta), P^*) - k_{eq}\,\delta = 0 \tag{4.22}$$

The root is found with a bracketed scalar solver (`fzero` in MATLAB) on the physically admissible interval between zero deflection and the deflection that would bring the actuator to its maximum usable contraction. Termination is controlled by the bracket, not by a residual tolerance: the iteration closes in on the root and stops when the bracket has collapsed to adjacent floating-point values, so the returned δ is converged to machine precision. Two safeguards act as secondary stop conditions. If the actuator already sits at

or beyond its maximum usable contraction at zero deflection, no root is sought and δ is set to zero; if the residual shows no sign change across the interval, the solver likewise returns $\delta = 0$ rather than a spurious root.

After solving Eq. 4.22, the corresponding force magnitude was used in Eq. 4.12 to compute the vector bracket deflection. This deflection was transformed back into the limb reference frame and used to update the BPA attachment location. The routed BPA length, force direction, moment arm, and torque were then recalculated using the updated attachment location. In the present implementation, the optimized stiffness terms χ_1 and χ_2 were treated as effective compliance parameters for the attachment system rather than direct measurements of an isolated bracket. This allowed unmodeled compliance in the surrounding hardware to be represented by the reduced-order stiffness model.

The actuator path can carry compliant hardware at both ends: the insertion-side bracket of the preceding development together with a second, origin-side bracket. Figure 4.5 distinguishes the three-dimensional displacement of each bracket from the scalar displacement along the BPA force path. For bracket j, the displacement resolves into components $\eta_{1,j}$, $\eta_{2,j}$, and $\eta_{3,j}$ along the local bracket-frame basis, whereas only its projection $\delta_{br,j}$ along the corresponding force direction $\hat{\mathbf{u}}_j$ changes the actuator length. The two bracket projections act in series with the tendon elongation δ_{tendon}:

$$\begin{aligned} \boldsymbol{\eta}_{br,j} &= \sum_{i=1}^{3} \eta_{i,j} \hat{\mathbf{e}}_i^{(j)}, \\ \delta_{br,j} &= \hat{\mathbf{u}}_j^T \boldsymbol{\eta}_{br,j}, \qquad j \in \{1, 2\}, \\ \delta_{\text{series}} &= \delta_{br,1} + \delta_{\text{tendon}} + \delta_{br,2}. \end{aligned} \tag{4.23}$$

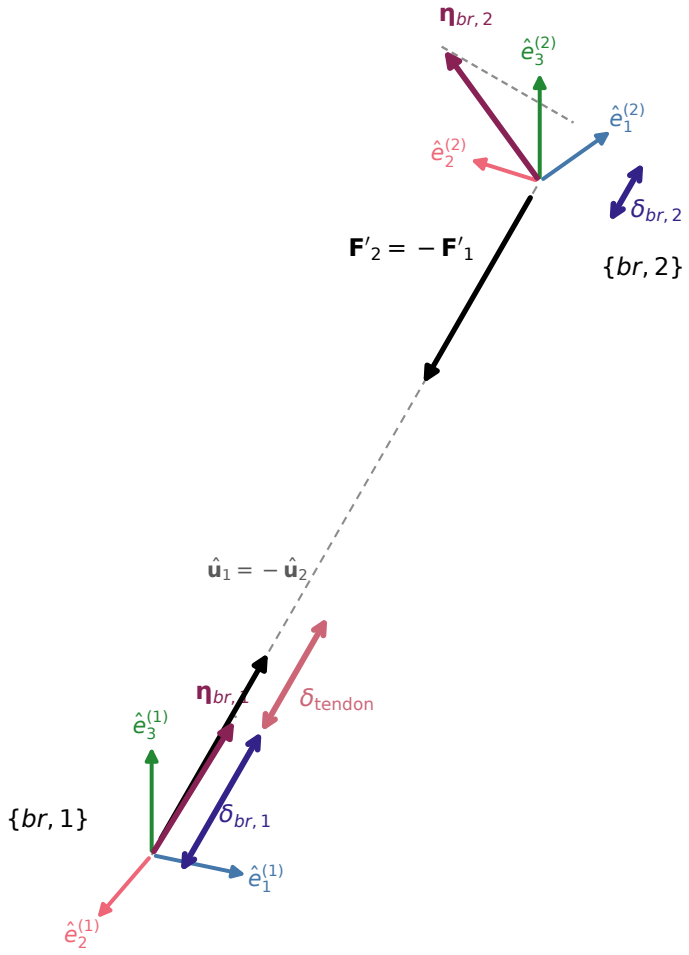


Figure 4.5: Two-bracket deflection and projection along the BPA force path. The force vectors at $\{br, 1\}$ and $\{br, 2\}$ point toward one another and satisfy $\mathbf{F}_2 = -\mathbf{F}_1$. The bracket projections $\delta_{br,1}$ and $\delta_{br,2}$ are offset from the corresponding coordinate frames for clarity, and tendon elongation δ_{tendon} continues collinearly from $\delta_{br,1}$. Colors follow a color-vision-deficiency-safe palette and are redundant with labels and line weights.

BPA Fictive Contraction

The extensor configuration required one further correction, for a loss with a different physical origin than the compliance terms. As the knee flexes, the extensor BPA bends over the front of the joint and makes contact with it, so part of the actuator follows the joint surface instead of a straight line. Bent over the joint this way, the BPA behaves like a beam: the pressurized wall on the inside of the bend carries less tension than the wall on the outside, which both unbalances the actuator's force and consumes part of the wrapped arc length that would otherwise shorten the path. The wrapped section therefore stops contributing usable contraction length, and the actuator behaves as if it were more contracted than it physically is. This excess is termed fictive strain: the additional contraction the force model must see to reproduce the observed force, beyond the actuator's physical contraction. A wrapping-loss term χ_3 was introduced to scale the length that this fictive strain removes, such that

$$\Delta l = \begin{cases} \chi_3 \; R \; |\theta_{\text{wrap}} - \theta_{\text{k}}| \; \zeta^2 & \theta_{\text{k}} \leqslant \theta_{\text{wrap}} \\ 0 & \theta_{\text{k}} \geqslant \theta_{\text{wrap}} \end{cases} \tag{4.24}$$

where Δl is the change in usable length of the BPA, R is the wrapping radius, $|\theta_{\text{wrap}} - \theta_{\text{k}}|$ is the angle through which the BPA is bent once contact begins, and θ_{wrap} is the knee flexion angle at which the BPA starts to bend over the knee joint, determined from solid modeling. ζ is the additive complement to relative strain, such that

$$\zeta = 1 - \varepsilon^* \tag{4.25}$$

so the wrapping loss grows quadratically as the actuator approaches its resting length and vanishes at full contraction. Combining (4.25) into (4.24), and subtracting Δl from (4.5) yields

$$l_{\mathrm{m}} = l_{LMT} - 2\, l_{fitting} - \chi_0 - \Delta l \tag{4.26}$$

Multiobjective Identification Algorithm

Multiobjective optimization using `gamultiobj` in MATLAB was used to determine the length offset (χ_0) and the stiffness terms (χ_1 and χ_2). The error between the model and the experimental torque was minimized across the pinned-knee flexor tests, with the $l_0 = 45.7\,\mathrm{cm}$ test held out for validation.

The optimization algorithm was again `gamultiobj` in MATLAB. This time, the muscle origins bracket was chosen. The bracket frame is shown in Fig. 4.4**B**. Since the bracket forms an arch shape and extends for 75 mm in the Z direction, for this configuration, we set χ_2 in the X_{br} direction, and χ_1 in the Y_{br} and Z_{br} directions. The optimization algorithm for this set solves for the best average goodness-of-fit (GoF) measures RMSE, FVU, and maximum absolute residual for two BPAs, while the third one was held out. Three independent rounds of optimization were performed, so that one of each of the BPAs could be held out as validation. The solution was then selected by the shortest distance between the average normalized GoF and the validation data, subject to improved GoF

values for all the individual BPAs, with the exception of maximum absolute residual on the $l_0 = 48.0\,\text{cm}$ BPA.

This initial identification was subsequently extended to the full test sets in two stages. First, the flexor model was given a second compliant bracket at the origin-side attachment (a hip-frame cantilever carrying the BPA origin), so that the compliance of both ends of the actuator path is represented; the stiffness pair (χ_1, χ_2) is shared by the two brackets in fixed axis assignments (Appendix C, Table C6), leaving a three-term fit (χ_0, χ_1, χ_2) cross-validated over the four remaining flexor tests, two of them held out per fold; the encoder-corrected $l_0 = 47.9\,\text{cm}$ test was not used in the two-bracket identification. Second, the extensor refit locked (χ_1, χ_2) at the flexor-identified values, enforcing a single stiffness pair across the flexor and extensor configurations, and solved only the extensor-specific terms (χ_0, χ_3) on a three-test training set with three tests held out for validation, with the extensor length offset allowed to take negative values. The bounds and the adopted values of both stages are recorded in Appendix C (Tables C7 and C8); the fit outcomes are summarized in Tables 4.2 and 4.3.

The goodness-of-fit metrics used to judge the optimization results are the root mean square error (RMSE), the fraction of variance unexplained (FVU), and the maximum absolute residual, each computed between the measured and predicted isometric torque over the tested range of motion.

Each metric isolates a different aspect of disagreement between the two curves. RMSE reports the overall magnitude of the error and, when the curve shapes already agree, is dominated by the remaining level offset. FVU normalizes the error variance by the variance of the measured torque, making it scale-free: it asks whether the prediction

reproduces the variation of the measurement rather than whether it is close in absolute units, is the complement of the Nash–Sutcliffe model efficiency [Nash and Sutcliffe, 1970], and takes values below one when the prediction outperforms the mean of the measurement itself. The maximum absolute residual is a worst-case local measure. A small shift of the predicted curve along the knee-angle axis produces large pointwise residuals where the torque gradient is steep and small residuals elsewhere, so this metric is sensitive to phase misalignment while responding only weakly to uniform level offsets.

This interpretation follows the standard decomposition of curve-comparison error into offset, amplitude, and pattern components [Taylor, 2001]: the mean square error between two curves separates into the squared mean offset, an amplitude mismatch, and a residual shape term. An equivalent three-term formulation, splitting waveform similarity into offset, amplitude, and shape, underlies the linear fit method developed for clinical gait waveform comparison [Iosa et al., 2014]. The metrics used here map onto this decomposition, with RMSE carrying the level disagreement, FVU the shape and amplitude agreement, and the maximum absolute residual serving as the phase-sensitive worst case. FVU is treated as the primary criterion because it is the only one of the three that is directly comparable across tests with different torque scales.

4.1.5 Optimization Algorithms and Cost Functions

Optimization enters this chapter in several places, and in every one of them it replaces a calculation that cannot be carried out any other way. The correction terms of the improved torque model are quantities that no instrument measures directly: they are hidden

inside an implicit force balance, and their best values are the ones that make the model's predicted torque match the measured torque. The route redesign is the mirror problem: the attachment locations are free design variables, and their best values are the ones that make the redesigned routes meet or exceed human torque capability. The cam-profile variant of the extensor route extends the redesign once more, searching over the shape of the tibial-side routing itself. In each case a search algorithm proposes candidate values, the physics model evaluates them, and the search proceeds until a stopping rule is met. This section states what is searched for, which solvers were chosen and why, and when each search stops.

Two families of optimization problem underpin the results of this dissertation and the design framework built on them. The first is the identification of the correction terms of Section 4.1.4 (χ_0, χ_1, χ_2, and χ_3) from the pinned-knee test data. The second is the redesign of the biomimetic knee route geometry, meaning the BPA origin $\mathbf{p}_1$, insertion $\mathbf{p}_{\text{end}}$, resting length, and tendon length (Appendix C), so that the flexor and extensor configurations meet or exceed human knee torque capability over the range of motion, together with its cam-profile extension, in which the tibial-side route points are free to trace a cam profile (Section 4.3.2). Both are carried out in MATLAB [The MathWorks, Inc., 2025].

The correction-term identification is a genuine multiobjective problem: candidate parameter sets are judged simultaneously on RMSE, FVU, and maximum absolute residual (Section 4.1.4). It is solved with the genetic multiobjective solver `gamultiobj`, an implementation of the elitist nondominated-sorting genetic algorithm NSGA-II [Deb et al., 2002, The MathWorks, Inc., 2025], which returns a Pareto front of parameter sets from which the adopted solution is selected by the holdout-validation procedure described above. The

production runs use adaptive-feasible mutation and scattered crossover at a fraction of 0.8, with MATLAB's default Pareto fraction of 0.35; the population settings are stage-specific. The flexor two-bracket fit evolves a population of 150 for up to 600 generations and terminates early once the geometric average relative change in the spread of the Pareto front falls below 4×10^{-3}; the extensor refit uses a population of 50 for up to 150 generations with a spread tolerance of 1×10^{-3}. In both stages the parameters are searched in a normalized space, with length offsets in centimeters and stiffnesses on a base-ten logarithmic scale, so that the genetic search samples the stiffness bounds, which span an order of magnitude, uniformly by decade, and the initial population is seeded around the expected solution region. The search bounds and the adopted values are recorded in Appendix C.

The route redesign replaces the multiobjective formulation with a single scalar cost assembled from weighted penalty terms, because the route prediction (contact activation, wrapping, and compliance correction) is expensive to evaluate and only moderately smooth. The cost has the form

$$J = w_w J_{\text{worst}} + w_t J_{\text{torque}} + \textstyle\sum_a J_a + J_{\text{geom}} + J_{\text{len}}, \tag{4.27}$$

where J_{worst} is the largest fraction by which robot torque falls short of the human target at any angle across the range of motion, weighted heavily ($w_w = 10^5$); J_{torque} is the mean-squared shortfall fraction ($w_t = 10^3$), weighted two orders of magnitude below the worst-angle term so that no shortfall at a single angle can be traded away for a better average; the J_a are configuration-specific auxiliary penalties described below; and J_{geom} and J_{len} are regularizers that keep the endpoints and actuator lengths near their seed

values. Human torque targets are computed from the Gait2392 lower-extremity model [Delp et al., 2007, Seth et al., 2018]: the flexor is graded against the biceps femoris short head and the extensor against vastus medialis. Each target is a maximum-isometric torque curve: OpenSim moment arms, obtained as the partial derivative of muscle-tendon length with respect to the knee coordinate, $r(\theta_k) = \partial l_{\mathrm{MT}} / \partial \theta_k$, are evaluated across the range of motion and multiplied by each muscle's maximum isometric force (provenance and exported tables in Appendix A). Hard constraints (bone clearance, wrap feasibility, actuator strain and contraction limits, tendon geometry) are enforced alongside the cost, and any candidate whose prediction fails is rejected with a prohibitive penalty.

The solution pipeline is common to both configurations. Each run begins with a full sanity evaluation of the seed design and a small smoke test of the solver setup, then executes `surrogateopt`, MATLAB's surrogate-assisted global solver for expensive black-box objectives, which builds a spatial interpolation surrogate over sampled candidate points and searches it adaptively [The MathWorks, Inc., 2025], in parallel under per-configuration evaluation budgets of 3,000 for the extensor route and 1,000 for the flexor route (the later cam variant of Section 4.3.2 uses 1,000 surrogate and 1,500 refinement evaluations), and finally refines the best feasible point with `patternsearch`, a derivative-free mesh-adaptive direct search method [Audet and Dennis, 2006, The MathWorks, Inc., 2025], for 15,000 further evaluations. The surrogate budgets sit well past the observed plateau of a few hundred evaluations, so they cap run time rather than define convergence. Each stage stops on its evaluation budget; the pattern search also stops early when its mesh collapses below the mesh tolerance with the constraints satisfied, which is the usual exit once the design has converged. The entry scripts and their cost functions are archived

with the code.

Although both configurations share this pipeline, their cost functions differ in ways that reflect the mechanics. The flexor BPA routes around a single moving wrap point, so clearance from the tibia and femoral condyles is enforced by continuous signed-distance constraints, and its cost adds an off-axis torque penalty (torques about the joint axes other than the knee axis, relative to the original single-BPA route) and a penalty if the wrap fails to release cleanly as the knee extends. The extensor BPA wraps over the front of the joint through as many as seven sequential contacts, so clearance is enforced instead by the contact-elimination rule that governs the route, and its cost penalizes torque overshoot above the human curve and excessive actuator contraction, which risks bottoming out at high flexion. The flexor and extensor targets alike carry a required torque margin of 5 % above the human curve, so that a route that merely touches the target at any angle is rejected. With two BPAs per joint, the flexor route is mirrored about the mid-sagittal plane and the two actuator forces are summed, while the extensor model scales the force of a single shared route by the actuator count.

4.2 Results

With the actuator force model validated at the actuator level, the same model was used to predict isometric torque about the knee joint for the test stands described in Sections 4.1.1 and 4.1.3. Measured torque is compared against three predictions: the original rigid-body torque model of Section 2.4, a prediction using the improved BPA force characterization alone (Chapter 3), and the improved torque model of Section 4.1.4 after optimization. The

hybrid torque method (Section 4.1.3) is used to isolate the sources of error.

4.2.1 Flexor on Pin Joint

Figure 4.6 shows the measured flexor torque on the pinned knee next to the pre-optimization prediction series for the four cross-validation tests. Both pre-optimization force models overpredicted the magnitude of the measured flexor torque, and on the angles where the two predictions are both defined the improved BPA characterization fit the measurements better than the original rigid-body prediction on every test, although it degraded near full extension, where the rigid-body prediction is undefined altogether (Table 4.1). The back-calculated (hybrid) torque reproduced the measurements closely, which localizes the misprediction to the modeled kinematics, the muscle length l_{m} and the moment arm r, rather than to the force model.

The initial identification, fitted on the $l_0 = 48.5\,\mathrm{cm}$ test, yielded a length offset $\chi_0 = 11.9\,\mathrm{mm}$ and effective stiffnesses $\chi_1 = 102\,\mathrm{kN\,m^{-1}}$ and $\chi_2 = 10.6\,\mathrm{kN\,m^{-1}}$. The corrected model at these terms followed the measured torque across the range of motion, including the loss of torque near full extension that the rigid-body prediction did not produce, and Figure 4.7 shows this corrected-family behavior at the adopted two-stage terms. The $l_0 = 45.7\,\mathrm{cm}$ actuator, configured to operate over a different joint range of motion, served as the validation test of the initial fit; the corrected prediction reproduced the measured torque there without refitting. At these initial terms, the corrected prediction for the training test had a root-mean-square error (RMSE) of $1.616\,\mathrm{N\,m}$, a fraction of variance unexplained (FVU) of 0.0958, and a maximum absolute residual of $3.2527\,\mathrm{N\,m}$; validation

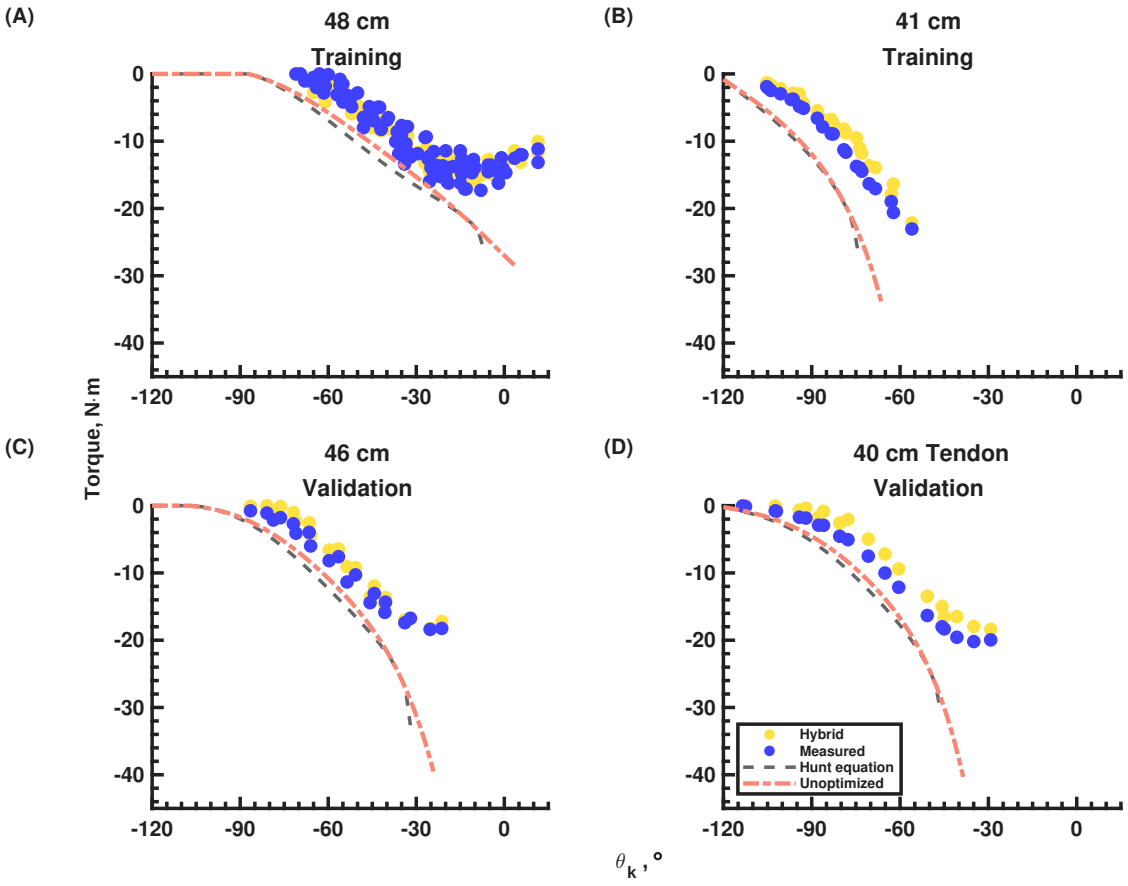


Figure 4.6: Pre-optimization pinned-knee torque versus knee angle for the four cross-validation flexor tests. Knee angle is negative in flexion, and flexion torque is negative. Gold circles are the hybrid calculation, which combines the measured muscle length and moment arm with the modeled force; indigo filled circles are experimentally measured torque; the gray dashed line is the rigid-body prediction; and the light orange dash-dot line is the unoptimized prediction, the improved BPA force characterization applied to the same rigid kinematics. Panel titles give the nominal resting length; the actual resting lengths are $l_0 = 48.5\,\text{cm}$ **(A)**, $l_0 = 41.0\,\text{cm}$ **(B)**, $l_0 = 45.7\,\text{cm}$ **(C)**, and $l_0 = 40.6\,\text{cm}$ with tendon **(D)**. Panels **(A)** and **(B)** are the training tests and panels **(C)** and **(D)** the validation tests of the adopted cross-validation fold.

Table 4.1: Pre-optimization pinned-knee flexor torque-prediction goodness of fit per test, comparing the three pre-optimization torque predictions of Figure 4.6: the rigid-body prediction, the unoptimized prediction of the improved BPA characterization, and the back-calculated (hybrid) torque. The rigid-body prediction is undefined near full extension, where its modeled strain falls below zero, so the rigid-body and improved-characterization columns are scored head-to-head on the measured points of each test at which the rigid-body prediction is defined (75 of 90, 17 of 20, 14 of 19, and 15 of 22 points, in panel order); the back-calculated torque is scored on all measured points. RMSE and maximum absolute residual (Max.) are in $\mathrm{N} \cdot \mathrm{m}$, and FVU denotes the fraction of variance unexplained. Training and validation follow the adopted cross-validation fold.

	Rigid body			**Improved char. alone**			**Back-calculated (hybrid)**		
Test	**RMSE**	**FVU**	**Max.**	**RMSE**	**FVU**	**Max.**	**RMSE**	**FVU**	**Max.**
$l_0 = 48.5\,\mathrm{cm}$ (training)	5.779	1.1761	9.798	4.871	0.8354	9.865	1.635	0.0991	3.842
$l_0 = 41.0\,\mathrm{cm}$ (training)	6.633	3.2313	11.409	6.074	2.7090	9.262	2.187	0.1184	4.243
$l_0 = 45.7\,\mathrm{cm}$ (validation)	5.315	0.9408	9.991	4.465	0.6640	9.536	1.246	0.0409	2.275
$l_0 = 40.6\,\mathrm{cm}$, tendon (validation)	3.945	0.6750	7.527	3.226	0.4512	7.376	2.085	0.0755	3.073
Mean (training)	6.206	2.2037	10.604	5.472	1.7722	9.563	1.911	0.1088	4.043
Mean (validation)	4.630	0.8079	8.759	3.845	0.5576	8.456	1.666	0.0582	2.674

on the held-out actuator gave 2.1214 N m, 0.0878, and 5.1690 N m, respectively (Table 4.4).

Table 4.2 completes this comparison at the adopted correction terms of the full re-identification (Section 4.1.4). Across the four tests of the two-bracket identification, the corrected model reduces the training-mean RMSE, over the $l_0 = 48.5$ cm and 41.0 cm training tests, from 8.66 N · m under the zero-correction prediction, the same kinematics and force model with every correction term disabled, to 1.62 N · m, and the validation mean over the held-out $l_0 = 45.7$ cm and 40.6 cm tendon tests improves from 7.59 N · m to 1.49 N · m. The behavior of the original rigid-body torque model on these tests is shown graphically in Fig. 4.6.

Table 4.2: Pinned-knee flexor torque-prediction goodness of fit per test. The zero-correction prediction uses the improved BPA force characterization alone (rigid attachments, no length offset, and no wrapping loss); the corrected model adds the adopted correction terms of the two-stage identification (Section 4.1.4). RMSE and maximum absolute residual (Max.) are in N · m. In the adopted cross-validation fold the $l_0 = 48.5$ cm and 41.0 cm tests formed the training set and the $l_0 = 45.7$ cm and 40.6 cm tendon tests were held out for validation; the $l_0 = 47.9$ cm test was not used in the identification and is reported in Appendix C.

	Improved BPA char. alone			**Corrected model**		
Test	**RMSE**	**FVU**	**Max.**	**RMSE**	**FVU**	**Max.**
$l_0 = 48.5$ cm (training)	7.617	2.1278	25.531	2.165	0.1718	5.467
$l_0 = 41.0$ cm (training)	9.694	2.1149	21.076	1.084	0.0277	1.766
$l_0 = 45.7$ cm (validation)	8.803	1.9380	26.264	1.368	0.0468	3.614
$l_0 = 40.6$ cm, tendon (validation)	6.373	0.5935	16.859	1.611	0.0427	5.891
Mean (training)	8.656	2.1213	23.303	1.624	0.0998	3.616
Mean (validation)	7.588	1.2657	21.562	1.490	0.0447	4.752

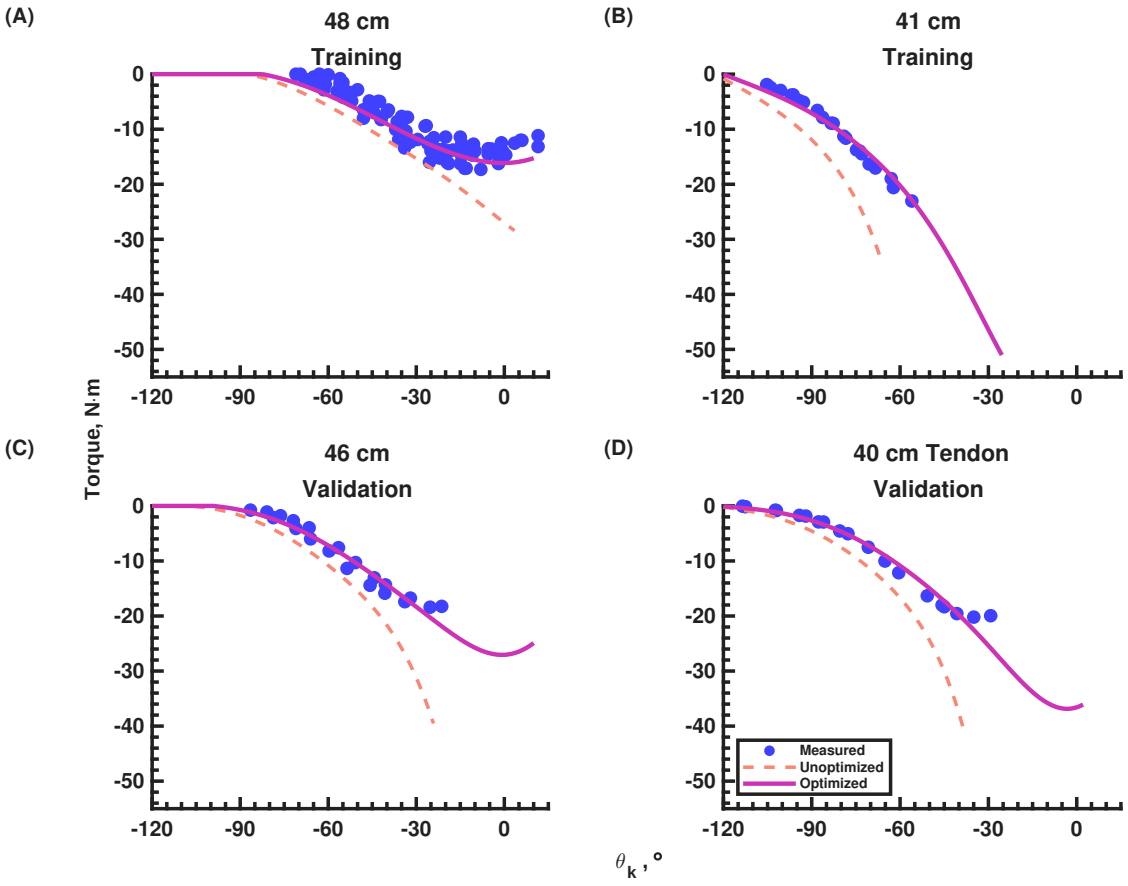


Figure 4.7: Optimized pinned-knee torque versus knee angle for the four cross-validation flexor tests, at the adopted correction terms (Table 4.5). Knee angle is negative in flexion, and flexion torque is negative. Blue filled circles are experimentally measured torque; the dashed light orange line is the unoptimized prediction of the improved BPA characterization; the magenta line is the optimized prediction of the corrected model. Panel titles give the nominal resting length; the actual resting lengths are $l_0 = 48.5\,\text{cm}$ **(A)**, $l_0 = 41.0\,\text{cm}$ **(B)**, $l_0 = 45.7\,\text{cm}$ **(C)**, and $l_0 = 40.6\,\text{cm}$ with tendon **(D)**. Panels **(A)** and **(B)** are the training tests and panels **(C)** and **(D)** the validation tests of the adopted cross-validation fold; the goodness-of-fit values are given in Table 4.2.

4.2.2 Extensor on Pin Joint

When the new stiffness and length offset model was used to predict torque for the knee extensor, measured torque was slightly higher than the original rigid-body model predicts for the 48.0 cm resting length BPA, slightly lower than expected for the 45.7 cm BPA, and much lower for the 41.5 cm BPA. Only the χ_1 and χ_2 terms from the previous optimization results were used with the extensor configuration. By inspection, the zero point of the measured torque was shifted to the right of the original model, indicating an offset opposite in sign to the χ_0 term in the previous section. The BPA with $l_0 = 45.7$ cm illustrates the pattern: the hybrid-method calculated torque correlates well with the originally predicted torque, while the actual measured torque diverges.

Results are shown in Fig. 4.8. The initial identification was fitted in cross-validated rounds over the 48.0 cm, 45.7 cm, and 41.5 cm actuators, holding each actuator out in turn, and the quoted round is the one holding out $l_0 = 41.5$ cm; it resulted in $\chi_0 = -10.9$ mm and $\chi_3 = 0.123$, the extensor length offset χ_0 being nearly equal in magnitude to the flexor offset but opposite in sign. The goodness-of-fit values of the initial round are found in Table 4.4. The two-stage extensor refit of Section 4.1.4 used a different six-test pool (Table 4.3) drawn from the same test campaign, in which the 41.5 cm and 48.0 cm tests of the initial set were excluded from the pool and the 45.7 cm test served as a training test of the adopted fold; Fig. 4.8 shows the six pool tests at the adopted terms.

At the adopted correction terms of the full re-identification (Section 4.1.4), Table 4.3 reports the per-test goodness of fit on the six extensor tests used by the identification, three for training and three for validation: the corrected model improves the RMSE of every

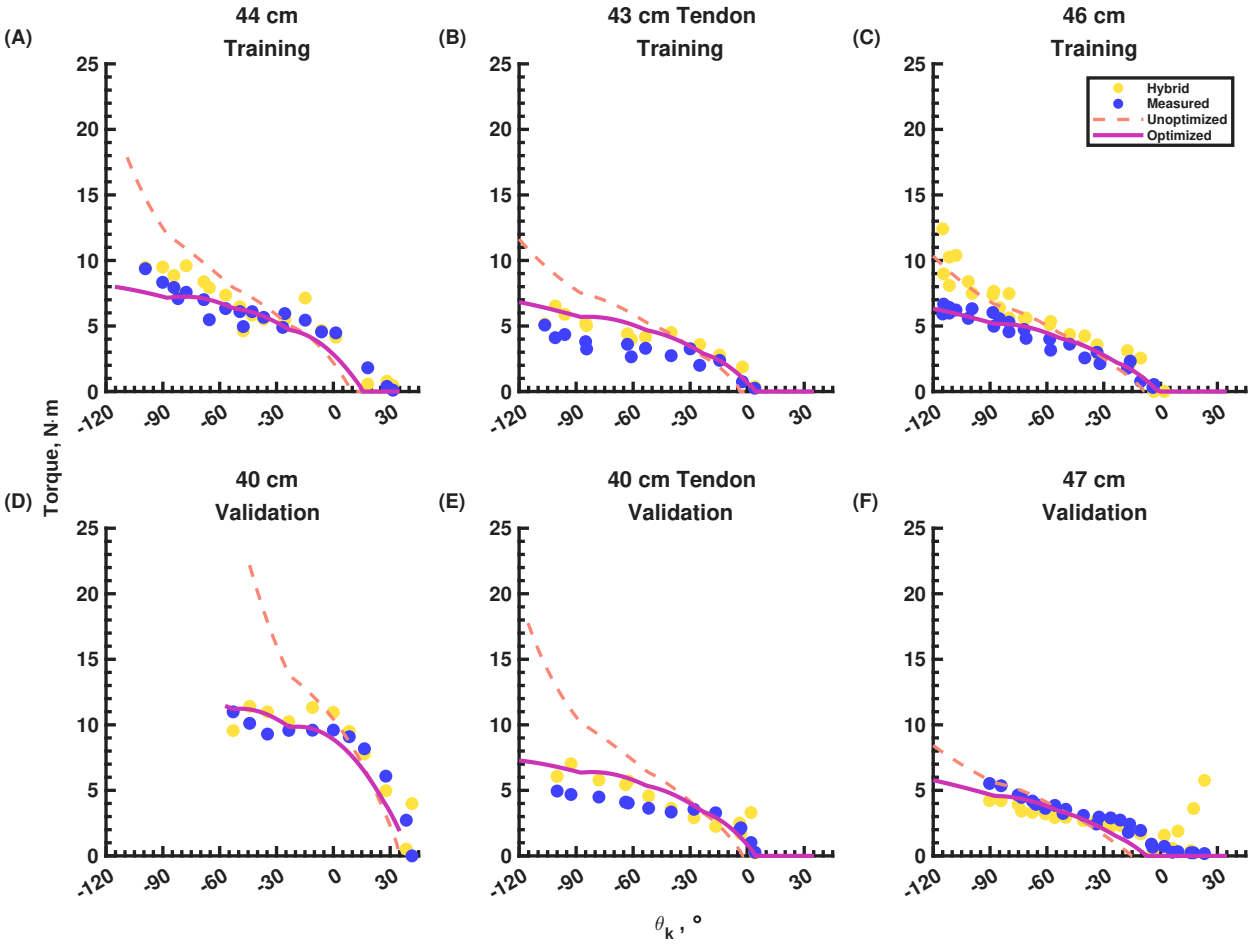


Figure 4.8: Pinned-knee torque versus knee angle for the extensor BPAs: the six identification-pool tests at the adopted correction terms, with the extensor-specific terms (χ_0, χ_3) on the shared stiffness pair (Table 4.5). Knee angle is negative in flexion, and extension torque is positive. Gold circles are the hybrid calculation; indigo filled circles are experimentally measured torque; the dashed light orange line is the unoptimized prediction of the improved BPA characterization; the magenta line is the optimized prediction of the corrected model. Panel titles give the nominal resting length; the actual resting lengths are $l_0 = 43.6\,\text{cm}$ **(A)**, $l_0 = 43.2\,\text{cm}$ with tendon **(B)**, $l_0 = 45.7\,\text{cm}$ **(C)**, $l_0 = 40.0\,\text{cm}$ **(D)**, $l_0 = 40.0\,\text{cm}$ with tendon **(E)**, and $l_0 = 46.5\,\text{cm}$ **(F)**. Panels **(A)–(C)** are the training tests and panels **(D)–(F)** the held-out validation tests of the adopted cross-validation fold; the legend is given in panel **(A)**.

used test over the zero-correction prediction and holds FVU below unity on all but the $l_0 = 43.2\,\text{cm}$ tendon test. The original rigid-body model on these tests is shown graphically in Fig. 4.8.

Table 4.3: Pinned-knee extensor torque-prediction goodness of fit per test, at the adopted correction terms (stiffness pair locked to the flexor-identified values). The zero-correction prediction uses the improved BPA force characterization alone; the corrected model adds the extensor-specific terms (χ_0, χ_3). RMSE and maximum absolute residual (Max.) are in $\text{N} \cdot \text{m}$. The extensor terms were trained on the $l_0 = 43.6\,\text{cm}$, $43.2\,\text{cm}$ tendon, and $45.7\,\text{cm}$ tests and validated on the $l_0 = 40.0\,\text{cm}$, $40.0\,\text{cm}$ tendon, and $46.5\,\text{cm}$ tests; the three remaining extensor tests were not used (Appendix C).

	Improved BPA char. alone			**Corrected model**		
Test	**RMSE**	**FVU**	**Max.**	**RMSE**	**FVU**	**Max.**
$l_0 = 43.6\,\text{cm}$ (training)	2.437	0.9618	4.284	0.978	0.1634	1.818
$l_0 = 43.2\,\text{cm}$, tendon (training)	2.878	4.7304	4.817	1.525	1.3270	2.481
$l_0 = 45.7\,\text{cm}$ (training)	2.178	1.0586	4.974	0.490	0.0535	1.206
$l_0 = 40.0\,\text{cm}$ (validation)	3.092	0.5638	4.842	1.244	0.1303	2.196
$l_0 = 40.0\,\text{cm}$, tendon (validation)	3.622	6.3919	7.845	1.350	0.8878	1.890
$l_0 = 46.5\,\text{cm}$ (validation)	0.949	0.3152	2.272	0.699	0.1714	1.559
Mean (training)	2.498	2.2502	4.692	0.997	0.5146	1.835
Mean (validation)	2.554	2.4236	4.986	1.098	0.3965	1.882

4.2.3 Extensor on Biomimetic Knee

The biomimetic knee was tested with a $\varnothing 10\,\text{mm}$ BPA in the extensor configuration. The bracket frame origin is the centroid of the cantilevered segment that holds the muscle origin bracket to the proximal femur (see Fig. 4.4**C**, right side).

The torque results for a $l_0 = 51.8\,\text{cm}$ BPA are shown in Fig. 4.9. The torque values measured are significantly smaller than both the original prediction and back-calculated

torque using the hybrid method. The results from the previous extensor configuration lead to a better torque prediction. GoF values can be found in table 4.4. Fig.4.10 shows the difficulty of modeling a BPA path that passes through the femoral condyles. Using the optimization results for extensor BPAs on the pinned knee joint, the $l_0 = 51.8\,\text{cm}$ BPA has an RMSE $= 0.8920$, FVU $= 0.3396$, and a maximum absolute residual of 1.4188.

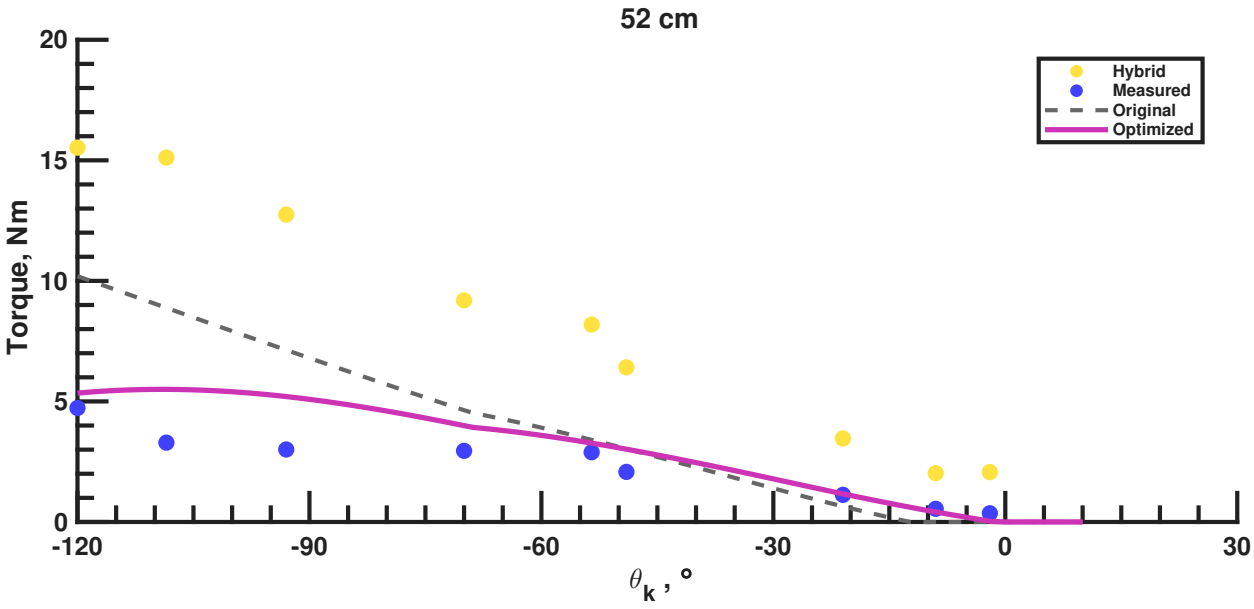


Figure 4.9: Measured versus expected isometric torque results using a ∅10 mm, $l_0 = 51.8\,\text{cm}$ extensor BPA on the biomimetic knee. Measured results are blue dots. Original prediction is a dashed black line. Optimized prediction is a solid magenta line. Hybrid calculation shown in yellow dots.

4.2.4 Flexor on Biomimetic Knee

Fig. 4.11 shows the biomimetic-knee flexor measurements with the flexor terms carried over unchanged from the pinned-knee identification; the bracket frame for the ∅10 mm

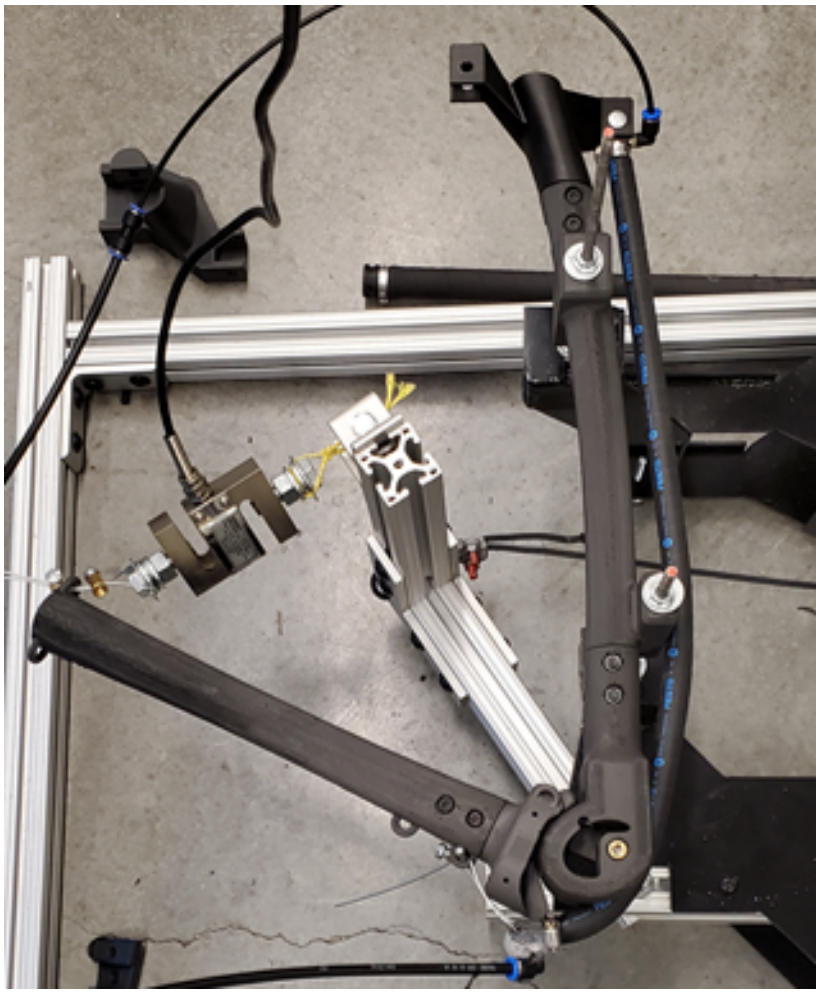


Figure 4.10: A configuration that is particularly hard to model. The BPA passes between the femoral condyles close to ICR.

BPA is shown in Fig. 4.4**C**, and for the ∅20 mm BPA in Fig. 4.4**D**. Panel **A** presents the ∅10 mm actuator at $l_0 = 41.5$ cm with a 21 mm artificial tendon (∅1.5 mm Shimano bicycle brake cable wrapped four times), pressurized to 604 kPa; the corrected model reached an RMSE of 0.3007 N m, an FVU of 0.0102, and a maximum absolute residual of 0.4647 N m. Panel **B** presents the ∅20 mm actuator at $l_0 = 41.5$ cm with a 14 mm artificial tendon made out of the same brake cable, wrapped three times so that six cable segments share the load, each wrap contributing two tendon lengths and cross-sectional areas. Panel **B** was tested at 620 kPa, for which the corrected model reached an RMSE of 1.8024 N m, an FVU of 0.0499, and a maximum absolute residual of 3.5678 N m.

The torque of the single ∅20 mm flexor actuator of panel **B** did not exceed the magnitude of the maximum human isometric flexion torque between 60° and 100° of knee flexion, and within a ±20% band about the human mean it still fell short of human-level torque between 78° and 103° of flexion. Because operating at 620 kPa across the full desired range of motion would load the printed femur beyond its ultimate strength, single points were also recorded at the reduced pressures 560 kPa, 421 kPa, 325 kPa and 281 kPa, which show that the desired range of motion is achievable at reduced pressure. This shortfall in flexion torque defines the target that the attachment-route redesign of Section 4.3.2 addresses.

4.3 Discussion

4.3.1 Sources of Error in Isometric Torque Generation

Our improved BPA characterization predicts torque more accurately than the original BPA model (see Fig. 4.6). This only accounts for part of the error. Many factors affect the torque

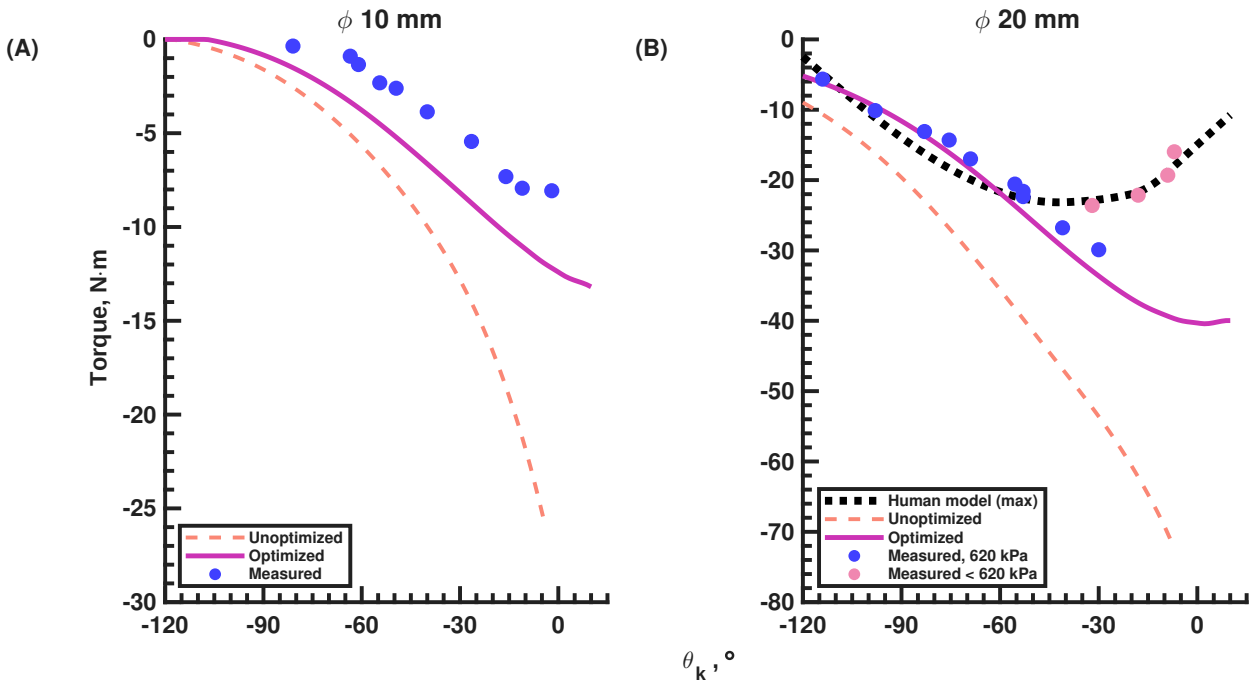


Figure 4.11: Biomimetic-knee torque versus knee angle for the flexor BPAs. Knee angle is negative in flexion, and flexion torque is negative. **(A)** ∅10 mm, $l_0 = 41.5$ cm actuator with a 21 mm artificial tendon at 604 kPa. **(B)** ∅20 mm, $l_0 = 41.5$ cm actuator with a 14 mm artificial tendon at 620 kPa. Indigo filled circles are measured torque; the magenta line is the optimized prediction of the corrected model using the flexor terms identified on the pinned knee; the dashed light orange line is the unoptimized prediction of the improved BPA characterization alone. In **(B)**, the black dotted line is the maximum human isometric torque, and the pink circles are measured points at the reduced pressures 560 kPa, 421 kPa, 325 kPa and 281 kPa (from left to right, respectively), recorded because full pressure would load the printed femur beyond its ultimate strength.

Table 4.4: Goodness of fit for the improved torque model across test configurations. RMSE and maximum absolute residual are in $\mathrm{N} \cdot \mathrm{m}$.

Configuration	**RMSE**	**FVU**	**Max. residual**
Flexor, pin joint, $l_0 = 48.5\,\mathrm{cm}$ (training)	1.616	0.0958	3.2527
Flexor, pin joint, $l_0 = 45.7\,\mathrm{cm}$ (validation)	2.1214	0.0878	5.1690
Extensor, biomimetic knee, $l_0 = 51.8\,\mathrm{cm}$	0.8920	0.3396	1.4188
Flexor, biomimetic knee, $\varnothing 10\,\mathrm{mm}$, $l_0 = 41.5\,\mathrm{cm}$	0.3007	0.0102	0.4647
Flexor, biomimetic knee, $\varnothing 20\,\mathrm{mm}$, $l_0 = 41.5\,\mathrm{cm}$ (620 kPa)	1.8024	0.0499	3.5678

results. Compliance in the testing system, positioning tolerances, flexibility of brackets, imperfect path modeling, and variation in individual BPAs are other factors.

Simplifying and testing the model revealed deficiencies in our previous analysis. The isometric system is not rigid. There is compliance in the winch cable, artificial tendons, and brackets that bend and stretch. We introduced a constant length offset term χ_0, axial stiffness χ_1, and bending stiffness χ_2. The bracket frame of reference was placed at the onset of the cantilevered section and rotated about Z. The optimization identified parameter values for a $\varnothing 10\,\mathrm{mm}$ BPA with $l_0 = 48.5\,\mathrm{cm}$ in the simple pinned knee configuration. These results were validated with a second $\varnothing 10\,\mathrm{mm}$ BPA with $l_0 = 45.7\,\mathrm{cm}$.

The modeling of extensor BPA paths that wrap around the knee joint is an approximation with inherent error (Fig. 4.10). When the moment arm is short, small changes to it can have a large effect on the resultant torque [Young et al., 2019]. It was assumed that wrapping the BPA around the joint would be easier to account for than adding a patella, deterministically designing its location as a function of knee angle, and accounting for friction. At greater knee flexion angles, shorter l_0 BPAs were observed to stretch and compress (see Fig. 4.1**C**). The same loading sometimes caused the BPA to slip off the anterior head

of the bolt that holds the femur to the knee. This produced a $\pm 20\,\text{mm}$ displacement in the Z direction, changing both the force vector and the torque. Fig. 4.8 shows a deviation between measured and originally predicted results that widens as expected torque increases. This indicates system compliance. However, the hybrid-calculated torque matches or exceeds the predicted torque, except at greater flexion angles for the $l_0 = 41.5\,\text{cm}$ BPA, indicating a loss of functional l_m; this is why the χ_3 term was created in (4.24). Fig. 4.8 shows the improved fit from running the optimization that includes this term.

The biomimetic knee joint with an extensor BPA was used to validate the extensor optimization results. The corrected prediction more closely matches the measured torque values, although it remains higher than the measurements.

The nonlinear biomimetic robotic system presented here replaces rigid-body simplifications commonly used in other robotic and biomechanical systems. Instead, progressing from simple to complex systems elucidated compliance mechanisms that can lead to more accurate first-iteration designs. Technologies have been developed to predict system bending and axial compliance, a constant length offset, and a loss of usable length that occurs as artificial muscle bends around a joint.

The quasi-static force characterization also fixes the coefficients that a dynamic actuator model needs. Writing the actuator's contraction as a second-order system, $m\ddot{\varepsilon}^* + c(\dot{\varepsilon}^*)\dot{\varepsilon}^* + k(\varepsilon^*)\varepsilon^* = F$, the stiffness term is the tangent slope of the normalized force surface of Equation (3.4), $k(\varepsilon^*, P^*) = \left.\frac{\partial F}{\partial \varepsilon^*}\right|_{P^*} = -c_0 c_1 e^{-c_1 \varepsilon^*} - 2c_2 \varepsilon^* P^* e^{-c_2 (\varepsilon^*)^2}$. Setting the mass and the external force to zero then yields the equivalent viscous damping coefficient, $c(\dot{\varepsilon}^*) = (k(\varepsilon^*)\varepsilon^*)/\dot{\varepsilon}^*$, the term that dynamic state-space actuator models, such as the pressure–force formulation of Elzein et al. [2025], pair with this stiffness.

4.3.2 Follow-Up Identification and Route Redesign

The coefficient values reported with the results figures belong to the initial single-bracket identification. Re-identifying the correction terms over the full test sets with the two-stage procedure of Section 4.1.4 yields the adopted values $\chi_0 = 3.94\,\mathrm{mm}$, $\chi_1 = 3.998 \times 10^4\,\mathrm{N\,m^{-1}}$, and $\chi_2 = 1.473 \times 10^4\,\mathrm{N\,m^{-1}}$ for the flexor, and $\chi_0 = -6.38\,\mathrm{mm}$ with $\chi_3 = 0.158$ for the extensor on that same locked stiffness pair (Table 4.5); the search bounds and the adopted values are recorded in Appendix C (Tables C7 and C8).

Table 4.5: Adopted correction terms of the two-stage identification. The extensor refit locked the effective-stiffness pair (χ_1, χ_2) to the flexor-identified values, so a single stiffness pair is shared across the flexor and extensor configurations, and only χ_0 and χ_3 were re-identified for the extensor. The pair was adopted from a family of near-equivalent fits by a requirement external to the identification, namely that both route-redesign configurations meet the required 5 % torque-overshoot target simultaneously. The stiffness terms are effective system stiffnesses that lump the compliance of the brackets, fixtures, and winch cable; χ_3 is unitless and scales the fictive-strain wrapping loss.

Parameter	Value	Search bounds	Configuration and role
χ_0	$3.94\,\mathrm{mm}$	$0\,\mathrm{mm}$ to $20\,\mathrm{mm}$	Flexor: constant effective-length offset
χ_1	$3.998 \times 10^4\,\mathrm{N\,m^{-1}}$	$3 \times 10^4\,\mathrm{N\,m^{-1}}$ to $1 \times 10^6\,\mathrm{N\,m^{-1}}$	Flexor: effective axial stiffness (bracket-frame x and z)
χ_2	$1.473 \times 10^4\,\mathrm{N\,m^{-1}}$	$5 \times 10^3\,\mathrm{N\,m^{-1}}$ to $2 \times 10^4\,\mathrm{N\,m^{-1}}$	Flexor: effective bending stiffness (bracket-frame y)
χ_0	$-6.38\,\mathrm{mm}$	$-20\,\mathrm{mm}$ to $5\,\mathrm{mm}$	Extensor: constant effective-length offset (opposite sign to the flexor value)
χ_3	0.158	0 to 1	Extensor: wrapping-loss (fictive-strain) scale factor

The pinned-flexor data do not uniquely determine the stiffness pair, because a family of pairs reproduces the training tests nearly equally well, so the adopted pair was selected by

a requirement external to the fit: both redesign configurations must meet the required 5 % torque-overshoot target simultaneously. What the correction terms do to the prediction is quantified in Tables 4.2 and 4.3. The corrected flexor model holds FVU at or below 0.18 on the four tests of the identification, the worst case being the 48.5 cm training test at 0.172, and the post hoc evaluation of the unused 47.9 cm test improves as well (Appendix C). On the extensor identification the corrected model improves the RMSE on every used test, cutting the training-triple mean from 2.50 N · m to 1.00 N · m and the validation-triple mean from 2.55 N · m to 1.10 N · m, and holds FVU below unity on every used test except the 43.2 cm tendon test. On the biomimetic extensor test ($l_0 = 51.8$ cm) the corrected prediction improves RMSE from 3.03 N · m to 1.17 N · m and FVU from 4.49 to 0.67, so the shared-pair model explains the bulk of that configuration's measured torque variance as well as its level. The genetic multiobjective solver is retained for the identification, where the Pareto front is required for holdout validation, and the surrogate-plus-pattern-search pipeline serves the route redesign (Section 4.1.5).

The route redesign built on these values addresses the single-muscle shortfall at high flexion angles documented in Section 4.2: each configuration now carries two 20 mm BPAs per route. The flexor solution meets the flexor torque target with a minimum margin of $+5.01$ % at -107°, under the modeling assumption that both actuators share one attachment route rather than a physically mirrored pair; making the mirrored pair explicit is the next design refinement. The direct extensor route meets or exceeds its human-scaled target with the required 5 % overshoot as the thinnest margin at every angle, $+5.02$ % at worst, while respecting the wrap-release schedule and contraction limits of the route-elimination rule (Appendix C); it was later superseded by the cam redesign of Section 4.3.2. These

redesign solutions are the model definitions carried into the neuromechanical simulation work of Sections 6.2–6.3 and Chapter 7.

Redesigned-Route Results on the Gait2392 Robot-Body Model

This subsection records the quantitative outcomes of the redesigned routes on the Gait2392-derived robot-body model, whose muscle force and torque targets define the redesign objectives. All three result families were produced by the surrogate-plus-pattern-search pipeline of Section 4.1.5 under the adopted correction terms (Appendix C, Table C8) with one shared actuator plan: two 20 mm BPAs per route, a 620 kPa pressure ceiling, and a maximum usable contraction of 0.255 in the force model. Each family minimizes its own objective function, a unitless weighted sum of normalized penalty terms in which zero corresponds to no violated penalty, led by the worst-angle and mean-square torque shortfall against the required 5 %-overshoot human envelope, followed by overshoot, rest-length, strain, contraction, and mild design-change penalties. The three functions differ in penalty families and normalizations (the flexor and pulley objectives add off-axis-excess, wrap-release, and second-BPA anchor terms, and the pulley objective evaluates its torque requirement on the insertion-side torque after the tackle's force loss), so their numerical values are not commensurable with one another. The human-scaled requirements are the OpenSim isometric torque profiles of the reference human muscles over the same design-grid angles, the vastus medialis for the extensor and the biceps femoris short head for the flexor.

The direct extensor route exceeds the human envelope in magnitude at every angle,

with zero shortfall points: across the design range the predicted two-BPA extensor torque spans 35.45 N · m to 109.4 N · m against the human-scaled requirement of 20.25 N · m to 50.32 N · m, and the required 5 % overshoot is consumed as the thinnest margin exactly as designed, at a minimum margin of +5.02 % at the worst-case angle of −20.2°.

The extensor route has since been redesigned once more by allowing the tibial-side route points to trace a cam profile, and this cam redesign is the current extensor design. The cam solution converts the razor-thin margin of the direct route into genuine design headroom. On the final geometry, which folds the optimizer winner together with a verified adjustment of the distal route point and tendon length, the minimum torque margin is +30.83 % with respect to the required envelope, the human target plus the 5 % overshoot, and +39.77 % with respect to the plain human target; the optimizer winner before that adjustment carried +31.30 % and +42.34 % at the same checks. The pressure required to meet the envelope spans 269 kPa–507 kPa across the range of motion, comfortably above the 150 kPa regulated floor of the VTUG valve manifold that supplies the hardware. The geometry confirms that the headroom is not bought with routing risk: the tibial wrap follows a bend radius of 96.9 mm, the guiding route point clears the femoral-condyle envelope by 43.1 mm at 10° of knee extension, and the moment arm spans 59.4 mm–102.8 mm (mean 72.4 mm) over the range of motion.

The flexor route, built on the same two-BPA arrangement sharing one attachment route, meets the flexor torque target with a minimum margin fraction of 0.0501 against the required 0.05. Its predicted knee-axis torque spans −81.23 N · m to −3.864 N · m against the human-scaled requirement of −23.16 N · m to −2.461 N · m (peak magnitude 23.16 N · m), again exceeding the human envelope at every angle. The flexor redesign applies the

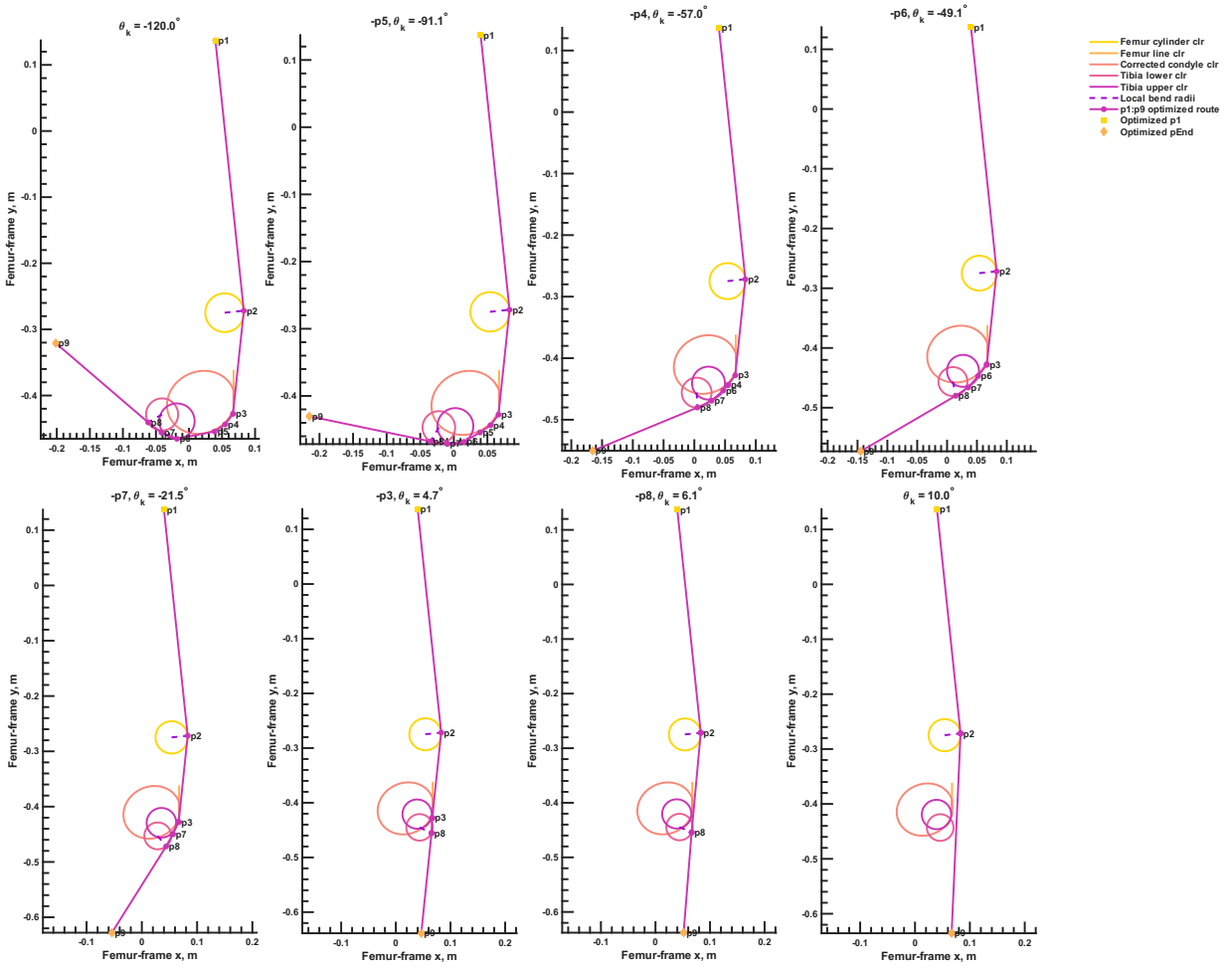


Figure 4.12: Route geometry of the direct extensor route redesign, drawn in the femur frame. Eight pose panels sweep the design range from full flexion ($\theta_k = -120^\circ$) through the frame immediately before each of the six contact-elimination events to full extension ($\theta_k = 10^\circ$); knee angle is negative in flexion. Each panel draws the active route over the femur-cylinder, femur-wall, condyle, and tibia clearance envelopes of the route-elimination rule, with the local bend-radius checks dashed. The square marker is the femur-side origin and the diamond marker is the tibial insertion.

adopted correction terms of Table C8, including the wrapping factor identified on the extensor configuration.

The Mono/Bi-pulley algorithms extend the route search space by allowing the artificial tendon to be reeved over pulleys before insertion. On the knee-flexor route, the pulley variant converged to a collision-feasible layout with zero bound violations and a peak predicted flexion torque of $85.56\,\mathrm{N}\cdot\mathrm{m}$ on the insertion-side measure, the torque applied at the insertion after the tackle's force loss; Section 4.3.3 quantifies what the redirect adds on the knee-axis measure and the off-axis torque it bounds.

The Bi-pulley batch applied the same reeving freedom to five muscle sites of the robot-body model's OpenSim leg, and all five runs failed the torque requirement, with the medial-gastrocnemius and biceps-femoris-short-head runs terminating at the infeasibility sentinel (Table 4.6). For the biceps femoris short head site, whose torque target is $40.2\,\mathrm{N}\cdot\mathrm{m}$ at the robot-body model's maximum isometric force of $804\,\mathrm{N}$, the pattern search exited with flag -2 and returned the sentinel objective of 101 from an initial objective of 1000; Section 4.3.3 reports the per-run objectives and the clearance constraints behind these outcomes.

The robot-body model itself carries one update from this work. The adductor magnus attachment point P2 was restored to its full-precision location $(-0.1357, -0.0929, 0.0591)$ m, displacing a rounded copy that was disqualified for producing a zero-length via point that nullified the flexion torque; the regenerated mirrored model passed its validation with zero non-mirrored pairs. At the restored location, the muscle delivers $34.28\,\mathrm{N}\cdot\mathrm{m}$ of flexion torque, $1.12\times$ the human target of $30.61\,\mathrm{N}\cdot\mathrm{m}$, and $27.62\,\mathrm{N}\cdot\mathrm{m}$ of adduction torque, $0.82\times$ the human target of $33.60\,\mathrm{N}\cdot\mathrm{m}$. These redesigned routes remain optimizer-stage

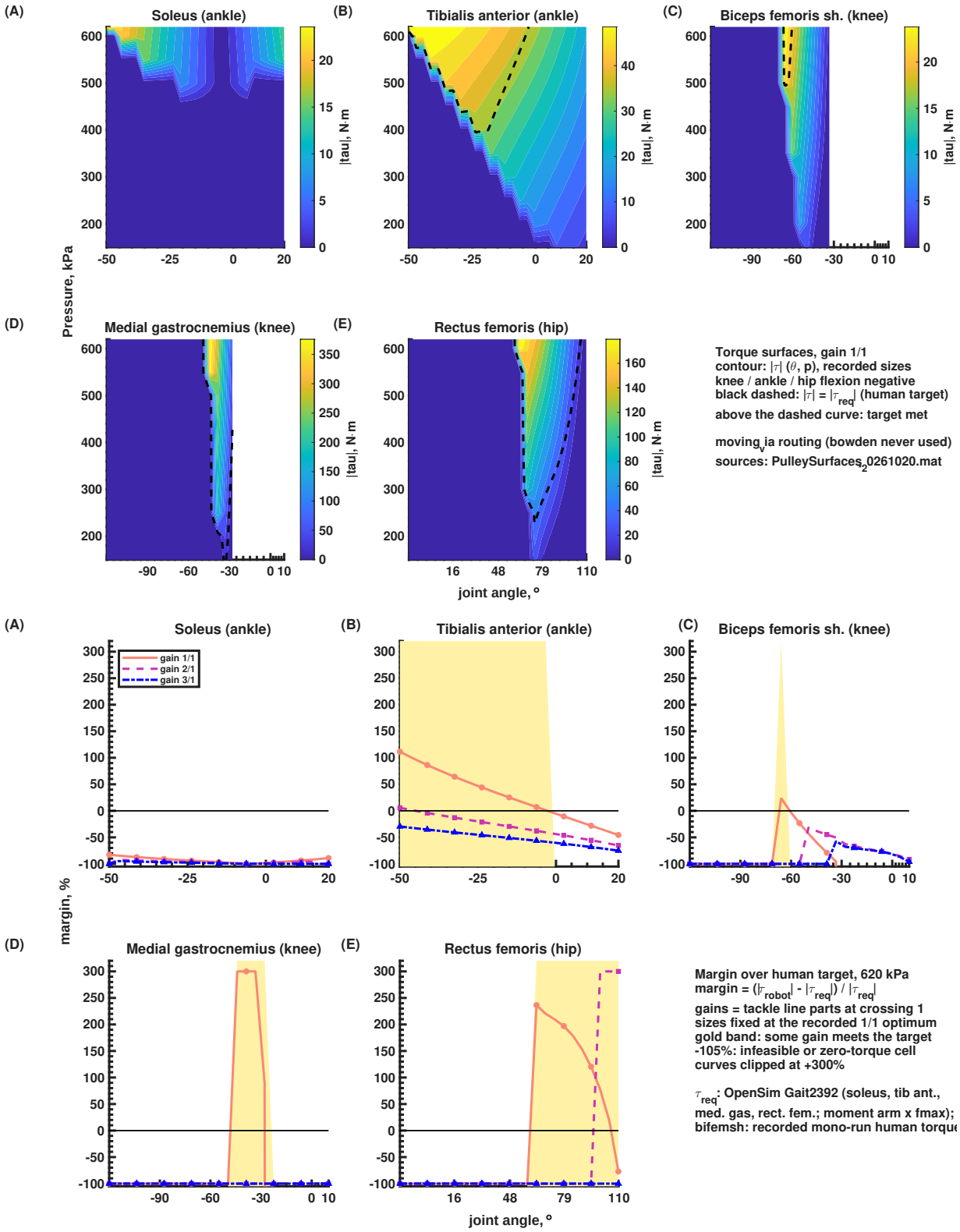


Figure 4.13: Pulley-position sweep for the reeved knee-flexor route. Top: the predicted torque surface over the two pulley positions along the tendon run; bottom: the corresponding torque-margin surface against the human-scaled flexor requirement, on the same position grid. The sweeps show where the reeving freedom pays, the feasible ridges of pulley placement, and where the margin collapses as the geometry approaches the

Table 4.6: Bi-pulley batch on five muscle sites of the robot-body model's OpenSim leg. Objective 101 is the batch's infeasibility sentinel; all five runs closed with a nonpositive torque margin. The soleus and tibialis-anterior runs converged feasibly (pattern-search exit flag 1), and their objectives are the normalized torque shortfalls; the remaining three runs exited with flag -2.

Muscle site	**Objective**	**Outcome**
soleus	0.9968	converged (flag 1), nonpos
tibialis anterior	0.9288	converged (flag 1), nonpos
rectus femoris	3.423 (worst constraint 0.2423)	nonpositive torque margin
medial gastrocnemius	101, infeasibility sentinel (worst constraint 10)	pattern-search exit flag -2
biceps femoris short head	101, infeasibility sentinel (worst constraint 10)	pattern-search exit flag -2

results: the robot-body OpenSim model contains no pulley entities, and no via points or markers were added to it for these routes (its eight pre-existing moving-path-point vias, the rectus-femoris and vastii patella-tracking points, one per muscle per leg, belong to the original muscle paths). The MuJoCo conversion used for the simulation work predates the pulley batch and the adductor-magnus restoration, so it carries one stale path point, the pre-restoration rounded adductor-magnus P2. Physical torque tests of the redesigned routes have begun (Section 4.3.3).

The same redesign pipeline was then extended from the knee to all 27 muscle-actuators of the robot-body model, and this whole-body extension is reported as work in progress rather than a completed result. The first pass reached 19 of the 27 actuators before the run stopped, and its chosen solutions showed that the penalty-based formulation itself was the binding limitation: all but one of the chosen routes violated the feasibility constraints, the sole exception being the biceps femoris long-head pair, whose worst-angle margin of $+0.03\,\%$ is indistinguishable from zero. The formulation was then restructured so that the

task envelope, the joint-angle window each route must serve, acts as a hard feasible set rather than a penalty. In the restructured pass, twelve of the 27 actuators terminated on a concatenation error in the route builder (the three lumbar-wall actuators, tibialis posterior, the seven distal-foot actuators, and adductor magnus), leaving fifteen finalized routes. Eight of the fifteen were chosen from constraint-feasible configurations: semimembranosus, biceps femoris long head, tensor fasciae latae, soleus, lateral gastrocnemius, tibialis anterior, iliacus, and psoas. Of those eight, six also met the torque requirement at every angle, with worst-angle margins spanning $+0.25\,\%$ (biceps femoris long head) to $+250.7\,\%$ (tensor fasciae latae); the semimembranosus and soleus routes satisfy the geometry but remain short of their torque targets. The crash set and the two torque-short feasible routes define the remaining engineering: the route builder needs a fix for the concatenation failure, and the short routes need either larger actuators or relaxed corridors.

4.3.3 Mono- and Bi-Pulley Tendon Routing

The route redesign of Section 4.3.2 placed both 20 mm actuators on direct attachment routes. The Mono/Bi-pulley algorithms extend that search space by allowing the artificial tendon to be reeved over pulleys before insertion, and the flexor pulley run quantifies what the added routing freedom is worth on the knee-flexor route. The pulley variant converged to a collision-feasible layout with zero bound violations (pattern-search exit flag 1) at an objective of 0.08465 under its own objective function, which differs from the direct route's objective of 74.47, so the two values are not commensurable. The best configuration on file reeves the two actuators over a single tackle-line part at a gain fixed at 1:1, so the pulley

acts as a redirect rather than a force multiplier; even so, the redirect alone deepened the peak predicted flexion torque from 81.23 N · m to 85.74 N · m, a gain of 4.51 N · m (5.6 %), while holding the off-axis torque to at most 0.803 N · m, under 1 % of the peak on-axis value; the predicted peaks of the direct and pulley routes exceed the human flexor torque scale of 23.16 N · m by factors of 3.51 and 3.70. The Bi-pulley batch establishes where that freedom runs out on anatomical routes. Across the five OpenSim muscle sites attempted (soleus, tibialis anterior, rectus femoris, medial gastrocnemius, and biceps femoris short head), every run closed with a nonpositive torque margin (objectives of 0.929 for tibialis anterior, 0.997 for soleus, and 3.42 for rectus femoris with a residual worst-constraint value of 0.242), and the medial-gastrocnemius and biceps-femoris-short-head runs terminated at the infeasibility sentinel (objective 101, worst-constraint value 10, exit flag -2). For the four runs that report a finite route-to-route clearance, the binding constraint was geometry rather than torque: minimum pair clearances spanned 4.26 mm to 19.14 mm against the 25 mm separation required between adjacent routes (two 10 mm actuator radii plus a 5 mm bundle margin); the medial-gastrocnemius run records no finite clearance, so its failure rests on the infeasibility sentinel alone. What the two algorithms buy is therefore reach and screening: the mono-pulley converts routing freedom into 4.51 N · m of additional predicted flexion torque without mechanical advantage and with off-axis torque below 1 % of the peak, and the bi-pulley sweep screens candidate anatomical sites, showing that its reeved routes do not yet fit these corridors and redirecting the design effort toward clearance-relaxed attachment sites. These pulley solutions are optimizer-stage results; they have not yet been propagated into the OpenSim or MuJoCo model trees used for the simulation work of Sections 6.2–6.3, and the hardware torque tests of the updated

compliant system have begun with the extensor route.

The first physical torque measurements of the redesigned extensor route have been collected. In the first test of the ∅20 mm extensor on the rebuilt route, three isometric points were held near full extension: measured knee torque of 1.163 N · m at a knee angle of 10°, 7.849 N · m at 11°, and 23.437 N · m at 10°, against hung loads of 3.7 N, 25.2 N and 76 N at gauge pressures of 388.5 kPa, 388 kPa and 592 kPa, respectively. The first and third points sit at essentially the same knee angle and span a twentyfold range of held torque, so they begin to map the route's response near extension as pressure and load vary. These three points are the start of the torque-versus-angle validation of the redesigned route against its required envelope; the full sweep, and the corresponding flexor measurements, are pending.

5 The Sensory Afferent Database

5.1 A Structured Afferent Resource

The synthetic nervous system of Chapter 6 is assembled from physiological parts: Ia, Ib, and II afferents, the reflex pathways they drive, and the spinal circuits those pathways modulate. Building that controller is a design problem, and design needs parameters, not only principles. For each sensory channel, the designer has to know what the receptor encodes, over what ranges of length, velocity, and force it operates and with what receptor dynamics, and what the afferent does at the spinal end: which interneuron populations it contacts, whether the action is excitatory or inhibitory, whether the target is ipsilateral or contralateral, and in which phase of the step cycle the pathway is open. Every element of that specification was established experimentally, one pathway and one preparation at a time, across a century of work in cat, mouse, rat, and human preparations, and the results are distributed over an enormous primary literature rather than collected anywhere. A controller designer is not the only reader stranded by that scatter. An animal researcher planning the next experiment wants to know which preparations established a pathway property, and how the property compares across species, before committing a preparation to it. A roboticist wants to know which receptor encoding a sensor should imitate. Those readers ask different questions of the same literature, and the literature answers none of them without being gathered, structured, and searchable first.

Reviews assemble part of this knowledge, but none can serve as the working resource

itself. The three this dissertation draws on illustrate both the value and the limit: Van de Crommert et al. [1998] organize the sensory control of the central pattern generator around the treadmill-training literature, Windhorst [2007] assigns functional roles to muscle proprioceptive feedback within a broader argument about spinal networks, and Windhorst [2022] weighs accumulated spinal-circuit model assumptions against the experimental record. Each synthesizes a slice of the literature; none can be filtered by afferent class, animal preparation, or pathway, none carries its demonstrating papers as attached and followable evidence, and each is fixed at its publication date. Extracting gain ranges, receptor dynamics, and phase-dependent feedback roles from reviews organized around a hypothesis means redoing the literature search the review was meant to spare the reader. When this work began, no structured compilation of afferent properties existed from which a controller designer, an animal researcher, or a roboticist could draw them; every reader redid the same search.

The Sensory Afferent Database (SADb) was built within the AARL's NeuroNex Communication, Coordination, and Control in Nervous Systems (C3NS) effort to close that gap, and we presented it at Neuroscience 2023 [Bolen et al., 2023]. It is a database rather than a bibliography: each paper carries distilled mechanism notes, each sensory feedback pathway is a first-class record linked to the papers that established it, and each computational model of afferent control is linked to its source paper and to the pathway records it implements. The database lives in a five-table Airtable base (Papers, Feedback, Models, Review Papers, and Recorders) backed by the laboratory's Zotero libraries, with the DOI as the canonical identifier for every record.

This chapter and the controller chapter are therefore built as one program. The database

informs the sensory channels of Chapter 6: the afferent classes the controller monitors, the signal encoding each channel carries, and the connection rules its circuit implements all trace to records held here, and Section 5.4 documents that mapping. The sections that follow describe how the corpus was assembled, audited, pruned, and packaged into its present form.

The audit and curation effort reported here began from a three-way reconciliation, because the database's three homes did not agree. The author's personal Zotero collection held 538 items, the AARL group library's Biology subtree held 911 items across 27 collections, and the Airtable base held 99 papers. Matching record by record on DOI and normalized title showed that the personal collection was essentially a subset of the group library (all 483 personal DOIs present) and that the Airtable base was a small curated subset rather than a mirror: 95 of its 99 papers resolved to records in the group library's Biology subtree, while 443 personal items had never been entered at all. The gap was imported in verified batches, with every written record re-read and checked after writing, bringing the Papers table to 532 records. The corpus then kept growing while curation caught up: citation mining in the style of Research Rabbit contributed a Crossref-verified digest of recent papers, newly saved references flowed in from the laboratory's Zotero libraries, and a bulk pass imported 372 papers auto-classified from the laboratory's staged collection of PDFs. By the close of the expansion the audited corpus stood at 943 papers, the full set against which every later decision in this chapter was made.

Curation then proceeded under rules designed to make every note citable. A note is written from the paper's own text, from the abstract at minimum and from the full text when available, never from the curator's recollection of the paper; papers for which no

text could be found were logged and left bare rather than guessed. Notes are one to three sentences of distilled mechanism, following the template of the laboratory's earliest records, and free-text fields are constrained by controlled vocabularies for animal preparations and afferent classes. Each feedback pathway is named by a fixed grammar of afferent, action, side, joint, and phase (for example, "Ib stance to swing"), so that pathway records compose into circuits the way the controller will use them. Ambiguous cases were flagged for a human ruling rather than decided silently.

The final curation wave added the fields that make the corpus answerable as a database: afferent class, animal-study potential, robot-and-simulation translation notes, and prune status fields, together with a corpus citation graph exposed as record-to-record links. Drafting was machine-assisted at scale, with 374 papers curated in fifteen batches; every batch passed automated vocabulary, abstract-paste, and duplicate gates before anything was written, and every record was verified against its grounding text after the write. Grounding remained the binding constraint throughout. A PDF-first rule recovered grounding for 47 of 55 flagged papers from the laboratory's own library, and a university-authenticated browser session recovered abstracts for 24 of the 70 papers that had no findable text; the rest stayed bare by rule.

Pruning followed, in three stages, each approved before execution, and together the stages removed 436 records: 382 papers, close to four hundred, plus 54 orphaned review, model, and pathway records that lost their anchors to the paper cuts. First, 95 records left as off-topic or orphaned: 88 papers removed as satellite topics (machine-learning and aerodynamics methods, flow sensing without neurophysiology, and fish swimming biomechanics), as duplicates, or as isolated records stranded by an earlier strict flagging

pass, plus seven orphaned pathway, model, and review records. Second, a cut list of 290 papers assembled from explicit rules for duplicates, monographs, and conference abstracts, and from a weighted rubric ranking the remainder by citation count, age, rating, network isolation, and record thinness, while 351 protected records (foundational authors, feedback-linked papers, highly cited work, and pre-1940 classics) were excluded from consideration. Third, a final round removed four papers and 47 review records whose links had been emptied by the earlier cuts. The prune was deliberate on two grounds. The working base had swollen past the laboratory's free-plan 1,000-record cap, and a curated resource that answers a design question in ten records beats a haystack of a thousand; the audit had shown the corpus contained both, and the satellite material was answering nothing. No record was destroyed: every deletion was snapshotted, and the packaged explorer described below carries the 382 cut papers under an archived flag, so the audited corpus remains reconstructible in full.

The audit and the prune are also why the laboratory built its own explorer app rather than adopting an existing tool. Airtable remains the working home of the data, but it is a curation tool, not a reader: it cannot travel past the laboratory's account, and a flat grid answers none of the questions this corpus exists to answer. An off-the-shelf bibliometric viewer was tried first and retired within weeks; it drew one crowded picture of the whole corpus, offered no drill-through from a cluster to the papers inside it, and let a handful of mega-cited classics dominate every view. The query patterns this resource must serve are specific: filter by afferent class, animal preparation, or pathway; follow a pathway record to the papers that established it in each species; walk the citation neighborhood of a candidate anchor paper; count the corpus by any two dimensions at once. A circuit designer wiring

a reflex, an animal researcher choosing a preparation, and a roboticist choosing a sensor analog ask different questions of the same records, and no existing tool answered any of them. The corpus itself was therefore packaged as a laboratory tool: a single-file, offline explorer embedding the audited corpus, the citation graph, and the named clusters, with searchable and pivotable tables, focus-mode citation neighborhoods drawn in the laboratory's own convention of papers as neurons with excitatory and inhibitory synapses, and a generated knowledge base with an index page plus cluster, pathway, afferent, and animal pages, all distributed as a zip that runs in a browser on any machine with no network connection.

5.2 Audited Corpus and Curation Coverage

At the close of the curation effort the working base held 957 records: 561 papers, 39 feedback-pathway records, 244 model records, 109 review records, and 4 recorder entries, 43 records below the laboratory's 1,000-record operating cap, with room to grow. Coverage of the retained papers is concentrated where the laboratory's design and experimental work needs it. Of the 561 retained papers, 523 carry curation notes, 426 carry animal-preparation tags, 240 carry notes assessing each paper's animal-study potential, 118 are linked to named feedback pathways, 86 carry an afferent classification, and 258 carry notes translating the paper's findings toward robot or simulation use; 550 carry DOIs, and 38 remain bare pending a findable text.

The citation graph was built over the full audited corpus of 943 papers, which includes the 382 records the prune later removed and which the packaged database retains under

an archived flag. Within that corpus, 730 papers are connected by 11,130 directed in-corpus citation pairs, and a Louvain community partition of the graph resolves 18 named topic clusters, from the classics of afferent locomotor control and spinal interneuron circuit modeling through human balance and posture to bioinspired flow sensing and deep-learning methods. The partition earned its keep in the prune: the satellite clusters it exposed, the flow-and-flight sensing, fish swimming, and neural-network method clusters, match the topic groups the first prune stage removed. Within that audited corpus, 833 papers carry curation notes and 318 carry attached PDFs, and the prune's 382 removed records remain in the package under an archived flag so the audited scope stays reconstructible in full. The corpus, its citation graph, its cluster assignments, and the flat corpus export are distributed together in the packaged explorer, and Figure 5.1 shows its table view as packaged. Figure 5.2 summarizes the same corpus at the topic-cluster level: the seven clusters the curated corpus retains, and the satellite clusters the prune removed in full.

5.3 AI-Assisted Literature Triage

Expanding a curated corpus is dominated by triage: deciding, paper by paper, what a record is about and whether it belongs. We evaluated AI tools for this task in a comparative study presented at Neuroscience 2024 [Bolen and Hunt, 2024]. The evaluation found the expected trade: the tools accelerated the population of the database, but their output still required substantial human oversight for accuracy.

The curation workflow reported here is the operational answer to that finding, a divi-

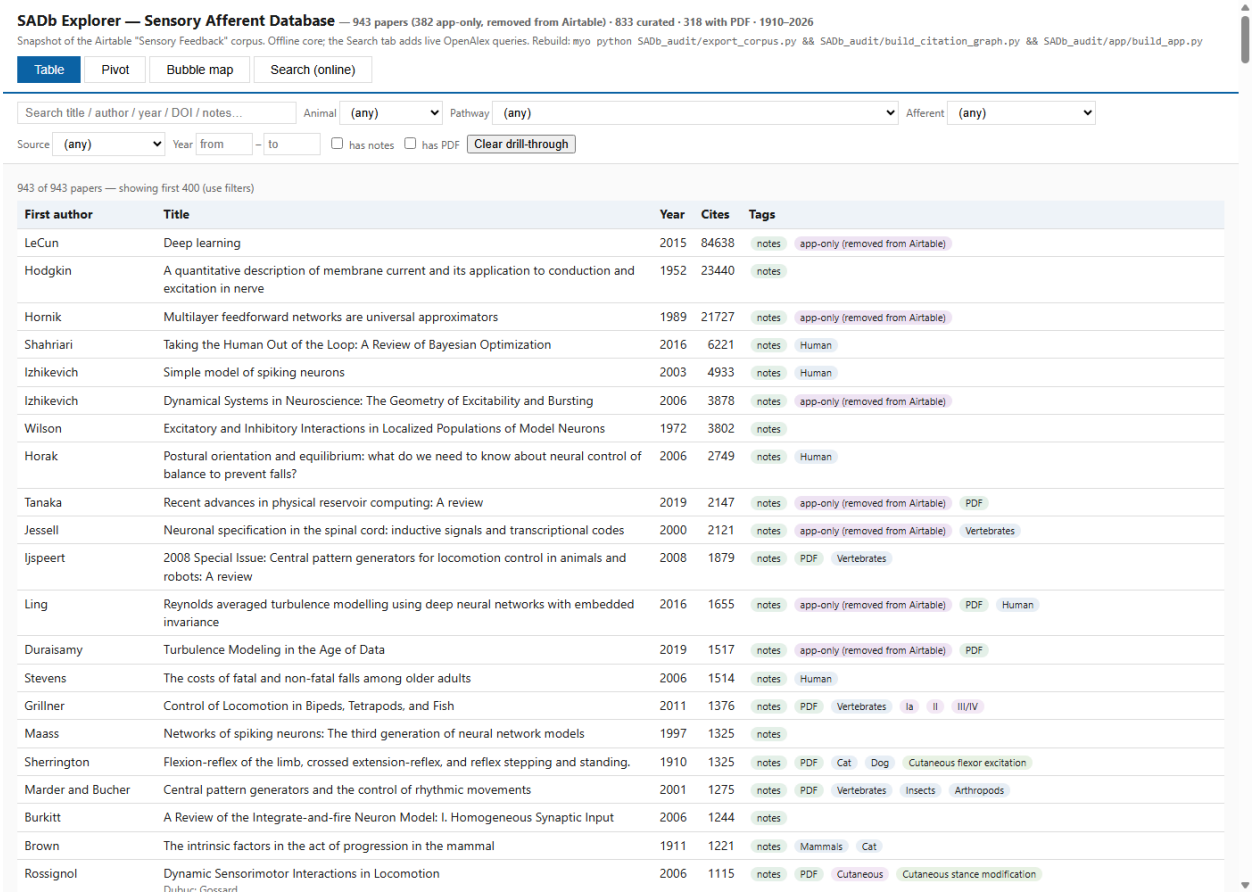


Figure 5.1: The SADb Explorer, the packaged offline interface to the audited corpus, shown on its table view: the Papers table sorted by citation count, with the tag column recording each paper's curation state, animal preparation, and afferent or pathway links. The application runs in a browser from a single file with no network connection required.

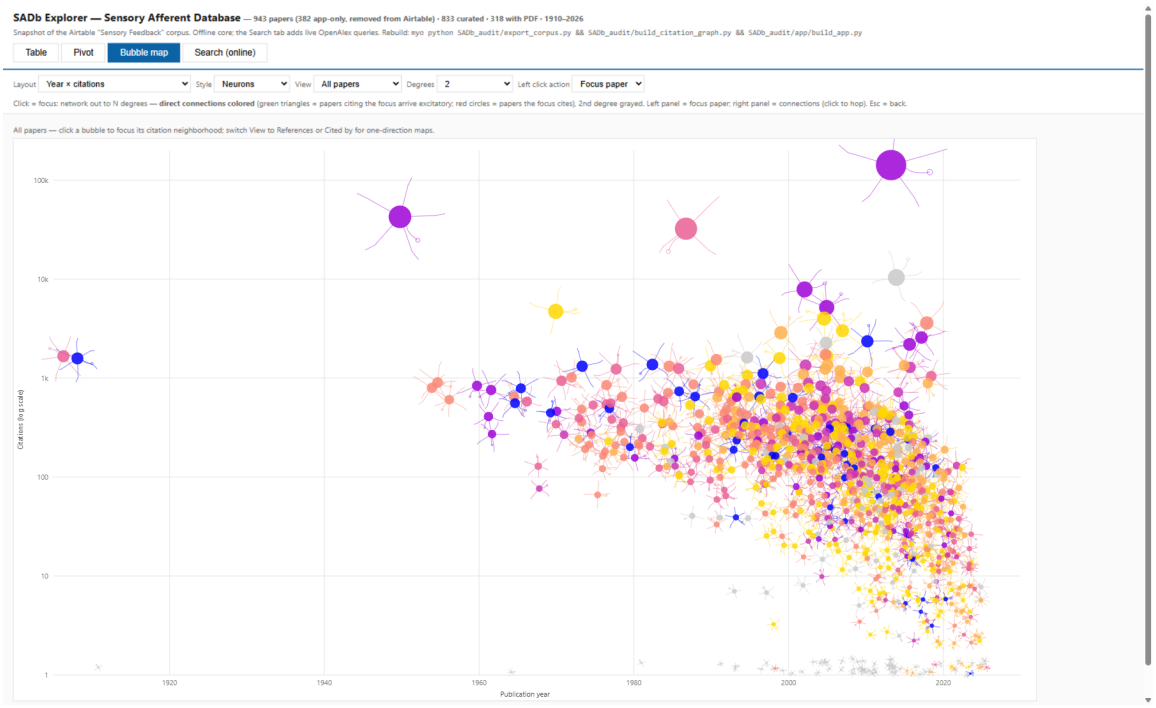


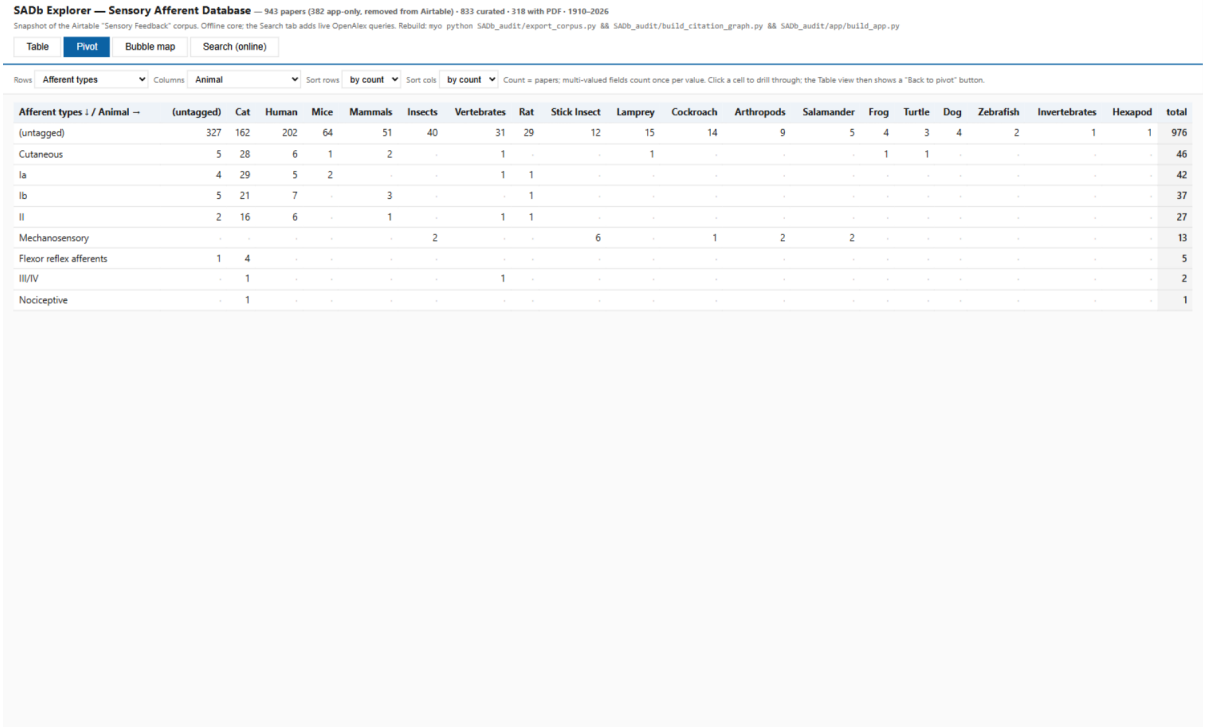
Figure 5.2: The SADb Explorer's bubble map view, the packaged citation graph of the full audited corpus: every paper is drawn as a neuron, its soma sized by citation count, with citation links rendered as excitatory and inhibitory synapses and cluster labels grouping the topic communities. Clicking a bubble focuses its citation neighborhood out to a chosen degree, with direct connections colored by direction. The view shows the curated core clusters, from the afferent-control classics and spinal-circuit modeling to neuromechanics and robot walking, alongside the satellite clusters the prune later removed.

sion of labor in which the machine drafts and the human rules. The machine contributed scale: batch drafting from grounded text, automated gates that rejected abstract pastes and duplicate records before writing, and per-record verification after it. The gates earned their place on caught failures, including a record that shipped another paper's abstract under its title until the grounding pass flagged it. The human held every decision that required judgment or carried risk: the curation vocabulary and its amendments, the 164 flagged judgment calls surfaced for ruling, and every single deletion in the prune, each stage of which waited for explicit approval. The gates and flags are not ceremony; they are the oversight the 2024 evaluation showed was necessary, made concrete.

5.4 Toward Robot and Simulation Use

The database exists to feed the sensorium of Chapter 6. Its pathway records enumerate the signal taxonomy the controller's sensory channels implement: Ia afferents carrying length and velocity, Ib afferents carrying force, II afferents carrying length, and contact receptors marking the stance–swing boundary (Section 6.2); Table 5.1 summarizes what the corpus establishes for each class. The pathway naming grammar maps directly onto circuit edges, since a record's afferent class fixes the signal encoding, its phase fixes the gating, and its action fixes the sign, so translating a pathway record into a controller connection is a lookup rather than a literature search; Table 5.2 lists the principal pathway records with the papers that demonstrate them. The Models table carries the lineage the controller instantiates, from the two-level rhythm-generator and pattern-formation architecture [Rybak et al., 2006] through muscle-reflex walking [Geyer and Herr, 2010] to

the two-layer neuromechanical walker the laboratory's controller descends from [Deng et al., 2019], and the 258 robot-and-simulation translation notes exist precisely to shorten the step from paper to wiring decision. Figure 5.3 shows the two-dimension count the explorer's pivotable tables produce over the retained corpus, the afferent classes against the animal preparations: the classical cat preparation carries the largest share of the Ia, Ib, II, and cutaneous records, and the arthropod records enter through the mechanosensory class.



Afferent types ↓ / Animal →	(untagged)	Cat	Human	Mice	Mammals	Insects	Vertebrates	Rat	Stick Insect	Lamprey	Cockroach	Arthropods	Salamander	Frog	Turtle	Dog	Zebrafish	Invertebrates	Hexapod	total
(untagged)	327	162	202	64	51	40	31	29	12	15	14	9	5	4	3	4	2	1	1	976
Cutaneous	5	28	6	1	2	·	1	·	·	1	·	·	·	1	1	·	·	·	·	46
Ia	4	29	5	2	·	·	1	1	·	·	·	·	·	·	·	·	·	·	·	42
Ib	5	21	7	·	3	·	·	1	·	·	·	·	·	·	·	·	·	·	·	37
II	2	16	6	·	1	·	1	1	·	·	·	·	·	·	·	·	·	·	·	27
Mechanosensory	·	·	·	·	·	2	·	·	6	·	1	2	2	·	·	·	·	·	·	13
Flexor reflex afferents	1	4	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	5
III/IV	·	1	·	·	·	·	1	·	·	·	·	·	·	·	·	·	·	·	·	2
Nociceptive	·	1	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	1

Figure 5.3: The SADb Explorer's pivot view, counting the corpus along any two dimensions at once; here rows are the afferent classifications and columns the animal preparations, so each cell counts the papers carrying both tags. The classical cat preparation dominates the Ia, Ib, II, and cutaneous records, and the arthropod records enter through the mechanosensory class; clicking a cell drills through to the underlying papers in the table view.

Table 5.1: Afferent classes represented in the Sensory Afferent Database, the signal each class carries, and the pathway roles the corpus records for it. Anchor references are corpus papers linked to the corresponding records; the signal summary for the Ia, II, and Ib rows follows the ensemble afferent recordings of Prochazka and Gorassini [1998].

Class	**Receptor**	**Signal carried**
Ia	muscle spindle primary	muscle length and velocity
II	muscle spindle secondary	static muscle length

Table 5.2: Principal feedback-pathway records of the database's Feedback table, named by the fixed grammar of afferent, action, and phase. Linked papers are counted over the full audited corpus of 943 papers; for every pathway listed, the count is unchanged under the retained corpus of 561. Demonstrating references are the corpus papers linked to each record, as curated in the database.

Pathway record	Class	Recorded action in the step cycle	Linked papers	Demonstrating references
Ia monosynaptic excitation	Ia	stretch excitation of the homonymous motor pool, with the pathway gain modulated over the step cycle	7	[Eccles and Lundberg, 1958, Capaday and Stein, 1986]
Ia reciprocal inhibition	Ia	inhibition of the antagonist motor pool through the Ia inhibitory interneurons, rhythmically active in fictive locomotion	6	[Jankowska et al., 1967, Pratt and Jordan, 1987]
Ib disynaptic inhibition	Ib	autogenic non-reciprocal group I inhibition of the loaded muscle's own pool, reduced during stance	5	[Jankowska et al., 1981, Stephens and Yang, 1996]
Ib disynaptic excitation	Ib	the locomotor reversal: group I input excites synergist extensor pools during the extensor phase, replacing the resting inhibition	6	[McCrea et al., 1995, Pearson et al., 1998]
Ib stance to swing	Ib	the load gate: Ib input from ankle extensors prolongs stance, so swing is not initiated while the leg is loaded	17	[Dietz and Duysens, 2000, Domínguez-Rodríguez et al., 2020]
Ia or II stance to swing	Ia, II	flexor-length gate on stance termination: lengthening of flexor muscles in late stance ends extension and resets the cycle	10	[Hiebert et al., 1996, Akay et al., 2014]
type II excitatory	II	stance-phase support of ankle-extensor activity through the midlumbar group II relay	5	[Edgley and Jankowska, 1987, Mazzaro et al., 2006]
Cutaneous flexor excitation	Cutaneous	flexion withdrawal and the stumbling corrective reaction, phase-modulated over the cycle	16	[Sherrington, 1910, Forssberg, 1979]

The 39 pathway records extend beyond the mammalian core of Table 5.2: paired excitation and inhibition records cover the chordotonal organs and trochanteral hair plates of insect legs, a load-feedback record covers the campaniform sensilla, and further records hold fictive locomotion without sensory feedback (12 linked papers), group III/IV fatigue, and one paper each for somatosensory, vestibular, and visual contributions to postural control. The controller of Chapter 6 implements the mammalian subset.

The corpus also reaches ahead to the sensorium extensions planned in Section 7.2.5. The human balance and posture cluster and the multisensory recalibration cluster are the entry points for the vestibular and ocular feedback work, already curated and already connected to the locomotor core through the citation graph. The knowledge base makes this reach practical for new students as well: the cluster, pathway, and animal pages compress a literature that took months to curate into an onboarding document measured in pages, and the animal pages let an experimentalist start from a preparation rather than a topic.

The honest accounting is that the resource is deeper than it is broad in places. The afferent classification covers 86 of 561 retained papers, concentrated in the pathway-rich core rather than spread across the corpus, and the flagged vocabulary gaps (vestibular and visual preparations among them) mark where the classification must grow before the extensions of Section 7.2.5 can be wired from the database alone. The growth policy follows from the prune: additions should be substantive records or clearly flagged stub twins, never bare rows, so that the corpus that the controller work, the next experiments, and the next students inherit stays as searchable as the one this chapter documents.

6 Synthetic Nervous System Simulation

The preceding chapters establish how braided pneumatic actuator force and knee torque can be predicted. To study how neural control interacts with these mechanics, I also needed a simulation environment that connects anatomically organized spinal circuits to the robot's muscle routing and BPA force model. The existing tools reviewed in Chapter 2 (Section 2.9) provide the neural, biomechanical, and physics components separately; using them for this robot required implemented interfaces and tests of the behavior preserved through conversion. The gap this chapter fills is that integration: no existing environment coupled an anatomically organized two-level controller, with Ia, Ib, group II, and contact feedback, to a BPA-actuated robot model with validated actuator forces.

In this chapter I develop and evaluate those interfaces in MuJoCo with SNS-Toolbox and in a graphical Simulink workflow, and I verify them rather than assume them: a controller ported between environments cannot be presumed equivalent, so each port is checked against its source implementation and, where the lineage's archived models allow, against the published behavior of the models it descends from. The complementary roles of the environments reflect the complementary needs of the program, fast scripted closed-loop simulation for tuning and evaluation, graphical circuit construction for assembly and inspection, and CAD-derived mechanics for the physical assemblies; Section 6.2.2 states which capability each environment was missing before this work. I also examine whether selected behaviors persist when graded neural components are replaced with spiking

components. The contribution is the integration and its verification, with the distinction from existing Simulink libraries established in Section 6.2.2 by direct comparison. The neural vocabulary, the two-level central pattern generator with rhythm-generator and pattern-formation layers, the graded neuron and synapse models, and the afferent channels, is defined in Sections 2.5–2.8 and used here without redefinition.

The demonstrations establish specific component and closed-loop capabilities, and they identify failures of behavior transfer and limits of the assembled models with equal weight. Supported stepping and progression are distinguished from unsupported walking, which has not been demonstrated. Section 6.1 records the AnimatLab walker whose hand-tuning motivated this verification-first procedure: manual tuning left the circuits' parameter sensitivity unmeasured, so the present procedure records performance under parameter and input perturbations instead. Section 6.2 is the methods section, Section 6.3 the results, Section 6.4 the spiking mirrors, and Section 6.5 the discussion.

6.1 The AnimatLab Bipedal Walker

As preliminary work of this dissertation, built before the simulation toolchain of this chapter existed, the author and Morrow constructed a neuromechanical model of a bipedal walker in AnimatLab, a graphical environment for building biomechanical systems actuated by synthetic neural systems [Cofer et al., 2010]. That model is the direct ancestor of the work reported here: its biomechanics and its two-layer controller architecture carry through every model in this chapter, and the limitations of its hand-tuned parameters are the reason the present toolchain emphasizes systematic verification.

The biomechanical model consisted of nine rigid bodies (pelvis, thigh, shank, foot,

and toe segments) connected by sagittal-plane hinge joints with human-like rotational capabilities and actuated by antagonistic pairs of Hill-type muscles at the hip, knee, and ankle. The toe joint, the author's addition to the Li design, carried a passive spring instead of a muscle pair, simplifying the system. The mechanics followed the suspended bipedal walking model of Li et al. [2017], whose pattern generators were entrained by heel and toe mechanosensory receptors and hip-angle position sensors standing in for the group-I loading, cutaneous, and hip-joint afferents of the biological scheme; muscle afferents were not modeled. The Li model also required support against falling: a system with only planar joints cannot actively balance side to side.

The synthetic nervous network was built from non-spiking, leaky-integrator neurons and followed the two-layer CPG organization of Deng et al. [2019]. A rhythm-generator layer set the timing of the system; a pattern-formation layer reshaped the timing signals into phase-appropriate activation of antagonist muscle groups; and a sensorimotor layer containing Renshaw-cell-like recurrent inhibition (motor-neuron collaterals exciting inhibitory cells that feed back onto the same pool) drove the motor neurons. The sensorimotor layer received Ia (length and velocity) and Ib (tension) afferent feedback from the muscles, but the feedback gains were never systematically tuned, and no ground-contact signals were modeled. Every applied current, synaptic weight, and muscle-tension parameter was set by hand: each was adjusted by trial and error, one at a time, until the three layers oscillated in phase and the motor neurons produced stepping-like activation of the muscle pairs. That hand-tuning consumed the bulk of the project's effort, left no record of the sensitivity of the behavior to individual parameters, and did not generalize when new feedback pathways were added; those limitations directly motivated the verification-first

approach of the remainder of this chapter. Alternating, stepping-like activity and motion were demonstrated with the walker suspended in the air, and the results were presented at the 2021 NIH BRAIN Initiative Investigators Meeting [Bolen and Hunt, 2021].

6.2 Methods: A Neuromechanical Simulation Toolchain

This section documents the laboratory in build order: the existing tools it stands on, the new tools built from them, each new tool's construction and fundamental demonstration, and the methodology by which the controllers are tuned and checked. Figure 6.1 diagrams the flow the toolchain implements, from model conversion through closed-loop simulation to the embedded hardware stack of Chapter 7. Quantitative results for every fundamental demonstration, the replications of published studies, and the experiments the tools enable follow in Sections 6.3 and 6.4.

6.2.1 Existing Tools and Data Resources

OpenSim supplies the musculoskeletal modeling side, the robot's source model, and the moment-arm and torque standards the conversions are audited against [Delp et al., 2007, Seth et al., 2018]. MuJoCo supplies the physics engine [Todorov et al., 2012], and MyoConverter performs the base OpenSim-to-MuJoCo conversion [Caggiano et al., 2022, Wang et al., 2022]. SNS-Toolbox builds and integrates the spinal networks [Nourse et al., 2023]. AnimatLab hosts the controller lineage's archived reference models and remains the graphical prototyping environment [Cofer et al., 2010]. Simulink and Simscape Multibody supply the controls-engineering environment into which the SNS component models and

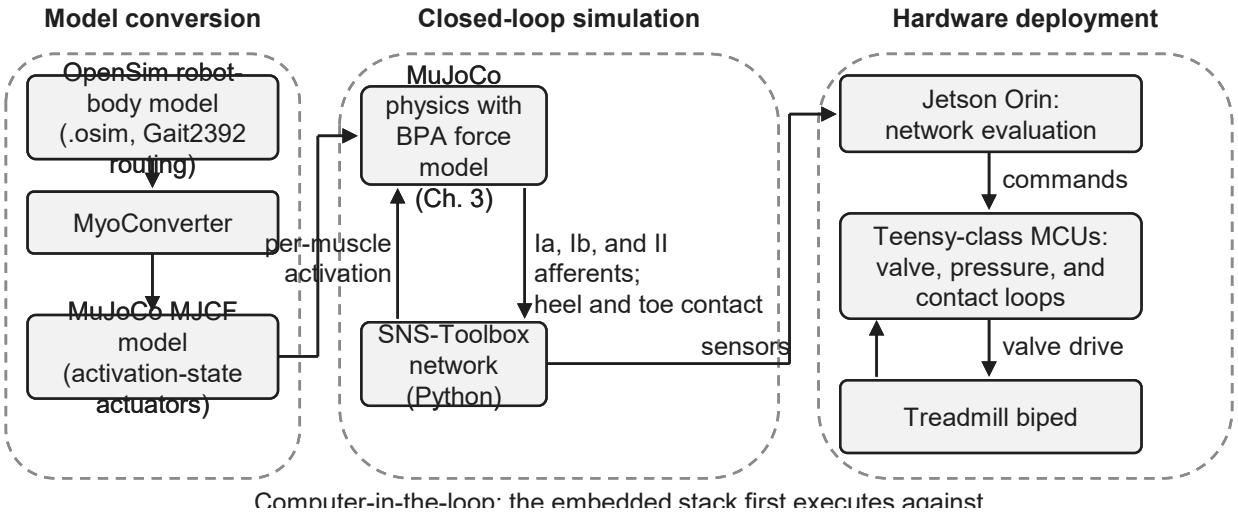


Figure 6.1: The simulation-to-robot pipeline in three phases. Model conversion carries the OpenSim robot-body model through MyoConverter to a MuJoCo MJCF model with activation-state muscle actuators. Closed-loop simulation couples that model, with actuator forces drawn from the BPA force model of Chapter 3 inside the loop, to an SNS-Toolbox network through per-muscle activation commands and Ia, Ib, and II afferent plus heel and toe contact feedback. Hardware deployment, the future-work stage of Chapter 7, moves the same loop onto Teensy-class microcontrollers for the per-joint valve, pressure, and contact loops and a Jetson Orin for network evaluation, computer-in-the-loop before embedded execution on the treadmill biped.

the CAD-derived bodies are ported. SCONE provides the reference point for mature predictive neuromuscular simulation [Geijtenbeek, 2019]: its published balance and gait demonstrations were reproduced headless as a practice baseline, and the standing-balance stage of the training curriculum (Section 6.5.2) is modeled on its balance tutorial.

The SADb of Chapter 5 is the data resource among the existing tools [Bolen et al., 2023]. Its segments are the five-table Airtable base (Papers, Feedback, Models, Review Papers, and Recorders), the laboratory's Zotero libraries that feed it, citation mining in the Research Rabbit style that expanded it, and the laboratory's own packaged explorer application that distributes it. The SADb framed this chapter's benchmarks and validation measurements: its pathway records fix what each afferent class encodes and what each connection rule does, so the sensory channels, the reflex topologies, and the acceptance phenomena against which the controllers are validated (the deafferentation and stimulation results cited in Section 6.3.6) are looked up rather than re-derived from the primary literature.

6.2.2 New Tools

Four tool families are new, and each exists because a needed combination did not. First, the full-lower-body pipeline: the robot's OpenSim musculoskeletal model converted to MuJoCo, audited against its source, and coupled to an SNS-Toolbox spinal network through a BPA force interface, with proprioceptive and heel and toe contact feedback closing the loop. Second, the SNS-Simscape library: the neuron, synapse, afferent-encoder, and actuator models of the SNS formalism as masked Simulink blocks, together with the imported SolidWorks and OpenSim bodies that give them real geometry to control. Third,

the controller ports: the Deng-organization two-level network, the contact-driven W2L core, and the Li walker re-implemented across NumPy, Simulink, MuJoCo, and Simscape Multibody and onto the converted models, so that the same controller runs under different realizations in different programs. Fourth, the verification instruments: the cross-engine harness, the circuit-structure audits, and the spiking-neuron mirrors that test whether conclusions depend on the graded-coding assumption.

Because a result that holds in only one software implementation is difficult to trust, the same contact-driven rhythm core was re-implemented in Simulink and checked against its NumPy reference, the archived AnimatLab models of the same lineage were re-run headless so that engine-level and network-level agreement could be separated, and spiking-neuron mirrors of all three simulation families test the coding assumption. The justification for every component is the experimental capability it adds, and the standard of proof for each is the fundamental demonstration that shows it works: a basic reflex pathway, the replication of a published rhythm or gait, or the removal of a single pathway class.

6.2.3 AnimatLab Body and Controller Organization

The walker representation and controller hierarchy are shown in Figures 6.2 and 6.4. The hierarchy separates rhythm generation, hip control, and knee–ankle pattern formation within left- and right-side neural subsystems. As a tool, this environment exists because the program's controller lineage is archived in it: the reference models whose replication is reported in Section 6.3 live as AnimatLab projects, and new circuit organizations are prototyped in the same format. The current AnimatLab working model was instrumented

for a new 10 s phase-1 run recording joint, neural, contact-sensory, and body-motion channels; that instrumented run is the tool's fundamental demonstration, and its recordings are reported in Section 6.3 and examined there without treating them as evidence of stable walking or effective ground-contact gating. The run is distinct from the earlier walker study of Section 6.1.

Since that run, the working model has been reorganized into a complete bilateral network. The right-side rhythm-generator, hip, and knee–ankle subsystems were mirrored from the left, inhibitory commissural pathways were added between the two rhythm generators, and each leg gained four biarticular muscle pathways (gastrocnemius, biceps femoris short head, semimembranosus, and rectus femoris) with human maximum isometric forces from the Gait2392 data set. The reorganized project opens in the AnimatLab environment without errors and exports a standalone simulation that runs headless. Its rhythm-generator populations hold a static latch under the tonic drive levels tested so far rather than oscillating, so conductance tuning toward robust oscillation is the active step; no locomotor claim is made for the reorganized model beyond its verified assembly.

Figure 6.3 illustrates the rhythm-generator, pattern-formation, and sensorimotor organization at the neuron and synapse level. These reference schematics complement the subsystem view by making the antagonist pathways and sensory connections visible.

6.2.4 MuJoCo Conversion and Synthetic BPA Knee

The next two tools operate in the MuJoCo engine: a conversion pipeline that moves the robot's musculoskeletal model from OpenSim into MuJoCo [Delp et al., 2007, Todorov

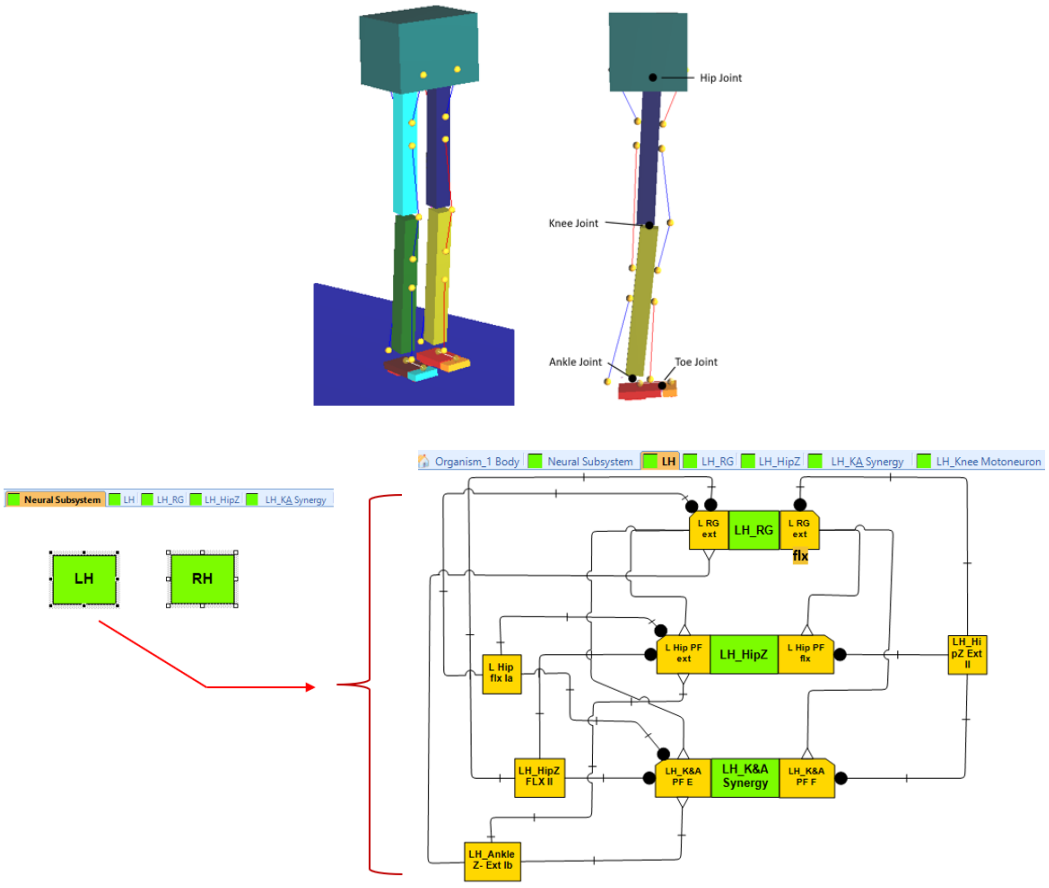


Figure 6.2: AnimatLab model and architecture illustrations. Top: the segmented walker and labeled hip, knee, ankle, and toe joints. Bottom: the left/right subsystem organization and an expanded view of the left-side rhythm-generator, hip, and knee–ankle modules. These architectural views describe model organization and are not snapshots of the phase-1 trajectory.

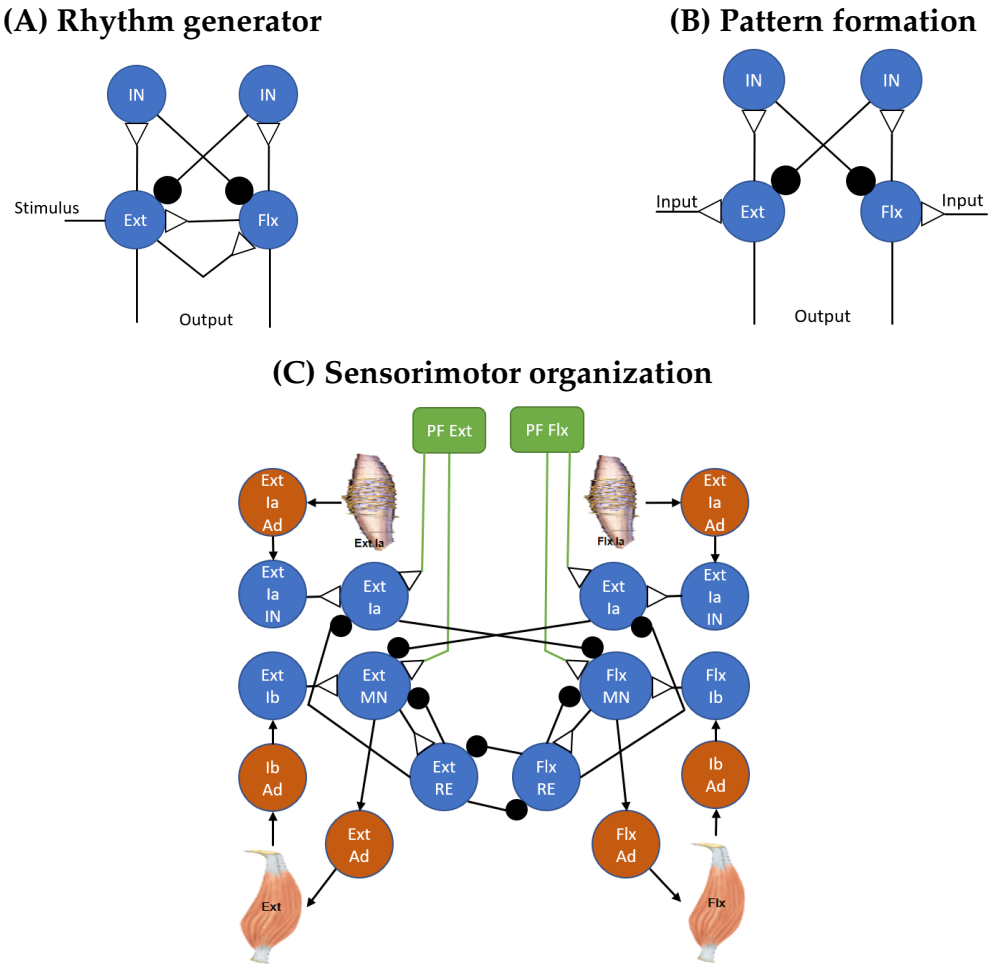


Figure 6.3: Reference schematics of **(A)** rhythm generation, **(B)** pattern formation, and **(C)** sensorimotor organization. Ext and Flx identify antagonist extensor and flexor pathways; IN denotes interneurons, MN motor neurons, and Ia and Ib muscle afferent pathways. Open triangles and filled circles distinguish excitatory and inhibitory synaptic connections. These schematics illustrate controller architecture and are not readouts from the phase-1 run.

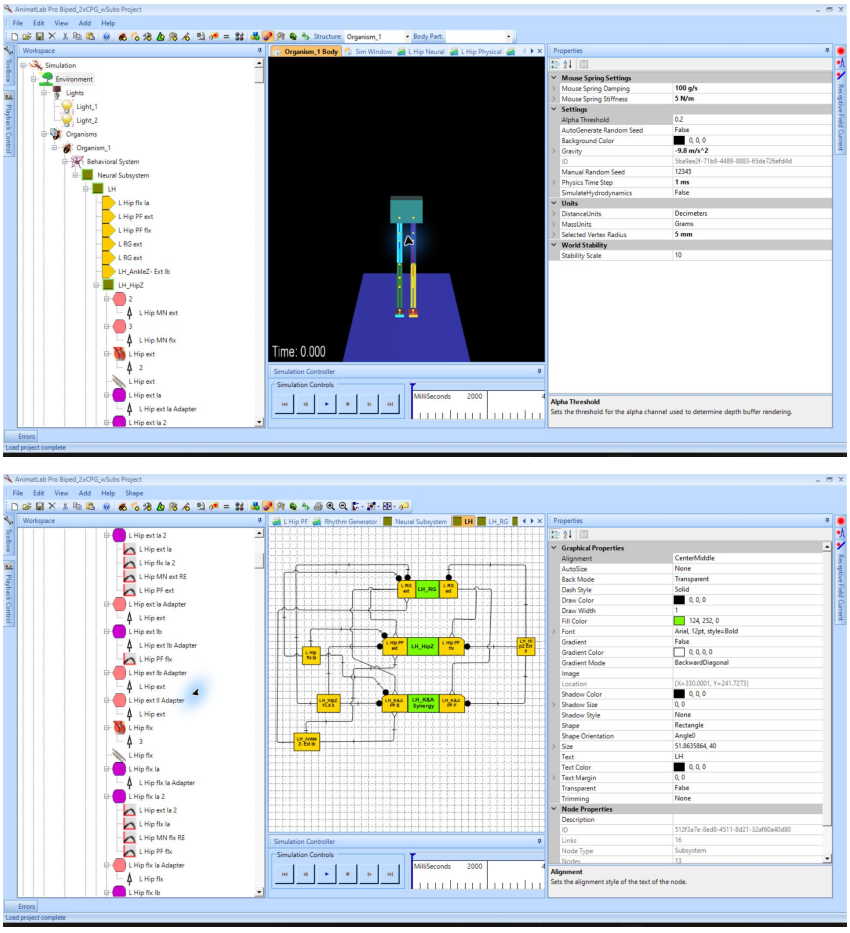


Figure 6.4: The saved AnimatLab working project opened in the application: the biomechanical body at initialization (top) and the expanded left-side controller subsystem (bottom). The workspace tree and property inspector expose the relationship between the body, sensory pathways, and neural modules. These interface captures were acquired without running a new simulation.

et al., 2012], and a force interface that makes the converted model's actuators behave as the BPAs of Chapter 3 rather than as generic muscles. The conversion exists because no physics engine carried the robot's custom muscle routing natively. The MyoConverter program performs the base conversion [Caggiano et al., 2022, Wang et al., 2022]; because an automatic conversion of a model with custom muscle routing introduces specific defects, among them muscle paths that are not mirrored correctly between the two legs and joint rotation axes with reversed sign, the conversion is followed by repair scripts, short Python programs that correct each defect and re-audit the converted model against the OpenSim source. The conversion runs in a recorded software environment, with the engine and library versions fixed, so that it can be re-run from the source model and reproduced. The force interface implements the quasi-static BPA force model of Chapter 3 in Python so that it runs inside the simulation loop: at each time step it reads every actuator's length and extension rate from the MuJoCo tendon paths and returns the actuator force, including the correction-factor force balance of Chapter 4. The fundamental demonstration of the pair is a synthetic knee: a minimal SNS with reciprocal inhibition driving a synthetic knee with antagonistic flexor and extensor BPAs (Figure 6.5, which distinguishes the converted robot from this separate knee demonstration). The demonstration establishes the controller–actuator–joint interface; it does not establish SNS control of the converted robot or validated correction-factor torque predictions in MuJoCo.

The conversion and the force interface are complete and verified. The converted model carries all 92 muscle-tendon actuators, with attachment geometry audited against the OpenSim source, and the joint sign conventions of the robot's modified muscle paths were verified dynamically: each actuator was activated on its own in simulation, and the sign of

the joint torque it produced identified its action at that joint. The check has to be dynamic because the attachment points of these paths move with joint configuration, so the routing cannot be read from a fixed table of moment arms. The Python implementation reproduces the MATLAB force calculation, including the correction-factor force balance. An SNS-Toolbox network couples to MuJoCo at a 2 ms neural step, matching the physics-engine timestep, with muscle force feedback entering as Ib afferent input. On this coupling, a 410-neuron two-level spinal network has been assembled for the converted model (per-leg rhythm-generator populations, four phase-shifted pattern-formation groups per leg, motoneuron pools for all actuators, and Ia, Ib, and II afferent populations), and its rhythm generator has been verified in isolation to oscillate at an approximately 0.9 s cycle with the two legs in antiphase under tonic drive. That isolated oscillation doubles as the rhythm-generator-level stimulus test: tonic drive with no phasic input evokes the alternating rhythm, the simulation-side analogue of the tonic central and sensory stimulation that facilitates involuntary air-stepping in humans [Selionov et al., 2009].

The sensory channels that close the loop trace to the SADb of Chapter 5 [Bolen et al., 2023]: the afferent classes the controller monitors, Ia carrying length and velocity, Ib carrying force, and group II carrying length, together with the heel and toe contact receptors that mark the stance–swing boundary, are the classes the database's curated records were assembled to cover, and the signal encoding and the connection rules each channel implements follow from those records (Tables 5.1 and 5.2) rather than from ad hoc literature searches at wiring time.

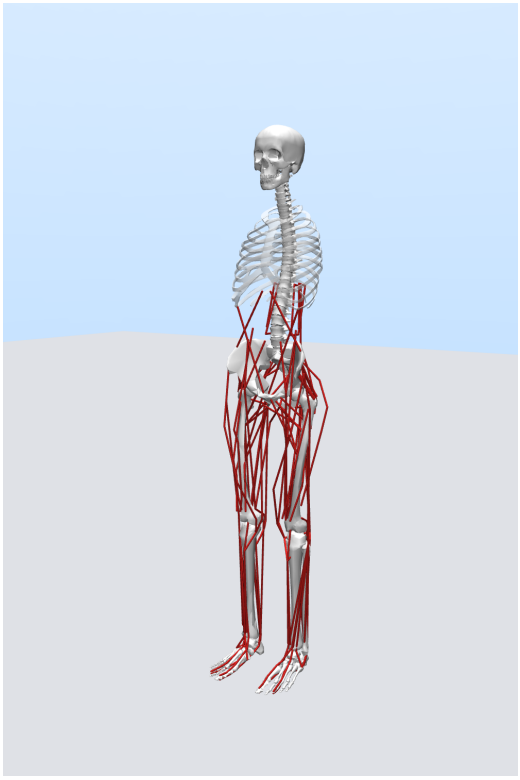
Full-body closed-loop co-simulation runs on this coupling: the four production walking configurations of Section 6.3.5 complete 16 s ground evaluations with gait cycles detected

and scored. Early full-body runs developed non-finite leg-joint states during longer simulations; those faults were traced to the converted model's moving-attachment followers, whose sampled range limits fought the coupling constraints that tie each follower to its path point, and repairing those constraints resolved them. The ground evaluations run under a simulated physical support harness, a body-weight-relief rig acting on the model's pelvis that mirrors the laboratory's overhead support rig, and unsupported walking is not among the claimed results (Section 6.5.5 states exactly what walks and what does not).

6.2.5 Simulink SNS Library and Reflex Topology

The Simulink tool is a block library, built so that synthetic nervous systems can also be assembled and inspected in a standard controls-engineering environment. Its seven core component blocks are a non-spiking (graded) neuron, a graded chemical synapse, a spiking leaky integrate-and-fire neuron, Ia and Ib afferent encoders, muscle activation dynamics, and a generic BPA force block. The blocks implement the same neuron and synapse equations used by the SNS-Toolbox networks of the preceding subsection, and their inventory is drawn in Figure 6.6 (top), which also shows the internal implementation of the non-spiking neuron block as a native Simulink export (bottom). The full library adds a synapse-sum junction for neurons receiving more than six synaptic inputs, three diameter-specific BPA force elements (10 mm, 20 mm and 40 mm) implementing the fitted force surfaces of Chapter 3, and a Hill-type biological muscle element, for twelve blocks before the spiking additions of Section 6.2.8. The library and its demonstrations run in base Simulink without additional toolboxes; the block icons derive their excitatory and

(A) Converted robot



(B) Synthetic BPA knee

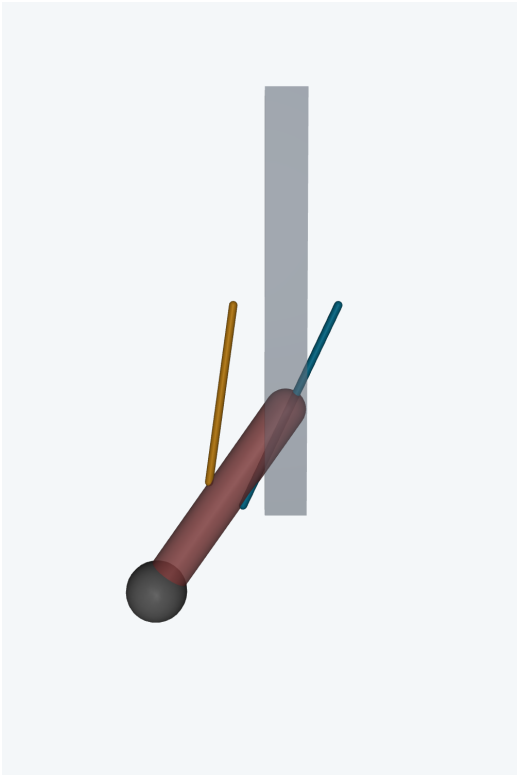


Figure 6.5: Native MuJoCo renders of **(A)** the converted Gait2392 robot at its saved reference pose, with muscle paths visible, and **(B)** the independent synthetic antagonist-BPA knee at an illustrative static angle of -35°. Cyan and gold identify its named flexor and extensor tendon routes; transparency exposes the routes through the view. The models are not shown to the same scale. These views document model construction, not SNS control of the converted robot or validated torque predictions.

inhibitory symbols automatically from the sign of each synapse, in a color-blind-safe palette, and the publication figures regenerate from scripts.

The library's fundamental demonstrations are two basic reflex pathways. In the first, the core blocks are assembled into a reduced-order knee-reflex model with antagonist actuation (Figure 6.7): Ia stretch feedback excites the motor neuron of the stretched muscle and inhibits that of its antagonist (reciprocal inhibition), while Ib force feedback inhibits the motor neuron of the muscle that generated the force (autogenic inhibition). In the second, an arm holds a beverage container while a pour loads it, with and without the stretch reflex enabled. Between them, the two demonstrations show the library reproducing the canonical spinal reflex pathways at the scale where their mechanisms are visible; their quantitative results are reported in Section 6.3.2 beside their spiking twins. To couple the controller to detailed mechanical CAD, the SolidWorks assembly is exported and imported into Simscape Multibody. The URDF import path is verified on a trial knee assembly, and the fuller Multibody-Link XML import of the knee test assembly exposes a documented defect: the exporter drops the SolidWorks hinge mates as unsupported, so the knee degree of freedom arrives welded, and the defined fix is to re-mate each hinge as a concentric and coincident pair before re-export. The tendon hardware is modeled in CAD but not yet inserted into the assembly. The demonstration model retains provisional mechanical parameters, the generic BPA force block remains a placeholder, and the saved runs are therefore not used to establish quantitative knee-regulation performance.

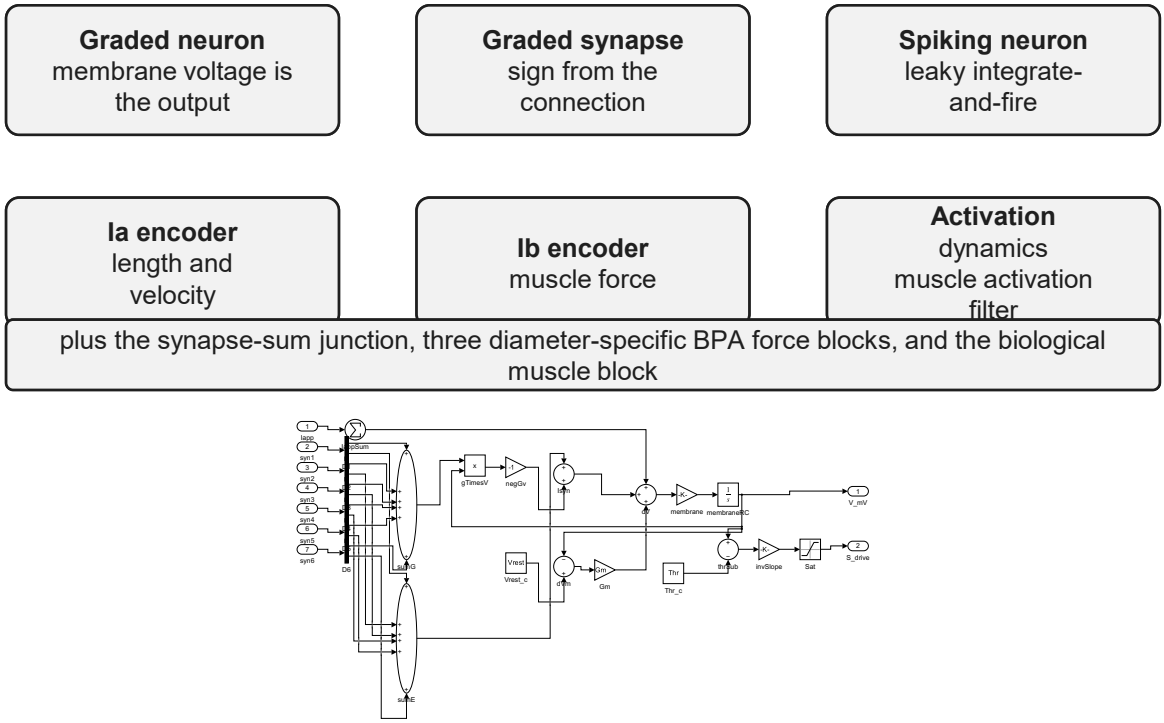


Figure 6.6: The core blocks of the SNS-Simscape library (top): the graded neuron and synapse, the spiking neuron, the Ia and Ib afferent encoders, and the activation dynamics block, each with its function summarized. The full library adds the synapse-sum junction, the diameter-specific BPA force blocks, and the biological muscle block. Below, the internal implementation of the non-spiking neuron block as a native Simulink export. The full library also carries the synapse-sum junction, the diameter-specific BPA force elements, and the biological muscle element described in the text. The generic BPA force block remains a placeholder. These implementation views do not establish quantitative knee regulation or a completed CAD-coupled Simscape Multibody model.

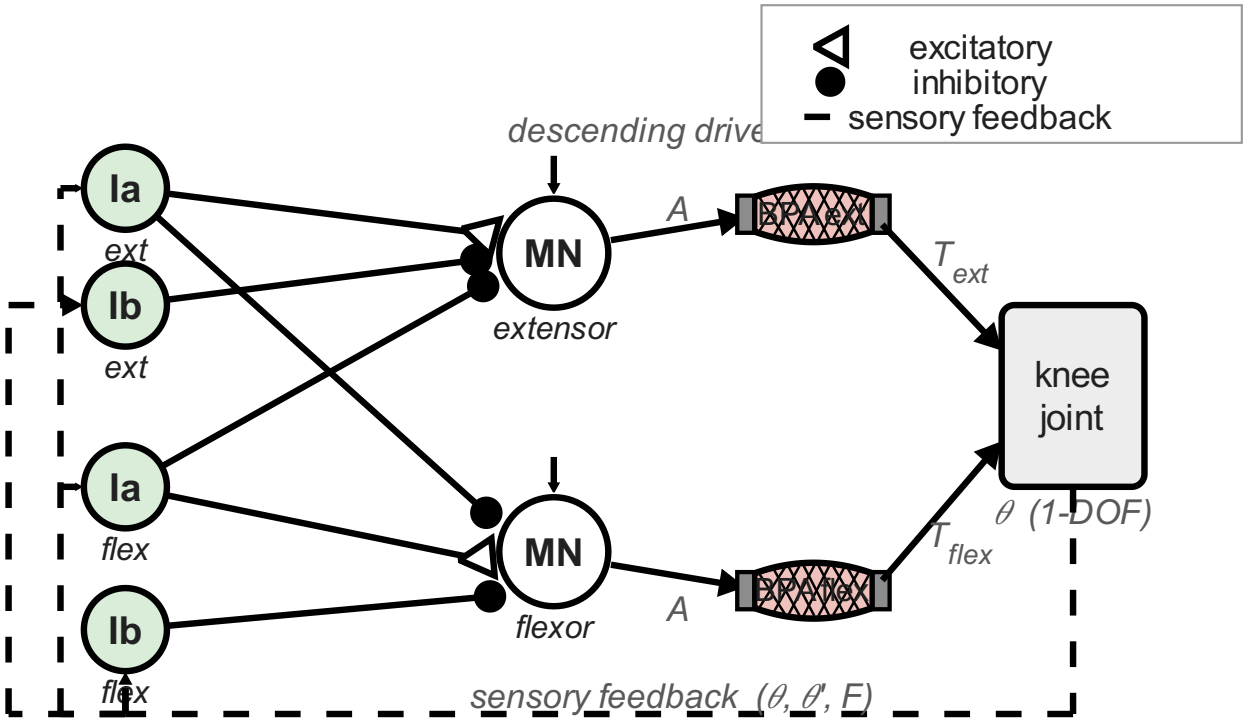


Figure 6.7: The reduced-order knee-reflex model drawn in the SNS library's marker language: triangular synaptic terminations mark excitatory inputs and filled round terminations inhibitory inputs. Ia stretch feedback excites the stretched muscle's motor-neuron pool and inhibits the antagonist pool (reciprocal inhibition); Ib force feedback inhibits the pool of the muscle that generated the force (autogenic inhibition). The motor-neuron pools drive the activation and BPA force blocks that act on the knee joint. The corresponding Simulink model assembles the library blocks one-for-one and simulates standalone. Mechanical parameters and the BPA force block remain provisional.

6.2.6 Controller and Body Ports into Simulink and Simscape Multibody

Making the controllers and bodies portable is the third new-tool family, and it exists because the program's models must run in more than one program. On the controller side, the Deng two-layer CPG was rebuilt as separate editable Simulink files, rhythm-generator and pattern-formation layers apart, with the published parameter tables carried over [Deng et al., 2019, Nourse et al., 2023]. One cell-level choice in that rebuild proved load-bearing for portability, a burst-termination property of the rhythm-generator cells whose membrane mechanism is recorded with the neuron equations in Appendix B; Section 6.5.1 returns to why it matters. The tuned production spinal network is exported in the other direction: a script turns the tuned network definition into an editable Simulink model built from the library blocks of the preceding subsection, one block per neuron and one per synapse, with every value living in an editable block mask so that the model regenerates whenever the tuning changes.

On the body side, two import routes bring real geometry into Simscape Multibody. The SolidWorks knee assembly travels the URDF and Multibody-Link XML routes of the preceding subsection. The OpenSim `gait2392_simbody` model travels OpenSim to MuJoCo to URDF to Simscape Multibody, with the massless muscle path-point slides preserved as links so that actuators can attach at the same physical points as in the source model; a URDF carries no muscles, so actuation re-enters as SNS library blocks driving the imported joints. Finally, a MuJoCo–Simulink bridge block set, rebuilt against the same engine version as the conversion, closes the loop from the Simulink side: muscle activation commands in, joint and actuator sensors out, with the model's saved reference

pose applied at initialization.

6.2.7 Cross-Platform Verification of the Contact-Driven Rhythm Core

The verification harness, in the software-testing sense of a fixed script suite that re-runs and checks the models, is a tool in its own right, and its existence is the chapter's answer to a trust problem: the toolchain spans several simulation engines, and a result that holds in only one software implementation is difficult to trust. Its fundamental demonstration is replication, in two forms; the same rhythm core rebuilt in two engines must reproduce the same rhythm, and the lineage's archived models must reproduce their published behavior in their original engine. The harness is defined around the contact-driven rhythm core of the Walker-2-Layer (W2L) bipedal architecture: four rhythm-generator populations, one extensor and one flexor population per leg, plus graded pattern-formation and sensory cells, driven by heel-contact mechanosensory feedback. The rhythm cells are spiking neurons that self-terminate each burst, so the mutually inhibiting extensor and flexor populations of a pair alternate [Deng et al., 2019]; the burst-termination mechanism is recorded with the neuron equations in Appendix B. The same core was implemented twice from a common connectivity specification, once as a NumPy reference integration through the SNS-Toolbox backend and once as native Simulink blocks from the library of Section 6.2, and both were driven with an identical protocol: a 12 s run at a 2 ms neural step, tonic drive with a brief kickoff pulse, and alternating 1 Hz heel-contact trains. Agreement was scored on rhythm-generator burst period, left–right antiphase correlation, burst counts, and full-trace correlation against the NumPy reference. The archived AnimatLab reference

models (the bilateral contact-driven model, the modern W2L model, and the Li walker) were additionally re-run headless from their standalone simulators with instrumented charts, so that engine-level agreement and network-level agreement could be separated. The protocol covers the rhythm core and the archived models only; full-network and whole-body behavior are reported as limitations alongside the results in Section 6.3.

6.2.8 Spiking-Neuron Mirrors of the Simulation Models

The final tool tests an assumption rather than adding a simulation capability. The controllers documented above all compute with graded (non-spiking) neurons, in which the membrane potential itself is the analog activity variable that drives the musculoskeletal model. To test that the conclusions drawn from those models are not artifacts of graded coding, spiking-neuron mirrors of all three simulation families were built, following one design rule: the motor-neuron layer stays non-spiking, because the musculoskeletal model consumes a continuous activation command, while the afferent encoders (the neurons that convert sensor signals into synaptic input), rhythm-generator populations, pattern-formation cells, interneurons, and commissural (cross-side) cells become leaky integrate-and-fire neurons operating at real membrane-potential ranges, so that the network carries spikes above the motor-neuron layer and analog voltages below it [Szczecinski et al., 2023, Nourse et al., 2023]. Event-based coding is expected to matter most in the afferent and rhythm layers, where mechanosensor onsets and phase transitions are discrete events, and least at the motor-neuron output, which the muscle activation dynamics low-pass filter anyway.

The MuJoCo–SNS-Toolbox mirror (`build_network_spiking.py`) reproduces the production spinal network of Section 6.2 population for population and connection for connection, and is selected at build time by a single environment-variable switch so that the default build path is untouched; that the non-spiking pipeline is unchanged was proven by re-running its defaults gate and baseline evaluation to bit-exact agreement. Because SNS-Toolbox 1.5.2 ships no spiking bursting-neuron class, the rhythm-generator populations are adapting integrate-and-fire cells: burst termination is carried by spike-frequency adaptation of the firing threshold, whose time constant reuses the corresponding slow-inactivation value of the non-spiking build so that tuned adaptation values transfer between the two. Spiking cells rest at $-70\,\mathrm{mV}$, spike at $-50\,\mathrm{mV}$, and reset to $-60\,\mathrm{mV}$, with a tonic bias current parking their equilibrium a few millivolts below threshold, the analog cells' near-threshold operating point. The network sub-steps at 0.5 ms (four sub-steps inside each 2 ms physics step) so that spike timing is resolved without changing the physics integration, and non-spiking readout taps are appended for the runner's stance gates and neural logging so that all existing analysis code reads analog levels unchanged.

The mirror's synaptic weights are set by a calibration script rather than by hand. Spikes onto the non-spiking motor neurons are calibrated by mean conductance, so that a steady presynaptic train reproduces the non-spiking steady-state motor-neuron voltage (calibration residuals at or below $3 \times 10^{-4}\,\mathrm{mV}$); synapses between spiking cells are calibrated by firing-rate transfer (a 40 Hz source drives its target at 40 Hz, and a calibrated inhibitory synapse silences a 40 Hz-driven cell); the graded-to-spiking command pathway is scaled from the measured afferent current-to-rate map. The full calibration, with the measured residual of every factor, is tabulated in Appendix B (Table B3).

In the Simulink family, three blocks were added additively to the SNS library of Section 6.2 (for fifteen blocks in total): a `SpikingNeuron` with the same internal synaptic summation as the non-spiking block, an event-wired `SpikingSynapse` whose conductance jumps by a fixed increment per presynaptic spike and saturates at its maximum, and a `HybridSpikingSynapse` that carries the same conductance dynamics onto a non-spiking postsynaptic cell. The blocks were verified against the analytic single-spike conductance decay (maximum deviation 6.2×10^{-7} µS), against a verbatim re-implementation of the SNS-Toolbox update equations on a hybrid two-neuron circuit (identical spike count and maximum spike-time deviation 11.75 µs), and against the SNS-Toolbox numpy backend itself (mean interspike-interval difference 0.19 µs). Spiking twins of the knee-reflex and beverage-holding demonstrations were then built by copy-and-patch: only the sensory interneuron layer and its synapses are replaced by spiking cells, while the motor neurons, muscles, joints, and measurements are identical to the committed demonstrations, and the twin is run at an operating point matched to the baseline gains.

In the AnimatLab family, spiking copies of the runnable standalone exports were produced by a conversion script that rewrites copies of the model files, leaving every original file untouched: neuron thresholds are moved from the graded regime (unreachably positive) into the spiking regime, every graded chemical synapse type is re-emitted as a per-spike synapse with the same identity and amplitude (interpreted as a per-spike conductance increment with 10 ms decay), and the per-connection class tag that AnimatLab stores independently of the synapse type is flipped to match. A calibrated variant sets each cell class's threshold just below its measured graded peak, and an ablation that converts the thresholds but keeps the graded synapses isolates which change does what. Each copy

passes a structural validator (neuron, connection, synapse-type, chart, and stimulus sets identical to its original, and every class tag consistent) and runs headless to its configured end. The 2023 spiking walker model of the same lineage, which was tuned as a rate code from birth, serves as the spiking reference baseline for this family.

Each mirror is verified against its non-spiking twin by a fixed gate set rather than by inspection: a topology gate requires the connection multiset to be identical between the two builds under three gain configurations; a rhythm gate requires self-sustained alternating rhythm under constant drive with the period within 20 % of the analog build; a basin gate requires the rhythm to persist under 1 % multiplicative jitter on every scalar parameter; an air-regime smoke gate compares cadence, duty, and joint ranges of the deafferented suspended-air preparation; and a regression gate requires every committed non-spiking artifact to reproduce its recorded values exactly with the spiking blocks present in the library. The mirrors carry mapped parameters only, with no new tuning, so their results (Section 6.4) establish what transfers structurally and what requires re-tuning, and any fresh tuning effort is future work.

6.2.9 Tuning, Robustness, and Reference Data

Above the physiologically set values, the free gains of each network are tuned by systematic search rather than by hand: a sequential model-based optimizer (a tree-structured Parzen estimator) proposes gain sets, each proposal is scored by a full simulation, and each stage's winner seeds the next. The design rule that matters here is that no hand-tuned parameter enters a production configuration; the failure mode of the prior walker (Section 6.1) was

precisely hand tuning that left no record of sensitivity. The stage structure and the scoring philosophy are the subject of Section 6.5.2.

Robustness is enforced during tuning rather than inspected afterward. Finalist configurations must retain their rhythm under 1 % multiplicative jitter on every scalar parameter (the basin gate of Section 6.2.8 applies the same test to the mirrors), survive stepping of the tonic drive about the tuned value, and, for standing configurations, absorb push perturbations without falling. The requirement exists because the tuned networks sit near bifurcations of their dynamics, a property this chapter makes explicit in its cross-engine and spiking results: a winner whose basin is narrow does not survive a solver change, let alone a hardware transfer.

The reference data for tuning and evaluation are inverse-kinematics cycles from many human subjects of different sizes and speeds rather than a single subject: a 43-cycle locomotion reference library, detailed in Section 6.3.5, comprising running trials from eight subjects at four speeds, walking references at prescribed speeds from $0.5\,\mathrm{m\,s^{-1}}$ to $2.0\,\mathrm{m\,s^{-1}}$ plus self-selected trials, and one predicted-walking cycle. Working against many references penalizes tuning that has memorized one subject's kinematics, in the same spirit as the perturbation gates. Circuit structure is treated as data under the same methodology: every network's connection multiset is compared programmatically against its intended topology (the topology gates), and the publication schematics are generated from the compiled network with each drawn edge asserted against the compiled connection list, so the drawings cannot drift from the circuits they document.

6.3 Results: Tool Demonstrations and Experiments

This section reports what the tools do, ordered by the questions that justify them. The AnimatLab instrumented run demonstrates the prototyping environment's recording capability. The Simulink library demonstrations and the cross-program ports establish the new Simulink-side tools at their fundamental scale. The cross-engine verification and the headless reruns replicate known behavior: the rhythm core must reproduce itself across engines, and the archived models must reproduce their published rhythms and gaits. The gait-validation sweep then measures how far the tuned networks remain from human reference locomotion, and the final subsection presents the pathway-removal and drive-perturbation experiments as the laboratory's deletion and stimulus experiments. The BPA-actuated walker that demonstrates the full MuJoCo chain end to end is discussed in Section 6.5.4, where the state of the walking program is consolidated.

6.3.1 AnimatLab Phase-1 Recordings

The phase-1 run described in Section 6.2 is the AnimatLab tool's fundamental demonstration: it exercises the instrumented model and the recording chain end to end. The run produced synchronized joint, neural, contact-sensory, and body-motion recordings over 10 s; basic checks confirmed finite values and a uniform 0.2 ms sampling interval in both exported charts, each containing 50,001 samples from 0 through 10 s with nine trailing zero-filled rows beyond the configured chart end excluded. Figure 6.8 shows bilateral hip, knee, and ankle angles. The contact-sensory channels record membrane voltage rather than contact force, and the model included an externally applied horizontal force during

the first second. The run demonstrates signal acquisition and alternating activity, not validated locomotor performance: neither stable walking nor effective ground-contact gating was established. The neural and contact-sensory recordings remain archived with the run.

6.3.2 Simulink Library Demonstrations and Cross-Program Ports

The SNS-Simscape library demonstrates the canonical reflex pathways at the scale where their mechanisms are visible. In the knee-reflex demonstration, the joint reads 15.79° at 0.05 s and settles at a mean of 43.5° over the last 2 s, with a mild irregular antagonist alternation about the mean. In the beverage-holding demonstration, the stretch reflex limits the maximum sag of the loaded arm to 2.00° against 7.85° with the reflex disabled; the committed values are tabulated beside their spiking twins in Table 6.5, Section 6.4. A third demonstration closes the loop onto the actuator model itself: a two-population CPG with burst-terminating adaptation drives the antagonist BPA pair of the one-degree-of-freedom knee through the library's diameter-specific force blocks, sweeping the joint through 9.7° to 48.0°.

The controller ports verify at the same standard. The Deng rhythm-generator and pattern-formation files, driven by a single 10 nA pulse of 20 ms, oscillate continuously at a 1.938 s period against the 1.94 s NumPy reference, with the extensor and flexor populations anticorrelated at $r = -0.86$. The exported full-network Simulink model reproduces its NumPy reference to a maximum deviation of 4.2×10^{-6} mV across all 410 neurons at 0.3 s; beyond roughly 0.4 s the network's chaos makes any two integrators diverge in

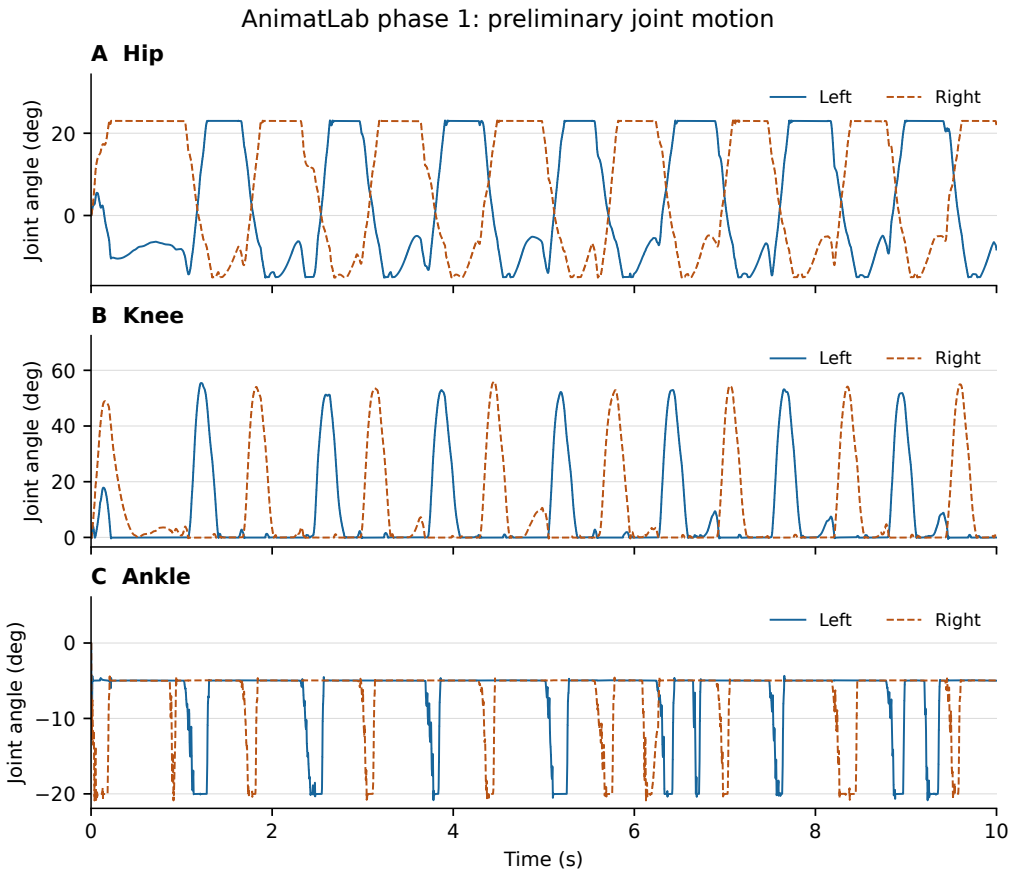


Figure 6.8: AnimatLab phase-1 joint motion: **(A)** hip, **(B)** knee, and **(C)** ankle angles in the model's recorded joint conventions. Blue solid and orange dashed traces denote left and right sides. All 50,001 samples within the configured 0–10 s chart window are retained without filtering. The model includes an external horizontal force during the first second. These traces document joint motion in an instrumented run; they do not establish stable walking or effective ground-contact gating.

phase (Section 6.5.1), so the verification window is the early transient and the faithful reproduction is the fixed-step Euler run at the production step size.

The MuJoCo–Simulink bridge was proven against an independent Python integration of the same engine version: single-muscle activation, two-rate control, and sensor-readback tests all pass, the readback bit-exact over the scored window. On the bridge, a single-muscle Ia/II/Ib reflex loop on the converted model's vastus medialis bounds the consequences of a postural collapse: without the reflex, the falling model drives transient muscle forces past 40 kN before the solver fails, and with it, the same loop holds muscle force at 469 N while the stretch response recruits the muscle. The full exported network then ran as a Simulink model block driving all 92 muscle actuators of the converted model, completing a 5 s closed-loop run with all drives finite and 96 % of the muscles recruited. That run also isolated a real property: under tonic drive alone, the Simulink realization settles to the tonic fixed point while the NumPy reference oscillates, the near-bifurcation behavior that Section 6.5.1 treats at length, and the production rhythm requires the full input schedule rather than drive alone.

The OpenSim-to-Simscape import completes the Simulink-side tool set. The `gait2392_simbody` model arrives in Simscape Multibody as a mechanism of 1,446 blocks and 149 joints, of which 20 are the anatomical revolute joints, 85 are the massless path-point slides that carry the muscle attachment points, and 44 are the fixed-link offset chains the conversion route requires; the imported mechanism compiles and simulates. What the route does not carry are the muscles themselves, which re-enter as SNS library blocks driving the imported joints, the division of labor the chapter's architecture assumes throughout.

6.3.3 Cross-Engine Verification of the Contact-Driven Rhythm Core

Between the NumPy reference and the native Simulink reimplementation of the W2L rhythm core, both legs' rhythm-generator burst periods matched at 1.000 s versus 1.000 s (0.0% error), the left–right antiphase correlation reproduced the reference exactly ($r = -0.525$), and all ten expected bursts occurred in the scored window. The full voltage traces cross-correlate at $r = 0.99996$–0.99998 at a -2 ms lag with an RMSE of 0.099–0.111 mV (Figure 6.9); the residual lag is consistent with the one-step delay of the discrete Simulink solver.

The same protocol also runs the chapter's first deletion experiment, and it surfaced an architectural fact rather than a defect: with the heel-contact trains removed, the full network holds a static latch under tonic drive with kickoff alone, in every implementation, so phasic contact feedback is part of the drive architecture of the W2L design, not an optional perturbation. The result is a load dependence that echoes the decerebrate-cat locomotion reviewed in Section 2.5, and it is the reason the pathway-removal experiments of Section 6.3.6 treat the contact channels as a pathway class rather than as a boundary condition.

6.3.4 Headless Validation of the AnimatLab Reference Models

The second half of the verification suite's fundamental demonstration is replication of published results in the original engine. The archived AnimatLab reference models were re-run headless from their standalone simulators with instrumented charts, and eleven targetable rhythm and gait metrics were extracted automatically; all eleven pass their

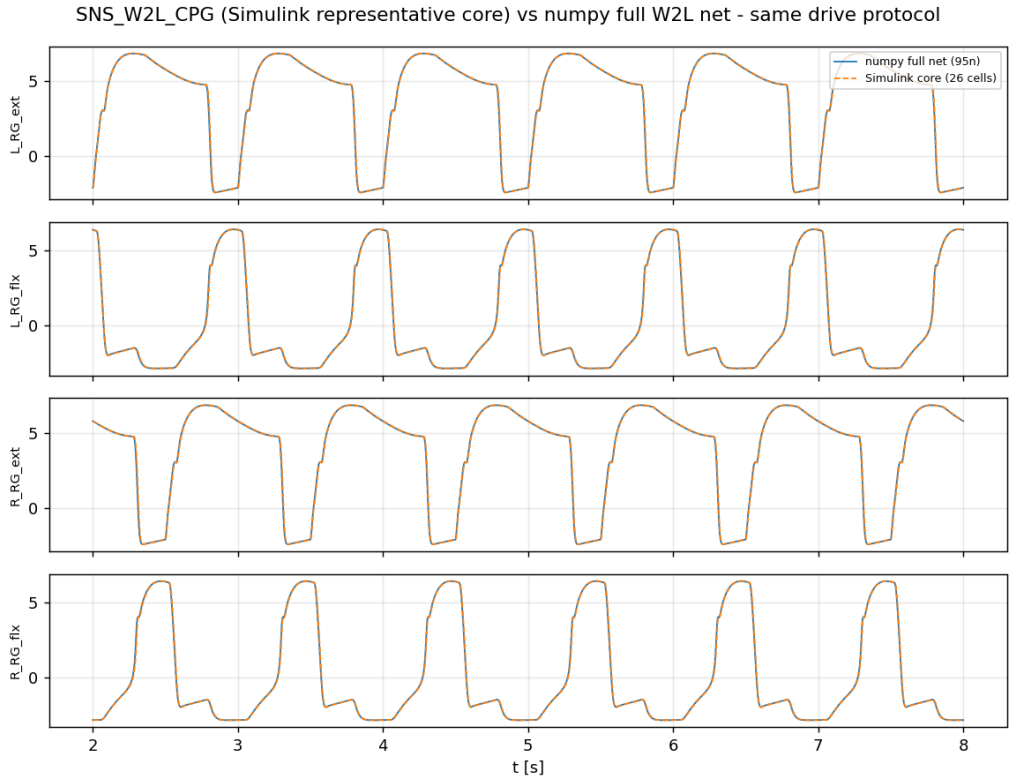


Figure 6.9: Left and right rhythm-generator extensor population voltages of the W2L contact-driven rhythm core under the shared verification protocol, from the NumPy reference (SNS-Toolbox backend) and the native Simulink reimplementation. Scored metrics (burst period, left–right antiphase correlation, and burst counts) agree exactly, and the full traces cross-correlate at $r \geqslant 0.99996$ at a -2 ms lag.

acceptance bands. In the bilateral contact-driven model, the rhythm generators oscillate at periods of 0.450 s (left) and 0.467 s (right) with a left–right lag of 0.529 cycle (cross-correlation $r = 0.976$), confirming alternating gait timing. In the modern W2L model, the joints oscillate at 2.25 Hz; its flexor populations peak at -56.4 mV and never cross the -55 mV threshold, so the rhythm is carried by clean extensor/flexor antiphase ($r = -0.996$) through the graded pathway rather than by flexor spiking. The Li-model rerun [Li et al., 2017] walks at $0.637\,\mathrm{m\,s^{-1}}$ with right and left step periods of $1.299(24)$ s and $1.333(80)$ s and duty factors of 0.443 and 0.555, reproducing the published behavior of the original model in its original engine.

The replication series has a defined extension, and its current boundary is stated plainly. The recent connectome-level circuit organizations of Shevtsova et al. [2016], Di Russo et al. [2023], and Shinohara et al. [2025] are specified in this program as circuit-rule translations, connection lists carrying per-edge citations, from which running models in SNS-Toolbox and SNS-Simscape would be built. On the current record they have not been built or verified, and no replication claim is made for them here; they are the next entries of the series this section's protocol defines.

6.3.5 Gait-Validation Sweep against the Locomotion Reference Library

As a scope check on the tuned networks rather than a demonstration of a tool, the four production walking configurations of the converted robot-body model were evaluated against a library of 43 human locomotion reference cycles: 32 running trials from eight subjects at four speeds, ten walking references (seven at prescribed speeds from $0.5\,\mathrm{m\,s^{-1}}$ to

2.0 m s^{-1}, plus three self-selected), and one further reference cycle, the predicted-walking cycle of the Falisse predictive-simulation dataset (`Case_40_motion` in the tables; period 1.11 s, duty factor 0.57). The four configurations span the two controller lineages of this program: the human-derived production configuration (`s3k`) and its pruned derivative (`s3kpruned`), the six-synergy variant (`syn6`), whose pattern-formation layers drive the motor pools through a six-synergy basis extracted from human gait data, and the W2L-layout variant (`w2lvar`), which restructures the rhythm and pattern-formation layers after the contact-driven W2L circuit of Section 6.2.7. Each was scored with the production pipeline's cycle-shape kinematic score, in which 0 denotes perfect agreement and -25 is the fixed no-cycle fallback, both on its tuning-reference evaluation and against every library cycle (Table 6.1).

Table 6.1: Gait-validation sweep of the four production walking configurations against the 43-cycle locomotion reference library. The score is the production pipeline's cycle-shape kinematic score (0 denotes perfect agreement); the tuning-reference score is each configuration's evaluation on the tuning pipeline's walking reference, and the library mean averages the same score over all 43 references (among genuine gait cycles, higher scores indicate closer agreement).

	Tuning-ref. score	**Library mean score (43 refs)**	**Most negative reference (score)**
`s3k`	-186.4	-8776	`Case_40_motion` (-8841)
`s3kpruned`	-176.1	-6924	`Case_40_motion` (-6986)
`syn6`	-200.4	-9961	`Case_40_motion` (-10040)
`w2lvar`	-144.5	-5808	`subject08_Run_20002` (-5896)

The sweep's reading is mostly negative. The ranking agrees across the two evaluations, with the W2L-layout variant closest to both evaluations and the six-synergy variant farthest (Table 6.1), while pruning moves the production configuration toward closer agreement.

But every configuration's library scores are one to two orders of magnitude more negative than its tuning-reference score, which quantifies how far a network tuned to a single walking reference transfers to the library's diverse reference cycles; the gap is not an artifact of the 32 running trials, because the production configuration's mean over the ten walking references alone (-8816) is slightly more negative than its running-only mean (-8761). For three of the four configurations the most negative single library score falls on the same reference, the predicted-walking cycle, while the W2L-layout variant's most negative reference is a running trial, `subject08_Run_20002`. All four configurations produce right-leg-only or asymmetric gait in these evaluations; the family-level reading of the ranking is taken up in Section 6.5.5.

6.3.6 Stimulus and Deletion Experiments on the Walker Networks

The locomotor physiology that this program's circuits imitate was established with two interventions, deletion and stimulation, and the laboratory's first experiments apply the same logic at the levels the two-level architecture defines. In the reference works, removing proprioceptive feedback degrades locomotor patterns in mice [Akay et al., 2014], proprioceptive input regulates and adapts locomotor activity [Lam and Pearson, 2002], and tonic central and sensory stimulation facilitates involuntary air-stepping in humans [Selionov et al., 2009]; the two-level CPG models themselves reproduce specific deletion phenomena, such as spontaneous deletions of motoneuron activity during which the locomotor period and phase are maintained [Rybak et al., 2006]. The experiments implemented here are the simulator-side analogues that the assembled tools make inexpensive: removal of an

entire pathway class from a fully tuned walker, re-evaluated on the tuning reference, and stepping of the tonic drive about its tuned value. They act at the rhythm-generator level (tonic drive, interleg commissural pathways, and the weak rhythm-generator mutual excitation), at the sensorimotor level (Renshaw cells), and at the afferent level (Ia, Ib, group II, and contact channels). Systematic stimulation series at the pattern-formation and motoneuron levels, and the motoneuron skipping-deletion phenomenon itself, are defined as verification tests in Chapter 7 and have not yet been run; the experiments below are therefore reported as pathway deletions and drive stimuli, labeled by the level at which each acts. The published models do not carry these pathway-class interventions as performed experiments, so fidelity for them rests on the receiving programs, which apply the same class-level deletions and drive-stepping protocols to the tuned networks.

The pathway-removal sweep (Table 6.2) removes individual pathway classes from each configuration and re-evaluates the tuning-reference score. In the production configuration, removing the Ia afferent pathway freezes the model outright: the evaluation detects no gait cycles and returns the fixed -25 fallback. Removing the Renshaw cells, the group-II afferents, or all afferent classes at once degrades the score to -267.3, -255.2, and -267.8, respectively, against the full model's -186.4, while removing the Ib, contact, or interleg commissural channels individually shifts the score by smaller amounts toward zero (-180.3, -183.9, and -174.9). The six-synergy variant behaves differently: with all afferents removed it scores -190.5 against its full model's -200.4 and still walks bilaterally, with ten cycles on each leg, a knee-flexion minimum of -71.2°, and a 0.684 s period, so its afferent complement is not what sustains its gait. The W2L-layout variant depends on both the Ia pathway (removed: -273.5) and the weak rhythm-generator mutual excitation

(removed: −279.9) against its full model's −144.5 with four right-leg and six left-leg cycles, while removing its Renshaw cells leaves a bilateral gait at −142.4.

Table 6.2: Pathway-removal sweep on the tuning-reference evaluation. The score is the production pipeline's cycle-shape kinematic score (0 denotes perfect agreement; −25 is the fixed no-cycle fallback, a failed evaluation that scores above every genuine gait in this table). The full-model rows repeat each configuration's unmodified production evaluation.

Configuration	**Removed pathway**	**Score**
`s3k` (full model)	none	−186.4
	Ia afferents	−25 (no cycles detected)
	all afferent classes	−267.8
	Renshaw cells	−267.3
	group-II afferents	−255.2
	Ib afferents	−180.3
	contact channels	−183.9
	interleg commissural pathways	−174.9
`syn6` (full model)	none	−200.4
	all afferent classes	−190.5
`w2lvar` (full model)	none	−144.5
	Ia afferents	−273.5
	weak rhythm-generator mutual excitation	−279.9
	Renshaw cells	−142.4

Read against the reference physiology, the sweep's structure is familiar even where its details are specific to these networks. Deleting all afferent classes from the production configuration degrades its gait without abolishing it, which is the qualitative outcome of the deafferentation experiments [Akay et al., 2014]; the network that matches that picture most closely is the six-synergy variant, which keeps a bilateral gait with every afferent class removed. The Ia-only removal freezing the production configuration is a stronger dependence than the animal experiments show, and it marks that network's

tuning as built around its length feedback. The degradation under group-II removal is consistent with proprioceptive regulation of the locomotor pattern [Lam and Pearson, 2002], and the equally large degradation under Renshaw-cell removal indicates that the recurrent-inhibition loop carries weight in the tuned production network. No experiment here reproduces the motoneuron-level deletion phenomena of the two-level CPG models [Rybak et al., 2006]; the deletions act at the afferent, interneuron, and rhythm-generator levels, and extending them to the pattern-formation and motoneuron levels is scheduled with the verification tests of Chapter 7.

The drive-perturbation and push tests are the stimulus side. The production configuration never fell in 15 stand and push evaluations (maximum push sway 0.123 m), and its walk survives drive perturbations of $+0.01$, -0.01, and -0.02 (scores of -201.9, -177.0, and -171.2), freezing only at $+0.02$; its pruned derivative survives the unperturbed evaluation (-176.1) and the -0.02 perturbation (-168.2). The six-synergy and W2L-layout variants fall under every ± 0.01 and ± 0.02 drive perturbation tested, with pelvis tilt reaching 39° to 40°. In stimulus terms, a step change of the tonic drive within this range leaves the production lineage walking while the two variant lineages fall. The sweep therefore identifies the production configuration and its pruned derivative as the robust pair of the family, and locates the fragility of the six-synergy and W2L-layout variants under drive perturbation.

6.4 Spiking-Neuron Mirror Results

The spiking-neuron mirrors of Section 6.2.8 were evaluated against their non-spiking twins in the same order in which the families were introduced: the MuJoCo spinal network, the

Simulink demonstrations, and the AnimatLab walker family. All numbers in this section come from the gated verification effort whose artifacts and reproduction commands are catalogued in Appendix B.

6.4.1 MuJoCo Spinal-Network Mirror

The non-spiking production baselines were first re-established on the recorded software environment: all three current production configurations reproduce their documented evaluation scores exactly, including the default build's kinematic score of -161.57 (the full-precision value is tabulated with the gate records in Appendix B), and after the mirror's selector was added the default build gate ($(410, 376, 1186)$ populations, inputs, and connections) and that same baseline score reproduce digit for digit, proving the non-spiking pipeline untouched.

The topology gate passes exactly: under the default, fully tuned, and joint-layer pattern-formation gain configurations, the spiking mirror's directed connection multiset is identical to the non-spiking build's (1,186, 7,908, and 7,762 connections respectively), with identical input lists, identical motor-neuron maps, and exactly the expected number of readout taps (12, 12, and 16). The mirror is structurally the same network.

Under constant drive (network alone, 20 s), the rhythm gate splits. The spiking mirror self-sustains clean bilateral antiphase, with rhythm-generator extensor and flexor cells firing at 4.9 Hz and 3.2 Hz per leg and the swing readout sustained in every window, but its envelope period is 0.621 s against the analog build's 0.399 s, a 55.6 % lengthening that misses the gate's 20 % tolerance (Figure 6.10). The basin gate passes in full: the rhythm

persists under 1 % multiplicative jitter on every scalar parameter in all 12 seeded trials, with a final-window swing minimum of 4.58 mV against the 5.13 mV baseline (margin $0.89\times$) and no divergent trials. In the deafferented suspended-air preparation, the mirror air-steps rhythmically with full-amplitude joint swings (knee range 106.0° against the twin's 117.9°), but its cadence is roughly ten times faster than the non-spiking twin's 0.47 Hz air-stepping regime in this preparation, so the air-regime gate fails on the cadence clause alone.

One untuned ground-walking evaluation was run as a stretch attempt: the spiking mirror at the non-spiking winner's mapped parameters, with no re-tuning. It scores worse overall (kinematic score -236.35 against -161.57), as expected for an untuned mapped starting point, but it produces a genuine bilateral gait (eight right-leg and nine left-leg cycles, a cycle-window knee flexion minimum of -79.8°, no non-finite states, and the model stays up), whereas the non-spiking winner's left leg is frozen (zero cycles) in every recorded run of its study, including this session's baseline (Figure 6.11, Table 6.3). This is a single run with no statistics, reported as a lead for the fresh tuning effort that the mirror's plan calls for, not as an established gait result.

6.4.2 Simulink Spiking Demonstration Twins

The three new library blocks were verified at the unit level before any demonstration was rebuilt. The single-spike conductance transient matches the exact exponential to 6.2×10^{-7} µS, saturating pulse-train accumulation matches to 4.6×10^{-5} µS, and a hybrid two-neuron circuit matches a verbatim re-implementation of the SNS-Toolbox update equations

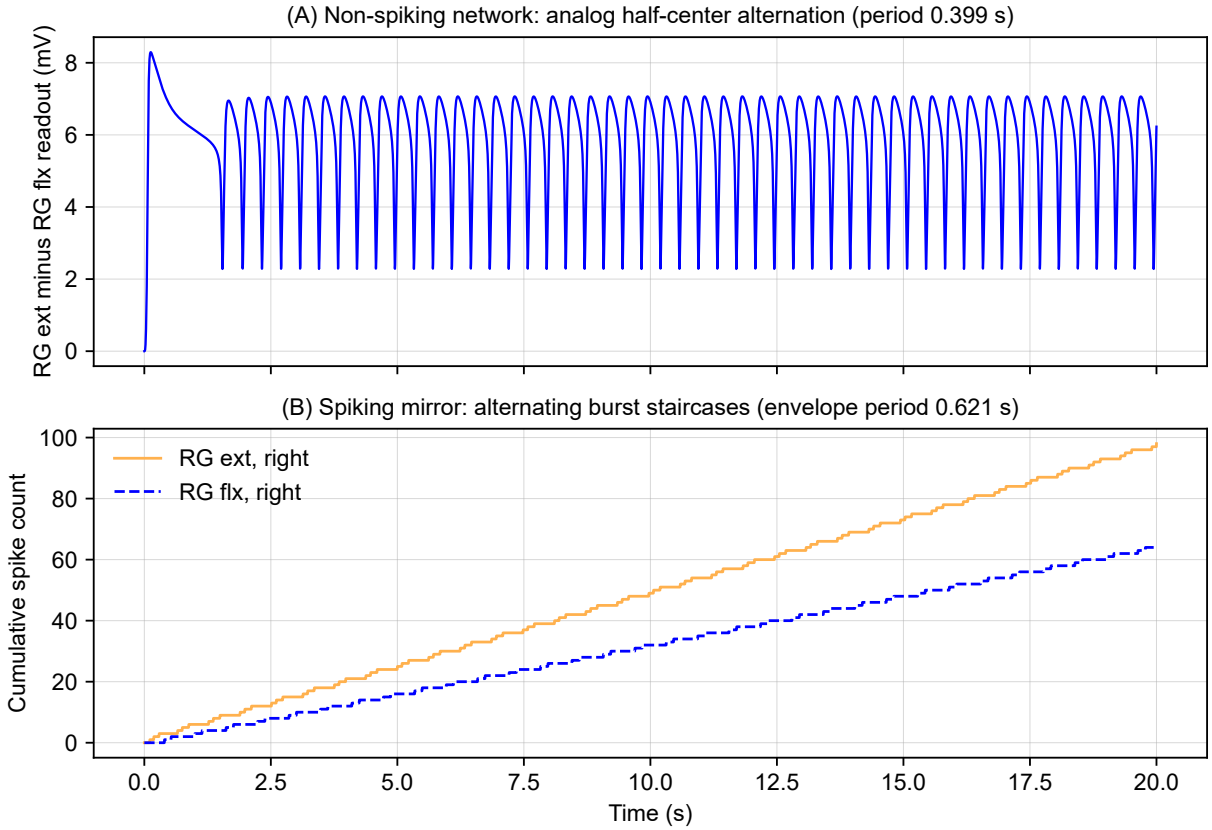


Figure 6.10: Constant-drive rhythm of the right rhythm-generator under network-only simulation. **(A)** The non-spiking network's extensor-minus-flexor readout voltage alternates irregularly about a 4.8 mV swing (larger only in the startup transient), with an average period of 0.399 s. **(B)** The spiking mirror's raw extensor (solid orange) and flexor (dashed indigo) cells fire in alternating bursts whose envelope period is 0.621 s, 55.6 % longer than the analog build.

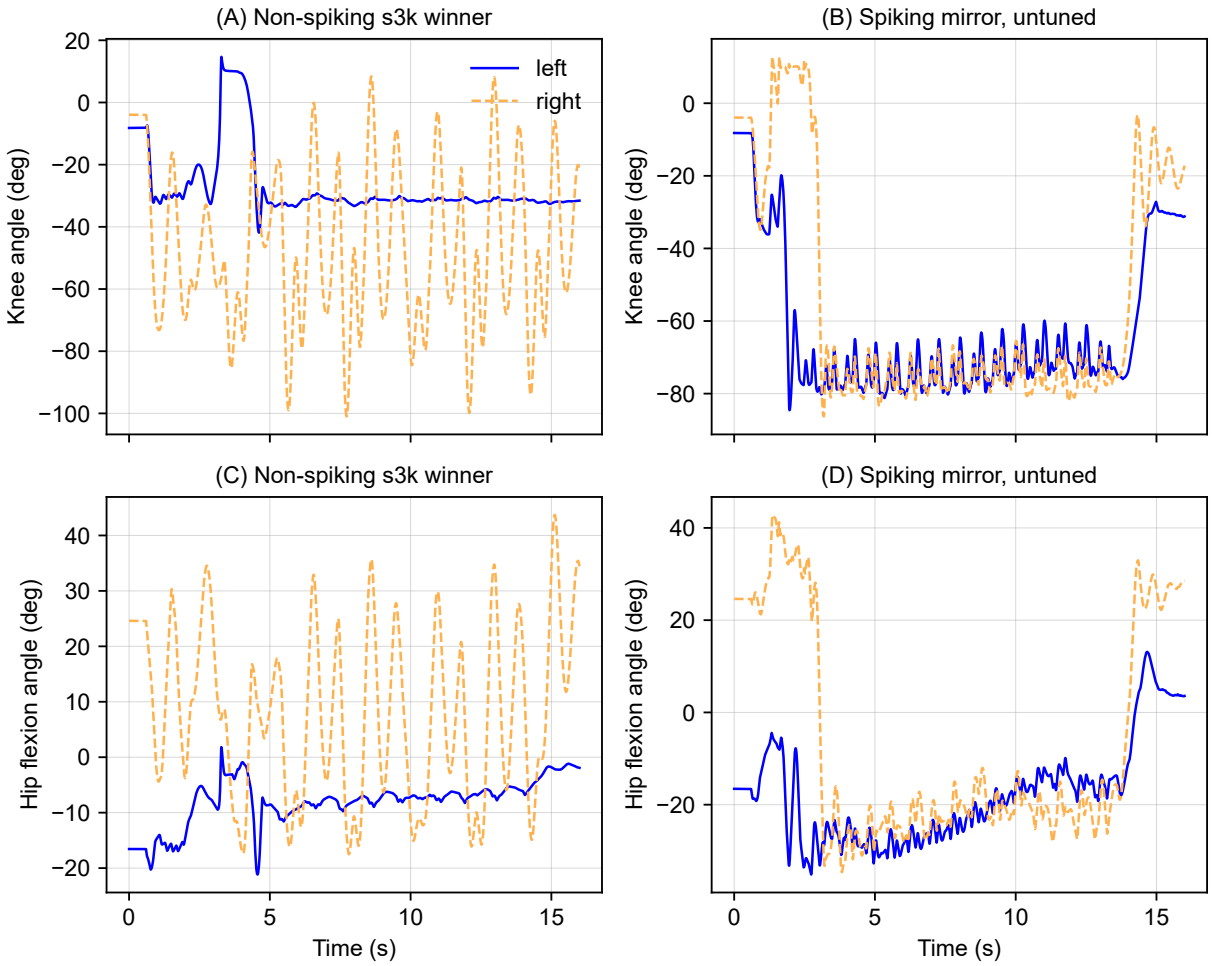


Figure 6.11: Ground-walking joint angles of the converted MuJoCo robot at the non-spiking winner's parameters. **(A, C)** The non-spiking winner steps with the right leg only; the left knee moves through a single slow excursion without cycling. **(B, D)** The untuned spiking mirror cycles both legs with deep knee flexion. Solid indigo traces denote the left leg, and dashed orange traces denote the right leg.

Table 6.3: Untuned spiking mirror versus its non-spiking twin at the same winner parameters, ground evaluation (16 s). The cycle knee minimum and stance fractions are computed over detected gait cycles; the kinematic score is the cycle-shape error metric of the production pipeline (0 denotes perfect agreement; higher scores indicate closer agreement). The spiking mirror carries mapped parameters only, with zero re-tuning.

Metric	**Non-spiking winner**	**Spiking mirror (untuned)**
Kinematic score	-161.57	-236.35
Gait cycles, right / left	9 / 0 (left frozen)	8 / 9 (bilateral)
Cycle knee minimum	-64.3°	-79.8°
Contact duty, right / left	0.637 / 0.985	0.799 / 0.856
Maximum pelvis tilt	22.1°	19.2°
Non-finite states / fell	no / up	no / up

spike for spike (62 of 62 spikes, maximum spike-time deviation 11.75 µs, maximum voltage deviation 1.2×10^{-2} mV). The same circuit run through the SNS-Toolbox numpy backend agrees with the Simulink copy to 0.19 µs in mean interspike interval and 0.81 µV in final voltage. With the spiking blocks present in the library, every committed demonstration and units test reproduces its recorded value exactly, including the committed two-neuron units test at 6.07×10^{-4} mV.

At the matched operating point, the spiking twin of the knee-reflex demonstration is indistinguishable from the baseline in its transient and within one degree in its settle behavior (Table 6.4, Figure 6.12): both read 15.79° at 0.05 s, and the twin settles 0.9° higher with a slightly hotter, faster antagonist alternation, because the rate-coded pathway adds the synaptic filter time constant that graded synapses do not have. The spiking twin of the beverage-holding demonstration likewise holds the cup at essentially the same level, with the reflex-off sag identical to the baseline and the reflex modulation within 0.02° (Table 6.5).

Table 6.4: Knee-reflex demonstration, committed baseline versus spiking twin at the matched operating point (5 s runs; settle statistics over the last 2 s).

Metric	**Baseline**	**Spiking twin**
Knee angle at 0.05 s	15.79°	15.79°
Settle mean (last 2 s)	43.5°	44.4°
Settle range	41.6°–44.9°	41.9°–46.7°
Extensor / flexor activation (window means)	0.53 / 0.53	0.58 / 0.56
Antagonist alternation	0.5 Hz–2 Hz	7.4 Hz (dominant peak)
Ia interneuron firing at settle	graded, subthreshold	95 Hz; Ib 0 Hz

Table 6.5: Beverage-holding demonstration, committed baseline versus spiking twin (10 s pour; sag measured after 2 s).

Metric	**Baseline**	**Spiking twin**
Maximum sag, reflex on	2.00°	2.02°
Maximum sag, reflex off	7.85°	7.85°
Reflex modulation (off minus on)	5.85°	5.83°
Biceps activation over the pour	0.392 → 0.417	0.392 → 0.447
Triceps activation over the pour	0.400 → 0.316	0.400 → 0.254
Biceps Ia firing rate (reflex on)	graded	166.6 Hz (mean)

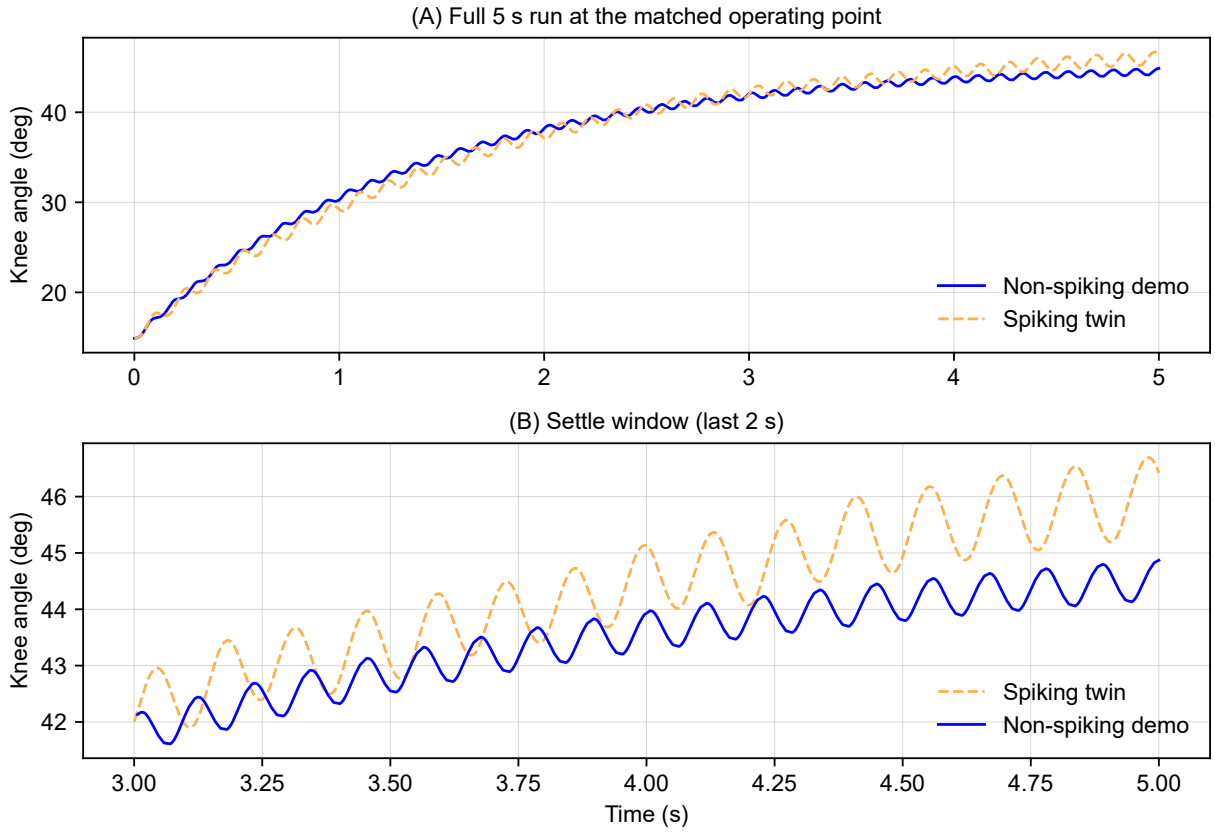


Figure 6.12: Knee angle of the Simulink knee-reflex demonstration: committed non-spiking baseline (solid indigo) versus spiking twin (dashed orange) at the matched operating point. **(A)** Full 5 s run. **(B)** Settle window (last 2 s); the spiking twin's alternation band is slightly wider.

A representative rhythm-generator-to-motor-pool subnetwork demonstrates the hybrid motif end to end: the spiking rhythm-generator populations drive, through a chain of inhibitory interneurons, a non-spiking three-motor-neuron pool whose voltage stays graded over a 5 mV depth (-52.0 to -46.9 mV) and tracks its rhythm generator's rate envelope. Burst alternation in that subnetwork is partial: the extensor population bursts four times in 12 s while the flexor population keeps a low-rate floor through the extensor bursts. This incompleteness is reported as is, and the full 410-population Simulink spiking conversion is scoped as future work for the same reason.

6.4.3 AnimatLab Spiking Conversions

The AnimatLab family produces the clearest negative result of the spiking comparison, and it is reported as such. The graded baselines were re-established first: the two-layer walker's rhythm generator oscillates at 0.443 s (2.26 Hz), the bilateral variant at 0.462 s with the two legs folded near antiphase (lag -0.42 cycle, $r = 0.869$), and the already-spiking 2023-lineage model, the family's spiking reference, bursts at 1.305 s with folded antiphase of $+0.494$ cycle ($r = 0.703$), reproducing its documented period. The current reorganized project's standalone export remains latched under the tonic drives examined, consistent with Section 6.2.

No spiking copy of the three graded models preserves the rhythm (Table 6.6). The two-layer conversions die at every threshold and decay tried (two or zero rhythm-generator spikes in 5 s); the bilateral variant latches with one side's populations saturated at 442 Hz–450 Hz; and the reorganized project's copy converts its latched baseline into strictly syn-

chronous 11.8 Hz co-bursting with no alternation: the left extensor and flexor populations emit identical 58-spike trains whose spike-count envelopes correlate at $r = 1.000$ at zero lag. The thresholds-only ablation, which keeps the graded synapses, preserves near-baseline joint oscillations, isolating the synapse conversion as the change that breaks the rhythm (Figure B1): per-spike conductances at the graded amplitude either starve the reciprocal loop of inhibitory charge or saturate it, whereas the graded synapses read the membrane potential continuously and keep oscillating through subthreshold voltages. A spiking version of this family therefore needs its synaptic amplitudes and decays re-tuned as a rate code, which is how the 2023 spiking reference was built in the first place, rather than copied parameter by parameter.

Table 6.6: AnimatLab spiking conversions versus their graded baselines (5.1 s headless runs; joint ranges in degrees where the rhythm survives). The thresholds-only ablation converts the neurons but keeps the graded synapses.

Variant	**Baseline (graded)**	**Spiking copy**
Two-layer walker	0.443 s rhythm; hip -15 to $+23$, knee -1 to $+60$	dead (2 spikes; hip pinned near $+6$)
Bilateral variant	0.462 s; antiphase -0.42 cycle	latched; one side at 442 Hz–450 Hz
Reorganized project (standalone export)	latched (tonic drives)	synchronous 11.8 Hz co-bursting; same-leg envelopes $r = 1.000$ at lag 0
Thresholds only (graded synapses)	hip -15 to $+23$, knee -1 to $+60$	oscillates: hip -15.1 to $+23.2$, knee -0.8 to $+47.9$

The spiking conversions of the two-layer family, the calibrated-threshold variants, and the decay probe are catalogued variant by variant, with neuron, synapse-type, and connection counts and the structural-validation outcome of each, in Appendix B (Table B5), together with the full gate table of the MuJoCo mirror (Table B4).

6.5 Discussion

This section consolidates what the results mean for the program: the spiking comparison first, because its conclusions constrain the rest; then the training curriculum and its scoring behind the tuned configurations; then the integration of the BPA force model, the standing of the robot models under their different controllers, and the BPA-actuated robot; and finally the new tools themselves as the chapter's contribution.

6.5.1 What Spiking Changes and Does Not

The spiking-mirror comparison (Sections 6.2.8 and 6.4) allows the dissertation's neural-coding assumption to be examined directly, and its outcome divides cleanly into what the coding regime changes and what it leaves alone.

What spiking does not change is the behavior of the controlled model, provided the design rule and the calibration hold. Because the motor-neuron layer stays non-spiking and the spikes onto it are calibrated by mean conductance, the interface that the actuator and torque models of this dissertation attach to, a continuous activation command per muscle, is unchanged by construction. The Simulink twins make the point quantitatively: with only the sensory interneuron layer converted, the knee-reflex transient is identical, the settle band moves by 0.9°, and the beverage-holding reflex modulation moves by 0.02°. To the musculoskeletal model, a calibrated 40 Hz train and the graded voltage it replaces are the same command; the activation dynamics of the muscle model filter the spike train the same way the biological membrane would. The actuator-level and joint-level conclusions of Chapters 3–4 therefore do not depend on the neural coding chosen above

the motor-neuron layer.

What spiking changes first is timing. The mirror's self-sustained rhythm runs 55.6 % slower than the analog build at the same constant drive, and its air-stepping cadence is roughly ten times faster than the non-spiking twin's in that preparation, deviations in opposite directions in the two tests, which is itself informative, since a single scaling error would move both the same way. The comparison traced the constant-drive period to two identifiable knobs, the threshold-increment magnitude and the adaptation time constant, and demonstrated alternation/period trade-offs along them; the untuned starting point, not a structural failure, is the honest reading. Any period-matched spiking controller is therefore a short tuning effort away, and the mirror's mapped parameters are the correct starting point for it.

What spiking changes second is robustness, in the mirror's favor. The analog network's constant-drive rhythm sits close to a bifurcation, a parameter region where an exact re-implementation in a different integration stack can settle to the tonic fixed point instead of oscillating, so marginal winners risk not surviving a solver change. The spiking rhythm-generator populations passed the perturbation basin gate in full, with the rhythm persisting under 1 % jitter on every scalar parameter including the spiking-specific ones, at $0.89\times$ of the baseline swing. For the hardware-transfer program of Chapter 7, where the network will be re-implemented on embedded hardware with finite-precision arithmetic, a rhythm whose basin is wide is the safer substrate, and the sub-stepping cost of resolving spikes (0.5 ms network steps inside the unchanged 2 ms physics step) is modest.

What spiking does not tolerate is a naive parameter copy at the synapse level. The AnimatLab conversions show that converting graded synaptic amplitudes to per-spike

increments of the same magnitude breaks the reciprocal rhythm-generator loop outright (the loop is either starved of inhibitory charge or saturated), while converting only the thresholds changes nothing, and the 2023 spiking walker of the same lineage walks because it was tuned as a rate code from birth. Synaptic strength in a spiking network is a rate-code quantity that must be derived from the presynaptic firing rate and the conductance decay, exactly as the MuJoCo mirror's calibration script derives it, not copied from the graded value. This is the one place where the negative results should temper expectations: spiking conversion is an exercise in re-derivation, and the calibration-first, gate-everything procedure of Section 6.2.8 is the transferable method.

Finally, the one place where the untuned spiking mirror outperforms its tuned non-spiking twin, producing a bilateral ground gait where the winner's left leg is frozen, is a single run with no statistics, and it is claimed only as a lead. But it points at the mechanism the design rule was after: the mirror's afferents are event-coded, its adaptation is spike-history-based, and its inhibitory charge arrives in discrete quanta, so the phase relationships that the analog build locks in around a frozen leg are not the only stable operating regime of the same topology. Establishing whether that regime survives tuning, and whether event-coded contact afferents are what unlock it, is precisely the closed-loop CPG comparison program defined in Chapter 7, and the mirror is now the instrument for running it.

6.5.2 The Five-Stage Training Curriculum and Its Scoring

The tuned configurations of this chapter were not tuned all at once. The training proceeds through five stages (Figure 6.13), each adding one capability and each seeded with the previous stage's winner so that only the newly activated pathway gains are searched: deafferented air stepping first, in which the rhythm populations' oscillation is verified with the legs suspended and every sensory channel silent; afferented air stepping second, with heel, toe, and load feedback active in the same suspended preparation; standing balance third, the stage modeled on the SCONE balance tutorial (Section 6.2.1) and scored on maintaining posture under push perturbations; ground walking fourth; and ground walking with the contact-driven pathway variant fifth. Standing before walking is the design rule: gait tuning begins only after a configuration holds posture, and the failures the curriculum surfaces early are the cheap ones.

The scoring is stage-appropriate and gated. The air stages are scored on rhythm and swing depth with the objective rhythm-gated, so a configuration that freezes cannot win by holding a static pose. The balance stage is scored on posture maintenance under push. The walking stages use the production pipeline's cycle-shape kinematic score against a human reference cycle, in which 0 denotes perfect agreement and a fixed fallback marks the absence of detectable gait cycles, a value that sits below every genuine gait so that a frozen model cannot out-score a walker. Scores are never read alone: finalists must pass the perturbation and basin requirements of Section 6.2.9, because a score achieved on a narrow basin is a property of the solver rather than of the circuit, a lesson the cross-engine results of this chapter turn into a gate.

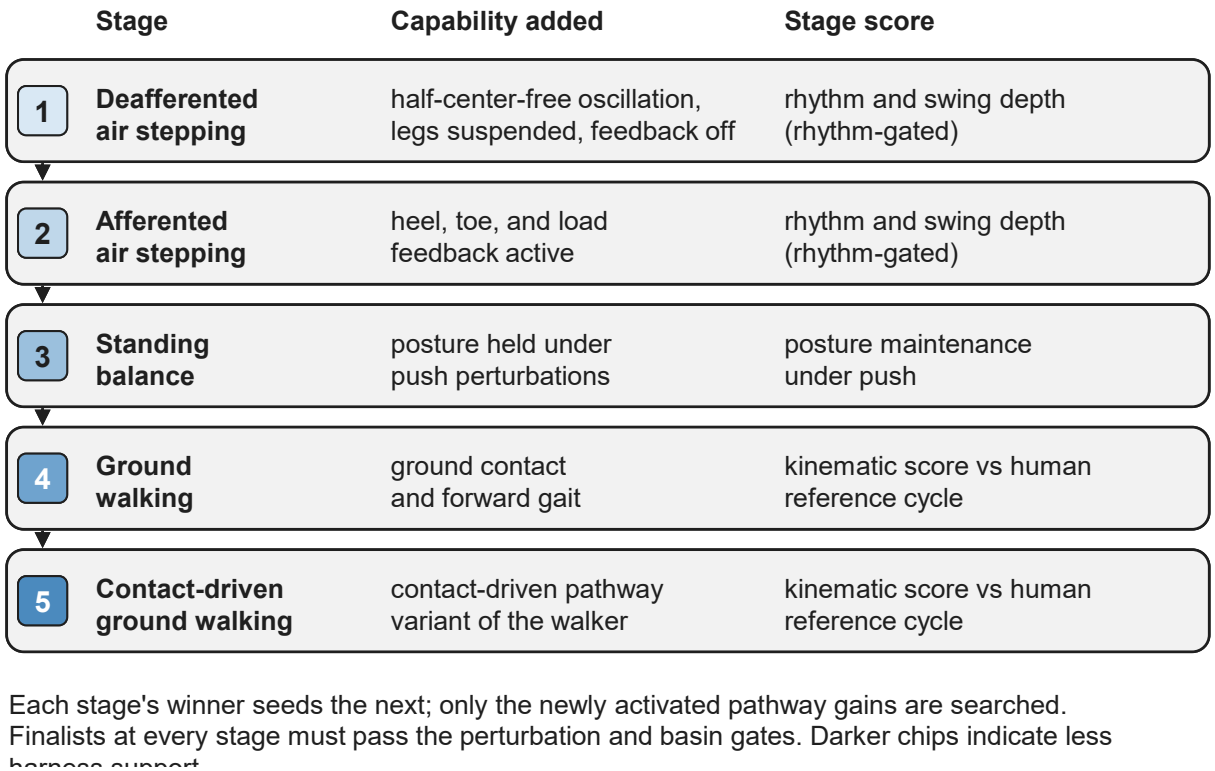


Figure 6.13: The five-stage training curriculum. Each stage adds one capability and is seeded with the previous stage's winner, so only the newly activated pathway gains are searched; the chip fill tracks the preparation's support level from suspended air stepping through stance to ground. Scores are stage-appropriate and rhythm-gated where rhythm is the capability under test, and finalists at every stage must pass the perturbation and basin gates before the score is read.

6.5.3 Integration of the BPA Force Model

The actuator model travels with the controller. Inside the MuJoCo loop, actuator force comes from the quasi-static BPA model of Chapter 3, including the correction-factor force balance of Chapter 4, rather than from a generic muscle approximation; in Simulink, the library's three diameter-specific force blocks implement the same fitted force surfaces, so a circuit assembled there commands the same actuators as its MuJoCo counterpart. The demonstrations that close the loop onto the actuator model, the synthetic knee of Section 6.2, the BPA-CPG leg demonstration of Section 6.3.2, and the BPA-actuated robot below, all draw their forces from this one validated source, which is what keeps actuator-level conclusions consistent across engines. The BPA-actuated robot's peak forces sit at 60% of the pressure ceiling, so the demonstrated gaits operate inside the characterized pressure range rather than extrapolating past it. What the integration does not yet carry is actuator dynamics: the force model is quasi-static, and the pressure–force dynamics under pulse-modulated valve control [Elzein et al., 2025] enter the program as future work (Chapter 7).

6.5.4 The BPA-Actuated Robot in MuJoCo

The end-to-end demonstration of the MuJoCo chain, conversion pipeline plus force interface plus network builder, is the BPA-actuated robot, and it belongs in this discussion because it is the tooling's standing answer to the question the chapter opened with: what the laboratory is for. The walker geometry of Section 6.1 was exported to MuJoCo and driven by antagonistic BPA actuator pairs under the split contact-driven network, with

actuator forces computed by the BPA model of Chapter 3. In the air-stepping preparation, with the harness carrying 70% of the 411 N body weight, the walker produced clean knee-flexion excursions at a 1.542 s period with a knee flexion range of 63.5°, peak BPA forces of 880 N (left) and 1372 N (right) at a peak pressure of 372 kPa (60% of the 620 kPa ceiling), and 7 and 5 heel-contact episodes on the left and right. This run demonstrates the complete chain from the characterized BPA force model to a multi-joint walker stepping under synthetic neural control, with actuator sizing operating inside the characterized pressure range.

The same evaluation series reported its failures with equal weight. On the ground, the W2L gate model under its untuned default gains stands (full-support stance) but does not walk: heel duty is zero and forward displacement is $+0.001$ m over the run. With the left and right drives separated and the feedback gains re-tuned, the same robot alternates in antiphase (inter-leg phase 0.487) and progresses 1.72 m over a 30 s window (0.066 m s^{-1}) under the harness protocol, with the neural drive weaned to 60 % of its bootstrap value and the result robust under perturbation (12 of 12 basin re-runs). Below 60 % drive the gait dies, the harness itself is not weaned, and foot duty remains below the strict walking gate, so the result is supported progression rather than validated walking. The MuJoCo re-implementation of the Li model produced no stance episodes in any of seven drop-protocol variants. Together with the latch result of Section 6.3.3, these negatives localize the open problem precisely: the contact-driven rhythm architecture itself transfers across engines intact, but translating it to unsupported ground locomotion requires the balance and weight-bearing layers defined in Chapter 7, and those layers are not present in these models.

6.5.5 Controller Comparison on the Robot Models and the Walking Status

The robot models carry more than one controller, and the comparison is a result of the tooling rather than an afterthought. On the converted robot-body model, the four production walking configurations span the controller lineages (Section 6.3.5), and scored with the same cycle-shape kinematic score against the 43-reference library, the family-level fact is that the robot-walker-lineage configuration, not the human-derived ones, agrees most closely with the library, while the six-synergy configuration, the most strongly human-derived in its pattern formation, agrees least closely; closing that gap is a tuning-program question rather than a tool question, and it is not pursued further here. On the physical-model side, the SolidWorks knee assembly is imported into Simscape Multibody with its degree-of-freedom defect documented and its fix defined (Section 6.2), and the imported `gait2392_simbody` mechanism awaits its muscle blocks; neither physical model yet carries a tuned controller, so their controller comparisons remain the MuJoCo-side results above.

The walking status of the tuned configurations deserves a precise statement, because the chapter's records support exactly this and no more. Each tuned configuration was brought through the curriculum's air and balance stages: deafferented and afferented air stepping with the legs suspended, and standing balance that absorbs push perturbations without falling (the production configuration survived 15 stand-and-push evaluations, Section 6.3.6). On the ground, all walking evaluations run under the support harness of the tuning pipeline, where the configurations produce gross walking-like progression, with cycles detected and scored but with right-leg-only or asymmetric gait patterns (Section 6.3.5). A subsequent drive-weaning study re-tuned the feedback gains while forcing

the hand-drive down a ladder of scales. Its finalist improves the default build's evaluation score from -161.57 to -145.07 with a bilateral gait of ten contact cycles per leg (contact duty 0.71 and 0.65, no non-finite states), survives a closed-loop basin check in eight of eight perturbed evaluations, and degrades gracefully against independent walking references; with its feedback held fixed, bilateral stepping persists at every drive scale down to zero hand-drive, at which the drive pathway is absent from the network entirely, whereas the un-retuned configuration collapses below 85 % of full drive. The gait remains in-place alternation under the support rig, without forward progression. Unsupported, harness-free walking has not been demonstrated in this program's records: the BPA-actuated robot's ground model stands without walking under its untuned default gains and reaches only supported progression at weaned drive (Section 6.5.4), and the archived reference walkers do not walk unsupported either. Whether any configuration walks without the harness is therefore an open question, and the balance and weight-bearing layers that would answer it are the first item of the simulation program defined in Chapter 7.

6.5.6 The New Tools and the Program They Enable

The chapter's contribution is the tool set. The OpenSim–MyoConverter–SNS-Toolbox pipeline couples the full-lower-body converted model to a two-level spinal network with Ia, Ib, group II, and contact feedback, and it runs the full neuromechanical loop faster than real time. The SNS-Simscape library assembles the same component models in a standard controls environment, with the reflex demonstrations, the diameter-specific BPA force blocks, the exported production network, and the imported OpenSim and

SolidWorks bodies behind it. The controller ports put the Deng-lineage and Li-lineage circuits into NumPy, Simulink, MuJoCo, and Simscape Multibody realizations on the same bodies, and the verification instruments (the cross-engine harness, the structure audits, the perturbation and basin gates, and the spiking mirrors) are what make the ports trustworthy. Every one of these tools is reusable by a researcher who did not build them, and the loop's speed is what makes deletion and stimulus experiments cheap enough to repeat.

The laboratory's limits are on record with equal weight. The archived walkers do not walk unsupported on the ground; the tuned networks transfer poorly to diverse human reference cycles and walk right-leg-only or asymmetrically under the harness, or alternate in place under the support rig at weaned drive; the systematic stimulation series at the pattern-formation and motoneuron levels, and the motoneuron-level deletion phenomena of the reference models, are not yet reproduced; the SolidWorks knee arrives in Simscape Multibody with its joint welded pending re-mating; and the Shinohara, Di Russo, and Shevtsova replications remain circuit-rule translations rather than running models. What the tools are for is the program they define next, in Chapter 7: the balance and weight-bearing layers that unsupported walking requires, the systematic stimulation series at the motoneuron, pattern-formation, and rhythm-generator levels against which the circuits' signs, gains, and phase structure will be validated, and the transfer of the whole stack toward real-time hardware.

7 Discussion

This chapter reads the preceding studies as one program, states what that program established, and delimits the work the results open. The overview leads with the design factors that make improved first-pass robot design achievable, then attributes them to the developments behind them, ordered from the actuator outward to the whole body. The future-work subsections that follow preserve that order, and each is written for the researchers who will take it up.

7.1 The Work as a Whole

The outcome that organizes this discussion is that improved first-pass design of a BPA-actuated biomimetic leg is now achievable: a designer can predict actuator force, joint torque, and range of motion accurately enough to commit to an actuator set, a resting length, and a route before the hardware exists (Figure 7.1). At the center of that outcome are the identified design factors that remove the rigid-body assumptions from the torque model (Chapter 4): a constant length offset χ_0, effective stiffnesses χ_1 and χ_2 that lump the series compliance of the attachment brackets, fixtures, and artificial tendons into the bracket frame's principal directions, and a wrapping-loss factor χ_3 that removes usable actuator length as an extensor bends over the joint. The terms were identified on the pinned knee, and they hold up beyond it: carried unchanged to the biomimetic knee with its reinforced frame, the corrected model explains the bulk of the measured torque variance there as well, cutting the biomimetic extensor's fraction of variance unexplained

from 4.49 to 0.67 with the shared terms (Section 4.3.2). Whether the set is complete is itself now an experimental question: a fourth term χ_4, a stiffness acting along the force path independently of force orientation and presently absorbed by the stiffness pair (χ_1 and χ_2), is arguably the next candidate, and the identification machinery delivered here is the instrument for settling it.

Improved BPA characterization is the first enabler (Chapter 3). The maximum isometric force of Festo BPAs was shown to depend strongly on resting length, a previously unreported characteristic that corrects the manufacturer's tool at exactly the short lengths a leg design requires, and a nondimensional force surface predicts isometric force from relative strain and relative pressure with only three diameter-dependent coefficients. Because resting length is a primary degree of freedom in actuator selection, this model is what makes that selection rational. The quasi-static surface is complemented at the dynamic end by the state-space, optimization-based model of pressure and force dynamics in pulse-modulated BPAs [Elzein et al., 2025], which supplies the actuator behavior that the isometric maps cannot.

Attachment and routing optimization is the second enabler, and it is tiered. The placement optimization of Morrow, Bolen, and Hunt found artificial muscle attachment locations that reproduce human torque profiles and ranges of motion about the trunk and leg joints [Morrow et al., 2020], converting the earlier finding that transplanted human attachment locations do not transfer directly to a BPA-actuated robot [Bolen, 2019, Bolen and Hunt, 2019] into a usable design tool; the route redesigns of Section 4.3.2 rerun that machinery under the corrected torque model. At the route-selection tier, with two 20 mm BPAs per route and routes chosen with the actuators' bending in mind, the redesigned

flexor and extensor each meet or exceed the magnitude of their human-scaled torque targets across the full knee range of motion, and the cam-profile extensor converts the direct route's thin margin into genuine design headroom (Section 4.3.2). Stated as a design program, the generalization is one or more BPAs replacing one or more human muscles across one or more joints, serving the two degrees of freedom that matter most at each joint, with routes grown outward from positions close to the bone and held to the spatial constraints of the candidate actuators and of the BPAs already mounted; the first whole-body pass over the robot's actuator set, with its successes and its failures, is reported in Section 4.3.2. Where the attachments are fixed and only the route and the tendon length can change, the cam-shape solver applies: the tibial-side route points trace a designed cam profile, trading biomimicry for torque-profile optimization and reduced component forces.

The Sensory Afferent Database (SADb) is the third enabler (Chapter 5). It organizes the sensory-afferent literature into a structured afferent resource [Bolen et al., 2023]: of the 561 papers retained after the audit and the prune, 523 carry structured curation notes, 86 carry an afferent classification, and 258 carry notes translating each paper toward robot or simulation use, and the full audited corpus of 943 papers, including the 382 pruned records preserved under an archived flag, remains reconstructible in the packaged tool. The toolset matters as much as the corpus. Its detailed table drills from any filter down to a paper's full notes; its pivot view counts the corpus along any two dimensions at once; its bubble map draws every paper as a neuron, soma sized by citations, with citation links rendered as excitatory and inhibitory synapses in the laboratory's convention and a focus mode that walks the citation neighborhood; and the monthly skimmer–curator workflow sweeps the recent literature for candidate records and routes every one of them to a human

ruling. The evaluation of AI-assisted literature triage established the division of labor that makes expansion affordable [Bolen and Hunt, 2024]: the machine drafts, and the human rules.

The fourth enabler is the set of novel neuromechanical simulation tools (Chapter 6): the audited OpenSim-to-MuJoCo conversion of the robot's musculoskeletal model, the force interface that computes actuator forces inside the loop from the validated BPA model, the two-level spinal networks built in SNS-Toolbox, the Simulink SNS library for assembly in a controls-engineering environment, the cross-engine verification harness, and the spiking-neuron mirrors that test the coding assumption. On these tools the controller program has matured from assembly into use. Production walking configurations have been tuned and evaluated in full-body ground simulations, the first pathway-deletion and drive-stimulus experiments have been run, the gait-validation sweep has measured how far the tuned networks remain from the diversity of human reference locomotion, and the contact-driven rhythm core has been shown to transfer between independent implementations with exact period agreement and trace correlations above 0.9999. In hardware, the companion balance-platform studies demonstrated that a BPA-actuated limb holds itself upright under pseudo-vestibular inertial feedback [Bolen et al., 2024b,a,c]. What remains unfinished is equally specific, and it organizes the future work below: the balance and weight-bearing layers that unsupported bilateral walking requires, wider transfer across reference cycles, the supraspinal sensorium, and embedded real-time execution.

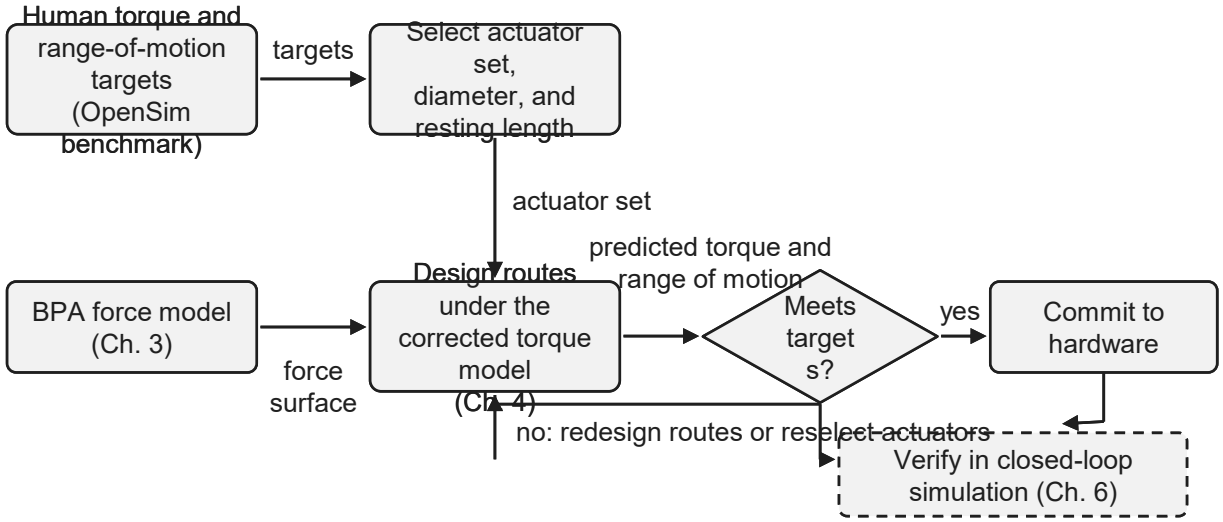


Figure 7.1: The first-pass design workflow the delivered enablers make possible. Human torque and range-of-motion targets from the OpenSim benchmark and the validated BPA force model of Chapter 3 drive actuator selection; route design runs under the corrected torque model of Chapter 4; the predicted torque and range of motion are checked against the targets, with route redesign or actuator reselection as the loop when they fall short; and the committed configuration is verified in closed-loop simulation with the tools of Chapter 6 before hardware is cut.

7.2 Future Work

The subsections that follow are options for the researchers who take this program up, ordered from the actuator outward to the whole body. Each draws directly on the delivered models, the database, and the simulation tools.

7.2.1 Expanding the BPA Force Characterization

The isometric characterization of Chapter 3 leaves one actuator curve unidentified and one regime unmeasured, and both are tractable next experiments. The unidentified curve is the maximum-contraction function: ε_{620} could not be pinned to a predictive relationship with resting length, so each actuator's free contraction is recorded individually before use. Controlled tests designed for that relationship, across multiple product batches of the 10 mm, 20 mm and 40 mm diameters, would either produce general fit surfaces or establish batch variation as the binding uncertainty. The 20 mm actuators characterized here came from a single product batch, so batch-to-batch variation is unquantified and those fits in particular carry that limitation.

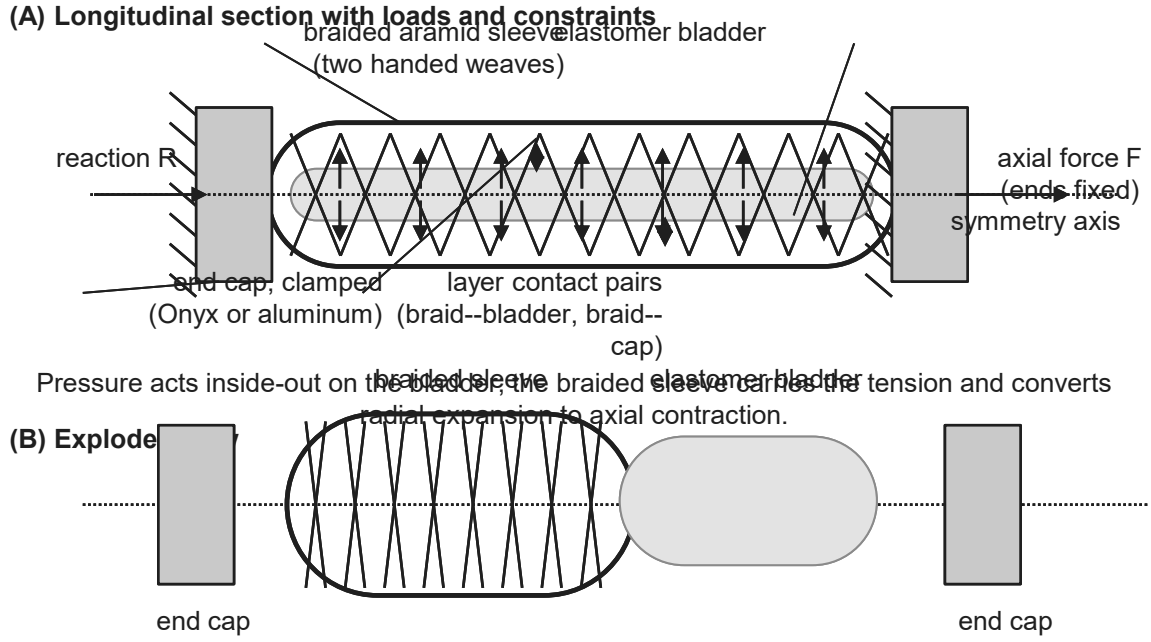
The unmeasured regime is everything off the isometric plateau. A biological muscle model separates active force from passive force and couples both to activation and contraction dynamics [Thelen, 2003, Millard et al., 2013], and Appendix A, which collects the Thelen and Millard formulations used as torque targets in this work, is the natural home for a BPA counterpart: extending it with the remaining equations of those models, the Gaussian active-force shape factors, the activation and deactivation time constants, and the tendon force curves, would let the BPA model be decomposed into active pressurized-

braid force, passive resistance of the unpressurized actuator and sleeve, and contraction dynamics. Isokinetic and isotonic or isobaric characterizations could then be assimilated into the same framework rather than fitted separately, with the pressure–force dynamics model of Elzein et al. [2025] supplying the actuator-side dynamics.

Two experimental programs would complete the mechanical picture. Instron tensile testing of actuators with their Onyx and aluminum end caps, unpressurized and pressurized, would measure the end-effect region that the characterization identifies as the mechanism behind the resting-length dependence of maximum force. A finite-element program sketched from the construction disclosed in Festo's patent literature would give the actuator a first-principles model to check the empirical surfaces against [Festo AG & Co. KG, 2026]: a symmetry half-model of the actuator, a half circle fixed at the outer and inner diameter at one end with the diameter free to expand at the other and an optional lateral force or fixity, assembled from thin-shell or membrane layers in inter-layer contact, with the two handed aramid weaves of the braided sleeve represented as beam or truss elements. Figure 7.2 shows the construction such a model would represent.

7.2.2 Expanding Attachment and Routing Design

Each routing tier of Section 7.1 leaves a named next step. The knee's flexor redesign assumes both actuators share one attachment route; making the mirrored pair explicit is the next refinement. The bi-pulley sweep showed that reeved routes do not yet fit the anatomical corridors, where geometry rather than torque was the binding constraint, so clearance-relaxed attachment sites are where that freedom should pay. The whole-body



Assembly axis shown as the dash-dot centerline; components separate along it.

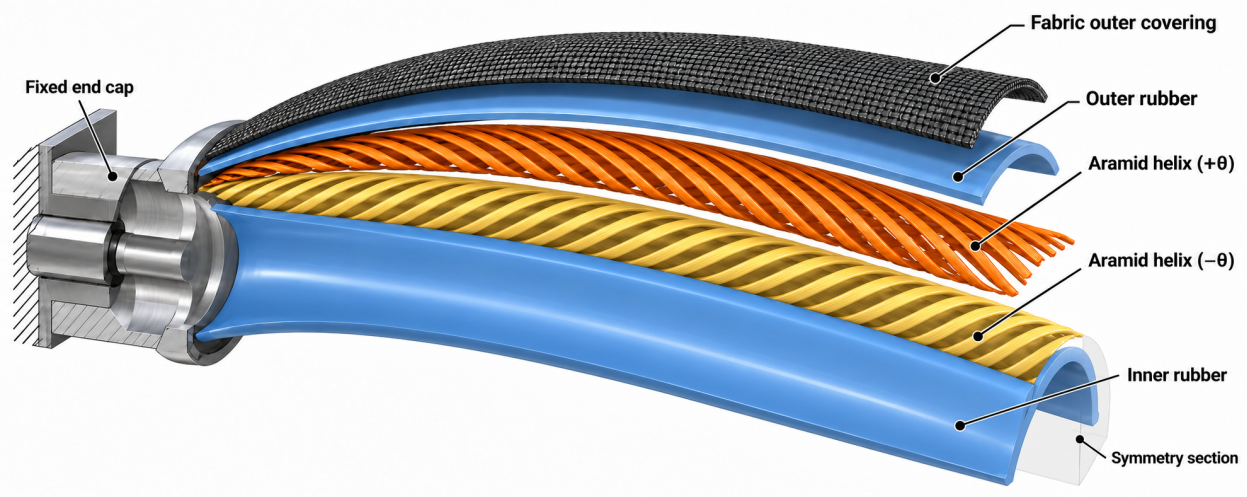


Figure 7.2: The construction the finite-element program of this section would represent. (A) Longitudinal section with loads and constraints: internal pressure acts inside-out on the elastomer bladder; the braided aramid sleeve, woven in two handed weaves, carries the tension and converts radial expansion to axial contraction; layer contact pairs act between bladder, sleeve, and caps; with the ends clamped in the Onyx or aluminum caps the pressurized actuator develops an axial force reacted at the fixed end; and the dash-dot centerline is the symmetry axis of the half-model. (B) Exploded half-cutaway of the layered construction, from inside out: the inner rubber tube, the two aramid helical reinforcement families of opposite handedness, the outer rubber, and the fabric covering, anchored at the sectioned end cap, with the open semicircular section marking the symmetry boundary of the half-model. Not to scale.

pass needs its route-builder failure fixed and either larger actuators or relaxed corridors on the two routes that satisfy geometry but fall short of torque.

The design element this program has so far declined is a patella. The torque model wraps the extensor BPA over the joint surface and accounts for the loss with χ_3, which was easier than adding a patella, deterministically designing its location as a function of knee angle, and accounting for friction (Chapter 4). A designed sesamoid is the biomimetic answer: an artificial patella, its location optimized over the range of motion the way route points now are, would buy back moment arm at high flexion and shield the actuator from the joint, at the cost of a friction model and a contact schedule the optimizer must respect. The cam-profile solver is the proof that a moving routing surface can be a design variable; a patella is its biomimetic limit case.

7.2.3 Expanding the Sensory Afferent Database

The database's growth now has two engines: the monthly skimmer–curator workflow that sweeps the recent literature, and machine-assisted drafting under human ruling. The next step the corpus itself suggests is prompt-driven passes over the existing records, designed to discover the notes and rules latent in the corpus, candidate pathway records, missing cross-links, controlled-vocabulary amendments, and gaps in the afferent classification, and to draft entries for the flagged stubs and for the vocabulary gaps that the extensions of Section 7.2.5 will need first, vestibular and visual preparations among them. Two constraints from the curation rules carry over unchanged. The passes must be economical and their accuracy measured against human rulings rather than assumed, and a human still

rules on every write; the triage evaluation found acceleration with substantial oversight required [Bolen and Hunt, 2024], and nothing in the pipeline removes the evaluator. What the passes buy is coverage at the corpus's edges, at a cost a laboratory can afford.

7.2.4 Closing the Loop in Simulation

The simulation laboratory of Chapter 6 is assembled, audited, and in active use. The converted robot model carries all of its muscle-tendon actuators with audited geometry, and the BPA force interface computes actuator forces from the validated model inside the loop; both are documented as complete in the chapter's methods (Section 6.2). The 410-neuron SNS-Toolbox instantiation of the spinal network, with per-leg rhythm-generator half-centers, phase-shifted pattern-formation groups, motoneuron pools for every actuator, and Ia, Ib, and II afferent populations, couples to the converted model at a 2 ms neural step, and production walking configurations have been tuned against a walking reference and evaluated on the ground (Section 6.3.5). A drive-weaning study has since produced a feedback-sustained bilateral configuration whose stepping persists with the hand-drive withdrawn entirely, still in place under the support rig, and the robot-side model progresses under its harness at weaned drive (Section 6.5.5). What those same evaluations localize as missing defines the work remaining here: the balance and weight-bearing layers that unsupported, bilateral walking requires; transfer from a single tuning reference to the diversity of the locomotion reference library; and the systematic verification series that follows.

The verification series comes first because it is cheap and decisive. The circuits' signs,

gains, and phase structure are checked against the experimental literature with the stimulus protocols the physiology itself uses: a tonic stimulus applied at the motoneuron, pattern-formation, and rhythm-generator layers, with the resulting outputs compared against the tonic central and sensory stimulation that facilitates involuntary air-stepping in humans [Selionov et al., 2009] and the deletion and stimulation phenomena reproduced by the two-level CPG models and their neuromechanical implementations [Rybak et al., 2006, Markin et al., 2010, Shevtsova et al., 2016]. The first of these tests is complete at the rhythm-generator level, where tonic drive evokes an approximately 0.9 s antiphase cycle in the isolated network (Section 6.2); the motoneuron-level and pattern-formation-level series remain to be run, together with the motoneuron skipping-deletion phenomenon of the reference models.

The laboratory's hypothesis-testing program is stimulus, deletion, and addition. The stimulus and deletion families have their first sweeps (Section 6.3.6). Pathway addition is their complement and the one the tools have not yet exercised: adding a candidate circuit to a tuned configuration, a commissural class, a sensory gain, or a modulatory layer, and re-tuning, which turns connectome hypotheses into falsifiable experiments. The closed-loop comparison of spiking and non-spiking controllers of the same topology belongs to the same program, with the spiking mirror's untuned bilateral ground gait as the standing lead (Section 6.5.1).

The tools also compose across programs, and the composition is now cheap enough to be a standard workflow. A full-body OpenSim model with arms and trunk, of the kind MyoSuite distributes [Caggiano et al., 2022], can be brought into the SNS-Simscape library for graphical assembly and inspection of the controller, and the same network imported

back into SNS-Toolbox on MuJoCo [Nourse et al., 2023] for the speed that tuning and long evaluations require. A controller built in that loop inherits the conversion audit, the cross-engine checks, and the deletion apparatus of Chapter 6 without rebuilding any of them.

7.2.5 Extending the Sensorium

Once stepping is stable, three supraspinal contributions will be added, and a fourth is in line behind them (Figure 7.3). First, vestibular feedback will be simulated with IMU-like sensing of body orientation and angular velocity, extending the balance-platform work, in which an inverted pendulum with pseudo-vestibular IMU feedback maintained balance on an oscillating platform [Bolen et al., 2024b,a,c] and analytic neural networks reproduced human balance dynamics [Hilts et al., 2019]. Second, ocular feedback will be modeled to investigate how visual stabilization cues can help reset IMU drift, a well-known practical problem for inertial sensing on robots, by treating visual measurements of the horizon and surroundings as a low-frequency correction to the vestibular estimate. Third, a cerebellar layer will be explored as an adaptive gateway between higher-level locomotion commands and the CPG: a structure that can gate between gaits or movement patterns (e.g., walking, running, and turning) and tune the phase relationships among them, in line with the cerebellum's established role in adaptation and coordination. Multilevel spinal-plus-cerebellar synthetic nervous systems have already been demonstrated for insect-like legs [Grummer et al., 2024], and hierarchical cortico-cerebellar-spinal models have been proposed for adaptive step-length control in bipedal locomotion [Chen et al.,

2025]. The fourth addition, basal-ganglia circuits realizing the inhibitory and excititory gating that selects among actions and locomotor modes, would give the controller a substrate for choosing between behaviors rather than only executing one. The database already holds the entry points for this program: the human balance and posture cluster and the multisensory recalibration cluster are curated and connected to the locomotor core through the citation graph (Chapter 5), and the proprioceptive–vestibular interaction literature will guide how the signals are weighted [Akay and Murray, 2021].

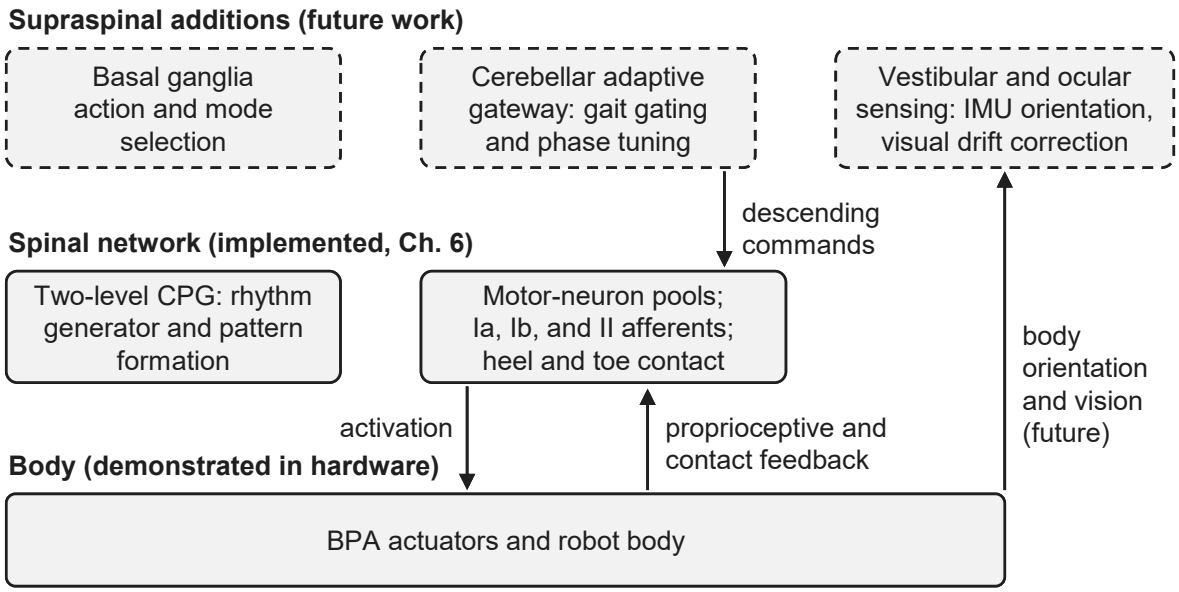


Figure 7.3: The control stack with the planned supraspinal additions. The spinal network tier, the two-level central pattern generator with motor-neuron pools and Ia, Ib, and II afferent plus heel and toe contact feedback, and the BPA-actuated body tier are implemented and demonstrated (Chapter 6), with the balance-platform studies as the demonstrated precursor of the vestibular channel. The dashed supraspinal tier is the extension program of this section: basal-ganglia circuits for action and mode selection, a cerebellar adaptive gateway between high-level commands and the pattern generator, and vestibular and ocular sensing closing body-orientation and visual feedback onto the same loop.

7.2.6 Real-Time Hardware Realization

The final stage returns to hardware, and its near version is a step-by-step integration problem for which the laboratory's existing equipment already stocks the parts. The stack is fixed: Teensy-class microcontrollers handle the low-latency per-joint loops (valve control, pressure sensing, and contact sensing), and an NVIDIA Jetson Orin serves as the companion computer for SNS network evaluation and activation computation. The integration steps then run in order. Execute the SNS-Toolbox network in real time on the treadmill biped. Close feedback from the sensors the lab already uses: the joint encoders, the actuator pressure sensors, the liquid-wire strain gauges, the pressure manifolds, and the IMUs. Compute the afferent signals from those streams exactly as the simulation tools compute them, Ia length and velocity and II length from the encoders, and Ib force from pressure-based force estimates checked against the strain gauges. Tune computer-in-the-loop, with the embedded stack executing the network against the simulated body before it executes against the physical one, so that gain changes are tested at simulation speed and validated on hardware only when they survive. Keep the serial connection to the host computer for data logging and high-level commands only (e.g., run, walk, or stimulate a named neural pathway), so the control loop stays fully embedded. The wiring schematics, the neural-pathway schematics, and the step-by-step commissioning procedure that this integration produces belong in a hardware appendix rather than in this chapter; the connectome figures of Appendix B already serve as the pathway schematics for the spinal network.

The precedent and the margins both support feasibility. Walking control of a physical

musculoskeletal biped with McKibben muscles has been demonstrated with spring-loaded inverted-pendulum controllers [Li et al., 2024], and the AARL's quadruped has already traveled part of this road: a MuJoCo model of the BPA quadruped was coupled to SNS-Toolbox, produced closed-loop coordinated stepping, and was transferred to hardware with potentiometer and pressure feedback on a treadmill [Lutz, 2025]; the present program extends that approach from the quadruped to a leg built on human anatomy. An actuation-level design lever also remains open: the BPA's limited contraction can force long musculotendon routings, a burden visible in the extensor redesign bounds of Appendix C and encountered in the author's master's routing work [Bolen and Hunt, 2019]. A block-and-tackle "reverse pulley" arrangement, demonstrated on the AARL's self-balancing inverted-pendulum platform, addresses this by trading transmitted force for travel: paired BPAs pulled through compound pulleys on a tendon coupled to the limb, increasing range of motion at the cost of reduced transmitted force and increased reaction force on the structure [Bolen et al., 2024b,a,c]. Because joint torque is computed in the body frame, configurations that turn that reaction force to advantage may exist, making the arrangement a candidate for joints whose optimized routes demand impractically long actuators.

Because the static force and torque models of Chapters 3 and 4 are computationally inexpensive lookup-table candidates, they will serve as the feedforward model inside the real-time controller, while the dynamic pressure–force behavior under pulse-modulated control will draw on the state-space model of Elzein et al. [2025]. Embedded execution of the complete CPG-plus-reflex network on this stack is the intended demonstration of the dissertation project: a robot that walks with human-like biomechanics under a biologically grounded neural controller, and whose nervous system, unlike a living animal's, can be

instrumented, stimulated, and lesioned at will.

7.2.7 Toward Dynamic Whole-Body Maneuvers

Beyond steady walking, the long-term goal of this line of research is dynamic, whole-body maneuvering: transitions between gaits, reactive balancing, and ultimately ballistic maneuvers that require coordinating the full actuator set through brief, precisely timed activation bursts. The author's aspiration for this program is a robot that can perform a kickflip, a maneuver that, like its skateboard namesake, demands explosive coordinated extension, precise timing, aerial body orientation, and controlled landing. It is exactly the kind of behavior that exercises every layer of the architecture described above, from contact receptors and spinal pattern generation to vestibular orientation sensing and cerebellar coordination, and it would demonstrate that biomimetic actuation and neuromechanical control scale from steady locomotion to dynamic whole-body movement.

These subsections are options, not prerequisites; each can be pursued on the delivered framework independently of the others. What they all draw on, the actuator and torque models, the routing machinery, the afferent database, and the simulation laboratory, is consolidated in Chapter 8.

8 Conclusion

This dissertation moved the biomimetic humanoid robot project from an unvalidated design concept to a quantitatively characterized actuator set and a validated knee-level biomechanical design, and it defined the path from those experimental foundations to neuromechanical control.

At the actuator level, the analysis in Chapter 3 produced novel equations for calculating isometric force in Festo 10 mm and 20 mm-diameter BPAs. The work elucidated the relationship between maximum BPA force at 620 kPa and resting length, $F_{620}(l_0)$, a previously unreported characteristic of these actuators that the manufacturer's tool predicts with the opposite trend, and it produced a nondimensional force surface, $F^*(\epsilon^*, P^*)$, with only three diameter-dependent coefficients. Taken together, these fits predict isometric BPA force across resting lengths from a small number of measurements, correct the large overpredictions of the manufacturer's tool at short resting lengths, and outperform published single-length models. In a design workflow, these equations make it possible to select actuator diameter and resting length and to evaluate whether a proposed muscle routing can meet force and range-of-motion requirements, before committing to a prototype.

At the joint level (Chapter 4), the improved actuator model was combined with a compliance- and wrapping-corrected musculotendon kinematics model and validated on two knee designs. The hybrid torque calculation method isolated the sources of error between prediction and measurement, and the resulting correction terms (a constant length

offset, an equivalent-stiffness representation of series compliance, and a wrapping-loss term) substantially improved torque prediction on the pinned knee and transferred to the biomimetic four-bar knee. The biomimetic knee driven by optimized BPAs was shown to meet or exceed human isometric knee torque over most of the range of motion, confirming that the optimized muscle routing of the author's earlier work is realizable in hardware and identifying the high-flexion regime where additional flexor capacity is needed.

Beyond the specific models, the dissertation's methodological contribution is the actuator-to-joint design chain itself: characterize the actuator, correct for the compliance and geometry of the transmission, validate against human biomechanics, and only then close the loop with a controller. The neural layer that completes the chain is assembled rather than absent: the converted MuJoCo model, the SNS-Toolbox coupling, and the two-layer spinal network with its verified antiphase rhythm generator are already in place. The portability of these neural primitives was verified directly: the contact-driven rhythm core reproduces with exact period agreement and trace correlations above 0.9999 between independent NumPy and Simulink implementations, the archived AnimatLab reference models reproduce their published rhythms headless, and a BPA-actuated walker in MuJoCo air-steps under synthetic neural control, while the honest negatives of the same evaluation series localize the remaining gap in the balance layer, not in the rhythm, the actuators, or the engines.

Completing that layer is the future work scoped in Chapter 7: from balance and bilateral coordination through vestibular, ocular, and cerebellar extensions to embedded hardware controlling a treadmill-walking robot.

The biomimetic platform that results will permit neuromechanical experiments (neural

lesions, sensory deletions, and pathway stimulation) that are impossible in living animals, closing the cycle between better robots and better biological understanding, and opening the way toward dynamic whole-body maneuvers.

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APPENDICES

Appendix A: Muscle Force and Joint Torque Calculations

This appendix documents the reference implementation of the force, torque, and compliance equations used in Chapters 3 and 4. When present in MATLAB code, file names are given so that any number in this dissertation can be traced to the code that produced it.

A.1 Human muscle force calculations

A.1.1 OpenSim Thelen2003 and Millard2012 Formulations

OpenSim's Thelen2003Muscle comprises a contractile element (first-order activation dynamics with distinct rise and fall time constants; active force–length curve; Hill-type force–velocity curve) in series with a compliant tendon, plus a parallel passive fiber element, with fiber pennation. Unlike the MuJoCo actuator, the fiber length is *not* the musculotendon length: at every instant the model solves the force equilibrium $F_{\text{tendon}}(L_{\text{mt}} - L_{\text{fiber}} \cos\alpha) = F_{\text{fiber}}(L_{\text{fiber}}, \dot{L}_{\text{fiber}}, a)$, so tendon compliance filters and delays force transmission and stores energy. The tendon is a quadratic-then-linear "strip" curve anchored at the slack length L_s (force zero until L_{mt} stretches the tendon beyond L_s). Millard2012EquilibriumModel, OpenSim's newer default, retains the same equilibrium structure but replaces the curve shapes (an asymmetric Gaussian active-force–length curve with a plateau, an exponentially stiffening tendon) and adds fiber damping; it is numerically better behaved at extreme lengths. Both OpenSim models therefore differ from the

converted MuJoCo model in the same single respect: series-elastic compliance.

A.1.2 OpenSim Moment Arms and Human Torque Targets

The human torque targets of Section 4.1.5 are OpenSim tendon-excursion moment arms: for each muscle, the moment arm about the knee is the partial derivative of muscle-tendon length with respect to the knee coordinate, $r(\theta_k) = \partial l_{\mathrm{MT}} / \partial \theta_k$, evaluated across the range of motion. The exported analysis tables archived with the test data (`OpenSim_Vasti_MomentArm.txt`, `OpenSim_Vasti_Results.txt`, `OpenSim_Bifem_MomentArm.txt`, `OpenSim_Bifem_Results.txt`) carry the per-muscle moment arms and the corresponding maximum-isometric torque curves (moment arm multiplied by each muscle's maximum isometric force) for vastus intermedius, lateralis, and medialis, and for the biceps femoris short head; the optimization loads the vastus medialis and biceps femoris short-head columns as its targets (`buildKneeExtContext20mm.m` lines 119–140; `Knee_Flexor_Data_20mm.m` line 546). This derivative definition and the geometric perpendicular-foot construction of Equation (A4) below are equivalent for a straight muscle path crossing a planar revolute joint; the derivative form remains correct for the wrapping and moving-path-point geometry of the biological knee, which is why the human targets are computed in OpenSim rather than by the robot-side construction.

A.2 Muscle Model Formulations and Cross-Platform Force Comparison

A.2.1 MuJoCo Muscle Actuator (as converted)

The conversion from OpenSim to MuJoCo re-expresses each Thelen2003 muscle as a MuJoCo `general` actuator with the built-in muscle dynamics. Activation is a first-order process with distinct rise and fall time constants (0.01 s and 0.04 s in the converted model). Force is computed from the actuator's normalized length $\bar{L} = (L - L_{\min})/(L_{\max} - L_{\min})$, where the length range is taken from the converted geometry, as

$$F = F_{\max} \left[a\, f_L(\bar{L})\, f_V(\dot{L}) - f_P(\bar{L}) \right], \tag{A1}$$

with $a \in [0, 1]$ the activation, f_L a force–length curve that peaks inside the normalized range, f_V a force–velocity curve, and f_P a passive element. The peak force scale $F_{\max}$ (actuator parameter `gainprm[2]`) is set by the converter such that the peak force over the model's range of motion matches the OpenSim `max_isometric_force`; audited across all 84 active muscles, the deviation is $\leqslant 15\%$ (Sec. A.2.2). Two properties matter for interpretation: (i) the tendon is *rigid*, so there is no series-elastic element and no tendon slack length, and force at the joint equals the contractile-element force immediately; and (ii) all curve parameters are shared defaults of the converted model rather than per-muscle fits.

A.2.2 Cross-Platform Force Comparison

Both models were driven with identical kinematics and identical activations along the subject01 walking trajectory (the same IK frames used for the static-optimization back-solve).

The OpenSim side is the StaticOptimization solution's own muscle forces (full Thelen dynamics with compliant tendon); the MuJoCo side replays the identical joint trajectory with those activations. Table A1 reports, per muscle, the coefficient of determination R^2 between the two force time series, the shape correlation, and the best-aligning lag [Provenance: `Code/MuJoCo_SNS/spinal/_muscle_force_compare.py`; per-muscle values for all twelve analyzed muscles are archived in `muscle_force_compare.csv` beside Figure A1].

Table A1: MuJoCo vs. OpenSim muscle force along the subject01 walk at matched activations. Lag in analysis frames ($\approx$ 17 ms each).

Muscle	R^2	shape corr^2	lag [frames]
soleus	0.742	0.800	1
med. gastrocnemius	0.888	0.894	2
lat. gastrocnemius	0.884	0.887	2
tibialis anterior	0.945	0.945	0
vastus lateralis	0.941	0.945	1
rectus femoris	-1.15	0.384	0
semimembranosus	0.565	0.621	0
biceps femoris short	-0.26	0.868	0
psoas	0.927	0.934	1
gluteus maximus (2)	0.926	0.943	1
tensor fasciae latae	0.933	0.935	0
peroneus brevis	0.977	0.989	0

For eight of the twelve representative muscles the two formulations agree with $R^2 =$ 0.87–0.98 and mean-force ratios within 10%, with *no* systematic phase shift (best-aligning lag 0–2 frames, $\leqslant$ 34 ms); soleus and semimembranosus agree moderately ($R^2 = 0.74$ and 0.57, respectively). The absence of a phase lead in the rigid-tendon model indicates that, along physiological trajectories, missing tendon slack compliance does not desynchronize force transmission; its signature is instead visible in waveform *shape*: for the soleus (the

largest series-elastic energy store in human gait), the OpenSim compliant tendon carries substantial force through mid-stance while the MuJoCo rigid-tendon force follows the activation dips. This is the expected smoothing of the series-elastic element, visible in the soleus panel of Figure A1. The two remaining outliers are anatomically explained: rectus femoris was rerouted in the conversion over the vastii's patella-tracking via point, and its moment-arm polyfit deviates from OpenSim's path (the same coupler artifact noted for the vastii); biceps femoris short head agrees in shape ($r^2 = 0.87$) but carries a scale offset. We conclude that the converted model reproduces OpenSim muscle force generation along physiological trajectories to a degree sufficient for neural-control studies, and that if tendon-elasticity phenomena (energy storage, force smoothing) become material to a claim, the appropriate upgrade is a compliant-tendon formulation rather than parameter retuning of the rigid-tendon model.

A.3 BPA Force Model Implementation

A.3.1 Goodness-of-Fit Metrics

All fitted quantities in this dissertation are scored by the shared routine `Functions/Go_OfF.m`: the root-mean-square error RMSE $= \sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2}$, the adjusted fraction of variance unexplained FVU $= \frac{n-1}{n}\ \sum_i (y_i - \hat{y}_i)^2 \ /\ \sum_i (y_i - \bar{y})^2$ (so that $R^2 = 1 - \text{FVU}$), and the maximum absolute residual $\max_i |y_i - \hat{y}_i|$ (`Go_OfF.m` lines 45–52). Lower is better for all three, which makes the triplet directly usable inside optimization objectives.

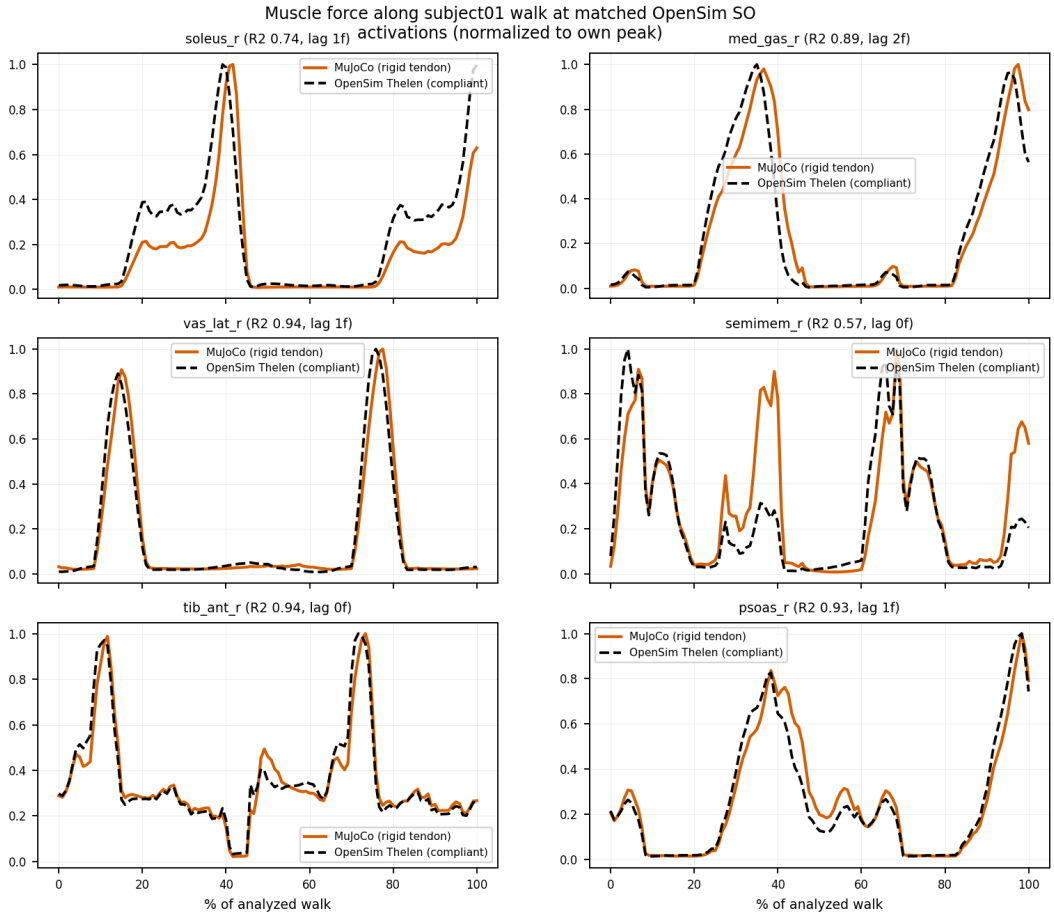


Figure A1: Normalized muscle force along the analyzed walk at matched OpenSim static-optimization activations (MuJoCo: rigid tendon; OpenSim: Thelen compliant tendon). Soleus illustrates the signature of the missing series compliance: OpenSim force is carried through mid-stance by tendon stretch while MuJoCo force follows the activation.

A.3.2 Strain Measures

The musculotendon path length l_{LMT} is the sum of straight segments between route via points, transformed into a common frame at each frame crossing. The contractile strain used by the force model subtracts the artificial tendon and both end-cap fittings from the path length before normalization by the resting length,

$$\varepsilon \;=\; \frac{l_0 - \left(l_{LMT} - l_{ten} - 2\,l_{fitting} - \chi_0\right)}{l_0}, \tag{A2}$$

where χ_0 is the constant length offset of Section 4.1.4 (zero in the uncorrected model). [Provenance: `MonoPamDataExplicit.m` get.Contraction, lines 210–217; `minimizeFlxPin.m` lines 415–426. End-cap fitting length $l_{fitting} = 25.4$ mm is a constructor default, `MonoPamDataExplicit.m` line 69.]

The maximum free-load contraction at 620 kPa is $\varepsilon_{620} = (l_0 - l_{620})/l_0$, where l_{620} is the BPA length at 620 kPa with no external load (code variable `KMAX`; `MonoPamDataExplicit.m` line 278). The relative strain is $\varepsilon^* = \varepsilon/\varepsilon_{620}$ (line 281).

A.3.3 Normalized Force Surface

The normalized isometric force surface of Equation (3.4) is implemented in closed form in `Functions/f_festo.m` (lines 1–26),

$$F^* = c_0\left(e^{-c_1\varepsilon^*} - 1\right) + P^*\,e^{-c_2(\varepsilon^*)^2}, \qquad F^* \leqslant 0 \;\Rightarrow\; 0, \tag{A3}$$

with $P^* = P/620\,\mathrm{kPa}$. The zero-clamp $F^* \leqslant 0 \Rightarrow 0$ is applied by the callers (e.g. `Functions/festo4.m` line 35), not inside `f_festo.m` itself. The coefficients were fitted with `fittype` nonlinear least squares and least-absolute-residual robustness (`Testing_Data/bpaFits.m`, lines 45–60 and 115–140, lower bounds $[0, 0, 0]$) and are reported in Table 3.2. The code additionally holds a 40 mm fit not used in this dissertation ($c_0 = 0.1224$, $c_1 = 10.47$, $c_2 = 2.023$; `f_festo.m` lines 17–20).

At runtime the torque model may alternatively evaluate a spline interpolation of the manufacturer's datasheet curves (`Functions/festo4.m`, backed by `FestoLookup.mat`), with force clamped to zero for $\varepsilon^* > 1$ or $F^* < 0$. Equation (A3) replaced this lookup in the corrected model.

A.3.4 Maximum Force

The resting-length-dependent maximum force of Equation (3.5) is implemented in `Functions/maxBPAforc` (10 mm: line 46; 20 mm: line 57; duplicate for 20 mm in `Functions/Fmax20.m`) with the fixed 7.5 mm end-contact offset. The fitted 20 mm coefficients carry confidence intervals $a_1 \in [1.4745, 1.5011]\,\mathrm{N\,kPa^{-1}}$ and $a_2 \in [0.0238, 0.0258]\,\mathrm{kPa^{-1}\,m^{-1}}$ (`Fmax20.m` lines 26–27). Total force is the product $F = F^*(\varepsilon^*, P^*)\, F_{620}(l_0)$, resolved into a vector along the force direction (`MonoPamDataExplicit.m` lines 283–287).

A.4 Muscle Path Length and Moment Arm

Path length accumulates the Euclidean norm of each segment between via points, applying the body-frame transformation whenever the path crosses between the femur and tibia

frames (`MonoPamDataExplicit.m` lines 78–95, total at lines 122–133).

The vector moment arm is implemented as the perpendicular foot of the joint origin onto the line of action,

$$\mathbf{r} \;=\; \mathbf{p} \;-\; \hat{\mathbf{u}}\,(\hat{\mathbf{u}}\cdot\mathbf{p})\,, \tag{A4}$$

where $\mathbf{p}$ is the vector from the joint origin to the muscle-path crossing point and $\hat{\mathbf{u}}$ is the unit force direction (`MonoPamDataExplicit.m` lines 156–168; identical implementations in `MonoMuscleData.m` lines 111–123, the `Mom` function of `minimizeFlxPin.m` at line 539, and `MonoPam_mult.m` lines 186–201). This is the same quantity as the cross-product projection of Equation (4.2); the planar scalar moment arm about the joint axis is the magnitude of the in-plane components, computed as `hypot(mA(:,1), mA(:,2))` (`predictKneeFlexor20mm.m` line 217). Torque about the joint z-axis is $\mathbf{m} = \mathbf{r}\times\mathbf{f}$, evaluated only where the BPA is in tension (values are discarded for $\varepsilon < -0.02$, the excessive stretching region; `MonoPamDataExplicit.m` line 308).

A.5 Compliance Model Implementation

The compliance-corrected muscle length entering Equation (4.5) is computed as $l_m = l_{LMT} - \chi_0 - l_{ten} - 2\,l_{fitting}$ (`minimizeFlxPin.m` lines 415–426; identically `minimizeExtX3.m` line 345). In the two-bracket fit of record, the insertion-bracket stiffness matrix and the projected scalar stiffness along the force direction are

$$\mathbf{K}_{br} = \mathrm{diag}(\chi_1,\,\chi_2,\,\chi_1), \qquad k_{\hat{u},br} = \left(\hat{u}^{\,T}\mathbf{K}_{br}^{-1}\hat{u}\right)^{-1}, \tag{A5}$$

matching Equation (4.8), and the origin-side second bracket contributes an additional scalar compliance c_{b2} in series (Appendix C, Table C6). As executed by the evaluator of record, both bracket frames are assembled pitch-only, at `minimizeFlxPin.m` line 289 (insertion) and line 306 (origin), with the two-rotation assemblies carried as commented alternatives; the insertion array is $[X_1, X_2, X_1]$ (line 387) and the origin array $[X_1, X_1, X_2]$ (line 400). The two-rotation-capable twin of the same block is `minimizeFlxPin2brk.m` lines 387–418. An earlier arm-1 array $[X_1, X_2, X_2]$ is superseded and survives only in the provenance comment at line 387, and the legacy single-bracket evaluator uses $[X_1, X_2, X_1]$ (`minimizeFlxPinX3.m` line 267).

The artificial tendon is modeled as an axial spring with stiffness

$$k_{ten} \;=\; \frac{n\,A_c\,E}{L_{ten}}, \tag{A6}$$

with $n = 2$ parallel cable strands ($\times 6$ for the 20 mm assembly), $A_c = 1.51 \times 10^{-6}\,\mathrm{m}^2$ (19-strand cable), and $E = 193 \times 10^{9}$ Pa (`MonoPam_mult.m`, function `Spr()`, lines 627–637; the evaluators obtain the spring rate through `Spr(bpa_i)`, `minimizeFlxPin.m` line 156). Series combination with the projected bracket compliance gives $k_{eq} = \left(k_{\hat{u},br}^{-1} + k_{ten}^{-1}\right)^{-1}$ (`minimizeFlxPin.m` lines 394–409).

The scalar compliance displacement δ of Equation (4.22) is found numerically with `fzero` on the bracket $[0,\, l_m - l_{620}]$,

$$F_{620}(l_0)\,F^*(\varepsilon^*(\delta), P^*) - k_{eq}\,\delta = 0, \tag{A7}$$

with $\delta \equiv 0$ when the uncorrected relative strain already exceeds unity (`minimizeFlxPin.m`

lines 434–441). The vector bracket deflection is then $\boldsymbol{\epsilon}_{br} = \mathbf{K}_{br}^{-1}\,(\|\mathbf{f}\|\,\hat{\mathbf{u}})$ (line 461), the cable stretch is $\|\mathbf{f}\|/k_{ten}$ (line 478), and the attachment point, path, moment arm, and torque are recomputed. For the two-BPA flexor grouping, the coupled balance shares one common force between both muscles and is solved analogously on $[0, F_{620}]$ (`MonoPam_mult.m` lines 556–612).

A.6 Wrapping-Loss Implementation

For the pinned-knee extensor, Equation (4.24) is implemented segment-wise over the wrap arc with radii $R_1 = 12\,\text{mm}$ and $R_2 = 40\,\text{mm}$ and segment boundaries $\theta_1 = 27^\circ$ and $\theta_2 = -23^\circ$ (`minimizeExtX3.m`, nested `Contraction`, lines 327–370). For the biomimetic-knee 20 mm cell, the arc-length factor is the routed bend measure $b(\theta_k)$ (arc length swept by the wrap, built with the `wRap` bend radius in `buildKneeFlexorRoute20mm.m` lines 265–268), giving $\Delta l = \chi_3\, b(\theta_k)\,(1-\varepsilon^*)^2$ (`MonoPam_mult.m` lines 398–441), with a hardcoded-geometry fallback for degenerate routes (lines 428–434). The loss enters only the force-producing strain; the kinematic (prediction) strain excludes it (`minimizeExtX3.m` lines 236–240).

Appendix B: Neuron Model Equations

This appendix documents the neuron and synapse models used by the synthetic nervous systems in this project, both in the AnimatLab walker model and in the SNS-Toolbox implementation of the MuJoCo pipeline (Chapter 6). The authoritative model state is the working AnimatLab project file (`Biped_2xCPG_wSubs/Biped_2xCPG_wSubs.aproj`); after the network reorganization described in Section 6.2, a fresh standalone export (`Biped_2xCPG_wSubs_Standalone.asim`) runs headless and carries the rhythm-generator charts, and the earlier phase-1 instrumented export (`walk new new tester added 2 axis_phase1.asim`) records the run reported in Section 6.3.

B.1 Spiking Neuron Model

Each spiking neuron is a conductance-based leaky integrator with threshold and adaptation,

$$C_m \dot{V} \;=\; -\,g_{\text{leak}}\,(V - E_{\text{rest}}) \;+\; I_{\text{syn}} \;+\; I_{\text{app}}, \tag{B1}$$

where V is the membrane potential, C_m the membrane capacitance ($g_{\text{leak}} = C_m/\tau_m$ with membrane time constant τ_m), E_{rest} the resting potential, I_{syn} the summed synaptic current, and I_{app} any applied (tonic stimulation) current. A spike is emitted when V exceeds the firing threshold, after which V resets and an after-hyperpolarization (AHP) of amplitude A_{AHP} with time constant τ_{AHP} is applied; the threshold itself accommodates with the firing history (relative accommodation r_{acc}, time constant τ_{acc}). Optional calcium channels are

defined but disabled in the current model ($G_{\text{max,Ca}} = 0$).

The parameter values shared by the neurons in the walker model are given in Table B1.

Table B1: Spiking neuron parameters in the AnimatLab bipedal walker model (`Biped_2xCPG_wSubs`). All neurons share these values except the motor neuron threshold noted below.

Parameter	**Symbol**	**Value**
Resting potential	E_{rest}	$-60\,\text{mV}$
Membrane time constant	τ_m	$5\,\text{ms}$
Initial firing threshold	V_{th}	$-55\,\text{mV}$ (motor neurons: $-59\,\text{mV}$)
Relative accommodation	r_{acc}	0.3
Accommodation time const	τ_{acc}	$10\,\text{ms}$
AHP amplitude	A_{AHP}	$1\,\text{mV}$
AHP time constant	τ_{AHP}	$3\,\text{ms}$

B.2 Non-Spiking (Graded) Chemical Synapses

Pattern-formation and motoneuron-drive connections in the walker use graded chemical synapses, whose conductance is a clipped linear ramp of the presynaptic potential,

$$g_{\text{syn}}(V_{\text{pre}}) \;=\; g_{\max}\;\text{clamp}\!\left(\frac{V_{\text{pre}} - V_{\text{th}}}{V_{\text{sat}} - V_{\text{th}}},\,0,\,1\right), \qquad I_{\text{syn}} = g_{\text{syn}}\left(V_{\text{post}} - E_{\text{syn}}\right), \tag{B2}$$

where V_{th} and V_{sat} are the synaptic threshold and saturation potentials and E_{syn} the reversal potential. The walker model instantiates three non-spiking synapse types:

- **Nicotinic ACh (excitatory drive):** $E_{\text{syn}} = -10\,\text{mV}$, $g_{\max} = 0.5\,\mu\text{S}$, $V_{\text{th}} \to V_{\text{sat}} = -65\,\text{mV} \to -20\,\text{mV}$.

- **Hyperpolarizing IPSP (inhibition):** $E_{\text{syn}} = -70\,\text{mV}$, $V_{\text{th}} \rightarrow V_{\text{sat}} = -55\,\text{mV} \rightarrow -40\,\text{mV}$.
- **Depolarizing IPSP:** $E_{\text{syn}} = -55\,\text{mV}$, $V_{\text{th}} \rightarrow V_{\text{sat}} = -60\,\text{mV} \rightarrow -40\,\text{mV}$.

Spiking chemical synapses and electrical synapses (rectifying coupling weight 0.1, non-rectifying 0.2) are also defined in the model and are available for the sensory afferent pathways of the MuJoCo/SNS implementation.

B.3 Relation to the SNS-Toolbox Implementation

These are the standard AnimatLab firing-rate/conductance models. SNS-Toolbox [Nourse et al., 2023] implements corresponding classes of non-spiking and spiking neurons with conductance-based synapses, and the AnimatLab parameters above served as the physically motivated initialization for the first SNS-Toolbox instantiation of the spinal network: a 410-neuron network with per-leg rhythm-generator half-centers, phase-shifted pattern-formation groups, motoneuron pools for all 92 muscle-tendon actuators of the converted model, and Ia, Ib, and II afferent populations. In that implementation the non-spiking membrane and graded-synapse equations take the same forms as Equations (B1) and (B2) with the spiking threshold and accommodation terms removed, and the afferent resting-tone offsets are zeroed so that Ia and Ib signals are pure activity measures; the rhythm-generator layer has been verified to oscillate at a human-like period with the two legs in antiphase under tonic drive (Section 7.2.4). An explicit parameter-mapping and validation study between the two implementations remains future work; the tonic stimulation tests of Section 7.2.4 correspond to constant I_{app} injections into Equation (B1) at the motoneuron,

pattern-formation, and rhythm-generator layers respectively.

B.4 Spiking-Mirror Implementation, Calibration, and Verification

This appendix documents the numeric details behind the spiking-neuron mirrors of Section 6.2.8, in the same file-traceable form as Appendix A: the cell and synapse equations as implemented, the calibration factors, the full verification-gate table of the MuJoCo mirror, and the variant-by-variant AnimatLab conversion table. The artifacts live under `Code/MuJoCo_SNS/spinal/reports_spiking_20261002/` (reports, gate scripts, logs, and metric files) and `Code/MuJoCo_SNS/spinal/` (the mirror builder `build_network_spiking.py` and its calibration script `calibrate_spiking.py`); the Simulink blocks are built by `Code/Matlab/SNS_Sims` and tested by `sns_units_test_spiking.m` in the same folder.

B.4.1 Adapting Integrate-and-Fire Cells and Hybrid Synapses

Between spikes, every spiking cell of the MuJoCo mirror obeys the membrane equation

$$C_m \dot{V} \;=\; -\,g_m\,(V - V_{\text{rest}}) \;+\; I_{\text{syn}} \;+\; I_{\text{bias}}, \tag{B3}$$

with a spike emitted when V reaches the firing threshold θ, after which V resets to V_{reset} and θ increases by the increment $\Delta\theta$; between spikes θ decays back toward its initial value with time constant τ_θ, bounded by the adaptation floor. The tonic bias I_{bias} parks the equilibrium a few millivolts below threshold, reproducing the analog cells' near-threshold operating point. Table B2 lists the per-class values as coded (SNS-Toolbox's `SpikingNeuron` class;

the toolbox's membrane update is dimensionally consistent only at $g_m = 1\,\mu\text{S}$, which the mirror therefore uses throughout). The rhythm-generator adaptation time constant reuses the analog network's persistent-sodium inactivation value, so that tuned adaptation values transfer between builds; at the rhythm gate's operating point it is 0.35 s. The mechanism behind that value, recorded here rather than in the chapter body, is as follows. In the analog rhythm cells of the contact-driven cores, burst termination is carried by a persistent sodium current, a slowly inactivating inward current; in every port of this dissertation its inactivation time constant is held fixed rather than voltage-dependent, because the voltage-dependent form of the AnimatLab original quenches burst termination and leaves the network latched in a static state, the failure documented with the archived-model results of Chapter 6.

Table B2: Spiking cell parameters of the MuJoCo mirror (real millivolts; `build_network_spiking.py`, `SPIKE` table, with the rhythm-generator increment superseded by the calibration of record). All cells rest at $-70\,\text{mV}$ with membrane conductance 1 µS and the tonic bias of Table B3.

Cell class	θ **(mV)**	V_{reset} **(mV)**	$\Delta\theta$ **(mV/spike)**	τ_θ **(s)**
Rhythm generator E/F	-50	-60	4.0 (calibrated)	0.35 (analog value)
Pattern-formation	-50	-60	0.3	0.15
Interneurons, commissurals	-50	-60	—	—
Afferent encoders	-52	-58	—	—
Renshaw	-52.5	-58	—	—

Spiking chemical synapses, in both the toolbox class used in the MuJoCo mirror and the Simulink `SpikingSynapse` block, carry a conductance that decays between events and

jumps by a fixed increment per presynaptic spike, saturating at its maximum,

$$\dot{g} \;=\; -\,g/\tau_{\text{syn}} \;\text{ between spikes,} \qquad g \leftarrow \min(g_{\max},\, g + g_{\text{inc}}) \;\text{ per spike,} \tag{B4}$$

with $I_{\text{syn}} = g\,(E_{\text{syn}} - V_{\text{post}})$ as for the graded synapse of Equation (B2). The Simulink `HybridSpikingSynapse` implements the same dynamics with the spike detected from the presynaptic voltage crossing a detection threshold that must sit strictly between the presynaptic resting and firing potentials (a detection threshold equal to the firing potential fires nothing, because the presynaptic voltage resets at threshold). Reversal potentials are 0 mV (excitation, above rest) and -70 mV (inhibition, shunting at rest) in the spiking frame, and $+8/-5$ mV in the motor neurons' 0–5 mV frame. The graded-to-spiking command pathways (descending drive, posture, and bias surrogates) keep the graded synapse of Equation (B2).

B.4.2 Calibration of Record

Table B3 lists every calibration factor produced by `calibrate_spiking.py` (`spiking_calibration.json`), with the measured residual of each. The spike-to-motor-neuron factors are bisected so that a steady 40 Hz presynaptic train reproduces the non-spiking steady-state motor-neuron voltage; the spike-to-spiking factors are rate-calibrated (a 40 Hz source drives a relay cell at 40 Hz, and a calibrated inhibitory factor silences a 40 Hz-driven cell); the readout-tap factor maps a 50 Hz train to the full-scale analog level. The rhythm-generator threshold increment was set by an isolated-pair sweep followed by a full-network sweep over alternation quality and period target.

Table B3: Calibration factors for the spiking mirror (`spiking_calibration.json`). Residuals are the measured deviation at each factor's calibration target.

Factor	Value	Calibrated against	Residual
Graded saturation reference	0.9567	analog network measurement	—
Spike-to-MN, excitatory ($k_{sn,exc}$)	5.4087	MN steady voltage	1.0×10^{-4} n
Spike-to-MN, inhibitory ($k_{sn,inh}$)	1.2284	MN steady voltage	2.6×10^{-4} n
Graded-to-spiking drive (k_{ns})	0.35	relay firing rate	-0.67 Hz
Spike-to-spiking, excitation	1.8754	40 Hz rate transfer	0.0 Hz
Spike-to-spiking, inhibition	0.9996	silences 40 Hz cell	0.0 Hz
Readout tap (g_{inc})	1.7093	50 Hz → full scale	5.00 mV che
RG threshold increment	4.0 mV/spike	pair + full-net sweeps	period targ
Tonic bias current	16 nA	equilibrium -54 mV	—
Synaptic time constants	5 ms (exc), 20 ms (inh)	analog values	—

B.4.3 Verification Gates of the MuJoCo Mirror

Table B4 records the complete gate set of Section 6.2.8 with verdicts and headline numbers. The two failures are reported as failures: the constant-drive period misses its tolerance with the tuning knobs identified, and the air-regime cadence clause misses while every structural clause passes. Gate scripts, logs, and machine-readable outputs are archived in the reports folder (`tools/`, `logs/`, and the `json` metric files respectively).

B.4.4 AnimatLab Conversion Variants

Table B5 catalogues every spiking copy built from the AnimatLab standalone exports. The conversion rules, applied by `tools/make_spiking.py` to copies (no original file touched), are: thresholds of neurons with unreachable (positive) thresholds move into the spiking regime, either -55 mV with -59 mV for motor-neuron-named cells in the v1 copies, or

Table B4: Complete verification-gate table of the spiking mirror. Baseline scores reproduce digit for digit; "split" denotes a gate whose clauses pass and fail in part.

Gate	Verdict	Headline numbers
Pipeline compile	PASS	runner, curriculum, parameters, and network builder all compile.
Defaults gate	PASS	(410, 376, 1186) populations, inputs, and connections after the selector edit; variant build (888, 382, 7430).
Non-spiking bit-identity	PASS	default baseline score -161.56754173676563 identical before and after the edit (all 17 digits); other baselines -144.46440311078322 and -200.37955834081458.
Topology mirror	PASS	directed edge multisets identical in three gain configurations (1,186 / 7,908 / 7,762); inputs and motor-neuron maps identical; readout taps 12 / 12 / 16.
Constant-drive rhythm	SPLIT	sustained bilateral antiphase (E 4.9 Hz, F 3.2 Hz per leg; swing 5.13 mV in every window) = PASS; period 0.621 s versus analog 0.399 s ($+55.6\,\% > 20\,\%$) = FAIL.
Perturbation basin	PASS	12/12 trials under 1 % jitter on every scalar; final swing minimum 4.58 mV (margin $0.89\times$); zero explosions.
Air-regime smoke	FAIL (cadence)	finite, 53 envelope rises, knee range 106.0° of twin's 117.9°, hip 50.4° of 71.1°; period 0.215 s versus 2.108 s.
Untuned ground evaluation	ran	score -236.35; bilateral (8 right / 9 left cycles); cycle knee minimum -79.8°; no non-finite states; stayed up. Single run, no statistics.
Simulink regression	PASS	all committed demonstrations and units tests reproduce exactly with the spiking blocks in the library (two-neuron test 6.07×10^{-4} mV).
AnimatLab baselines	PASS	five headless baseline runs, exit 0, periods and antiphase reproducing the records (0.443 s, 0.462 s, 1.305 s).

per-class calibrated values from the measured graded peaks in the v2 copies (rhythm-generator/pattern-formation/interneuron $-53.5\,\mathrm{mV}$, motor neurons $-44\,\mathrm{mV}$, Renshaw $-44.5\,\mathrm{mV}$); relative accommodation is zeroed; every graded chemical synapse type is re-emitted as a per-spike type with the same identifier and amplitude, 10 ms decay, and maximum conductance $\max(5,\, 2\times\text{amplitude})$; and each connection's own class tag is flipped to the spiking enum. That last rule is a documented format trap: the per-connection `<Type>` tag is stored independently of the synapse type, and converting only the type kills the run at load with no diagnostic detail. Structural validation confirms for every copy that neuron, connection, synapse-type, chart, and stimulus sets are identical to the original and that every class tag matches its synapse type.

Figure B1 shows the ablation result at the joint level for the two-layer family: with graded synapses kept, spiking-regime thresholds leave the joint oscillation essentially intact, while the full conversion pins the joints. This is the visual form of Table 6.6's conclusion that the synapse conversion is the breaking change.

The spiking copies are headless-run artifacts only: none has been opened in the AnimatLab graphical environment, and graphical verification (or an equivalent acceptance ruling) remains outstanding for every `.asim` listed in Table B5. The mirror work also scoped, but did not attempt, two further conversions: the full 410-population Simulink spinal network (whose every one of 1,392 gains is tuned for graded voltages, making a spiking conversion a full re-tune rather than a parameter port) and the spiking mirrors of the alternative MuJoCo network variants (separate architectures whose fresh tuning is future work).

Table B5: AnimatLab spiking-conversion variants of the spiking mirror. "Connections" counts flipped per-connection class tags; the ablation converts thresholds only. All copies validate structurally and run headless to their configured end; rhythm outcomes as in Table 6.6.

Variant	Thresholds	Neurons	Synapse types	Connections	Rhythm outcome
Two-layer v1	−55/−59 mV	80	17	150	dead (2 spikes / 5 s)
Two-layer v2	calibrated	80	17	150	dead
Two-layer v3	v1, 30 ms decay	80	17	150	dead
Bilateral v1	−55/−59 mV	96	20	208	latch, one side 442 Hz–450 Hz
Bilateral v2	calibrated	96	20	208	latch
Project export v1	−55/−59 mV	122	50	243	synchronous 11.8 Hz co-bursti
Project export v2	calibrated	122	50	243	dead (passive settle)
Ablation	spiking, graded synapses	80	0	0	oscillates, near-baseline ranges
2023 reference (baseline)	spiking from birth	56	3 (spiking)	84	1.305 s bursts, antiphase +0.49

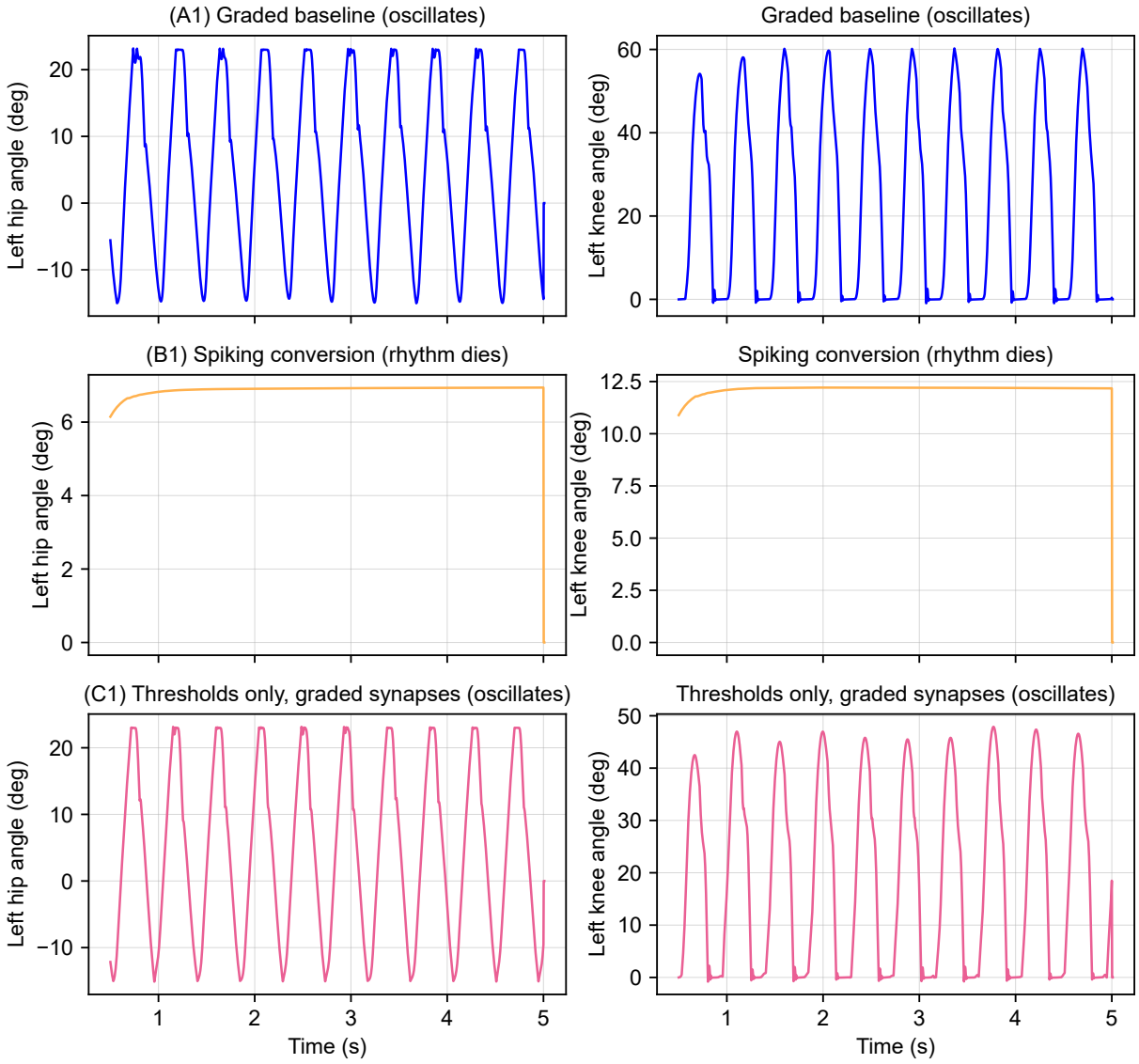


Figure B1: Left-leg joint angles of the AnimatLab two-layer walker family over the 0.5 s–5 s chart window: **(A, B)** graded baseline; **(C, D)** full spiking conversion (rhythm dies, joints pin); **(E, F)** thresholds-only ablation with graded synapses kept (oscillation preserved). Left column: hip angle. Right column: knee angle.

B.5 Connectomes of the Spinal-Network Models

The connectome of the production gait2392 spinal network (the 410-neuron SNS-Toolbox instantiation of Section 7.2.4) is documented in Figures B2 and B3. Both schematics are generated by `Code/MuJoCo_SNS/spinal/draw_circuit.py` from the compiled network at the fitted walking configuration (`fitted_walk_params.json`, the IK/NNLS back-solve refit, in which the pattern-formation and posture weights are refit against muscle activations back-solved from recorded walking kinematics, Section 6.2.9; 410 neurons, 201 per leg plus 8 shared, and 376 inputs; the eight shared cells are the single `DRIVE` and `POSTURE` descending neurons and the six balance cells `BAL_PF`, `BAL_DF`, `BAL_TRK_EXT`, `BAL_TRK_FLX`, `BAL_LAT_R`, and `BAL_LAT_L`), with every motoneuron-pool membership pulled live from `muscle_map.py` rather than hardcoded, so the drawings cannot drift from the built network. The caption of Figure B2 and its alt-text sidecar repeat these population counts, so prose, caption, and alt text cannot drift apart either. Synapse markers follow the project convention throughout: open circles are neurons, white triangles mark excitatory synapses, and solid black dots mark inhibitory synapses.

Figure B2 draws both legs in full. A vertical dashed midline separates the sides and a pale central corridor gathers the interleg commissural inhibition. Each side is organized in labeled layer bands: descending DRIVE (MLR surrogate), POSTURE, and balance inputs; the rhythm-generator half-centers with their mutual inhibition and adaptation cells; the phase-window pattern-formation cells; the Ia, II, and Ib afferent populations with the stance-gated Ib load-sharing interneurons; and the ten motoneuron pools per side (HIP-E 7, HIP-F 4, KNEE-E 4, KNEE-F 4, ANK-PF 9, ANK-DF 3, HIP-AB 6, HIP-AD 6, TRK-E

1, and TRK-F 2, that is 46 motoneurons per leg) driving their muscles. Figure B3 shows the same network for one leg as a DRIVE/POSTURE to rhythm-generator to pattern-formation to motoneuron hierarchy, with the solved standing bias injected into every pool, the sensory column of Ia, II, and Ib afferents, and a boxed key listing every pool abbreviation with its member muscles. The spiking mirror of Section B.4 replicates this topology exactly (Table B4, topology-mirror gate), so these two figures also serve as the connectivity reference for that model.

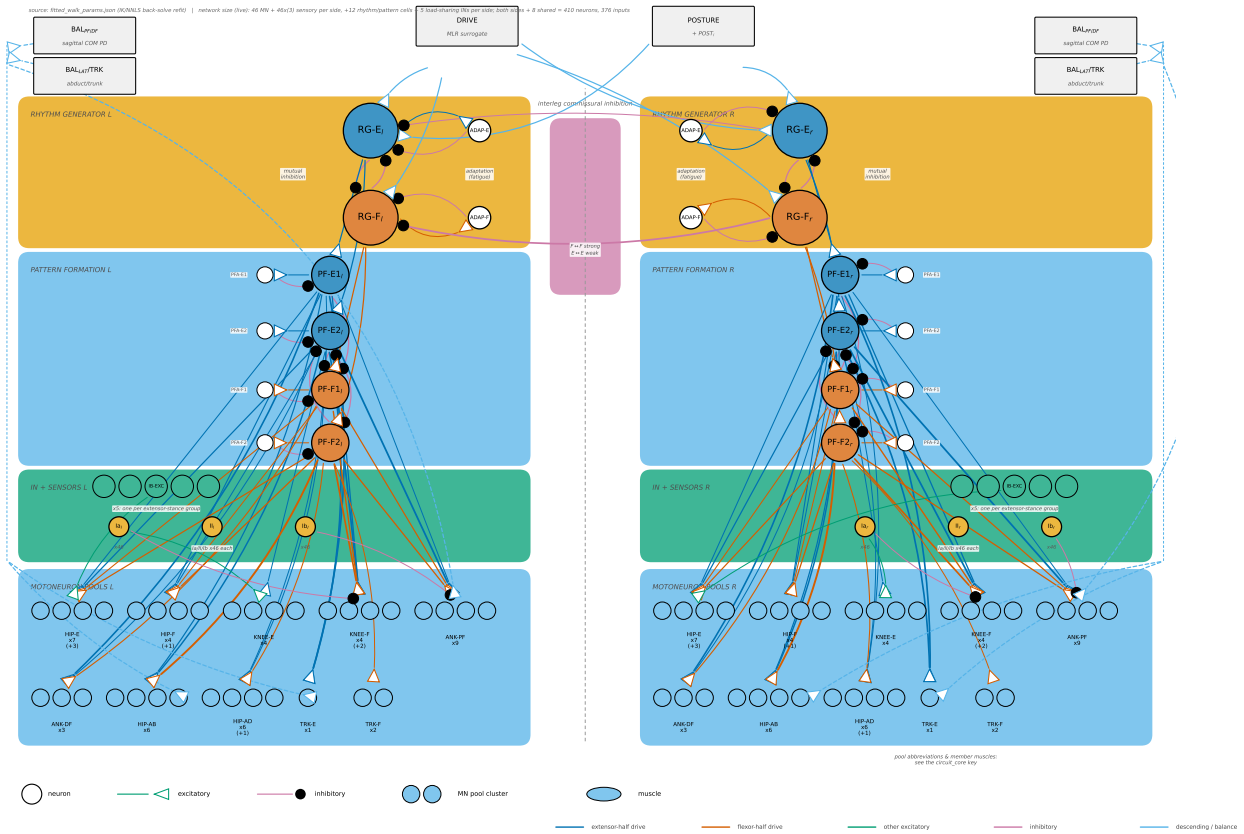


Figure B2: Bilateral connectome of the gait2392 spinal network rendered from the compiled network at the fitted walking configuration (410 neurons, 201 per side plus 8 shared; 376 inputs). Each side is organized in labeled layer bands: descending DRIVE (MLR surrogate), POSTURE, and balance inputs; rhythm-generator half-centers with mutual inhibition and adaptation cells; phase-window pattern-formation cells; Ia, II, and Ib afferents with the stance-gated Ib load-sharing interneurons; and the ten motoneuron pools per side driving their muscles. The central corridor gathers the interleg commissural inhibition across the dashed midline. Open circles are neurons, white triangles mark excitatory synapses, and solid black dots mark inhibitory synapses; the gray note names the source files.

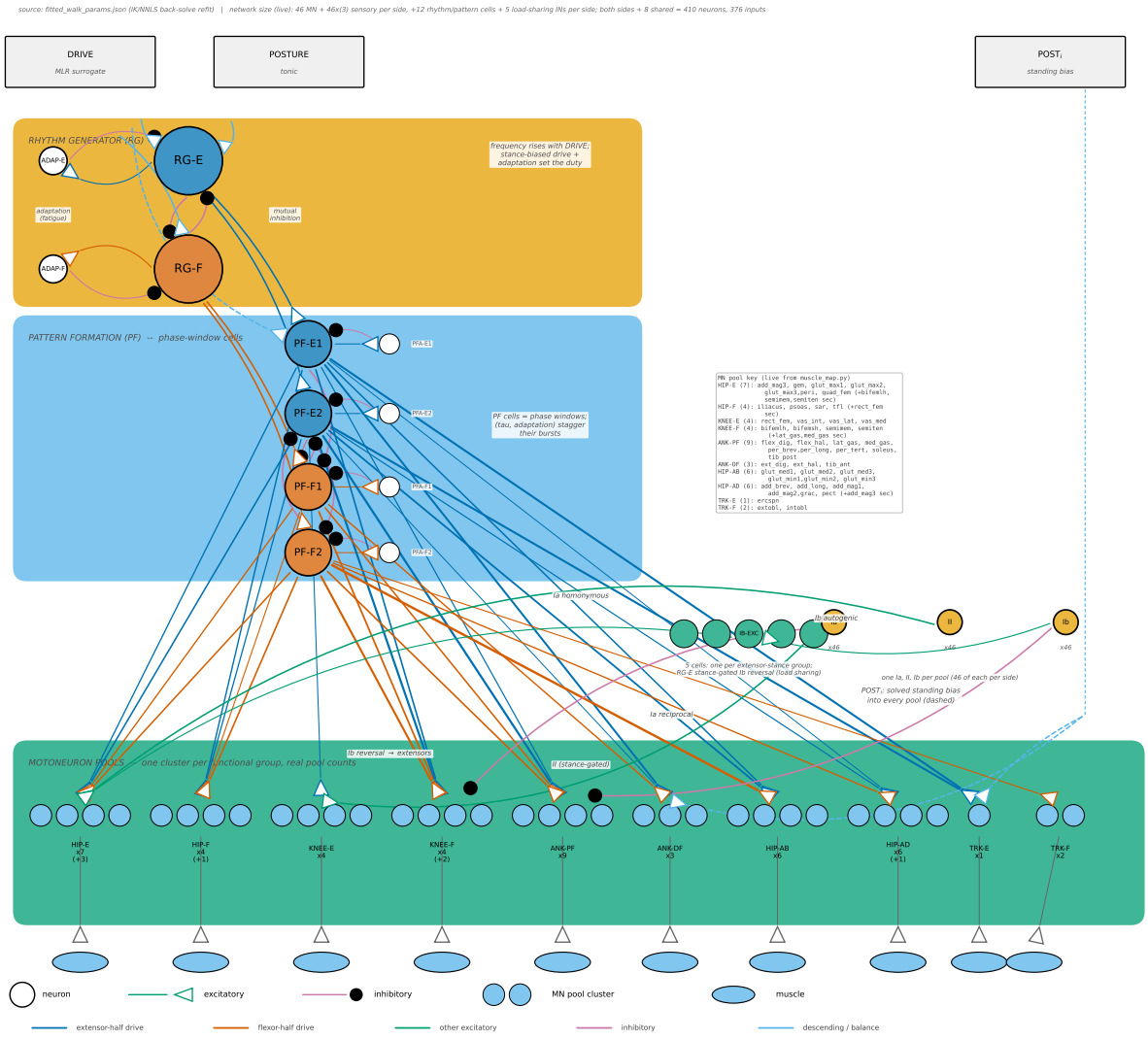


Figure B3: One-side hierarchy of the same gait2392 spinal network: descending DRIVE (MLR surrogate) and POSTURE inputs, the solved standing bias injected as dashed lines into every motoneuron pool, the persistent-sodium rhythm-generator half-centers, the phase-window pattern-formation cells, the Ia, II, and Ib afferent column, and the motoneuron pools (46 per leg) projecting to their muscles. A boxed key lists every pool abbreviation with its member muscles.

Appendix C: Model Geometry, Bracket Frames, and Compliance Parameters

This appendix documents the numeric geometry and compliance parameters behind the force and torque models of Chapters 3 and 4 as they exist in the project's MATLAB code, with file names given so that any value can be traced to the code that produced it. Route and bracket coordinates are coded in meters and reported here in millimeters; stiffnesses are coded and reported in newtons per meter. The bracket reference points are interpretation choices for the 3D-printed (Onyx FDM) attachment hardware, set from CAD by the author; moving a reference point changes the identified stiffness values, so the points are recorded here alongside the values they produced.

C.1 Static BPA Pathway Data

The static BPA model routes each actuator as a straight-segment polyline through via points stored in a per-test `Location` array of dimensions $M \times 3 \times N$, where N is the number of joint frames in the motion sweep and M the number of via points; the first row in the distal link frame is marked by a cross-point index. There are no binormal or surface-frame quantities: at each frame the force direction is the normalized vector from the cross-point row to the previous row transformed into the crossing frame,

$$\hat{\mathbf{u}} \propto \mathbf{R}^{-1}\,\mathbf{p}_{C-1} \;-\; \mathbf{p}_{C}, \tag{C1}$$

where $\mathbf{p}_C$ is the cross-point row, $\mathbf{p}_{C-1}$ the previous row, and $\mathbf{R}$ the frame-crossing rotation [Provenance: `minimizeFlxPin.m` lines 160 and 524–537 (`UD`) and 201–235 (`computeForceVector`); `minimizeExtX3.m` lines 687–700; BPA struct fields, `minimizeFlxPin.m` lines 28–35]. The pinned-knee test routes are archived as `FlxPinBPASet.mat` (5 flexor tests, $M = 2$, knee angle -120° to $+10^\circ$) and `ExtPinBPASet.mat` (9 extensor tests, $M = 7$, knee angle -125° to $+35^\circ$).

Table C1 lists the five pinned-knee flexor tests; their routes are two points, femur-frame origin $\mathbf{p}_1 = [-75.0,\ 100.0,\ 32.8]$ mm and tibia-frame insertion $\mathbf{p}_2 = [-50.1,\ -45.0,\ 32.3]$ mm (test 1; `FlxPinBPASet.mat`, kf(1), frame 1). Table C3 lists the extensor pinned-knee route of test 1 at $\theta_k = -125^\circ$; rows 1–6 are femur-frame and row 7 is tibia-frame, and as the knee extends, rows 2–6 release onto $\mathbf{p}_1$ (wrap released at $+35^\circ$). Table C4 lists the per-test constants of the nine extensor tests; the correction terms of Section 4.1.4 used tests {1, 2, 5, 6, 7, 8}, of which the adopted fold trained on {5, 6, 7} and held out {1, 2, 8} for validation; tests 3, 4, and 9 were not used in the evaluation [Provenance: fold columns of `minimizeExt10mmX3_results.mat`; `Dig_crossPredict.m` line 46].

The biomimetic flexor route (`buildKneeFlexorContext20mm.m`) is a three-point path (femur-frame origin $\mathbf{p}_1$, a moving wrap point $\mathbf{p}_{\text{wrap}}$ on the tibia, and tibia-frame insertion $\mathbf{p}_{\text{end}}$), with the wrap point held on a standoff surface of radius `bpaRb` about the joint and its z placed on the projected $\mathbf{p}_1$–$\mathbf{p}_{\text{end}}$ line; the optimizer seed is $\mathbf{p}_{1,0} = [-50.0,\ 35.0,\ 50.0]$ mm and $\mathbf{p}_{2,0} = [-12.24,\ -8.87,\ 27.87]$ mm with $l_0 = 415$ mm and 25 mm tendon [Provenance: `buildKneeFlexorContext20mm.m` lines 209–218; `buildKneeFlexorRoute20mm.m` lines 98–186]. The biomimetic extensor seed route is given in Table C2, including the release schedule of the route-elimination rule.

Table C1: Pinned-knee flexor test BPA constants, read directly from `FlxPinBPASet.mat` (kf(1)–kf(5)). Labels are the laboratory shorthand used in the fitting logs (`minimizeFlxPin10mm_2brk.m` line 51); l_0, tendon, and pressure are the values used by the model. All tests use ∅10 mm BPAs; the 47 cm test's measured angles carry the $+5.3^\circ$ encoder correction at load time (`minimizeFlxPin.m` lines 77–80). The test was not used in the two-bracket identification, and the corrected-model fit reported for it in Table C9 is a post hoc evaluation; of the remaining four, the adopted cross-validation fold trained on kf(1) and kf(5) and held out kf(2) and kf(4) for validation.

Test	**Label**	$\boldsymbol{l_0}$ (m)	Tendon (m)	P (kPa)	F_m (N)	**Use**
kf(1)	48 cm	0.485	0	605.7	443.4	training
kf(2)	46 cm	0.457	0	614.2	441.4	validation
kf(3)	47 cm	0.479	0	623.2	443.0	not used
kf(4)	40 cm-tendon	0.406	0.035	623.4	436.9	validation
kf(5)	41 cm	0.410	0	615.4	437.3	training

C.2 Joint Transformation Matrices

The knee transform is planar: every joint frame is a rotation about z by the knee angle θ_k composed with a translation fitted to robot CAD tables of the migrating instantaneous center of rotation. With $\mathbf{R} = \mathbf{R}_z(\theta_k)$, the robot (femur→ICR) transform is $\mathbf{T}_{\text{Pam}} = [\mathbf{R},\ [f_3(\theta_k),\ f_4(\theta_k),\ 0]^\top;\ \mathbf{0}^\top\ 1]$ with f_3, f_4 cubic-spline fits (ICR x: 23.30 to 10.47 mm; y: -416.65 to -395.66 mm over $\theta_k = 0.17$ to -2.62 rad), and the ICR→tibia transform is the pure translation $[f_{13}(\theta_k),\ f_{14}(\theta_k),\ 0]^\top$ (x: 29.66 to -7.58 mm; y: 25.97 to 22.33 mm), composed as $\mathbf{T}_{t1\to f} = \mathbf{T}_{t1\to\text{ICR}}\ \mathbf{T}_{\text{Pam}}^{-1}$ over 100 frames spanning $\theta_k \in [-120^\circ, +10^\circ]$ [Provenance: `buildKneeFlexorContext20mm.m` lines 34–80; `buildKneeExtContext20mm.m` lines 30–78; `Functions/ModernRobotics/RpToTrans.m`]. In the archived test sets the per-frame

Table C2: Biomimetic knee, extensor 20 mm BPA route seed (femur frame: p1–p5; tibia frame: p6–p9; `buildKneeExtContext20mm.m` lines 431–440, the author's fcec-ray construction; the prior SolidWorks seed is kept commented at lines 421–430). Release schedule of the verified elimination, read from the redesign of record (`Vas_Pam_20mm_Result_20260925.mat`, `routeInfo.eliminatedAngleD`): p5 at -89.8°, p4 at -55.7°, p6 at -47.8°, p7 at -20.2°, p3 at $+6.1^\circ$, p8 at $+7.4^\circ$.

Point	x (mm)	y (mm)	z (mm)	**Frame**	**Role**
p1	40.00	35.00	0	femur	BPA origin (attachment)
p2	83.90	−274.76	0	femur	BPA contacts mounting base
p3	64.06	−426.80	0	femur	femoral contact
p4	52.83	−441.88	0	femur	femoral condyle contact
p5	37.77	−452.14	0	femur	femoral condyle contact
p6	69.85	34.33	0	tibia	tibial contact (upper)
p7	73.50	19.13	0	tibia	tibial contact (mid)
p8	71.98	−12.23	0	tibia	tibial contact (lower)
p9	34.06	−66.87	0	tibia	patellar ligament ring (distal, pEnd)

Table C3: Pinned-knee extensor BPA route of test 1 at $\theta_k = -125^\circ$ (`ExtPinBPASet.mat`, ke(1), frame 1; mm).

Point	x (mm)	y (mm)	z (mm)
p1	30.00	−50.00	0
p2	65.00	−350.00	0
p3	42.40	−408.50	0
p4	28.20	−421.60	0
p5	10.50	−425.20	0
p6	−8.60	−422.80	0
p7	42.50	−75.90	0

Table C4: Pinned-knee extensor test BPA constants, read directly from `ExtPinBPASet.mat` (ke(1)–ke(9)). Labels are the laboratory shorthand used in the fitting logs (`Dig_resultsTable.m` line 32); l_0, tendon, and pressure are the values used by the model; the shorthand rounds several lengths (e.g. "42 cm" is $l_0 = 0.415$ m and "43 cm-tendon" is $l_0 = 0.432$ m). All tests use ∅10 mm BPAs. The correction terms of Section 4.1.4 used tests $\{1, 2, 5, 6, 7, 8\}$: the adopted fold trained on $\{5, 6, 7\}$ and held out $\{1, 2, 8\}$ for validation; tests 3, 4, and 9 were not used in the evaluation.

Test	**Label**	$\boldsymbol{l_0}$ (m)	Tendon (m)	P (kPa)	F_m (N)	**Use**
ke(1)	40 cm	0.400	0	624.9	436.3	validation
ke(2)	40 cm-tendon	0.400	0.040	618.4	436.3	validation
ke(3)	42 cm	0.415	0	605.2	437.8	not used
ke(4)	42 cm-tendon	0.415	0.040	605.2	437.8	not used
ke(5)	43 cm	0.436	0	621.8	439.7	training
ke(6)	43 cm-tendon	0.432	0.022	622.9	439.3	training
ke(7)	46 cm	0.457	0	605.9	441.4	training
ke(8)	47 cm	0.465	0	605.9	442.0	validation
ke(9)	48 cm	0.480	0	605.9	443.1	not used

transform maps tibia to femur/ICR and takes the form

$$\mathbf{T}_k(\theta_k) \;=\; \begin{bmatrix} \cos\theta_k & -\sin\theta_k & 0 & d_x(\theta_k) \\ \sin\theta_k & \cos\theta_k & 0 & d_y(\theta_k) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \tag{C2}$$

with rotation block $\mathbf{R}_z(\theta_k)$ and translation components d_x, d_y in meters. In the archived flexor test set the translation is constant across the sweep, $[d_x,\, d_y,\, 0] = [4.5,\, -395.8,\, 0]$ mm, and the rotation sweeps from $\mathbf{R}_z(-120^\circ)$ at full flexion to $\mathbf{R}_z(+10^\circ)$ at full extension (`FlxPinBPASet.mat`, kf(1)).

C.3 Bracket Reference Points and Frames

Table C5 lists the bracket reference points. Each compliant bracket carries an instantaneous body-fixed frame anchored at its reference point, constructed by two rotations (yaw about z, then pitch about the yawed y). With $\mathbf{p}_b$ the vector from the reference point to the bracket-to-BPA attachment target, the yaw and pitch angles are

$$\theta_{br} = \operatorname{atan2}(p_{by},\, p_{bx}), \qquad \phi_{br} = \operatorname{atan2}(p'_{bz},\, p'_{bx}) \quad \text{with} \quad \mathbf{p}'_b = \mathbf{R}_z(\theta_{br})^\top \mathbf{p}_b, \tag{C3}$$

and the bracket-frame pose is

$$\mathbf{R}_{kbr} = \mathbf{R}_z(\theta_{br})\, \mathbf{R}_y(\phi_{br})^\top, \qquad \mathbf{T}_{kbr} = [\mathbf{R}_{kbr},\; \mathbf{p}_{br};\; \mathbf{0}\;\; 1]\,, \tag{C4}$$

so the frame's x axis aims along the nominal load path. The angles are computed per configuration from the bracket geometry, not fixed [Provenance: `minimizeFlxPin2brk.m` lines 252–298; `minimizeExtX3.m` lines 461–483]. As executed, however, the flexor evaluator of record assembles *both* of its bracket frames from the yaw rotation alone (the pitch-only, one-transform pose $\mathbf{T}_{kbr} = [\mathbf{R}_z(\theta_{br}), \mathbf{p}_{br}; \mathbf{0}\ 1]$): insertion bracket at line 289 and origin bracket at line 305 of `minimizeFlxPin.m`, with the matching pitch-only point reconstructions at lines 318 and 331; the two-rotation assembly of Eq. (C4) is computed but left commented at lines 288, 304, and 330. For the pinned extensor the two-rotation assembly is live (transMode default `2trans`; `minimizeExtX3.m` lines 3–6 and 479–482). Force is transformed into the bracket frame by $\mathbf{RowVecTrans}(\mathbf{T}_{kbr}^{-1}, \mathbf{F}_k)$ and deformed points are mapped back by $\mathbf{RowVecTrans}(\mathbf{T}_{kbr}, \cdot)$. Because the insertion array $[X_1, X_2, X_1]$ is y-symmetric, the assembly choice is numerically immaterial there; for the origin bracket it is not (see Table C6). The pitch-only (yaw-only) variant that the earlier one-transform fit used survives in `minimizeFlxPinX3.m` lines 200–209.

Table C5: Bracket reference points (mm, as coded; divided by 1000 in code). All are author-set CAD interpretation points; frame denotes the frame the vector is expressed in.

Point	x	y	z	**Configuration / meaning**
Pbri	−48.11	−107.81	13.80	pinned-knee flexor, insertion bracket (knee-ICR frame)
Pbr2	−52.61	0	75.06	pinned-knee flexor, origin-side 2nd bracket (hip frame)
Pbr	−3.84	−46.44	62.50	pinned-knee extensor, rib midpoint, medial side (hip frame)
Pbr	8.38	20.75	25.10	biomimetic extensor 10 mm, cantilever centroid
Pbr	9.48	−33.38	33.90	biomimetic flexor 20 mm, bolt-pattern centroid
Pbr	−19.00	22.00	27.60	biomimetic flexor 10 mm, cantilever centroid

C.4 Stiffness Arrays and Compliance

Every configuration models bracket compliance as a diagonal Hooke-law spring in the bracket frame, $\mathbf{K}_{br} = \mathrm{diag}(k_1, k_2, k_3)$ newtons per meter. The scalar compliance along the force path and its series combinations are

$$c_b = \hat{\mathbf{u}}\,\mathbf{K}_{br}^{-1}\hat{\mathbf{u}}^{\top}, \qquad c_{\mathrm{eff}} = c_b + c_{\mathrm{ten}}\ (+\,c_{b2}\ \text{for the second bracket}), \tag{C5}$$

exactly as in Equations (4.8)–(4.18); when an artificial tendon is present its axial spring ($\times 2$ cables, effective steel area 1.51×10^{-6} m^2, $E = 193$ GPa) adds in series, and the force balance $F_{\max} F^*(\varepsilon^*(\delta), P^*) = k_{\mathrm{eff}}\,\delta$ is solved per frame by `fzero` on the contraction interval [Provenance: stiffness arrays: `minimizeFlxPin.m` lines 387 and 400; `minimizeExtX3.m` lines 575–584; `minimizeExt.m` line 352; `minimizeFlx.m` line 314. `fzero` force balance: `minimizeFlxPin.m` line 438; `minimizeExtX3.m` line 611; `minimizeExt.m` line 389; `minimizeFlx.m` line 352]. The per-configuration stiffness arrays and their series combinations are given in Table C6.

The wrapping-loss term X_3 (Equations 4.24–4.26) is evaluated as $X_3\,R\,|\Delta\theta|\,\zeta^2$ per contact arc. On the pinned extensor the two physical contact arcs use $R_1 = 12$ mm (27° to −23°) and $R_2 = 40$ mm, with arc-length normalizers $27.75/(ang_1 - ang_2)$ and $92.46/(ang_2 + 120^\circ)$; on the biomimetic configurations the bend measure is $w_{\mathrm{rap}}|\theta_{\mathrm{turn}}|$ with $w_{\mathrm{rap}} = 25$ mm [Provenance: `minimizeExtX3.m` lines 333–359; `buildKneeFlexorRoute20mm.m` line 268; `MonoPamDataExplicit_balanceX3.m` lines 489–494]. Table C7 records the optimizer parameterization and bounds of the adopted identification: the two-bracket flexor fit solves all three terms, and the extensor refit locks (X_1, X_2) to the adopted flexor-front values (the

Table C6: Bracket stiffness arrays by configuration ($K = [k_1, k_2, k_3]$ in bracket-frame axes, N/m; optimizer solves X_1, X_2 assigned as shown), as executed by the evaluators of record. The flexor evaluator has no frame switch: both brackets use the pitch-only assembly (see the Bracket Reference Points and Frames section above). The pinned-extensor evaluator carries a `transMode` argument (default `2trans`, env `EXTX3_TRANS`); its axis assignment was revised by the author, with the previous assignment kept commented in the file.

Configuration	$\boldsymbol{K}$ array	**Source**
Pinned flexor, single bracket (legacy)	$[X_1,\ X_2,\ X_1]$	`minimizeFlxPinX3.m` line 267
Pinned flexor, insertion bracket (pitch-only frame)	$[X_1,\ X_2,\ X_1]$	`minimizeFlxPin.m` line 387 (frame li
Pinned flexor, origin bracket (pitch-only frame)	$[X_1,\ X_1,\ X_2]$	`minimizeFlxPin.m` line 400 (frame li
Pinned extensor, 2trans (with X_3)	$[X_1,\ X_2,\ X_2]$	`minimizeExtX3.m` line 575
Biomimetic extensor	$[X_1,\ X_2,\ X_1]$	`minimizeExt.m` line 352
Biomimetic flexor	$[X_1,\ X_2,\ X_2]$	`minimizeFlx.m` line 314

extensor front was generated with the pair fixed to the adopted values of Table C8 and only (X_0, X_3) searched), enforcing a single stiffness pair across configurations.

Table C7: Optimizer parameterization and bounds of the adopted identification. Length offsets are carried in centimeters and stiffnesses on a base-10 logarithmic scale; the initial population of the flexor run was drawn from $X_0 \in [5, 15]$ mm, $X_1 \in [5 \times 10^4, 5 \times 10^5]$ N/m, $X_2 \in [7 \times 10^3, 1.5 \times 10^4]$ N/m.

Identification	**Solved terms**	**Bounds**
Two-bracket pinned flexor	X_0, X_1, X_2	$X_0 \in [0,\ 20]$ mm; $X_1 \in [3 \times 10^4,\ 10^6]$, $X_2 \in [5 \times 10^3,\ 2 \times 10^4]$
Pinned extensor refit	X_0, X_3	$X_0 \in [-20,\ +5]$ mm; $X_3 \in [0,\ 1]$

[Provenance: `minimizeFlxPin10mm_2brk.m` lines 82–83 (bounds) and 424–432 (parameterization); `minimizeExt10mmX3.m` lines 23–26 (lock load) and 67–68 (bounds); `Collect_ExtPinX3_sweep.m` lines 59–60.]

C.5 Adopted Correction-Term Values

Table C8 records the correction-term values adopted at the time of writing. The initial single-bracket identification reported with the results figures (Section 4.2) was superseded by the two-stage identification described in Section 4.3.2: a two-bracket flexor identification over the four tests of Table C1 other than the 47 cm test, cross-validated with two tests held out per fold (the adopted fold trained on the 48 cm and 41 cm tests and validated on the 46 cm and 40 cm-tendon tests), followed by an extensor refit that locks (X_1, X_2) to a member of the same flexor Pareto front and solves only the extensor-specific terms (X_0, X_3) on its three-test training set with three tests held out for validation. The pinned-flexor data do not uniquely determine (X_1, X_2) (the cross-validation fronts contain a family of stiffness pairs of comparable fit quality), so the adopted flexor row and the locked extensor refit are two operating points tied to one and the same stiffness pair; Table C8 and the paragraph that follows it state the adopted values and their front-row provenance. The adopted pair was chosen because it is the pair for which both redesign configurations meet the required 5 % torque-overshoot target simultaneously: in the adoption sweep the flexor redesign reached a minimum margin of $+5.00$ % and the locked extensor redesign $+5.043$ % at -20.2° (`Dig_out/frommat_flx_20260920.log` and `Dig_out/Opt_flxr77_20260920.log`); the refined redesigns improve this to $+5.01$ % (flexor) and $+5.02$ % (extensor).

The flexor values are pooled-front row 77 of `minimizeFlxPin10_results.mat` (212 members); the extensor values are row 32 of `minimizeExt10mmX3_results.mat`, a front generated with (χ_1, χ_2) locked to that same row-77 pair (mat fields `LOCKSRC`, `xi1lock`, `xi2lock`, `flxRow`) with only (χ_0, χ_3) searched. The redesign context builders load exactly

Table C8: Adopted correction-term values of record. The flexor fit solves (χ_0, χ_1, χ_2); the extensor refit solves (χ_0, χ_3) with the stiffness pair locked to the flexor values.

	χ_0 (mm)	χ_1 (N/m)	χ_2 (N/m)	χ_3
Flexor	$+3.94$	3.998×10^4	1.473×10^4	
Extensor	-6.38	3.998×10^4	1.473×10^4	0.158

these values: `buildKneeFlexorContext20mm.m` pick 77 at lines 171–182, with χ_3 from the extensor front's row 32 at lines 188–191, and `buildKneeExtContext20mm.m` pick 32 at lines 164–171.

Table C9 reports the per-test goodness of fit at the adopted values. The corrected model improves on the uncorrected rigid model in every extensor fit-set test, and the fraction of variance unexplained stays below unity on every fit-set test except the 43 cm-tendon test (FVU 1.33, against a rigid-baseline FVU of 4.73). The three excluded tests (3, 4, 9) are not part of the fit of record. On the biomimetic extensor test ($l_0 = 51.8$ cm) the same values give RMSE 1.17 N m, FVU 0.67, and maximum residual 2.21 N m, against a rigid baseline of RMSE 3.03 N m, FVU 4.49, and maximum residual 5.59 N m; the shared-pair correction now explains the bulk of that configuration's measured torque variance as well as its level [Provenance: `Dig_out/xi_audit_tmp/reeval_gof_20260926.m` and `reeval_bio_20260926.m`; extensor rigid baseline from the `baselineScores` field of the extensor results mat].

Table C9: Per-test torque-prediction goodness of fit at the adopted correction terms of Table C8: RMSE and FVU of the corrected model, with the uncorrected rigid-model RMSE for the extensor tests. The 47 cm flexor test was not used in the identification; its corrected-model fit improves over the zero-correction prediction but is a post hoc evaluation.

Configuration	**Test**	**RMSE** (N m)	**FVU**	**Rigid RMSE** (N m)
Pinned flexor	48 cm (training)	2.16	0.17	
	46 cm (validation)	1.37	0.05	
	47 cm (not used)	2.29	0.16	
	40 cm-tendon (validation)	1.61	0.04	
	41 cm (training)	1.08	0.03	
Pinned extensor identification	43 cm (training)	0.98	0.16	2.44
	43 cm-tendon (training)	1.52	1.33	2.88
	46 cm (training)	0.49	0.05	2.18
	40 cm (validation)	1.24	0.13	3.09
	40 cm-tendon (validation)	1.35	0.89	3.62
	47 cm (validation)	0.70	0.17	0.95

C.6 Redesigned Extensor Route Geometry

This section documents the direct redesigned extensor route, the first closed redesign of the extensor configuration. It remains the reference geometry for the route-elimination machinery and the seed from which the later work proceeded, but it is no longer the extensor design of record: when hardware testing of the direct route began, the printed and machined parts started breaking under the route's higher actuator forces, so the operating force had to be lowered, and the cam-profile redesign of Section 4.3.2, which converts the direct route's thin torque margin into headroom at reduced force, was developed in response and supersedes it as the current design. The geometry below is recorded as saved by the optimization results file (`Vas_Pam_20mm_Result_20260925.mat`, the run

cited in Table C2). Figure C1 reproduces the optimizer's tiled route-geometry view from that results file: eight pose panels sweep the 100-frame solution, drawing full flexion ($\theta_k = -120.0^\circ$), the solver frame immediately before each of the six contact-elimination events, and full extension ($+10.0^\circ$). The panel angles are the last poses at which the released row is still active; the release angles of Table C2 are the first poses at which each row is off, one pose spacing of 1.31° later. Beyond the release schedule, the results file also carries a top-level `removedText` field reading '`-p8`'. This field is a plotting leftover rather than a route statement: the plotting block of `Opt_run_Ext.m` (lines 730–736) computes `removedText` only as part of each interior pose-panel title, naming the point released in the next frame, and the value left in the results file is the one from the last interior panel, the p8 release-event panel at $+6.06^\circ$ in Figure C1. The route of record does contain p8: like p3, released at $+6.1^\circ$, p8 stays active from full flexion through 97 of the 100 design-grid frames until its release at $+7.4^\circ$ near full extension (Table C2), so the seed table and release schedule document the route as run. Each panel draws the active p1:p9 route in the femur frame (tibia-side rows transformed through the ICR) over the femur-cylinder, femur-wall, corrected-condyle, and tibia clearance envelopes of the route-elimination rule, with the local bend-radius checks at p2 and p8 drawn dashed. The optimized endpoint values of this design are $\mathbf{p}_1 = [40.0,\ 137.0,\ 0]$ mm (femur frame; the square marker of Figure C1) and $\mathbf{p}_{\mathrm{End}} = [34.4,\ -197.1,\ 0]$ mm (tibia frame; the diamond marker), with resting length 827.7 mm and 36.8 mm of tendon.

Figure C2 places the same optimized route on the OpenSim bone-surface point clouds (`Open_Sim_Bone_Geometry`) at exactly zero knee angle: the full skeleton (femur, tibia, pelvis and hip, sacrum, lumbar spine, and foot) is drawn behind the three human vastus reference

paths in light orange, the optimized p1-to-pEnd route is drawn in magenta with its endpoints circled, and the optimized route carries a total BPA-plus-tendon path length of 0.784 m at this pose. Animated sweeps of both redesigned routes over the full flexion range are archived with the optimization results (`Knee_Extensor_20mm.gif`, 2.3 MB, and `Knee_Flexor_20mm.gif`, 4.7 MB, in `Code/Matlab/Mesh_Optimization/Results`).

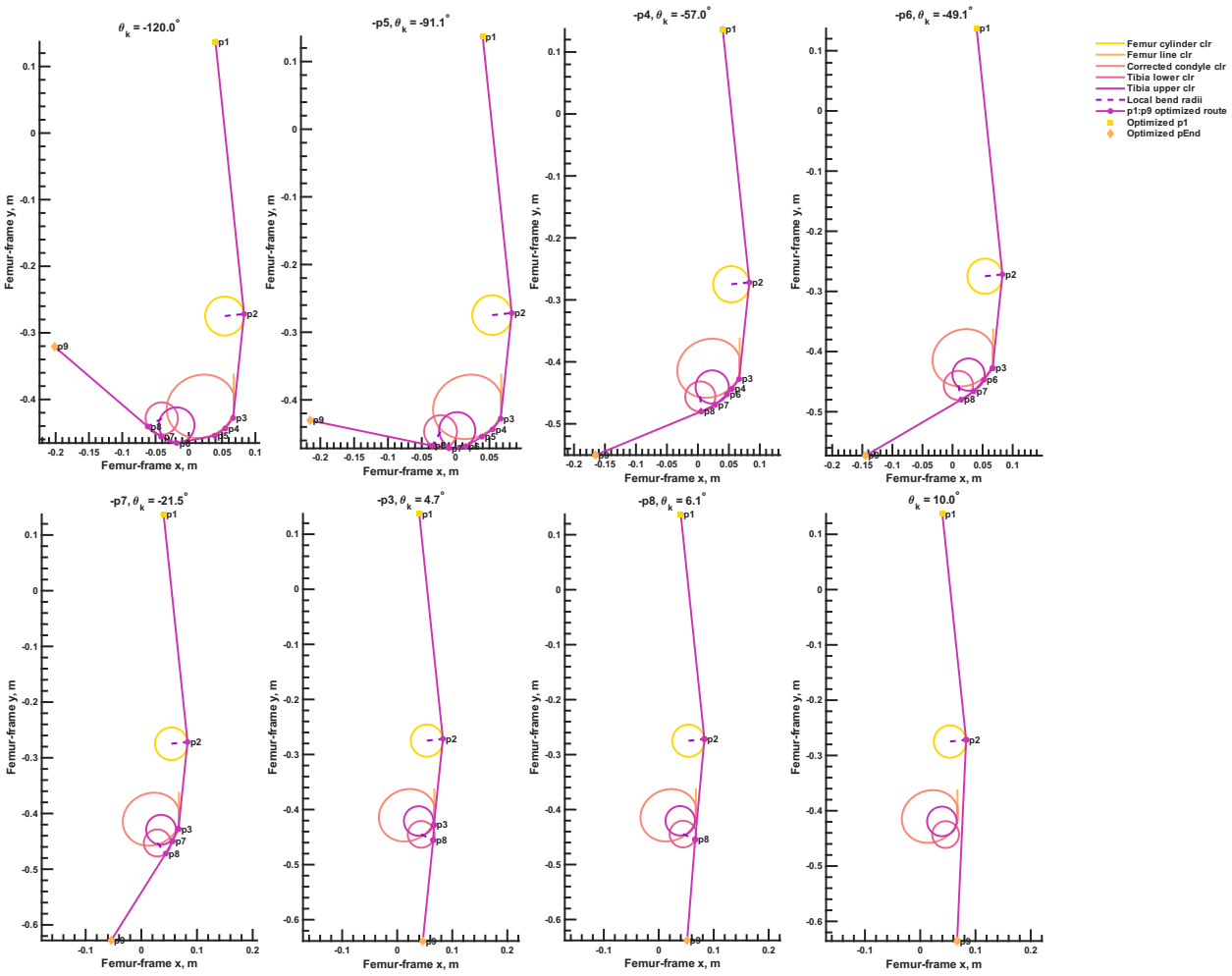


Figure C1: Eight-panel route-geometry view of the optimized 9-point extensor route of record (`Vas_Pam_20mm_Result_20260925.mat`), drawn in the femur frame. Regenerated by `Export_AppendixRouteGeo.m`, which ports the plotting block of `Opt_run_Ext.m` verbatim.

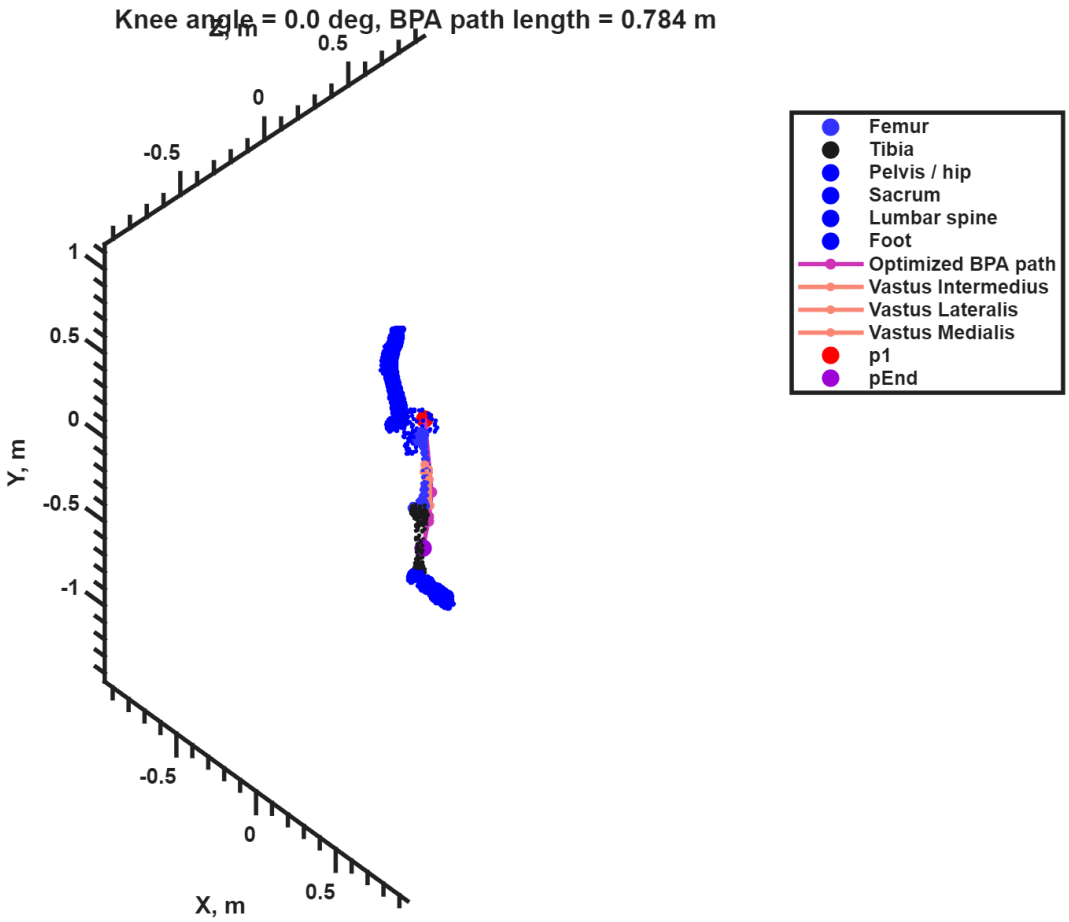


Figure C2: Static bone-mesh view of the optimized 20 mm extensor route at $\theta_k = 0^\circ$, rendered by `Export_AppendixBoneMesh.m`.